ULTIMATE CYBERSECURITY MASTER BLUEPRINT 2026-2027

Start Here: Easy Resource Guide

Exact websites, exact playlist order, lecture SOP and zero-confusion daily workflow

Edition: Start Here
Resource verification date: 2026-10-27
Plan window: 2026-10-27 to 2027-10-26 (shift dates if you start later)
System: 365 days, 12 academic months, 52 weeks + 1 graduation day
Profile: BS CS student in Pakistan; free/high-value resources; safe authorized labs

Academic-style self-study manual. No employment, salary, visa, certification or top-1% guarantee.

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Start Here: exact websites, exact order, zero confusion

Purpose. Ye short guide bats ata hai ke roz kis website par jana hai, kis button/learning path ko follow karna hai, lecture kaise dekhna hai, aur time waste kaise rokna hai. Poori detailed 365-day plan Volume 2-4 mein hai; yahan sirf operating system hai.

> **Roman Urdu rule:** "Video dekh li" is not study. Har lecture ke baad ek command, ek lab result, ek retrieval answer, aur ek commit hona chahiye. Agar 1 hour hai to bhi "Core track" complete karo; baaki optional extension Day 7 ya weekend par.

1. One-minute decision tree

Aaj ki situation	Exact action
Bilkul new din aur naya topic hai	Volume 2-4 mein aaj ka `MxxWxDy` card kholo → Primary resource link par jao → exact section parho/dekho → Core practical task karo → evidence file save karo → retrieval question ka answer likho.
Website samajh nahi aa rahi	Section 3 "Website use karne ka exact tarika" mein uss website ka row parho.
Video dekhne ka mann hai but lab nahi karna	Video ko 20-25 minute ke chunk mein dekho; har chunk ke baad pause karke command/code reproduce karo. Lab skip allowed nahi.
Lab error aa gaya	Error ka exact text copy karo → error log mein likho → primary docs/man page search karo → ek variable change karke retest karo → fix aur root cause dono likho.
Time kam hai	Core track (~1h): 20 min resource, 30 min minimum lab, 10 min retrieval + log. Standard (~2h) adds one variation. Intensive (3-4h) adds independent extension.
Assessment fail hua (<70/100)	Scorecard mein lowest category mark karo → primary resource ka exact section dobara parho → supplementary alternate explanation use karo → same exercise nayi input par dobara karo → naya mini-test do → agle din retest. Advanced topic par tab tak na jao jab tak critical prerequisite weak hai.
Motivation kam hai	Day 7 ka light revision/catch-up karo; streak todne se behtar hai 30-minute honest retrieval. Burnout ko data ki tarah treat karo, guilt ki tarah nahi.

2. Daily study SOP (the 5/25/45/15/10 method)

1. **Preview — 5 min.** Aaj ke card se topic, subtopics, objective aur completion criteria parho. Notebook mein likho: "Mujhe kya prove karna hai?"
2. **Input — 25 min.** Primary lecture ya docs ka assigned section. Video ho to exact playlist/lesson title match karo; timestamps tabhi use karo jo video page par actually dikhte hon. Kisi PDF mein likhe hue invented timestamp par trust na karo.
3. **Hands-on — 45 min (Core: 25-30).** Exact practical task karo. Har command/code apne VM/container/browser lab mein type karo; copy-paste ke baad bhi ek variation khud likho.
4. **Retrieval — 15 min.** Book/notes band karke aaj ka revision question answer karo. Apne words mein 5-8 lines. Phir answer compare karo.
5. **Evidence — 10 min.** Expected output file/screenshot/command log banao, hash ya timestamp add karo, Git commit ya journal entry karo.

Standard track (~2h) adds: one modified exercise, one documented error, and one extra retrieval question. **Intensive track (~3-4h)** adds: an independent extension from a blank file/VM, a mini teach-back paragraph, and one primary-documentation lookup. Har din 4 hours zaroori nahi.

3. Website use karne ka exact tarika

Website / platform	Account? / cost	Kahan jana hai	Exact workflow	Common mistake
Harvard CS50x / CS50 Cyber	Free OCW; edX certificate optional paid	https://cs50.harvard.edu/x/ or https://cs50.harvard.edu/cybersecurity/	Weeks → choose current week → Lecture/Notes/Problem Set → submit where free gradebook available. YouTube playlist sirf lecture dekhne ke liye.	Sirf YouTube dekhna aur problem sets skip karna.
YouTube playlists	Free	Exact playlist links in Volume 5 and the "Playlist control panel"	Playlist open karo → video title aur course objective match karo → 20-25 min chunk → pause → reproduce → notes → lab.	Channel homepage se random video hopping.
Jeremy's IT Lab CCNA	Free YouTube; optional paid course	https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ and course index https://courses.jeremysitlab.com/p/ccna	Course index se exact "Day N" title lo → playlist mein woh lecture dhundo → Packet Tracer lab file download karo → lab complete karo → Anki cards review.	Subnetting sirf dekhna; daily 20 questions na karna.
Professor Messer	Free videos; paid notes optional	Course index https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/	Objective code (e.g., 1.2) se exact video title dhoondo → video dekho → usi objective ka hands-on lab karo. certification version check karo: SY0-701 V7 vs upcoming SY0-801 V8.	Purane SY0-601 playlist ko current samajhna.
Linux Journey	Free	https://linuxjourney.com/	Lesson parho → apne Ubuntu VM mein har command dobara type karo → result aur error command notebook mein likho.	Browser lesson complete karna but VM practice na karna.
OverTheWire Bandit	Free	https://overthewire.org/wargames/bandit/	Official level page parho → SSH se connect karo → level solve karo → **concept** note likho; passwords/flags publicly publish na karo.	Walkthrough copy karna; ye independent problem-solving ko destroy karta hai.
PortSwigger Web Security Academy	Free account	Learning Paths https://portswigger.net/web-security/learning-paths → current path → topic → Apprentice lab	Read topic → launch lab → Burp scope set karo → solve without walkthrough → root cause + remediation note likho.	Lab solve karna but defensive fix na likhna.
OWASP projects	Free	Top 10 https://owasp.org/www-project-top-ten/, ASVS https://owasp.org/www-project-application-security-verification-standard/, Cheat Sheets https://cheatsheetseries.owasp.org/	Standards ko checklist ki tarah use karo; har requirement ko apne local project ke test se map karo.	OWASP page ko step-by-step course samajhna; ye reference hai.
TryHackMe	Freemium	https://tryhackme.com/path/outline/presecurity then Cyber Security 101	Free account → free rooms filter → browser AttackBox ya local VM → notes. Paid subscription tabhi jab aapne free equivalent labs complete kiye ho.	Subscription lena aur phir bhi sirf videos dekhna.
Hack The Box Academy	Freemium	https://academy.hackthebox.com/	Free modules filter → module resources download → isolated VM/HTB environment → notes.	Cubes/subscription ko skill samajhna.
pwn.college /	Free	Dojo index https://pwn.college/dojos or https://www.cylabacademy.org/	Core Material order follow karo → module lecture → browser workspace → challenge → concept journal.	Software Exploitation dojo directly try karna

Website / platform	Account? / cost	Kahan jana hai	Exact workflow	Common mistake
CyLab Academy				before Linux/C/ assembly.
CyberDefenders / BTLO / Malware Traffic Analysis	Freemium/ Free datasets	https://cyberdefenders.org/, https://blueteamlabs.online/, https://malware-traffic-analysis.net/	Dataset download karo → hash karo → disposable VM mein analyze karo → questions ka answer evidence ke saath likho.	Malware samples ko host machine par run karna. Files data hain; execute mat karo.
MITRE ATT&CK / NIST / CISA	Free	https://attack.mitre.org/, https://www.nist.gov/cyberframework, https://www.cisa.gov/known-exploited-vulnerabilities-catalog	Exact technique/control/CVE search karo → apne lab evidence se map karo → limitations likho.	Framework ko law ya certification samajhna.
Microsoft Learn / AWS Skill Builder	Free content; exams paid	https://learn.microsoft.com/en-us/training/, https://skillbuilder.aws/	Learning path → module → knowledge check → optional sandbox → free practice assessment. AWS hands-on se pehle Free Tier terms aur billing alarms set karo.	Credit card add karke paid resources enable karna bina budget alert ke.
Docker / Kubernetes docs	Free	https://docs.docker.com/, https://kubernetes.io/docs/home/	Get Started → exact concept/task → local Docker/kind/minikube → cleanup.	Vulnerable container ko public port par expose karna.
GitHub / Git	Free	https://git-scm.com/book/en/v2, https://docs.github.com/en/code-security	Repo → README → branch → commit → PR → CI → secret scan.	.env, private log, API key ya real IP/ public identifier commit karna.

4. Playlist control panel: channel + exact link + kab use karein

> Exact dates: YouTube par individual video ka upload date dekho. Neeche "course year" ya first-publication year diya gaya hai jahan official page/search result se verify hua. Kisi bhi video timestamp ko PDF mein invent nahi kiya gaya; video ke apne chapters use karo.

Order	Channel / provider	Exact playlist or full-course link	Kab aur kaise	Cost	Age / status warning
1	CS50 (Harvard)	https://www.youtube.com/playlist?list=PLhQjrBD2T380hlTqAU8HfvVepCcjCqTg6	Month 1, Week 0-9 ke concepts; official course page ke saath	Free	2026 edition; current
2	CS50 (Harvard)	https://www.youtube.com/playlist?list=PLhQjrBD2T3839iqxqUdw7NWOCboh00vsV	CS50 Sections; jab lecture ka example samajh na aaye	Free	2026; optional
3	CS50 (Harvard)	https://www.youtube.com/playlist?list=PLhQjrBD2T383Cqo5I1oRrbC1EKRAKGKUE	Month 5 core security ke saath	Free	2023 lectures; old but gold
4	freeCodeCamp.org	https://www.youtube.com/watch?v=eWRfhZUzrAc	Month 4 Python fundamentals; 25-min chunks	Free	2022; concepts still valid
5	Jeremy's IT Lab	https://www.youtube.com/playlist?	Month 2 networking; Day 1 se sequential	Free	

Order	Channel / provider	Exact playlist or full-course link	Kab aur kaise	Cost	Age / status warning
		list=PLxbwE86jKRgMpuZuLBivzIM8s2Dk5lXBQ			Started 2020; v1.1 additions; **old but gold**, current exam blueprint verify karo
6	Professor Messer	https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8	Month 2 targeted Network+ objective review	Free	N10-009; 2024 onward
7	Professor Messer	https://www.youtube.com/playlist?list=PLG49S3nxzAnl4QDVqK-hOnoqcSKEIDDuv	Month 5 core security objective review	Free	SY0-701 2023-24; V8 launch Nov 2026
8	MIT OpenCourseWare	https://www.youtube.com/playlist?list=PLUl4u3cNGP62K2DjQLRxDNRi0z2IRWnNh	Year 2+ systems security research; selected lectures	Free	Fall 2014; old but gold, update tools
9	Stanford Online / Dan Boneh	https://www.coursera.org/learn/crypto	Month 5 optional rigorous crypto; free audit/lectures where available	Free mium	Classic; primitives durable, current protocols verify karo
10	pwn.college	https://www.youtube.com/c/pwncollege	Year 2+ dojo lecture support; use dojo index first	Free	Channel dates vary; current dojo first
11	OpenSecurityTraining2	https://www.youtube.com/watch?v=11KJYHCwcrE	Year 2+ Arch1001 intro; full course on OST2 site	Free	2022 intro; old but gold
12	Microsoft Azure Data Explorer	https://www.youtube.com/playlist?list=PLKHIhZJiQ5AkOMEC54mQQcAkfg4UXT8uK	Month 8 KQL essentials	Free	Dates vary; docs first
13	Microsoft exam prep	https://www.youtube.com/playlist?list=PLahhVEj9XNTcHNtYH-inn7xFJactEM7zz	Month 8/10 SC-200 exam-prep supplement	Free	Objectives refreshed 2026; official study guide first
14	AWS	https://www.youtube.com/playlist?list=PLRFRM3JNuCkvtHYbxjeEA4vx3G4NrqTn3	Year 2+ AWS Security Specialty concept supplement; verify current SCS-C03 guide	Free	Exam content changes; official guide first
15	SANS Digital Forensics	https://www.youtube.com/playlist?list=PLrY_AbzZGqt9MJiLjyH-zTrypGLs154e4	Month 8/Year 2 DFIR artifact webcasts	Free	Mixed dates; webcast ≠ SANS course
16	SANS DFIR	https://www.youtube.com/playlist?list=PL4mjNXFbBRJmippWE8BYMVx0ERb9fdBcP	Network forensics practice with sample captures	Free	Older webcasts; technically useful
17	SANS	https://www.youtube.com/playlist?list=PLfouvuAjspTp23StynIAowjTK2n-5i1QG	Threat-analysis/CTI supplement after Month 8	Free	Dates vary; primary advisories first
18	picoCTF / CyLab	https://www.youtube.com/playlist?list=PLJ_vkrXdcgH9HTNVNIXco0urgrLt349VN	picoCTF 2026 competition/ lecture archive	Free	2026 competition; practice site changed to CyLab Academy
19	KQL community playlist	https://www.youtube.com/playlist?	Extra KQL examples after official docs	Free	Publication dates vary; verify syntax with Microsoft docs

Order	Channel / provider	Exact playlist or full-course link	Kab aur kaise	Cost	Age / status warning
		list=PLWf6TEjiiuIDDXxYJQryn0KxZ5OjdG83n			
20	PortSwigger community demos	https://www.youtube.com/playlist?list=PLDVnRJQ0p1WUP6zgjrlrDq5wixceuE5Do	Optional visual demo only; Academy labs are primary	Free	Unofficial playlist; never replace official labs

5. Time-waste control rules

1. **One primary resource per day.** Supplementary sirf alternate explanation ya gap ke liye.
2. **Do not collect playlists.** Har week ke liye sirf assigned lessons; baaki "backlog" folder mein, not in your study queue.
3. **No walkthrough on first attempt.** 20-30 min struggle karo, phir hint, phir documentation, aakhir mein solution. Solution ke baad problem ko naye input se dobara solve karo.
4. **No tool tourism.** Nmap, Burp, Wireshark, Docker, Sigma aur KQL ko gehra seekho; har naya tool tabhi add karo jab project mein zarurat ho.
5. **Certificate is not curriculum.** Exam voucher tabhi kharido jab official objectives + free practice + portfolio evidence ready ho.
6. **No unauthorized testing.** PortSwigger labs, TryHackMe/HTB, pwn.college, local VMs aur explicitly authorized ranges ke alawa kisi public system ko scan/exploit mat karo.
7. **Weekly stop rule.** Agar 2 hafte se zyada backlog hai to naya module shuru mat karo; pehle Day 7 catch-up protocol use karo.

6. Your first two hours (today)

1. Folder banao: `cybersecurity-blueprint/`, `journal/`, `evidence/`, `projects/`, `scripts/`.
2. VM install karo: VirtualBox + Ubuntu LTS (official links: https://www.virtualbox.org/manual/, https://help.ubuntu.com/).
3. Git repo banao: `git init`, README, `.gitignore`, first commit.
4. Open Volume 2, Day `M01W1D1`, primary resource F01, and complete the Core track.
5. Stop after evidence file + retrieval answer. **Don't watch five extra videos.** Kal ka card kal.



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Volume 1 — Master Roadmap

Assumptions, milestones, dependency map, study system and navigation

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Volume 1 — Master roadmap and learning system

1. Assumptions and scope

This blueprint assumes:

- You are a BS Computer Science student with some HTML/CSS/JavaScript exposure but no professional security experience.
- Python, Bash, PowerShell, Linux, Windows internals, networking, cloud and security tooling need structured development.
- You can study roughly 1, 2, or 3-4 hours per day depending on university workload; the default design target is **2 hours/day**.
- Hardware is modest: an 8 GB RAM laptop can complete most work with browser labs and lightweight VMs; 16 GB+ enables Windows/Kubernetes labs. No paid cloud lab is required.
- You are in Pakistan. Skills are internationally portable, but hiring, work authorization, taxation and exam prices vary by country. No salary or visa promise is made.
- You have intermittent internet. Download only legitimate free course files/docs when allowed; most work should be local and reproducible.
- You will not target systems you do not own or lack written authorization to test.

Calendar-free numbering. The plan uses `M01W1D1 ... M12W4D28`, then `W49D1 ... W52D7`, then `D365`. Optional real dates assume a start of **2026-10-27** and end **2027-10-26**. If you start later, preserve sequence numbers and shift dates.

Duration arithmetic. 12 academic months × 4 weeks × 7 days = 336 days. Final consolidation = 4 weeks × 7 days + 1 graduation day = 29 days. Total = **365 days = 52 weeks + 1 day**. Every academic month has four complete weekly plans; the extra four weeks are a defined examination, portfolio and next-year transition period, not vague "keep practising".

2. Workload tracks

Track	Approx. daily time	Non-negotiable minimum	Best for
Core	~1 h	20 min input, 30 min one practical task, 10 min retrieval/ evidence	Exam weeks, internships, low-energy days
Standard	~2 h	Core + full practical task + one variation + error log + journal	Default target
Intensive	~3-4 h	Standard + independent extension + 150-word teach-back + primary-doc lookup	Holidays, project sprints

Do not do 3-4 hours every day by default. The 30/50/20 allocation is a starting point: **30% theory, 50% hands-on, 20% revision/assessment/documentation**. During project weeks use 20/60/20. During exam weeks use 20/40/40.

Daily pass condition: at least 70/100 on that day's completion criteria, evidence saved, retrieval answer attempted, and no safety violation. Watching a lecture alone = 0 practical evidence.

3. Seven levels of competence

Level	What it means	Evidence	What it does not mean
1. Course completion	You watched/read the material and finished quizzes	Progress page, notes	You can use it under pressure
	You met a vendor's scored objectives	Certificate	

Level	What it means	Evidence	What it does not mean
2. Certification exam pass			You can run investigations or build systems independently
3. Guided-lab competence	You can reproduce known steps in a designed environment	Lab badges, walkthrough-free notes	You can solve unfamiliar ambiguity
4. Independent problem-solving	You can diagnose an unknown fault using docs, hypotheses and tests	Time-boxed practical, error log, new test	You have production judgement
5. Original research/ engineering	You create a new tool, detection, experiment or defensible finding with limitations	Original repo/report, tests, peer critique	You are trusted with real incidents
6. Trusted professional	You repeatedly deliver safe, documented work under real constraints and feedback	Internship/job/OSS maintainer history, incident reports	You are top 1% globally
7. Exceptional specialized expertise	Deep impact in a narrow peer group over years; peers rely on your work	Publications, high-quality OSS, advisories, architecture, reproducible research	A fixed number of years or a certificate

Top 1% is not a 12-month outcome. Twelve months can produce strong foundations, intermediate practical competence and an early specialization portfolio. Years 2-5 build independent engineering/research; beyond year 5, impact and peer recognition depend on problems solved, not time served.

4. Measurable milestones

Milestone	Realistic outcome	Required evidence
Month 1	Set up safe lab; explain hardware/OS/CLI/HTTP/SQL/Git; solve basic CS problems	PRJ-01 home lab, 70/100 assessment, clean VM snapshot, command notebook
Month 3	Configure/troubleshoot Linux and Windows; interpret basic logs; write simple Bash/PowerShell	PRJ-05/06, M3 milestone gate 70+, 6 Bandit levels independently, event log timeline
Month 6	Build and secure a small web/API app; solve Apprentice web labs; write defensive fixes	PRJ-08/09, 20+ tests, authorized local assessment + retest, 70+ gate
Month 9	Review cloud/container/K8s/CI security using local evidence; no surprise cloud bill	PRJ-13/14, hardened Dockerfile/workflow, Kubernetes RBAC/NetworkPolicy tests, 70+ gate
Month 12	Pass integrated practicals; defend capstone; publish a coherent portfolio; choose Year 2 path	PRJ-15, three final practicals, capstone defence 75+, portfolio audit, Year 2 plan
Years 2-3	Deepen one specialization; original projects; read primary sources; reproduce one result; internship/entry-level or OSS feedback	2-3 substantial repos, one technical paper reproduction, one accepted OSS contribution or equivalent supervised work
Years 3-5	Investigate unfamiliar problems; build tools/detections; document original findings; get peer review	Repeatable research method, public technical writing, maintained project, signals of real-world use
Beyond year 5	Exceptional expertise may emerge through sustained impact, architecture, research, mentorship and peer review	Publications/advisories/standards/major OSS/incident leadership; no guarantee from years alone

Competence bands. Foundational = can explain and reproduce with guidance. Intermediate = can build/ troubleshoot familiar systems independently. Advanced specialization = can design experiments and reason about failure modes in one domain. Expert research/engineering = repeatedly creates new, defensible work that changes practice in a defined peer group.

5. Curriculum dependency map (Modules A-K)

Module	Domain	Must come before	Covered mainly in	Advanced/optional extensions
A	Computer Science and IT Foundations	Everything	M1; revisited M4, M6, M11	Nand2Tetris, OSTEP security chapters, MIT 6.858
B	Networking	Security monitoring, AD, cloud, IR	M2; daily subnetting W5-W8	RFC 9293/8446, Zeek/Suricata, advanced packet analysis
C	Linux and Windows	Automation, blue team, AD, cloud	M3	Linux internals, Windows internals, AD security, OST2
D	Programming and Automation	Log parsing, secure coding, detection engineering	M4; used M6-M12	Secure software engineering, tool development, fuzzing
E	Core Security Principles	All security modules	M5	Applied cryptography, Security Engineering, AI risk
F	Web and API Security	Ethical testing, AppSec specialization	M6	GraphQL/request smuggling, advanced auth, LLM appsec
G	Ethical Security Testing	Authorized assessment capstone	M7	Pentest certs, vulnerability research, disclosure
H	Blue Team, SOC and Incident Response	Detection engineering specialization	M8	DFIR, EDR, threat hunting, malware traffic analysis
I	Cloud, Containers and Infrastructure Security	Cloud security specialization	M9	Kubernetes security, IaC security, cloud architecture
J	Governance, Risk and Compliance	Professional reporting, security architecture	M5, M8-M12	ISO lead-auditor concepts, privacy engineering, legal nuance
K	Advanced Specialization Topics	Years 2-5	M10-M12, Volume 7	RE, malware analysis, vuln research, AI appsec, security research

Sequence changes from the requested starting point. The original monthly sequence is preserved, but prerequisites are interleaved: Month 1 includes minimal HTTP/SQL/Git; Month 2 introduces TLS/Wireshark before Month 3 logs; Month 3 includes AD foundations; Month 4 emphasizes secure coding habits before Month 6; Month 5 revisits cryptography and GRC; Month 10 is a measured specialization tasting sprint rather than a vague choice; Weeks 49-52 add an explicit integration period.

6. Twelve-month overview

Academic month	Main outcome	Weeks	Deliverable
M1	Systems, OS, CLI, VM lab	1-4	PRJ-01 home lab baseline
M2	Networking, subnetting, packets, troubleshooting	5-8	PRJ-03 topology + PRJ-04 Wireshark report
M3	Linux/Windows admin, logs, hardening	9-12	PRJ-05 Linux audit + PRJ-06 Windows investigation
M4	Python, Bash, PowerShell, Git, testing	13-16	PRJ-07 log parser / audit utility
M5	Security principles, crypto, IAM, risk, GRC	17-20	PRJ-02 security control baseline
M6	Web/API security and secure coding	21-24	PRJ-08 secure API + PRJ-09 assessment/ remediation
M7	Authorized testing methodology and reporting	25-28	PRJ-10 scoped assessment
M8	SOC, logs, SIEM, IR, detection engineering	29-32	

Academic month	Main outcome	Weeks	Deliverable
			PRJ-11 incident report + PRJ-12 detection rule
M9	Cloud, Docker, Kubernetes, CI/CD, IaC	33-36	PRJ-13 cloud review + PRJ-14 container review
M10	Specialization tasting and focused sprint	37-40	Specialization decision + advanced detection/cloud artifacts
M11	Integrated capstone and independent investigation	41-44	PRJ-15 capstone + report/demo
M12	Practical exams, revision, portfolio, certification decision	45-48	Final practicals, CV, Year 2 plan
Final consolidation	Independent scenarios, research reproduction, teaching, career setup	49-52 + D365	Completion certificate + 90-day sprint

7. Daily, weekly and monthly study system

Daily learning checklist (print this)

- [ ] Read today's card and completion criteria.
- [ ] Open only the assigned primary resource; supplementary only if needed.
- [ ] Take active notes: 3 facts, 2 questions, 1 command.
- [ ] Perform the practical task and save evidence.
- [ ] Deliberately cause or observe one error; record root cause.
- [ ] Answer retrieval question without notes.
- [ ] Commit evidence or write journal entry.
- [ ] Mark Core / Standard / Intensive and actual minutes.
- [ ] Schedule weak cards for days 1, 3, 7, 30, 90.

Weekly review (Day 7)

1. Score six rubric categories: concept, practical, independent, troubleshooting, documentation, safety.
2. Rework every error marked conceptual or safety-related.
3. Pick one old topic and explain it in 10 sentences without notes.
4. Run one modified exercise from earlier in the week.
5. Clean lab snapshots; remove unnecessary files and secrets.
6. Update backlog: keep only three active items; move rest to "later".
7. Choose next week's first repair task based on evidence, not mood.

Monthly assessment

Use the 100-point rubric in Volume 6. Passing threshold = **70/100**, with safety ≥9/10 and practical ≥18/25. If any critical prerequisite is below 60%, pause advancement and run the six-step repair protocol.

Spaced repetition and active recall

Create cards from why/how questions, not acronyms alone. Review on days 1, 3, 7, 30, 90. A card is retired only after two correct recalls in different contexts. Each month's Day 28 is explicitly light to preserve recovery.

Technical journals

Maintain five logs: **command notebook** (command, purpose, example, pitfall); **error log** (symptom, hypothesis, test, fix, lesson); **technical journal** (explanations in your own words); **project log** (decisions and trade-offs); **reading/research log** (claim, evidence, confidence, reproduction idea).

8. Assessment gates and progression rules

Milestone gates occur after M3, M6, M9 and M12. A failed gate triggers: (1) gap identification by rubric category; (2) revisit exact primary lesson/section; (3) use a different medium as alternate explanation; (4) repeat the failed exercise with a new input; (5) complete a novel mini-exercise; (6) retest under a timebox with new questions. Do not proceed to advanced exploitation, DFIR or cloud architecture while networking, OS, programming or ethical prerequisites remain seriously weak.

9. Primary specialization recommendation

Based on the profile and first-year portfolio opportunities, the recommended **primary** is **Detection Engineering & Security Automation** and the **secondary** is **Cloud/Container Security**. Reasons: it uses Python, Linux/Windows logs, networking and cloud telemetry; it has strong free primary sources (Sigma, KQL, ATT&CK, NIST, Microsoft Learn, CISA); it produces credible portfolio artifacts without paid labs; and it is complementary to both SOC and security engineering roles. This is a recommendation, not a promise of employment. Month 10 contains a two-sprint falsification test; if your evidence strongly favours web/API security or DFIR, switch using the decision framework in Volume 7.

10. Navigation and reading order

1. **00 Start Here** — use daily and whenever a website/playlist is confusing.
2. **Volume 1** — master strategy, milestones, trackers.
3. **Volumes 2-4** — execute all 365 daily cards in order.
4. **Volume 5** — use as resource directory and verification register; do not browse randomly.
5. **Volume 6** — set up lab, build projects, score assessments, prepare certifications/career.
6. **Volume 7** — after Month 12, start the multi-year specialization blueprint.
7. **COMPLETE bundle** — all volumes in one file for search/backup; volumes remain easier to print.

11. Glossary (core terms)

Term	Working definition
Active recall	Retrieving an answer from memory before checking sources
Allow-list	Explicitly permitted inputs/actions; everything else denied
Asset	Anything of value that needs protection
Attack surface	All points where an attacker can attempt to influence a system
Authorization (permission)	What an authenticated identity is allowed to do
Availability	Property that authorized users can access systems/data when needed
Blue team	Defensive practitioners who monitor, detect, respond and harden
CIDR	Classless addressing notation using prefix length, e.g., /26
Confidentiality	Preventing unauthorized disclosure
Control	Technical, administrative or physical measure that reduces risk
CVSS	Standardized vulnerability severity scoring; not full risk
Defense in depth	Multiple independent controls so one failure is not fatal
Detection engineering	Designing, testing, deploying and maintaining detections as code
Evidence integrity	Preserving data so it can be trusted and reproduced
Exploit	Technique/code that takes advantage of a vulnerability

Term	Working definition
Framework	Voluntary structured guidance, not automatically law
Hash	Fixed-length digest from data; not encryption and not password KDF
IAM	Identity and access management lifecycle and controls
Incident response	Coordinated process to manage and recover from security events
Integrity	Preventing unauthorized or accidental modification
Least privilege	Granting only the rights needed for a task, only as long as needed
PCI DSS / GDPR / HIPAA	Examples of contractual/regulatory regimes; applicability is jurisdiction-specific
Red team	Authorized adversary simulation
Risk	Effect of uncertainty on objectives; combines context, likelihood and impact
SIEM	Platform that collects, normalizes, correlates and queries security telemetry
Sigma	Vendor-neutral detection rule format
Threat model	Structured reasoning about assets, trust boundaries, threats and mitigations
TLS	Protocol providing authenticated, encrypted transport
Vulnerability	Weakness that can be exploited; not automatically an incident
Zero trust	Design approach that verifies explicitly and assumes no implicit network trust

12. Final completion checklist

- [ ] 365 daily cards have a journal entry and evidence link.
- [ ] 15 projects are published or privately archived with READMEs.
- [ ] Monthly scorecards M1-M12 are complete.
- [ ] M3/M6/M9/M12 milestone gates are passed.
- [ ] Three final practicals and capstone defence are complete.
- [ ] Portfolio audit found no secrets, personal data or unauthorized exploit material.
- [ ] Certification decision cites current official pages and a real reason.
- [ ] Year 2 plan has a narrow specialization, peer group, 90-day sprint and review date.
- [ ] Lab is restored to a clean, isolated state; no billable cloud resources are running.



ULTIMATE CYBERSECURITY MASTER BLUEPRINT 2026-2027

Volume 2 — Daily Curriculum: Months 1-4

112 daily cards with exact resources, labs, evidence, retrieval and completion criteria

Edition: Volume 2
Resource verification date: 2026-10-27
Plan window: 2026-10-27 to 2027-10-26 (shift dates if you start later)
System: 365 days, 12 academic months, 52 weeks + 1 graduation day
Profile: BS CS student in Pakistan; free/high-value resources; safe authorized labs

Academic-style self-study manual. No employment, salary, visa, certification or top-1% guarantee.

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Month 01 — Computer systems, IT foundations, operating systems and command line

Days D001-D028. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Baseline diagnostic and binary number systems D2: Computer organization and the fetch-decode-execute cycle D3: Operating-system architecture and abstraction D4: Virtualization and lab safety D5: Command-line navigation and file operations D6: Users, groups, ownership and permissions D7: Home lab baseline project and evidence workflow
Week 2	D1: Weekly retrieval, error-log cleanup and catch-up D2: Filesystem hierarchy, inodes and links D3: Processes, threads, scheduling and signals D4: Services, units, timers and system logs D5: Client-server architecture and sockets D6: HTTP requests, responses and status codes D7: Databases and SQL fundamentals
Week 3	D1: Git, branching and merge conflicts D2: Weekly review and mini practical examination D3: Memory, stack, heap and addresses D4: Binary logic and CPU instruction foundations D5: Storage, filesystems, journaling and backups D6: HTTP sessions, cookies and authentication basics D7: Troubleshooting methodology and technical journaling
Week 4	D1: Month 1 project completion and portfolio publication D2: Month 1 practical assessment and milestone review D3: APIs, JSON and structured data D4: Identity, sessions and access control foundations D5: Virtual machine networking troubleshooting D6: CS50-style problem solving and decomposition D7: Spaced revision, recovery and Month 2 preview

> **Gate:** score the Month 1 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 01 · Week 1 · daily cards

D001 · M01W1D1 · Baseline diagnostic and binary number systems

Phase: Academic month · **Optional date:** 2026-10-27 · **Domain resources:** F01 / F06

1. Day / sequence	Global Day 1 of 365; calendar-free M01W1D1; optional date 2026-10-27.
2. Exact topic	Baseline diagnostic and binary number systems
3. Exact subtopics	Decimal, binary and hexadecimal; bits and bytes; signed/unsigned integers; character encodings
4. Learning objective	Convert values between bases and explain why computers use binary
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x Week 0, Scratch and binary representation lesson; do not skip the problem set. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run a 20-question self-diagnostic on number systems, programming and networking; create a baseline score file. **2.** Standard (~2 h): Run a 20-question self-diagnostic on number systems, programming and networking; create a baseline score file. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run a 20-question self-diagnostic on number systems, programming and networking; create a baseline score file. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	diagnostic.csv and a one-page learning contract; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is 255 the largest unsigned decimal value in one byte, and where is this relevant to IPv4? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score logged, errors classified, and learning contract committed to the journal Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x Week 0, Scratch and binary representation lesson; do not skip the problem set"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D002 · M01W1D2 · Computer organization and the fetch-decode-execute cycle

Phase: Academic month · **Optional date:** 2026-10-28 · **Domain resources:** F07 / F06

1. Day / sequence	Global Day 2 of 365; calendar-free M01W1D2; optional date 2026-10-28.
2. Exact topic	Computer organization and the fetch-decode-execute cycle
3. Exact subtopics	CPU, control unit, ALU, registers, cache, RAM, storage, buses, clock and instruction cycle
4. Learning objective	Trace the path of one instruction and distinguish volatile from persistent storage
5. Primary lecture / reading + direct URL	Nand2Tetris / The Elements of Computing Systems `https://www.nand2tetris.org/` **Type:** Free course and software · **Access:** Free software and course materials; book optional paid
6. Exact section / lesson locator	Nand2Tetris Part I, Project 1 overview and Project 2 arithmetic introduction. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a labelled block diagram of a simple computer from input to output; annotate each component's security relevance. **2.** Standard (~2 h): Build a labelled block diagram of a simple computer from input to output; annotate each component's security relevance. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a labelled block diagram of a simple computer from input to output; annotate each component's security relevance. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	architecture.svg and a 150-word explanation; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would an attacker gain from direct memory access, and which layer prevents it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram has all required components and the explanation uses correct terminology Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Nand2Tetris Part I, Project 1 overview and Project 2 arithmetic introduction"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Operating Systems: Three Easy Pieces (OSTEP); write a five-line correction; retest next morning with a new 10-minute problem.

D003 · M01W1D3 · Operating-system architecture and abstraction

Phase: Academic month · **Optional date:** 2026-10-29 · **Domain resources:** F06 / F01

1. Day / sequence	Global Day 3 of 365; calendar-free M01W1D3; optional date 2026-10-29.
2. Exact topic	Operating-system architecture and abstraction
3. Exact subtopics	Kernel mode versus user mode, system calls, device drivers, virtual memory, process isolation, multitasking
4. Learning objective	Explain how an OS provides abstractions and enforces isolation
5. Primary lecture / reading + direct URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/` **Type:** Free textbook · **Access:** Free online chapters; paid print optional
6. Exact section / lesson locator	OSTEP Chapter 2 Introduction and Chapter 4 Processes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Observe process creation on Linux or Windows, record PID/PPID and map five OS abstractions in a table. **2.** Standard (~2 h): Observe process creation on Linux or Windows, record PID/PPID and map five OS abstractions in a table. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Observe process creation on Linux or Windows, record PID/PPID and map five OS abstractions in a table. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	process-observation.md with real command output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can one crashed application usually be terminated without rebooting the OS? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Observed evidence matches the explanation and all five abstractions are named correctly Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OSTEP Chapter 2 Introduction and Chapter 4 Processes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x 2026 OpenCourseWare; write a five-line correction; retest next morning with a new 10-minute problem.

D004 · M01W1D4 · Virtualization and lab safety

Phase: Academic month · **Optional date:** 2026-10-30 · **Domain resources:** F12 / F13

1. Day / sequence	Global Day 4 of 365; calendar-free M01W1D4; optional date 2026-10-30.
2. Exact topic	Virtualization and lab safety
3. Exact subtopics	Hypervisors, virtual machines, snapshots, NAT/host-only/internal networking, resource allocation, attack surface
4. Learning objective	Create an isolated Ubuntu VM with a clean snapshot and a documented network mode
5. Primary lecture / reading + direct URL	Oracle VirtualBox User Manual `https://www.virtualbox.org/manual/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	VirtualBox User Manual, Chapters 1-3, 5-6 and 8 on setup, storage, networking and snapshots. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Ubuntu Desktop Guide `https://help.ubuntu.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Install VirtualBox and Ubuntu LTS in a VM; create NAT plus host-only/internal adapters; take a clean snapshot. **2.** Standard (~2 h): Install VirtualBox and Ubuntu LTS in a VM; create NAT plus host-only/internal adapters; take a clean snapshot. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Install VirtualBox and Ubuntu LTS in a VM; create NAT plus host-only/internal adapters; take a clean snapshot. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	VM screenshots, network settings export and lab-setup.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why must vulnerable services not be bridged to a public or shared network? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	VM boots, has internet only through NAT where intended, and a restore point exists Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "VirtualBox User Manual, Chapters 1-3, 5-6 and 8 on setup, storage, networking and snapshots"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Windows Server documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D005 · M01W1D5 · Command-line navigation and file operations

Phase: Academic month · **Optional date:** 2026-10-31 · **Domain resources:** L01 / F08

1. Day / sequence	Global Day 5 of 365; calendar-free M01W1D5; optional date 2026-10-31.
2. Exact topic	Command-line navigation and file operations
3. Exact subtopics	Shell prompt, absolute/relative paths, globbing, ls/cd/pwd/mkdir/touch/cp/mv/rm, wildcards and quoting
4. Learning objective	Navigate and manipulate files safely without a graphical file manager
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, Command Line and Filesystem Hierarchy sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 25 navigation and file operations in the VM using only the terminal; include one deliberate quoting error and its fix. **2.** Standard (~2 h): Complete 25 navigation and file operations in the VM using only the terminal; include one deliberate quoting error and its fix. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 25 navigation and file operations in the VM using only the terminal; include one deliberate quoting error and its fix. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	commands.log and errors.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the difference between rm -r and rm -rf, and why does quoting matter? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All required operations succeed and the error is documented with a safe correction Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, Command Line and Filesystem Hierarchy sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D006 · M01W1D6 · Users, groups, ownership and permissions

Phase: Academic month · **Optional date:** 2026-11-01 · **Domain resources:** L01 / F06

1. Day / sequence	Global Day 6 of 365; calendar-free M01W1D6; optional date 2026-11-01.
2. Exact topic	Users, groups, ownership and permissions
3. Exact subtopics	Discretionary access control, rwx bits, umask, sudo, principle of least privilege, ownership
4. Learning objective	Create users and groups and justify a least-privilege file policy
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, Users and Groups and Permissions sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create two users and one group, set directory permissions for a shared lab folder, and verify access as each user. **2.** Standard (~2 h): Create two users and one group, set directory permissions for a shared lab folder, and verify access as each user. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create two users and one group, set directory permissions for a shared lab folder, and verify access as each user. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	permission-matrix.csv and terminal transcript; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does execute permission on a directory mean traversal rather than program execution? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each test user can perform only the intended operations Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, Users and Groups and Permissions sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D007 · M01W1D7 · Home lab baseline project and evidence workflow

Phase: Academic month · **Optional date:** 2026-11-02 · **Domain resources:** F09 / F12 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 7 of 365; calendar-free M01W1D7; optional date 2026-11-02.
2. Exact topic	Home lab baseline project and evidence workflow
3. Exact subtopics	Asset inventory, baseline configuration, diagram, evidence integrity, README and Git commit
4. Learning objective	Produce the first reproducible home-lab deliverable
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git, Chapters 1-2: getting started and Git basics. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Oracle VirtualBox User Manual `https://www.virtualbox.org/manual/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a repository for the lab, add topology, VM details, setup steps, hashes of exported settings, and a checklist; commit in small units — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a repository for the lab, add topology, VM details, setup steps, hashes of exported settings, and a checklist; commit in small units — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a repository for the lab, add topology, VM details, setup steps, hashes of exported settings, and a checklist; commit in small units — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-01 repository with README, diagram, evidence and clean commit history; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which three facts must another person know to reproduce your VM network? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Repository has no secrets, builds from clean instructions, and contains a tagged v0.1 release Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git, Chapters 1-2: getting started and Git basics"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

Month 01 · Week 2 · daily cards

D008 · M01W2D1 · Weekly retrieval, error-log cleanup and catch-up

Phase: Academic month · **Optional date:** 2026-11-03 · **Domain resources:** F01 / F06

1. Day / sequence	Global Day 8 of 365; calendar-free M01W2D1; optional date 2026-11-03.
2. Exact topic	Weekly retrieval, error-log cleanup and catch-up
3. Exact subtopics	Number systems, OS abstractions, VM isolation, CLI, users and permissions; spaced repetition
4. Learning objective	Repair gaps found by the diagnostic and prepare for Week 2
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x Week 0 notes and Week 1 C introduction; use the official notes page. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Do 15 closed-book retrieval questions, re-run three failed commands, and rewrite two explanations from memory. **2.** Standard (~2 h): Do 15 closed-book retrieval questions, re-run three failed commands, and rewrite two explanations from memory. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Do 15 closed-book retrieval questions, re-run three failed commands, and rewrite two explanations from memory. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	weekly-review-M1W1.md with question scores and corrected answers; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Explain the chain from CPU instruction to OS process to user file access without notes Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70 percent and all previous unsafe lab mistakes have corrections Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x Week 0 notes and Week 1 C introduction; use the official notes page"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D009 · M01W2D2 · Filesystem hierarchy, inodes and links

Phase: Academic month · **Optional date:** 2026-11-04 · **Domain resources:** L01 / L04

1. Day / sequence	Global Day 9 of 365; calendar-free M01W2D2; optional date 2026-11-04.
2. Exact topic	Filesystem hierarchy, inodes and links
3. Exact subtopics	Root filesystem, mount points, inodes, hard links, symbolic links, file metadata, ext4 basics
4. Learning objective	Explain how paths resolve to storage and why files persist after a name is removed
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, Filesystem Hierarchy and Filesystem sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux man-pages project `https://man7.org/linux/man-pages/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Use stat, df, mount, ln and ln -s on benign files; delete a hard link and observe link counts. **2.** Standard (~2 h): Use stat, df, mount, ln and ln -s on benign files; delete a hard link and observe link counts. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Use stat, df, mount, ln and ln -s on benign files; delete a hard link and observe link counts. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	filesystem-notes.md with command output and diagrams; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a symbolic link break while a hard link count changes? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All observations match the predicted link and inode behaviour Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, Filesystem Hierarchy and Filesystem sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D010 · M01W2D3 · Processes, threads, scheduling and signals

Phase: Academic month · **Optional date:** 2026-11-05 · **Domain resources:** F06 / L01

1. Day / sequence	Global Day 10 of 365; calendar-free M01W2D3; optional date 2026-11-05.
2. Exact topic	Processes, threads, scheduling and signals
3. Exact subtopics	Process lifecycle, PIDs, parent/child relationships, threads, context switching, SIGTERM versus SIGKILL
4. Learning objective	Observe process state and explain how the CPU shares time
5. Primary lecture / reading + direct URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/` **Type:** Free textbook · **Access:** Free online chapters; paid print optional
6. Exact section / lesson locator	OSTEP Chapters 4-7: Processes, Process API, Direct Execution and CPU Scheduling. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Start a benign background process, inspect it with ps/top/pstree, send TERM and then repeat with KILL; record state changes. **2.** Standard (~2 h): Start a benign background process, inspect it with ps/top/pstree, send TERM and then repeat with KILL; record state changes. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Start a benign background process, inspect it with ps/top/pstree, send TERM and then repeat with KILL; record state changes. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	process-lab.md with timestamps and signal results; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is SIGKILL a last resort and what evidence might be lost? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Process tree, state transitions and signal results are correctly explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OSTEP Chapters 4-7: Processes, Process API, Direct Execution and CPU Scheduling"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x 2026 OpenCourseWare; write a five-line correction; retest next morning with a new 10-minute problem.

D011 · M01W2D4 · Services, units, timers and system logs

Phase: Academic month · **Optional date:** 2026-11-06 · **Domain resources:** L05 / L01

1. Day / sequence	Global Day 11 of 365; calendar-free M01W2D4; optional date 2026-11-06.
2. Exact topic	Services, units, timers and system logs
3. Exact subtopics	systemd units, systemctl, journalctl, dependencies, restart policy and log levels
4. Learning objective	Create, inspect and remove a benign service and timer safely
5. Primary lecture / reading + direct URL	systemd.unit manual `https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	systemd.unit manual, unit structure and dependency sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a harmless timer that writes the time to a local file, inspect journal logs, then disable and remove the unit. **2.** Standard (~2 h): Create a harmless timer that writes the time to a local file, inspect journal logs, then disable and remove the unit. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a harmless timer that writes the time to a local file, inspect journal logs, then disable and remove the unit. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	unit-file.txt, journal excerpts and cleanup proof; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the difference between a service failing and a service being inactive? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Service ran, was observed, and no residual unit remains Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "systemd.unit manual, unit structure and dependency sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Linux man-pages project; write a five-line correction; retest next morning with a new 10-minute problem.

D012 · M01W2D5 · Client-server architecture and sockets

Phase: Academic month · **Optional date:** 2026-11-07 · **Domain resources:** F10 / N14

1. Day / sequence	Global Day 12 of 365; calendar-free M01W2D5; optional date 2026-11-07.
2. Exact topic	Client-server architecture and sockets
3. Exact subtopics	Clients, servers, listeners, localhost, loopback, IP address, port, socket, bind/listen/accept/connect
4. Learning objective	Map a local client-server interaction to ports and sockets
5. Primary lecture / reading + direct URL	MDN HTTP Overview `https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview` **Type:** Official web documentation · **Access:** Free
6. Exact section / lesson locator	MDN HTTP Overview, HTTP basics and client-server communication sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	IANA Service Name and Transport Protocol Port Number Registry `https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Start a local Python HTTP server on loopback, connect with curl/browser, and identify local and remote addresses and ports. **2.** Standard (~2 h): Start a local Python HTTP server on loopback, connect with curl/browser, and identify local and remote addresses and ports. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Start a local Python HTTP server on loopback, connect with curl/browser, and identify local and remote addresses and ports. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	client-server.md and ss/netstat output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can two different services use port 80 on different hosts but not the same IP simultaneously? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Loopback service worked and all socket fields are labelled correctly Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MDN HTTP Overview, HTTP basics and client-server communication sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MDN HTTP documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D013 · M01W2D6 · HTTP requests, responses and status codes

Phase: Academic month · **Optional date:** 2026-11-08 · **Domain resources:** F10 / W04

1. Day / sequence	Global Day 13 of 365; calendar-free M01W2D6; optional date 2026-11-08.
2. Exact topic	HTTP requests, responses and status codes
3. Exact subtopics	Methods, URL, request headers, response headers, status code families, content types, redirects
4. Learning objective	Read a raw HTTP exchange and explain each field's purpose
5. Primary lecture / reading + direct URL	MDN HTTP Overview `https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview` **Type:** Official web documentation · **Access:** Free
6. Exact section / lesson locator	MDN HTTP Overview and HTTP methods/status sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MDN HTTP documentation `https://developer.mozilla.org/en-US/docs/Web/HTTP`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Use curl -v or browser DevTools against a local test server; save one request/response pair and annotate headers. **2.** Standard (~2 h): Use curl -v or browser DevTools against a local test server; save one request/response pair and annotate headers. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Use curl -v or browser DevTools against a local test server; save one request/response pair and annotate headers. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	http-annotated.txt; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the difference between 301, 302, 401 and 403 in practical terms? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All headers and status codes in the capture are correctly interpreted Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MDN HTTP Overview and HTTP methods/status sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MDN HTTP documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D014 · M01W2D7 · Databases and SQL fundamentals

Phase: Academic month · **Optional date:** 2026-11-09 · **Domain resources:** F11 / P07 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 14 of 365; calendar-free M01W2D7; optional date 2026-11-09.
2. Exact topic	Databases and SQL fundamentals
3. Exact subtopics	Tables, rows, columns, primary keys, SELECT, INSERT, WHERE, ORDER BY and LIMIT
4. Learning objective	Query and insert synthetic security events in SQLite
5. Primary lecture / reading + direct URL	SQLBolt `https://sqlbolt.com/` **Type:** Interactive lessons · **Access:** Free
6. Exact section / lesson locator	SQLBolt Lessons 1-5: SELECT queries, filtering and sorting. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SQLite documentation `https://www.sqlite.org/docs.html`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a small auth_events table, insert ten synthetic rows and write five queries with expected outputs — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a small auth_events table, insert ten synthetic rows and write five queries with expected outputs — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a small auth_events table, insert ten synthetic rows and write five queries with expected outputs — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	schema.sql, queries.sql and results.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Why should production code use parameterized queries rather than string concatenation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All five queries run and results are verified against expected rows Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SQLBolt Lessons 1-5: SELECT queries, filtering and sorting"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Secure Coding Practices Quick Reference Guide; write a five-line correction; retest next morning with a new 10-minute problem.

Month 01 · Week 3 · daily cards

D015 · M01W3D1 · Git, branching and merge conflicts

Phase: Academic month · **Optional date:** 2026-11-10 · **Domain resources:** F09 / F01

1. Day / sequence	Global Day 15 of 365; calendar-free M01W3D1; optional date 2026-11-10.
2. Exact topic	Git, branching and merge conflicts
3. Exact subtopics	Repository initialization, staging, commits, branches, diffs, merge, conflict markers and .gitignore
4. Learning objective	Use version control to recover from a deliberate mistake
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git, Chapters 2-3: Git basics and branching. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a repository, make two branches that modify the same line, resolve the conflict, and add a .gitignore for lab secrets. **2.** Standard (~2 h): Create a repository, make two branches that modify the same line, resolve the conflict, and add a .gitignore for lab secrets. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a repository, make two branches that modify the same line, resolve the conflict, and add a .gitignore for lab secrets. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	git-log.txt and resolved conflict screenshot; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the difference between git reset, revert and checkout in your own workflow? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Clean history, conflict resolved, and secret placeholders ignored Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git, Chapters 2-3: Git basics and branching"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D016 · M01W3D2 · Weekly review and mini practical examination

Phase: Academic month · **Optional date:** 2026-11-11 · **Domain resources:** F11 / F09

1. Day / sequence	Global Day 16 of 365; calendar-free M01W3D2; optional date 2026-11-11.
2. Exact topic	Weekly review and mini practical examination
3. Exact subtopics	Filesystems, processes, services, sockets, HTTP, SQL and Git; active recall and troubleshooting
4. Learning objective	Demonstrate Month 1 Week 2 skills under a timed scenario
5. Primary lecture / reading + direct URL	SQLBolt `https://sqlbolt.com/` **Type:** Interactive lessons · **Access:** Free
6. Exact section / lesson locator	SQLBolt Lessons 6-10: joins, aggregation and NULL values. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 60-minute practical: inspect a process, start a loopback service, query a SQLite table, and commit evidence. **2.** Standard (~2 h): Complete a 60-minute practical: inspect a process, start a loopback service, query a SQLite table, and commit evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 60-minute practical: inspect a process, start a loopback service, query a SQLite table, and commit evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	timed-practical-M1W2.md with all command output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Given a broken service, which three logs or commands would you check first and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	At least 70 percent on the rubric and all evidence is reproducible Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SQLBolt Lessons 6-10: joins, aggregation and NULL values"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Secure Coding Practices Quick Reference Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D017 · M01W3D3 · Memory, stack, heap and addresses

Phase: Academic month · **Optional date:** 2026-11-12 · **Domain resources:** F01 / F06

1. Day / sequence	Global Day 17 of 365; calendar-free M01W3D3; optional date 2026-11-12.
2. Exact topic	Memory, stack, heap and addresses
3. Exact subtopics	Virtual memory, address space, stack frames, heap allocation, pointers, buffer boundaries
4. Learning objective	Describe how data and control information are laid out in memory
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x Week 4 Memory lecture and notes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run a safe C/Python memory visualization exercise; draw stack and heap regions for a simple function call. **2.** Standard (~2 h): Run a safe C/Python memory visualization exercise; draw stack and heap regions for a simple function call. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run a safe C/Python memory visualization exercise; draw stack and heap regions for a simple function call. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	memory-diagram.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the security consequence of writing beyond an array boundary? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram correctly locates code, globals, heap, stack and function frames Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x Week 4 Memory lecture and notes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D018 · M01W3D4 · Binary logic and CPU instruction foundations

Phase: Academic month · **Optional date:** 2026-11-13 · **Domain resources:** F07 / F02

1. Day / sequence	Global Day 18 of 365; calendar-free M01W3D4; optional date 2026-11-13.
2. Exact topic	Binary logic and CPU instruction foundations
3. Exact subtopics	Logic gates, truth tables, ALU operations, machine language, registers and program counter
4. Learning objective	Connect Boolean logic to a CPU instruction and a security decision
5. Primary lecture / reading + direct URL	Nand2Tetris / The Elements of Computing Systems `https://www.nand2tetris.org/` **Type:** Free course and software · **Access:** Free software and course materials; book optional paid
6. Exact section / lesson locator	Nand2Tetris Part I Projects 1-3: logic gates, arithmetic and sequential logic. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50x Lectures 2026 (official YouTube playlist) `https://www.youtube.com/playlist?list=PLhQjrBD2T380hlTqAU8HfvVepCcjCqTg6`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete one software simulator project or, if unavailable, write truth tables and explain a half-adder and flip-flop. **2.** Standard (~2 h): Complete one software simulator project or, if unavailable, write truth tables and explain a half-adder and flip-flop. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete one software simulator project or, if unavailable, write truth tables and explain a half-adder and flip-flop. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	logic-project files or truth-tables.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How does a branch instruction change control flow at the hardware level? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Project passes local checks or all truth tables are correct Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Nand2Tetris Part I Projects 1-3: logic gates, arithmetic and sequential logic"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Operating Systems: Three Easy Pieces (OSTEP); write a five-line correction; retest next morning with a new 10-minute problem.

D019 · M01W3D5 · Storage, filesystems, journaling and backups

Phase: Academic month · **Optional date:** 2026-11-14 · **Domain resources:** F12 / C27

1. Day / sequence	Global Day 19 of 365; calendar-free M01W3D5; optional date 2026-11-14.
2. Exact topic	Storage, filesystems, journaling and backups
3. Exact subtopics	Block devices, partitions, filesystem metadata, journaling, snapshots, 3-2-1 backup concept, restore testing
4. Learning objective	Explain why an untested backup is not a valid recovery control
5. Primary lecture / reading + direct URL	Oracle VirtualBox User Manual `https://www.virtualbox.org/manual/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	VirtualBox User Manual virtual storage and snapshots chapters. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-34 Revision 1 `https://csrc.nist.gov/pubs/sp/800/34/r1/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a benign file set in the VM, snapshot or copy it, corrupt/delete one file, and restore; record hashes before and after. **2.** Standard (~2 h): Create a benign file set in the VM, snapshot or copy it, corrupt/delete one file, and restore; record hashes before and after. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a benign file set in the VM, snapshot or copy it, corrupt/delete one file, and restore; record hashes before and after. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	backup-test.md with SHA-256 values; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the difference between a VM snapshot, a file backup and a disaster-recovery plan? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Restore is proven by matching hashes and cleanup is documented Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "VirtualBox User Manual virtual storage and snapshots chapters"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Windows Server documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D020 · M01W3D6 · HTTP sessions, cookies and authentication basics

Phase: Academic month · **Optional date:** 2026-11-15 · **Domain resources:** F04 / W17

1. Day / sequence	Global Day 20 of 365; calendar-free M01W3D6; optional date 2026-11-15.
2. Exact topic	HTTP sessions, cookies and authentication basics
3. Exact subtopics	Statefulness, cookies, Set-Cookie attributes, sessions, authentication factors and password hashing concept
4. Learning objective	Distinguish authentication from authorization and explain safe password storage
5. Primary lecture / reading + direct URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/` **Type:** University course · **Access:** Free OCW; paid edX verified certificate optional
6. Exact section / lesson locator	CS50 Cybersecurity Week 0 Securing Accounts and Week 1 Securing Data notes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Inspect a local app's cookie in DevTools; annotate Secure, HttpOnly, SameSite and explain password hashing versus encryption. **2.** Standard (~2 h): Inspect a local app's cookie in DevTools; annotate Secure, HttpOnly, SameSite and explain password hashing versus encryption. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Inspect a local app's cookie in DevTools; annotate Secure, HttpOnly, SameSite and explain password hashing versus encryption. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cookies-notes.md and a conceptual login-flow diagram; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is storing plaintext passwords worse than storing unsalted fast hashes? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All cookie attributes and identity terms are explained without confusing hashing with encryption Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50 Cybersecurity Week 0 Securing Accounts and Week 1 Securing Data notes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50's Introduction to Cybersecurity (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D021 · M01W3D7 · Troubleshooting methodology and technical journaling

Phase: Academic month · **Optional date:** 2026-11-16 · **Domain resources:** F01 / L01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 21 of 365; calendar-free M01W3D7; optional date 2026-11-16.
2. Exact topic	Troubleshooting methodology and technical journaling
3. Exact subtopics	Define symptom, establish baseline, form hypotheses, change one variable, collect evidence, rollback and document
4. Learning objective	Diagnose a deliberately broken local service using a repeatable method
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x Week 1 C problem-solving and debugging sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Break a local HTTP server configuration on purpose, diagnose it from logs, fix it, and write a postmortem — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Break a local HTTP server configuration on purpose, diagnose it from logs, fix it, and write a postmortem — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Break a local HTTP server configuration on purpose, diagnose it from logs, fix it, and write a postmortem — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	troubleshooting-postmortem.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which observation falsified each incorrect hypothesis? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Root cause is proven and rollback steps are included Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x Week 1 C problem-solving and debugging sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

Month 01 · Week 4 · daily cards

D022 · M01W4D1 · Month 1 project completion and portfolio publication

Phase: Academic month · **Optional date:** 2026-11-17 · **Domain resources:** F09 / F12

1. Day / sequence	Global Day 22 of 365; calendar-free M01W4D1; optional date 2026-11-17.
2. Exact topic	Month 1 project completion and portfolio publication
3. Exact subtopics	Problem statement, architecture, validation, limitations, screenshots, ethical scope and README standards
4. Learning objective	Publish a sanitized, reproducible home-lab baseline report
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapter 2 and GitHub Docs basic README conventions via F09's official site navigation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Oracle VirtualBox User Manual `https://www.virtualbox.org/manual/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Finalize PRJ-01, add verification tests, remove IP/MAC addresses or personal data, and tag v0.2. **2.** Standard (~2 h): Finalize PRJ-01, add verification tests, remove IP/MAC addresses or personal data, and tag v0.2. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Finalize PRJ-01, add verification tests, remove IP/MAC addresses or personal data, and tag v0.2. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	published PRJ-01 repository and report PDF/Markdown; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would a reviewer check first to verify this lab actually works? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Repository passes the Month 1 rubric and contains no secrets or personal data Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapter 2 and GitHub Docs basic README conventions via F09's official site navigation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D023 · M01W4D2 · Month 1 practical assessment and milestone review

Phase: Academic month · **Optional date:** 2026-11-18 · **Domain resources:** F01 / F04

1. Day / sequence	Global Day 23 of 365; calendar-free M01W4D2; optional date 2026-11-18.
2. Exact topic	Month 1 practical assessment and milestone review
3. Exact subtopics	Number systems, architecture, OS, virtualization, command line, HTTP, SQL, Git and troubleshooting
4. Learning objective	Demonstrate foundational competence through an independent 90-minute practical
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x Week 1-4 notes for algorithmic reasoning and memory. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Without notes, configure a fresh lightweight VM, create users, run a local service, query a synthetic database, commit evidence and answer 20 oral questions. **2.** Standard (~2 h): Without notes, configure a fresh lightweight VM, create users, run a local service, query a synthetic database, commit evidence and answer 20 oral questions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Without notes, configure a fresh lightweight VM, create users, run a local service, query a synthetic database, commit evidence and answer 20 oral questions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	assessment-M1.pdf with command evidence and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Explain how a request travels from browser to local server and where isolation occurs Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 with no safety violation and practical section at least 18/25 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x Week 1-4 notes for algorithmic reasoning and memory"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D024 · M01W4D3 · APIs, JSON and structured data

Phase: Academic month · **Optional date:** 2026-11-19 · **Domain resources:** F10 / P06

1. Day / sequence	Global Day 24 of 365; calendar-free M01W4D3; optional date 2026-11-19.
2. Exact topic	APIs, JSON and structured data
3. Exact subtopics	REST resources, HTTP verbs, status codes, JSON objects/arrays, serialization and schema validation
4. Learning objective	Read and design a simple API request/response pair with explicit validation
5. Primary lecture / reading + direct URL	MDN HTTP Overview `https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview` **Type:** Official web documentation · **Access:** Free
6. Exact section / lesson locator	MDN HTTP Overview and HTTP methods/status sections; F08 Python JSON tutorial section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Requests: HTTP for Humans `https://requests.readthedocs.io/en/latest/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Use a local mock server or Python requests against localhost; send valid, malformed and missing-field JSON; record status codes and errors. **2.** Standard (~2 h): Use a local mock server or Python requests against localhost; send valid, malformed and missing-field JSON; record status codes and errors. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Use a local mock server or Python requests against localhost; send valid, malformed and missing-field JSON; record status codes and errors. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	api-notes.md and sample request/response files; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why must a client validate server data even when it sent valid JSON? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All three cases are documented and status codes are interpreted correctly Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MDN HTTP Overview and HTTP methods/status sections; F08 Python JSON tutorial section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MDN HTTP documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D025 · M01W4D4 · Identity, sessions and access control foundations

Phase: Academic month · **Optional date:** 2026-11-20 · **Domain resources:** F04 / W17

1. Day / sequence	Global Day 25 of 365; calendar-free M01W4D4; optional date 2026-11-20.
2. Exact topic	Identity, sessions and access control foundations
3. Exact subtopics	Identification, authentication, authorization, accountability, session identifiers, logout and least privilege
4. Learning objective	Map a login workflow to identity controls and list three session risks
5. Primary lecture / reading + direct URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/` **Type:** University course · **Access:** Free OCW; paid edX verified certificate optional
6. Exact section / lesson locator	CS50 Cybersecurity Week 0 Securing Accounts notes and Week 3 Securing Software notes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a local login sequence, mark where identity is established and checked, and describe session fixation and timeout risks. **2.** Standard (~2 h): Draw a local login sequence, mark where identity is established and checked, and describe session fixation and timeout risks. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a local login sequence, mark where identity is established and checked, and describe session fixation and timeout risks. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	identity-flow.svg and risk notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a long-lived session identifier dangerous after a password change? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram separates authentication from authorization and identifies concrete controls Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50 Cybersecurity Week 0 Securing Accounts notes and Week 3 Securing Software notes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50's Introduction to Cybersecurity (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D026 · M01W4D5 · Virtual machine networking troubleshooting

Phase: Academic month · **Optional date:** 2026-11-21 · **Domain resources:** F12 / N08

1. Day / sequence	Global Day 26 of 365; calendar-free M01W4D5; optional date 2026-11-21.
2. Exact topic	Virtual machine networking troubleshooting
3. Exact subtopics	Virtual NIC types, NAT forwarding, host-only addressing, internal network, promiscuous mode, DNS and DHCP inside VMs
4. Learning objective	Diagnose VM connectivity without weakening host security
5. Primary lecture / reading + direct URL	Oracle VirtualBox User Manual `https://www.virtualbox.org/manual/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	VirtualBox User Manual Chapter 6 Virtual Networking and Chapter 8 VBoxManage/network troubleshooting. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	tcpdump manual page `https://www.tcpdump.org/manpages/tcpdump.1.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create three harmless VM network faults, diagnose each from guest and host, then restore the clean snapshot and document commands. **2.** Standard (~2 h): Create three harmless VM network faults, diagnose each from guest and host, then restore the clean snapshot and document commands. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create three harmless VM network faults, diagnose each from guest and host, then restore the clean snapshot and document commands. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	vm-network-troubleshooting.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is promiscuous mode risky on a shared network and when might it be needed? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each fault has evidence-based diagnosis and the clean snapshot is restored Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "VirtualBox User Manual Chapter 6 Virtual Networking and Chapter 8 VBoxManage/network troubleshooting"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Windows Server documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D027 · M01W4D6 · CS50-style problem solving and decomposition

Phase: Academic month · **Optional date:** 2026-11-22 · **Domain resources:** F01 / F08

1. Day / sequence	Global Day 27 of 365; calendar-free M01W4D6; optional date 2026-11-22.
2. Exact topic	CS50-style problem solving and decomposition
3. Exact subtopics	Abstraction, algorithms, pseudocode, inputs/outputs, edge cases, test cases and debugging
4. Learning objective	Solve a small unfamiliar programming problem by decomposing it before coding
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x Week 1 C lecture and problem-solving notes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write pseudocode and a Python implementation for a log-line classifier; test empty input, large input, malformed input and unexpected characters. **2.** Standard (~2 h): Write pseudocode and a Python implementation for a log-line classifier; test empty input, large input, malformed input and unexpected characters. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write pseudocode and a Python implementation for a log-line classifier; test empty input, large input, malformed input and unexpected characters. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	classifier.py and test-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which edge case caused a failure and how did you change the algorithm? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Code passes all planned tests and the write-up explains the decomposition Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x Week 1 C lecture and problem-solving notes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D028 · M01W4D7 · Spaced revision, recovery and Month 2 preview

Phase: Academic month · **Optional date:** 2026-11-23 · **Domain resources:** N16 / F06 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 28 of 365; calendar-free M01W4D7; optional date 2026-11-23.
2. Exact topic	Spaced revision, recovery and Month 2 preview
3. Exact subtopics	Error log, command notebook, flashcards, weak-topic repair, subnetting readiness and binary practice
4. Learning objective	Consolidate Month 1 and prepare binary arithmetic for IP addressing
5. Primary lecture / reading + direct URL	Subnetting Practice `https://subnettingpractice.com/` **Type:** Interactive practice · **Access:** Free tier; optional ads/premium features may appear
6. Exact section / lesson locator	Subnetting Practice, introduction and binary/IP basics; do only five diagnostic questions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Review all failed retrieval questions, convert 20 decimal values to binary, write five Anki cards, and plan the next week's lab — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Review all failed retrieval questions, convert 20 decimal values to binary, write five Anki cards, and plan the next week's lab — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Review all failed retrieval questions, convert 20 decimal values to binary, write five Anki cards, and plan the next week's lab — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	revision-M1.md and flashcards.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which Month 1 concept is still weakest, and what exact exercise would repair it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All weak topics have a scheduled repair action and at least 18/20 binary conversions are correct Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Subnetting Practice, introduction and binary/IP basics; do only five diagnostic questions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Practical Networking Subnetting Mastery; write a five-line correction; retest next morning with a new 10-minute problem.

Month 02 — Networking, subnetting, packet analysis and troubleshooting

Days D029-D056. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: OSI seven-layer model and TCP/IP four-layer model D2: Ethernet frames, MAC addresses and switching D3: ARP, ICMP and local address resolution D4: VLANs, trunks and broadcast domains D5: IPv4 header, addresses and private ranges D6: CIDR, subnet masks and network/broadcast/host calculations D7: VLSM, route aggregation and daily subnetting fluency
Week 2	D1: Weekly packet and subnetting review D2: Routing, default gateways and static routes D3: NAT, PAT and public versus private addressing D4: Dynamic routing and routing security concepts D5: TCP handshake, sequence numbers and reliability D6: UDP, ports and common services D7: DNS resolution and record types
Week 3	D1: DHCP, DHCPv6 and network configuration lifecycle D2: HTTP, HTTPS and TLS at the network layer D3: Weekly network services practical and retrieval D4: Firewalls, proxies, VPNs and segmentation architectures D5: Wireshark display filters and protocol dissection D6: Nmap host discovery and port scanning in an isolated lab D7: TCP retransmissions, MTU and fragmentation analysis
Week 4	D1: Network troubleshooting methodology and tool selection D2: Network-security project: topology and traffic report D3: Monthly network practical and daily subnetting test D4: Monthly review and gap repairs D5: IPv6 addressing and protocol differences D6: Wireless LANs, network monitoring and SPAN concepts D7: Spaced revision, recovery and Month 3 preview

> **Gate:** score the Month 2 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 02 · Week 1 · daily cards

D029 · M02W1D1 · OSI seven-layer model and TCP/IP four-layer model

Phase: Academic month · **Optional date:** 2026-11-24 · **Domain resources:** N01 / N04

1. Day / sequence	Global Day 29 of 365; calendar-free M02W1D1; optional date 2026-11-24.
2. Exact topic	OSI seven-layer model and TCP/IP four-layer model
3. Exact subtopics	Physical, Data Link, Network, Transport, Session, Presentation, Application; Network Access, Internet, Transport, Application; encapsulation and decapsulation
4. Learning objective	Map a web request to each layer and name its protocol data unit
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 3, The TCP/IP Model; use the exact Day 3 lecture title on the playlist. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Professor Messer N10-009 Network+ Training Course `https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a full encapsulation diagram from HTTP to bits and label PDUs at each layer; answer five mapping questions. **2.** Standard (~2 h): Draw a full encapsulation diagram from HTTP to bits and label PDUs at each layer; answer five mapping questions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a full encapsulation diagram from HTTP to bits and label PDUs at each layer; answer five mapping questions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	encapsulation.svg and answers.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Where is a MAC address used and where is an IP address used as data travels? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every layer has correct PDU, protocol and address type Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 3, The TCP/IP Model; use the exact Day 3 lecture title on the playlist"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D030 · M02W1D2 · Ethernet frames, MAC addresses and switching

Phase: Academic month · **Optional date:** 2026-11-25 · **Domain resources:** N01 / N02

1. Day / sequence	Global Day 30 of 365; calendar-free M02W1D2; optional date 2026-11-25.
2. Exact topic	Ethernet frames, MAC addresses and switching
3. Exact subtopics	Ethernet header, FCS, MAC address format, unicast/broadcast, switch MAC learning, collision domains
4. Learning objective	Explain how a switch forwards a frame and why MAC addresses are local to a link
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 5 and Day 6, Ethernet LAN Switching Parts 1 and 2. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Jeremy's IT Lab CCNA Course Index and Files `https://courses.jeremysitlab.com/p/ccna`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a small Packet Tracer LAN, inspect the switch MAC table before and after ping, and capture one frame if possible. **2.** Standard (~2 h): Build a small Packet Tracer LAN, inspect the switch MAC table before and after ping, and capture one frame if possible. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a small Packet Tracer LAN, inspect the switch MAC table before and after ping, and capture one frame if possible. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	switching-lab.pkt and annotated MAC table; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does the destination MAC change at each routed hop while IP endpoints may remain? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Switch table reflects learned addresses and the explanation is correct Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 5 and Day 6, Ethernet LAN Switching Parts 1 and 2"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D031 · M02W1D3 · ARP, ICMP and local address resolution

Phase: Academic month · **Optional date:** 2026-11-26 · **Domain resources:** N01 / N06

1. Day / sequence	Global Day 31 of 365; calendar-free M02W1D3; optional date 2026-11-26.
2. Exact topic	ARP, ICMP and local address resolution
3. Exact subtopics	ARP request/reply, ARP cache, gratuitous ARP, ICMP echo, TTL and basic error messages
4. Learning objective	Trace how a host finds the next-hop MAC address for an IP packet
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 12, Life of a Packet; Day 3 for ARP context. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Wireshark User's Guide `https://www.wireshark.org/docs/wsug_html_chunked/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Generate a ping on your own lab network or a Packet Tracer simulation; inspect ARP cache and packet sequence; explain each step. **2.** Standard (~2 h): Generate a ping on your own lab network or a Packet Tracer simulation; inspect ARP cache and packet sequence; explain each step. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Generate a ping on your own lab network or a Packet Tracer simulation; inspect ARP cache and packet sequence; explain each step. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	arp-icmp.md with Wireshark/Packet Tracer screenshots; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is ARP not routable and what security assumption does it make? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Sequence is complete from ARP to ICMP reply and all addresses are labelled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 12, Life of a Packet; Day 3 for ARP context"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D032 · M02W1D4 · VLANs, trunks and broadcast domains

Phase: Academic month · **Optional date:** 2026-11-27 · **Domain resources:** N01 / N03

1. Day / sequence	Global Day 32 of 365; calendar-free M02W1D4; optional date 2026-11-27.
2. Exact topic	VLANs, trunks and broadcast domains
3. Exact subtopics	Access ports, trunk ports, 802.1Q tags, native VLAN, broadcast domains and basic segmentation
4. Learning objective	Explain how VLANs isolate traffic and where tags appear
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Days 16-18, VLANs Parts 1-3. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Jeremy's IT Lab CCNA Course Files `https://jitl.jp/ccna-files`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Configure two VLANs in Packet Tracer, test intra/inter-VLAN behaviour, and document why a router or Layer 3 switch is needed between them. **2.** Standard (~2 h): Configure two VLANs in Packet Tracer, test intra/inter-VLAN behaviour, and document why a router or Layer 3 switch is needed between them. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Configure two VLANs in Packet Tracer, test intra/inter-VLAN behaviour, and document why a router or Layer 3 switch is needed between them. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	vlan-lab.pkt and segmentation-notes.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the security benefit and the main limitation of VLAN segmentation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Configuration works and the answer distinguishes isolation from strong security boundaries Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Days 16-18, VLANs Parts 1-3"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D033 · M02W1D5 · IPv4 header, addresses and private ranges

Phase: Academic month · **Optional date:** 2026-11-28 · **Domain resources:** N01 / N11

1. Day / sequence	Global Day 33 of 365; calendar-free M02W1D5; optional date 2026-11-28.
2. Exact topic	IPv4 header, addresses and private ranges
3. Exact subtopics	IPv4 header fields, network/host portions, classful history, RFC1918 ranges, loopback and link-local addresses
4. Learning objective	Read an IPv4 header and identify private, public, loopback and link-local addresses
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Days 7-8 and Day 10, IPv4 Addressing Parts 1-2 and The IPv4 Header. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RFC 791: Internet Protocol v4 `https://www.rfc-editor.org/rfc/rfc791`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Inspect an IPv4 packet in Wireshark; annotate version, IHL, DSCP, length, ID, flags, TTL, protocol, checksum and addresses. **2.** Standard (~2 h): Inspect an IPv4 packet in Wireshark; annotate version, IHL, DSCP, length, ID, flags, TTL, protocol, checksum and addresses. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Inspect an IPv4 packet in Wireshark; annotate version, IHL, DSCP, length, ID, flags, TTL, protocol, checksum and addresses. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ipv4-header.png with annotations; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is 192.168.1.10 not visible as a destination address on the public Internet? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All header fields are correctly mapped and address categories are identified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Days 7-8 and Day 10, IPv4 Addressing Parts 1-2 and The IPv4 Header"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D034 · M02W1D6 · CIDR, subnet masks and network/broadcast/host calculations

Phase: Academic month · **Optional date:** 2026-11-29 · **Domain resources:** N01 / N16

1. Day / sequence	Global Day 34 of 365; calendar-free M02W1D6; optional date 2026-11-29.
2. Exact topic	CIDR, subnet masks and network/broadcast/host calculations
3. Exact subtopics	Prefix length, wildcard mask, network address, broadcast address, first/last usable host, block size
4. Learning objective	Solve /8-/30 subnetting questions with a written method
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Days 13-15, Subnetting Parts 1-3. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Subnetting Practice `https://subnettingpractice.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 20 timed subnetting questions; show binary boundaries for five; record mistakes and repeat each missed type. **2.** Standard (~2 h): Complete 20 timed subnetting questions; show binary boundaries for five; record mistakes and repeat each missed type. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 20 timed subnetting questions; show binary boundaries for five; record mistakes and repeat each missed type. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	subnetting-M2D6.csv with score and corrections; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Given 192.168.10.77/26, what are the network, broadcast and usable range? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	At least 80 percent accuracy with all corrections shown Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Days 13-15, Subnetting Parts 1-3"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D035 · M02W1D7 · VLSM, route aggregation and daily subnetting fluency

Phase: Academic month · **Optional date:** 2026-11-30 · **Domain resources:** N17 / N16 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 35 of 365; calendar-free M02W1D7; optional date 2026-11-30.
2. Exact topic	VLSM, route aggregation and daily subnetting fluency
3. Exact subtopics	Variable-length subnet masks, route summarization, address conservation, longest-prefix match
4. Learning objective	Design an efficient addressing plan for a small topology
5. Primary lecture / reading + direct URL	Practical Networking Subnetting Mastery `https://www.practicalnetworking.net/stand-alone/subnetting-mastery/` **Type:** Course / articles · **Access:** Free articles/videos; optional paid resources
6. Exact section / lesson locator	Practical Networking Subnetting Mastery, parts on FLSM, VLSM and summarization. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Subnetting Practice `https://subnettingpractice.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Allocate subnets for five departments with different host counts, verify no overlap, then complete 20 timed questions — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Allocate subnets for five departments with different host counts, verify no overlap, then complete 20 timed questions — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Allocate subnets for five departments with different host counts, verify no overlap, then complete 20 timed questions — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	vlsm-plan.csv and quiz score; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which aggregate route covers your assigned subnets and what extra space does it include? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan has no overlap, meets host counts, and quiz accuracy is at least 80 percent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Practical Networking Subnetting Mastery, parts on FLSM, VLSM and summarization"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Subnetting Practice; write a five-line correction; retest next morning with a new 10-minute problem.

Month 02 · Week 2 · daily cards

D036 · M02W2D1 · Weekly packet and subnetting review

Phase: Academic month · **Optional date:** 2026-12-01 · **Domain resources:** N04 / N16

1. Day / sequence	Global Day 36 of 365; calendar-free M02W2D1; optional date 2026-12-01.
2. Exact topic	Weekly packet and subnetting review
3. Exact subtopics	OSI/TCP/IP, Ethernet, ARP, VLANs, IPv4, CIDR and VLSM; retrieval and error analysis
4. Learning objective	Consolidate Month 2 Week 1 and repair subnetting gaps
5. Primary lecture / reading + direct URL	Professor Messer N10-009 Network+ Training Course `https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8` **Type:** YouTube playlist · **Access:** Free
6. Exact section / lesson locator	Professor Messer N10-009, objectives 1.1-1.4; use exact video titles on the course index. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Subnetting Practice `https://subnettingpractice.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 15 mixed questions, rework all mistakes, and write a one-page explanation of encapsulation with a subnetting example. **2.** Standard (~2 h): Complete 15 mixed questions, rework all mistakes, and write a one-page explanation of encapsulation with a subnetting example. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 15 mixed questions, rework all mistakes, and write a one-page explanation of encapsulation with a subnetting example. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M2W1-review.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which layer or address changes at each hop and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70 percent and all corrections show working, not just answers Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer N10-009, objectives 1.1-1.4; use exact video titles on the course index"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer N10-009 Course Index; write a five-line correction; retest next morning with a new 10-minute problem.

D037 · M02W2D2 · Routing, default gateways and static routes

Phase: Academic month · **Optional date:** 2026-12-02 · **Domain resources:** N01 / N03

1. Day / sequence	Global Day 37 of 365; calendar-free M02W2D2; optional date 2026-12-02.
2. Exact topic	Routing, default gateways and static routes
3. Exact subtopics	Routing table, longest-prefix match, connected routes, default route, next hop, static route configuration
4. Learning objective	Predict and configure how a packet leaves its local network
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 11, Routing Fundamentals and Static Routing. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Jeremy's IT Lab CCNA Course Files `https://jitl.jp/ccna-files`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a three-router Packet Tracer topology, configure static routes, and verify with ping/traceroute from both ends. **2.** Standard (~2 h): Build a three-router Packet Tracer topology, configure static routes, and verify with ping/traceroute from both ends. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a three-router Packet Tracer topology, configure static routes, and verify with ping/traceroute from both ends. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	routing-lab.pkt and route tables; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What happens when two routes match a destination but have different prefix lengths? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	End-to-end connectivity works and route selection is explained by longest-prefix match Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 11, Routing Fundamentals and Static Routing"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D038 · M02W2D3 · NAT, PAT and public versus private addressing

Phase: Academic month · **Optional date:** 2026-12-03 · **Domain resources:** N04 / N15

1. Day / sequence	Global Day 38 of 365; calendar-free M02W2D3; optional date 2026-12-03.
2. Exact topic	NAT, PAT and public versus private addressing
3. Exact subtopics	NAT translation table, inside/outside/local/global terminology, port address translation, limitations of NAT as security
4. Learning objective	Explain how many private hosts share one public address and why NAT is not a firewall
5. Primary lecture / reading + direct URL	Professor Messer N10-009 Network+ Training Course `https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8` **Type:** YouTube playlist · **Access:** Free
6. Exact section / lesson locator	Professor Messer N10-009 objective on NAT and port forwarding; use the course index to locate exact video. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Cisco Packet Tracer course page `https://www.netacad.com/courses/packet-tracer`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Configure NAT/PAT in Packet Tracer or diagram a translation table for five local connections; test one external connection. **2.** Standard (~2 h): Configure NAT/PAT in Packet Tracer or diagram a translation table for five local connections; test one external connection. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Configure NAT/PAT in Packet Tracer or diagram a translation table for five local connections; test one external connection. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	nat-table.csv and topology diagram; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does NAT not guarantee inbound protection or application-layer security? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Translation entries correctly map addresses/ports and limitations are explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer N10-009 objective on NAT and port forwarding; use the course index to locate exact video"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer N10-009 Course Index; write a five-line correction; retest next morning with a new 10-minute problem.

D039 · M02W2D4 · Dynamic routing and routing security concepts

Phase: Academic month · **Optional date:** 2026-12-04 · **Domain resources:** N01 / N02

1. Day / sequence	Global Day 39 of 365; calendar-free M02W2D4; optional date 2026-12-04.
2. Exact topic	Dynamic routing and routing security concepts
3. Exact subtopics	OSPF areas and adjacencies, metrics, route propagation, authentication, route filtering and failure domains
4. Learning objective	Explain the difference between static and dynamic routing and one security risk of incorrect route advertisement
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Days 24-28, Dynamic Routing through OSPF Part 3; select the Day 26 lecture for a first pass. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Jeremy's IT Lab CCNA Course Index and Files `https://courses.jeremysitlab.com/p/ccna`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Configure single-area OSPF in Packet Tracer, break one link, observe convergence, and write a security caveat. **2.** Standard (~2 h): Configure single-area OSPF in Packet Tracer, break one link, observe convergence, and write a security caveat. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Configure single-area OSPF in Packet Tracer, break one link, observe convergence, and write a security caveat. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ospf-lab.pkt and convergence notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can an unauthorized routing update redirect traffic even without exploiting an application? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Neighborship forms, connectivity restores, and the risk is correctly described Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Days 24-28, Dynamic Routing through OSPF Part 3; select the Day 26 lecture for a first pass"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D040 · M02W2D5 · TCP handshake, sequence numbers and reliability

Phase: Academic month · **Optional date:** 2026-12-05 · **Domain resources:** N01 / N10

1. Day / sequence	Global Day 40 of 365; calendar-free M02W2D5; optional date 2026-12-05.
2. Exact topic	TCP handshake, sequence numbers and reliability
3. Exact subtopics	SYN, SYN-ACK, ACK; sequence and acknowledgment numbers; windowing, retransmission, FIN/RST, TCP state
4. Learning objective	Annotate a TCP connection and explain how reliability emerges from sequence numbers
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 30, TCP and UDP. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RFC 9293: Transmission Control Protocol `https://www.rfc-editor.org/rfc/rfc9293`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Capture a local TCP connection with Wireshark, label handshake, seq/ack progression, one data segment and teardown. **2.** Standard (~2 h): Capture a local TCP connection with Wireshark, label handshake, seq/ack progression, one data segment and teardown. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Capture a local TCP connection with Wireshark, label handshake, seq/ack progression, one data segment and teardown. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	tcp-handshake.png and field table; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is the initial sequence number unpredictable and security-relevant? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All packet fields are correctly labelled and the reliability explanation is coherent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 30, TCP and UDP"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D041 · M02W2D6 · UDP, ports and common services

Phase: Academic month · **Optional date:** 2026-12-06 · **Domain resources:** N01 / N14

1. Day / sequence	Global Day 41 of 365; calendar-free M02W2D6; optional date 2026-12-06.
2. Exact topic	UDP, ports and common services
3. Exact subtopics	UDP header, connectionless behaviour, DNS/DHCP/QUIC use cases, ephemeral ports, TCP versus UDP trade-offs
4. Learning objective	Decide when an application should use UDP and explain the security implications
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 30, TCP and UDP. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	IANA Service Name and Transport Protocol Port Number Registry `https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Capture a DNS query over UDP, compare its header to TCP, and build a table of ten common ports with protocol and purpose. **2.** Standard (~2 h): Capture a DNS query over UDP, compare its header to TCP, and build a table of ten common ports with protocol and purpose. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Capture a DNS query over UDP, compare its header to TCP, and build a table of ten common ports with protocol and purpose. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	udp-dns.md and ports.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is UDP spoofing often easier than TCP spoofing? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Capture and port table are correct and the trade-off explanation includes integrity/authentication Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 30, TCP and UDP"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D042 · M02W2D7 · DNS resolution and record types

Phase: Academic month · **Optional date:** 2026-12-07 · **Domain resources:** N13 / N01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 42 of 365; calendar-free M02W2D7; optional date 2026-12-07.
2. Exact topic	DNS resolution and record types
3. Exact subtopics	Recursive/iterative lookup, root/TLD/authoritative servers, A/AAAA/CNAME/MX/TXT/NS/SOA, TTL, caching and DNSSEC concepts
4. Learning objective	Trace a DNS answer and explain why DNS is both naming and a security control point
5. Primary lecture / reading + direct URL	Cloudflare Learning Center: What is DNS? `https://www.cloudflare.com/learning/dns/what-is-dns/` **Type:** Educational documentation · **Access:** Free
6. Exact section / lesson locator	Cloudflare Learning Center, What is DNS and DNS server/resolution sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Use dig or nslookup to query an example domain from your own machine, capture the exchange, and record each answer section and TTL — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Use dig or nslookup to query an example domain from your own machine, capture the exchange, and record each answer section and TTL — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Use dig or nslookup to query an example domain from your own machine, capture the exchange, and record each answer section and TTL — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	dns-trace.txt and annotated capture; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which party is authoritative for a name and why can a cached answer be stale? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Resolution chain is correct and capture fields are explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Cloudflare Learning Center, What is DNS and DNS server/resolution sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Suricata documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 02 · Week 3 · daily cards

D043 · M02W3D1 · DHCP, DHCPv6 and network configuration lifecycle

Phase: Academic month · **Optional date:** 2026-12-08 · **Domain resources:** N04 / N05

1. Day / sequence	Global Day 43 of 365; calendar-free M02W3D1; optional date 2026-12-08.
2. Exact topic	DHCP, DHCPv6 and network configuration lifecycle
3. Exact subtopics	DORA process, lease, scope, options, default gateway, DNS server, reservation, rogue DHCP risk
4. Learning objective	Explain how a host obtains configuration and what an attacker could do with a rogue DHCP server
5. Primary lecture / reading + direct URL	Professor Messer N10-009 Network+ Training Course `https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8` **Type:** YouTube playlist · **Access:** Free
6. Exact section / lesson locator	Professor Messer N10-009 objective on DHCP; use the exact objective video title in the course index. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Professor Messer N10-009 Course Index `https://www.professormesser.com/network-plus/n10-009/n10-009-video/n10-009-training-course/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Observe your own lab VM's DHCP lease and options; diagram the four-message exchange; list two mitigations for rogue DHCP. **2.** Standard (~2 h): Observe your own lab VM's DHCP lease and options; diagram the four-message exchange; list two mitigations for rogue DHCP. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Observe your own lab VM's DHCP lease and options; diagram the four-message exchange; list two mitigations for rogue DHCP. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	dhcp-lease.txt and DORA diagram; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why are DHCP options security-relevant beyond assigning an IP address? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Observed lease fields are identified and mitigations are technically plausible Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer N10-009 objective on DHCP; use the exact objective video title in the course index"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer N10-009 Course Index; write a five-line correction; retest next morning with a new 10-minute problem.

D044 · M02W3D2 · HTTP, HTTPS and TLS at the network layer

Phase: Academic month · **Optional date:** 2026-12-09 · **Domain resources:** F04 / C13

1. Day / sequence	Global Day 44 of 365; calendar-free M02W3D2; optional date 2026-12-09.
2. Exact topic	HTTP, HTTPS and TLS at the network layer
3. Exact subtopics	HTTP over TCP, TLS handshake, certificates, ALPN, HTTPS port 443, encrypted versus visible metadata, HSTS
4. Learning objective	Explain what TLS protects and what metadata remains visible
5. Primary lecture / reading + direct URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/` **Type:** University course · **Access:** Free OCW; paid edX verified certificate optional
6. Exact section / lesson locator	CS50 Cybersecurity Week 2 Securing Systems, HTTPS/TLS/certificates notes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RFC 8446: TLS 1.3 `https://www.rfc-editor.org/rfc/rfc8446`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Capture a browser connection to an authorized site or a local HTTPS test server; identify ClientHello, certificate, encrypted application data and visible IP/DNS metadata. **2.** Standard (~2 h): Capture a browser connection to an authorized site or a local HTTPS test server; identify ClientHello, certificate, encrypted application data and visible IP/DNS metadata. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Capture a browser connection to an authorized site or a local HTTPS test server; identify ClientHello, certificate, encrypted application data and visible IP/DNS metadata. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	tls-capture.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which parts of HTTPS are visible to a network observer and which are protected? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Capture is correctly annotated and the answer distinguishes encryption from anonymity Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50 Cybersecurity Week 2 Securing Systems, HTTPS/TLS/certificates notes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50's Introduction to Cybersecurity (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D045 · M02W3D3 · Weekly network services practical and retrieval

Phase: Academic month · **Optional date:** 2026-12-10 · **Domain resources:** N01 / N16

1. Day / sequence	Global Day 45 of 365; calendar-free M02W3D3; optional date 2026-12-10.
2. Exact topic	Weekly network services practical and retrieval
3. Exact subtopics	Routing, NAT, TCP, UDP, DNS, DHCP and TLS; 20 subnetting questions; service troubleshooting
4. Learning objective	Diagnose a broken small network using the OSI layer method
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Day 38, DNS, and Day 30, TCP and UDP, for recap. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Subnetting Practice `https://subnettingpractice.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: In Packet Tracer or a VM lab, fix three injected faults and complete 20 timed subnetting questions; write a layer-by-layer diagnosis. **2.** Standard (~2 h): In Packet Tracer or a VM lab, fix three injected faults and complete 20 timed subnetting questions; write a layer-by-layer diagnosis. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): In Packet Tracer or a VM lab, fix three injected faults and complete 20 timed subnetting questions; write a layer-by-layer diagnosis. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M2W2-practical.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which evidence allowed you to eliminate layers 1, 2 and 3 before checking DNS? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All faults are fixed, diagnosis is evidence-based, and subnetting score is at least 80 percent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Day 38, DNS, and Day 30, TCP and UDP, for recap"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D046 · M02W3D4 · Firewalls, proxies, VPNs and segmentation architectures

Phase: Academic month · **Optional date:** 2026-12-11 · **Domain resources:** C01 / C06

1. Day / sequence	Global Day 46 of 365; calendar-free M02W3D4; optional date 2026-12-11.
2. Exact topic	Firewalls, proxies, VPNs and segmentation architectures
3. Exact subtopics	Stateless/stateful rules, default deny, DMZ, forward/reverse proxies, VPN tunnels, microsegmentation and zero-trust concepts
4. Learning objective	Compare controls by what they inspect, where they operate and what they cannot see
5. Primary lecture / reading + direct URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/` **Type:** Course index · **Access:** Free videos; paid notes/practice exams optional
6. Exact section / lesson locator	Professor Messer SY0-701 Security+ objective 3.2 and 4.4 videos on network security and segmentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw three architectures for a small fictional organization and write a rule table for each; identify one bypass condition. **2.** Standard (~2 h): Draw three architectures for a small fictional organization and write a rule table for each; identify one bypass condition. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw three architectures for a small fictional organization and write a rule table for each; identify one bypass condition. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	firewall-architectures.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does a VPN not automatically make a compromised laptop trustworthy? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each control's scope and limitation is explicit and rule tables are internally consistent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer SY0-701 Security+ objective 3.2 and 4.4 videos on network security and segmentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer SY0-701 Security+ official playlist; write a five-line correction; retest next morning with a new 10-minute problem.

D047 · M02W3D5 · Wireshark display filters and protocol dissection

Phase: Academic month · **Optional date:** 2026-12-12 · **Domain resources:** N06 / N07

1. Day / sequence	Global Day 47 of 365; calendar-free M02W3D5; optional date 2026-12-12.
2. Exact topic	Wireshark display filters and protocol dissection
3. Exact subtopics	Capture filters versus display filters, conversations, endpoints, Follow TCP Stream, protocol hierarchy and expert info
4. Learning objective	Use Wireshark to answer questions from a legal sample capture without guessing
5. Primary lecture / reading + direct URL	Wireshark User's Guide `https://www.wireshark.org/docs/wsug_html_chunked/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Wireshark User's Guide, Chapters 4-6: capturing, display filters and packet dissection. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Wireshark SampleCaptures `https://wiki.wireshark.org/SampleCaptures`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Open a Wireshark sample HTTP/DNS capture; list top talkers, identify request/response pairs and export one structured field table. **2.** Standard (~2 h): Open a Wireshark sample HTTP/DNS capture; list top talkers, identify request/response pairs and export one structured field table. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Open a Wireshark sample HTTP/DNS capture; list top talkers, identify request/response pairs and export one structured field table. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	wireshark-M2D17.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which display filter isolates DNS responses and how is it different from a capture filter? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All questions are answered with packet-level evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Wireshark User's Guide, Chapters 4-6: capturing, display filters and packet dissection"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Wireshark SampleCaptures; write a five-line correction; retest next morning with a new 10-minute problem.

D048 · M02W3D6 · Nmap host discovery and port scanning in an isolated lab

Phase: Academic month · **Optional date:** 2026-12-13 · **Domain resources:** N09 / N08

1. Day / sequence	Global Day 48 of 365; calendar-free M02W3D6; optional date 2026-12-13.
2. Exact topic	Nmap host discovery and port scanning in an isolated lab
3. Exact subtopics	Written scope, target specification, host discovery, TCP SYN scan, connect scan, version detection, timing and output formats
4. Learning objective	Interpret scan results and explain how each probe appears on the wire
5. Primary lecture / reading + direct URL	Nmap Reference Guide `https://nmap.org/book/man.html` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Nmap Reference Guide, host discovery, port scanning techniques and service/version detection sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	tcpdump manual page `https://www.tcpdump.org/manpages/tcpdump.1.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Scan only a dedicated lab VM you own; save normal, XML and grepable output; capture packets during a small scan and map probes to results. **2.** Standard (~2 h): Scan only a dedicated lab VM you own; save normal, XML and grepable output; capture packets during a small scan and map probes to results. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Scan only a dedicated lab VM you own; save normal, XML and grepable output; capture packets during a small scan and map probes to results. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	scope.txt, nmap outputs and packet capture; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a closed port reply RST while a filtered port produce no response? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Written scope exists, targets are private/isolated, and results are reconciled with the capture Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Nmap Reference Guide, host discovery, port scanning techniques and service/version detection sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RFC 9293: Transmission Control Protocol; write a five-line correction; retest next morning with a new 10-minute problem.

D049 · M02W3D7 · TCP retransmissions, MTU and fragmentation analysis

Phase: Academic month · **Optional date:** 2026-12-14 · **Domain resources:** N10 / N07 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 49 of 365; calendar-free M02W3D7; optional date 2026-12-14.
2. Exact topic	TCP retransmissions, MTU and fragmentation analysis
3. Exact subtopics	Retransmission, duplicate ACK, window size, MTU, MSS, fragmentation flags and Path MTU Discovery
4. Learning objective	Diagnose slow or broken connections from packet-level symptoms
5. Primary lecture / reading + direct URL	RFC 9293: Transmission Control Protocol `https://www.rfc-editor.org/rfc/rfc9293` **Type:** Authoritative standard · **Access:** Free
6. Exact section / lesson locator	RFC 9293 sections on retransmission and flow control; use Wireshark's expert info alongside the standard. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Wireshark SampleCaptures `https://wiki.wireshark.org/SampleCaptures`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Analyze a provided sample capture with retransmissions; calculate one round-trip estimate and identify fragmentation or MTU clues — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Analyze a provided sample capture with retransmissions; calculate one round-trip estimate and identify fragmentation or MTU clues — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Analyze a provided sample capture with retransmissions; calculate one round-trip estimate and identify fragmentation or MTU clues — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	retransmission-analysis.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: How would you distinguish congestion loss from an MTU black hole using packet evidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Symptoms are supported by specific frames and confidence limits are stated Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "RFC 9293 sections on retransmission and flow control; use Wireshark's expert info alongside the standard"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Suricata documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 02 · Week 4 · daily cards

D050 · M02W4D1 · Network troubleshooting methodology and tool selection

Phase: Academic month · **Optional date:** 2026-12-15 · **Domain resources:** N05 / N08

1. Day / sequence	Global Day 50 of 365; calendar-free M02W4D1; optional date 2026-12-15.
2. Exact topic	Network troubleshooting methodology and tool selection
3. Exact subtopics	Layered hypothesis testing, ipconfig/ip/ifconfig, ping, traceroute, ss/netstat, dig, curl, logs, baseline and rollback
4. Learning objective	Resolve an unknown network issue with minimal changes and documented evidence
5. Primary lecture / reading + direct URL	Professor Messer N10-009 Course Index `https://www.professormesser.com/network-plus/n10-009/n10-009-video/n10-009-training-course/` **Type:** Course index · **Access:** Free
6. Exact section / lesson locator	Professor Messer N10-009 course index, Network Troubleshooting objective videos. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	tcpdump manual page `https://www.tcpdump.org/manpages/tcpdump.1.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Diagnose three instructor-created or self-created faults in the isolated lab; record command, prediction, result and conclusion for each step. **2.** Standard (~2 h): Diagnose three instructor-created or self-created faults in the isolated lab; record command, prediction, result and conclusion for each step. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Diagnose three instructor-created or self-created faults in the isolated lab; record command, prediction, result and conclusion for each step. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	troubleshooting-M2D20.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which observation changed your hypothesis and what alternative did it eliminate? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All faults are resolved and each step includes a prediction and result Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer N10-009 course index, Network Troubleshooting objective videos"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer N10-009 Network+ Training Course; write a five-line correction; retest next morning with a new 10-minute problem.

D051 · M02W4D2 · Network-security project: topology and traffic report

Phase: Academic month · **Optional date:** 2026-12-16 · **Domain resources:** N06 / N09

1. Day / sequence	Global Day 51 of 365; calendar-free M02W4D2; optional date 2026-12-16.
2. Exact topic	Network-security project: topology and traffic report
3. Exact subtopics	Asset inventory, trust boundaries, traffic flows, segmentation, capture evidence, risks and mitigations
4. Learning objective	Produce a professional network diagram and traffic-analysis report
5. Primary lecture / reading + direct URL	Wireshark User's Guide `https://www.wireshark.org/docs/wsug_html_chunked/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Wireshark User's Guide statistics and protocol hierarchy chapters. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Nmap Reference Guide `https://nmap.org/book/man.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Combine your lab topology, one packet capture, one authorized scan and a segmentation proposal into a single report. **2.** Standard (~2 h): Combine your lab topology, one packet capture, one authorized scan and a segmentation proposal into a single report. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Combine your lab topology, one packet capture, one authorized scan and a segmentation proposal into a single report. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-03 and PRJ-04 evidence package; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which traffic flow is riskiest and what control would reduce it without breaking function? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram, capture, scan and report are consistent and all identifiers are sanitized Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Wireshark User's Guide statistics and protocol hierarchy chapters"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Wireshark SampleCaptures; write a five-line correction; retest next morning with a new 10-minute problem.

D052 · M02W4D3 · Monthly network practical and daily subnetting test

Phase: Academic month · **Optional date:** 2026-12-17 · **Domain resources:** N01 / N16

1. Day / sequence	Global Day 52 of 365; calendar-free M02W4D3; optional date 2026-12-17.
2. Exact topic	Monthly network practical and daily subnetting test
3. Exact subtopics	OSI/TCPIP, Ethernet, ARP, VLANs, IPv4/CIDR, routing, NAT, TCP/UDP, DNS/DHCP/TLS, Wireshark, Nmap and troubleshooting
4. Learning objective	Demonstrate independent networking competence under timed conditions
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Course Index Days 1-30 for targeted revision; use exact Day titles only. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Subnetting Practice `https://subnettingpractice.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 90-minute practical with 20 subnetting questions, one unfamiliar capture, one broken topology and one scope assessment. **2.** Standard (~2 h): Complete a 90-minute practical with 20 subnetting questions, one unfamiliar capture, one broken topology and one scope assessment. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 90-minute practical with 20 subnetting questions, one unfamiliar capture, one broken topology and one scope assessment. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M2-practical.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How would you prove a DNS issue rather than a routing issue to a skeptical teammate? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, practical at least 18/25 and safety at least 9/10 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Course Index Days 1-30 for targeted revision; use exact Day titles only"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D053 · M02W4D4 · Monthly review and gap repairs

Phase: Academic month · **Optional date:** 2026-12-18 · **Domain resources:** N04 / N17

1. Day / sequence	Global Day 53 of 365; calendar-free M02W4D4; optional date 2026-12-18.
2. Exact topic	Monthly review and gap repairs
3. Exact subtopics	Error-log analysis, subnetting weak types, packet-analysis mistakes, firewall misconceptions and project fixes
4. Learning objective	Close Month 2 gaps and produce a prerequisites checklist for Month 3
5. Primary lecture / reading + direct URL	Professor Messer N10-009 Network+ Training Course `https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8` **Type:** YouTube playlist · **Access:** Free
6. Exact section / lesson locator	Professor Messer N10-009 Course Index, objectives 1-5; choose only the three weakest objectives. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Practical Networking Subnetting Mastery `https://www.practicalnetworking.net/stand-alone/subnetting-mastery/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework every failed assessment item with a new example; revisit the matching video; retake 20 subnetting questions; update project README. **2.** Standard (~2 h): Rework every failed assessment item with a new example; revisit the matching video; retake 20 subnetting questions; update project README. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework every failed assessment item with a new example; revisit the matching video; retake 20 subnetting questions; update project README. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M2-review.md and retest score; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which three network concepts are prerequisites for Linux and Windows security? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and each original gap has new evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer N10-009 Course Index, objectives 1-5; choose only the three weakest objectives"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer N10-009 Course Index; write a five-line correction; retest next morning with a new 10-minute problem.

D054 · M02W4D5 · IPv6 addressing and protocol differences

Phase: Academic month · **Optional date:** 2026-12-19 · **Domain resources:** N01 / N12

1. Day / sequence	Global Day 54 of 365; calendar-free M02W4D5; optional date 2026-12-19.
2. Exact topic	IPv6 addressing and protocol differences
3. Exact subtopics	IPv6 address notation, prefix, global/unicast/link-local, EUI-64/privacy addresses, ICMPv6, neighbour discovery and extension headers
4. Learning objective	Compare IPv4 and IPv6 address configuration and security assumptions
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Days 31-33, IPv6 Parts 1-3. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RFC 8200: Internet Protocol v6 `https://www.rfc-editor.org/rfc/rfc8200`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write ten IPv6 addresses in compressed and expanded form, configure a lab IPv6 address if supported, and compare an IPv6 header with IPv4. **2.** Standard (~2 h): Write ten IPv6 addresses in compressed and expanded form, configure a lab IPv6 address if supported, and compare an IPv6 header with IPv4. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write ten IPv6 addresses in compressed and expanded form, configure a lab IPv6 address if supported, and compare an IPv6 header with IPv4. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ipv6-notes.md and optional Packet Tracer file; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does IPv6 still need firewall policy despite its larger address space? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Address conversions are correct and the security answer is not based on address obscurity Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Days 31-33, IPv6 Parts 1-3"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D055 · M02W4D6 · Wireless LANs, network monitoring and SPAN concepts

Phase: Academic month · **Optional date:** 2026-12-20 · **Domain resources:** N01 / N06

1. Day / sequence	Global Day 55 of 365; calendar-free M02W4D6; optional date 2026-12-20.
2. Exact topic	Wireless LANs, network monitoring and SPAN concepts
3. Exact subtopics	802.11 association, SSID/BSSID, WPA2/WPA3 concepts, channel, switch port mirroring/SPAN, taps and flow records
4. Learning objective	Explain where to place a sensor and what it can and cannot observe
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA wireless lectures in the Day 50+ sequence; use the course index to locate current Day numbers. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Wireshark User's Guide `https://www.wireshark.org/docs/wsug_html_chunked/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw two monitoring placements for a small network, list visible/invisible traffic for each, and state privacy/authorization constraints. **2.** Standard (~2 h): Draw two monitoring placements for a small network, list visible/invisible traffic for each, and state privacy/authorization constraints. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw two monitoring placements for a small network, list visible/invisible traffic for each, and state privacy/authorization constraints. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	monitoring-design.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does encrypted traffic still yield useful metadata for defenders? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Design correctly addresses span, taps, encryption visibility and legal authorization Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA wireless lectures in the Day 50+ sequence; use the course index to locate current Day numbers"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D056 · M02W4D7 · Spaced revision, recovery and Month 3 preview

Phase: Academic month · **Optional date:** 2026-12-21 · **Domain resources:** N16 / L01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 56 of 365; calendar-free M02W4D7; optional date 2026-12-21.
2. Exact topic	Spaced revision, recovery and Month 3 preview
3. Exact subtopics	Retrieval practice, command notebook cleanup, OSI mapping, binary/CIDR fluency and lab snapshots
4. Learning objective	Preserve networking knowledge and prepare Linux/Windows system administration
5. Primary lecture / reading + direct URL	Subnetting Practice `https://subnettingpractice.com/` **Type:** Interactive practice · **Access:** Free tier; optional ads/premium features may appear
6. Exact section / lesson locator	Subnetting Practice mixed set; 15-question maintenance drill. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create ten flashcards from errors, clean VM snapshots, install/verify Linux and Windows lab tools, and rest or do a 30-minute light review — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create ten flashcards from errors, clean VM snapshots, install/verify Linux and Windows lab tools, and rest or do a 30-minute light review — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create ten flashcards from errors, clean VM snapshots, install/verify Linux and Windows lab tools, and rest or do a 30-minute light review — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M2-consolidation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Explain one full packet journey from DNS request to TLS session in ten sentences without notes Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Snapshot is clean, flashcards are scheduled, and the explanation is complete Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Subnetting Practice mixed set; 15-question maintenance drill"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Practical Networking Subnetting Mastery; write a five-line correction; retest next morning with a new 10-minute problem.

Month 03 — Linux, Windows, system administration and hardening

Days D057-D084. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Linux filesystem hierarchy and package management D2: Linux users, groups, sudo and file permissions D3: Processes, services and resource monitoring D4: Bash scripting fundamentals and quoting D5: Pipes, redirection and text processing D6: SSH keys, remote administration and secure configuration D7: Weekly Linux review and Bandit independent practice
Week 2	D1: Linux networking and firewall basics D2: Linux logs, auditd concepts and evidence integrity D3: Linux hardening and CIS controls D4: Systemd security and privilege boundaries D5: OverTheWire Bandit intermediate independent lab D6: Weekly Linux practical and troubleshooting D7: Windows architecture, accounts and permissions
Week 3	D1: Windows processes, services and Sysinternals D2: PowerShell fundamentals and safe scripting D3: Windows Event Viewer and event IDs D4: Sysmon installation and process telemetry D5: Active Directory architecture and identity boundaries D6: Weekly Windows and AD review D7: Group Policy and security baselines
Week 4	D1: Endpoint security controls and patch management D2: Windows image and event-log investigation project D3: Linux and Windows audit utility project D4: Month 3 practical examination and milestone gate D5: Monthly review, cleanup and Month 4 preview D6: Kerberos, NTLM and AD authentication flows D7: Spaced revision, recovery and portfolio update

> **Gate:** score the Month 3 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 03 · Week 1 · daily cards

D057 · M03W1D1 · Linux filesystem hierarchy and package management

Phase: Academic month · **Optional date:** 2026-12-22 · **Domain resources:** L01 / F13

1. Day / sequence	Global Day 57 of 365; calendar-free M03W1D1; optional date 2026-12-22.
2. Exact topic	Linux filesystem hierarchy and package management
3. Exact subtopics	FHS, absolute/relative paths, package repositories, apt, dpkg, dependencies and update hygiene
4. Learning objective	Navigate Linux and manage packages from the CLI with a security rationale
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, Filesystem Hierarchy and Package Management sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Ubuntu Desktop Guide `https://help.ubuntu.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: In the Ubuntu VM, locate ten configuration paths, install one benign package, inspect its files, then remove it and verify cleanup. **2.** Standard (~2 h): In the Ubuntu VM, locate ten configuration paths, install one benign package, inspect its files, then remove it and verify cleanup. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): In the Ubuntu VM, locate ten configuration paths, install one benign package, inspect its files, then remove it and verify cleanup. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	linux-inventory.md and package transcript; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should you avoid installing random security tools from untrusted repositories? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All paths are correctly explained and package state is verified before/after Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, Filesystem Hierarchy and Package Management sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D058 · M03W1D2 · Linux users, groups, sudo and file permissions

Phase: Academic month · **Optional date:** 2026-12-23 · **Domain resources:** L01 / L04

1. Day / sequence	Global Day 58 of 365; calendar-free M03W1D2; optional date 2026-12-23.
2. Exact topic	Linux users, groups, sudo and file permissions
3. Exact subtopics	/etc/passwd, /etc/shadow, /etc/group, UID/GID, umask, setuid/setgid, sticky bit, sudoers
4. Learning objective	Audit permissions and apply least privilege safely
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, Users and Groups and Permissions sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux man-pages project `https://man7.org/linux/man-pages/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Find setuid binaries in the VM, explain three of them, create a shared group directory, and check sudo configuration without editing it blindly. **2.** Standard (~2 h): Find setuid binaries in the VM, explain three of them, create a shared group directory, and check sudo configuration without editing it blindly. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Find setuid binaries in the VM, explain three of them, create a shared group directory, and check sudo configuration without editing it blindly. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	linux-permissions.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a world-writable setuid binary dangerous? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Audit is complete and no insecure test permissions remain Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, Users and Groups and Permissions sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D059 · M03W1D3 · Processes, services and resource monitoring

Phase: Academic month · **Optional date:** 2026-12-24 · **Domain resources:** L05 / L04

1. Day / sequence	Global Day 59 of 365; calendar-free M03W1D3; optional date 2026-12-24.
2. Exact topic	Processes, services and resource monitoring
3. Exact subtopics	ps, top, htop if available, nice, kill, systemctl, journalctl, cgroups and service dependencies
4. Learning objective	Explain process trees and control a benign service without harming the host
5. Primary lecture / reading + direct URL	systemd.unit manual `https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	systemd.unit manual, unit dependencies and targets. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux man-pages project `https://man7.org/linux/man-pages/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Start, stop, restart, enable and disable a harmless service; inspect logs and CPU/memory; map parent/child processes. **2.** Standard (~2 h): Start, stop, restart, enable and disable a harmless service; inspect logs and CPU/memory; map parent/child processes. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Start, stop, restart, enable and disable a harmless service; inspect logs and CPU/memory; map parent/child processes. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	linux-processes.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the difference between stopping and disabling a service? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Commands, unit state and logs agree with the written explanation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "systemd.unit manual, unit dependencies and targets"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Linux man-pages project; write a five-line correction; retest next morning with a new 10-minute problem.

D060 · M03W1D4 · Bash scripting fundamentals and quoting

Phase: Academic month · **Optional date:** 2026-12-25 · **Domain resources:** L03 / L01

1. Day / sequence	Global Day 60 of 365; calendar-free M03W1D4; optional date 2026-12-25.
2. Exact topic	Bash scripting fundamentals and quoting
3. Exact subtopics	Shebang, variables, expansion, quoting, exit codes, conditionals, loops, functions, getopts, shellcheck
4. Learning objective	Write a defensive script that fails safely on malformed input
5. Primary lecture / reading + direct URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide` **Type:** Free scripting guide · **Access:** Free
6. Exact section / lesson locator	Greg's Wiki BashGuide, Commands and Arguments, Quoting, Tests and Loops. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create an inventory script that collects hostname, kernel, users and listening ports; handle empty fields and filenames with spaces. **2.** Standard (~2 h): Create an inventory script that collects hostname, kernel, users and listening ports; handle empty fields and filenames with spaces. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create an inventory script that collects hostname, kernel, users and listening ports; handle empty fields and filenames with spaces. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	audit.sh and shellcheck output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does unquoted variable expansion break filenames with spaces and create security bugs? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Script runs with set -euo pipefail where appropriate and passes shellcheck for assigned issues Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Greg's Wiki BashGuide, Commands and Arguments, Quoting, Tests and Loops"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RegexOne; write a five-line correction; retest next morning with a new 10-minute problem.

D061 · M03W1D5 · Pipes, redirection and text processing

Phase: Academic month · **Optional date:** 2026-12-26 · **Domain resources:** L03 / P04

1. Day / sequence	Global Day 61 of 365; calendar-free M03W1D5; optional date 2026-12-26.
2. Exact topic	Pipes, redirection and text processing
3. Exact subtopics	stdin/stdout/stderr, pipes, grep, sed, awk, cut, sort, uniq, tee, regular expressions, process substitution
4. Learning objective	Compose safe command pipelines to analyze synthetic logs
5. Primary lecture / reading + direct URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide` **Type:** Free scripting guide · **Access:** Free
6. Exact section / lesson locator	Greg's Wiki BashGuide, Redirection and Arrays; P04 RegexOne Lessons 1-8. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RegexOne `https://regexone.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a pipeline that counts failed SSH logins by user/IP from a synthetic auth.log and explain each stage. **2.** Standard (~2 h): Build a pipeline that counts failed SSH logins by user/IP from a synthetic auth.log and explain each stage. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a pipeline that counts failed SSH logins by user/IP from a synthetic auth.log and explain each stage. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	log-pipeline.sh and results.txt; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should script input be treated as untrusted even if it came from a local log? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Pipeline output is verified against a hand-calculated sample Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Greg's Wiki BashGuide, Redirection and Arrays; P04 RegexOne Lessons 1-8"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RegexOne; write a five-line correction; retest next morning with a new 10-minute problem.

D062 · M03W1D6 · SSH keys, remote administration and secure configuration

Phase: Academic month · **Optional date:** 2026-12-27 · **Domain resources:** L01 / C08

1. Day / sequence	Global Day 62 of 365; calendar-free M03W1D6; optional date 2026-12-27.
2. Exact topic	SSH keys, remote administration and secure configuration
3. Exact subtopics	SSH protocol roles, host keys, key pairs, authorized_keys, sshd_config basics, known_hosts and MFA concept
4. Learning objective	Set up key-based SSH between two lab VMs without exposing services publicly
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, SSH and Networking sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-63B `https://pages.nist.gov/800-63-4/sp800-63b.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Generate an ed25519 key pair, install the public key, verify host-key fingerprint, disable password login only in a disposable VM, and document rollback. **2.** Standard (~2 h): Generate an ed25519 key pair, install the public key, verify host-key fingerprint, disable password login only in a disposable VM, and document rollback. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Generate an ed25519 key pair, install the public key, verify host-key fingerprint, disable password login only in a disposable VM, and document rollback. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ssh-lab.md and config diffs; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What does a host-key change warning mean and when could it be legitimate? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Key-based login works, host key is verified, and no production or host networking is exposed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, SSH and Networking sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D063 · M03W1D7 · Weekly Linux review and Bandit independent practice

Phase: Academic month · **Optional date:** 2026-12-28 · **Domain resources:** L02 / L01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 63 of 365; calendar-free M03W1D7; optional date 2026-12-28.
2. Exact topic	Weekly Linux review and Bandit independent practice
3. Exact subtopics	Filesystem, permissions, processes, Bash, pipes, SSH and troubleshooting; active recall
4. Learning objective	Demonstrate independent Linux problem-solving without walkthroughs
5. Primary lecture / reading + direct URL	OverTheWire Bandit `https://overthewire.org/wargames/bandit/` **Type:** CTF / command-line lab · **Access:** Free
6. Exact section / lesson locator	OverTheWire Bandit Levels 0-10; use only the official level pages and man pages for hints. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Solve at least six Bandit levels independently, write concept notes rather than passwords, and complete a 15-question retrieval quiz — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Solve at least six Bandit levels independently, write concept notes rather than passwords, and complete a 15-question retrieval quiz — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Solve at least six Bandit levels independently, write concept notes rather than passwords, and complete a 15-question retrieval quiz — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	bandit-notes.md and weekly-review-M3W1.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which command revealed the crucial fact in each level and why did it work? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Six levels are solved with original notes and quiz score is at least 70 percent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OverTheWire Bandit Levels 0-10; use only the official level pages and man pages for hints"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Linux Journey; write a five-line correction; retest next morning with a new 10-minute problem.

Month 03 · Week 2 · daily cards

D064 · M03W2D1 · Linux networking and firewall basics

Phase: Academic month · **Optional date:** 2026-12-29 · **Domain resources:** L01 / N08

1. Day / sequence	Global Day 64 of 365; calendar-free M03W2D1; optional date 2026-12-29.
2. Exact topic	Linux networking and firewall basics
3. Exact subtopics	ip addr/route/link, ss, resolvectl, ufw/nftables concepts, /etc/hosts, DNS resolution and loopback
4. Learning objective	Inspect and control local Linux network state safely
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey, Networking sections; L04 man pages for ip and ss. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	tcpdump manual page `https://www.tcpdump.org/manpages/tcpdump.1.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Configure a static address on an internal adapter in the VM, inspect routes and listening ports, apply one harmless firewall rule and test connectivity. **2.** Standard (~2 h): Configure a static address on an internal adapter in the VM, inspect routes and listening ports, apply one harmless firewall rule and test connectivity. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Configure a static address on an internal adapter in the VM, inspect routes and listening ports, apply one harmless firewall rule and test connectivity. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	linux-network.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a service be running but unreachable even when the firewall allows the port? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Network state, process state and connectivity evidence are all consistent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey, Networking sections; L04 man pages for ip and ss"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D065 · M03W2D2 · Linux logs, auditd concepts and evidence integrity

Phase: Academic month · **Optional date:** 2026-12-30 · **Domain resources:** L09 / L05

1. Day / sequence	Global Day 65 of 365; calendar-free M03W2D2; optional date 2026-12-30.
2. Exact topic	Linux logs, auditd concepts and evidence integrity
3. Exact subtopics	/var/log, journald, auth.log, syslog, log rotation, timestamps, time zones, hashes and chain of custody
4. Learning objective	Collect and preserve Linux logs without contaminating evidence
5. Primary lecture / reading + direct URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	Windows Event Log documentation for comparison; L04 man pages for journalctl and logger. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	systemd.unit manual `https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Generate benign failed logins, locate them in two log sources, hash the exported evidence and write an evidence-handling note. **2.** Standard (~2 h): Generate benign failed logins, locate them in two log sources, hash the exported evidence and write an evidence-handling note. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Generate benign failed logins, locate them in two log sources, hash the exported evidence and write an evidence-handling note. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	linux-log-evidence/ with SHA-256 manifest; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is time synchronization important for correlating logs from two hosts? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Exported logs are unchanged per hash and the note includes source, time and collector Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Windows Event Log documentation for comparison; L04 man pages for journalctl and logger"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Sysmon documentation and download; write a five-line correction; retest next morning with a new 10-minute problem.

D066 · M03W2D3 · Linux hardening and CIS controls

Phase: Academic month · **Optional date:** 2026-12-31 · **Domain resources:** L14 / L13

1. Day / sequence	Global Day 66 of 365; calendar-free M03W2D3; optional date 2026-12-31.
2. Exact topic	Linux hardening and CIS controls
3. Exact subtopics	Patch management, service minimization, SSH hardening, password policy, audit, firewall, file integrity and backups
4. Learning objective	Apply and validate five low-risk hardening controls in a snapshot
5. Primary lecture / reading + direct URL	CIS Benchmarks `https://www.cisecurity.org/cis-benchmarks` **Type:** Consensus benchmarks · **Access:** Free PDFs may require registration; some tools paid
6. Exact section / lesson locator	CIS Benchmarks, Ubuntu or Debian profile; select five Level 1 controls. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Windows security baselines `https://learn.microsoft.com/en-us/windows/security/operating-system-security/device-management/windows-security-configuration-framework/windows-security-baselines`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Take a snapshot, record baseline, apply five controls, test functionality, then restore one control and document impact. **2.** Standard (~2 h): Take a snapshot, record baseline, apply five controls, test functionality, then restore one control and document impact. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Take a snapshot, record baseline, apply five controls, test functionality, then restore one control and document impact. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	hardening-baseline.md and before/after evidence; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control created a compatibility problem and how would you communicate that risk? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each control has a test, result and rollback note; VM remains functional Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CIS Benchmarks, Ubuntu or Debian profile; select five Level 1 controls"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Windows security baselines; write a five-line correction; retest next morning with a new 10-minute problem.

D067 · M03W2D4 · Systemd security and privilege boundaries

Phase: Academic month · **Optional date:** 2027-01-01 · **Domain resources:** L05 / L04

1. Day / sequence	Global Day 67 of 365; calendar-free M03W2D4; optional date 2027-01-01.
2. Exact topic	Systemd security and privilege boundaries
3. Exact subtopics	Unit sandboxing, capabilities, namespaces, DynamicUser, ProtectSystem, NoNewPrivileges and cgroup limits
4. Learning objective	Explain how service privileges can be reduced without rewriting an application
5. Primary lecture / reading + direct URL	systemd.unit manual `https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	systemd.unit manual and systemd.exec security options; use current manual search for exact directives. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux man-pages project `https://man7.org/linux/man-pages/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a benign unit that writes to one directory, add three sandbox directives, test failure/success, and explain each restriction. **2.** Standard (~2 h): Create a benign unit that writes to one directory, add three sandbox directives, test failure/success, and explain each restriction. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a benign unit that writes to one directory, add three sandbox directives, test failure/success, and explain each restriction. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	systemd-hardening.service and test log; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a file write fail even when traditional Unix permissions appear correct? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Service works under the intended policy and failures are explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "systemd.unit manual and systemd.exec security options; use current manual search for exact directives"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Linux man-pages project; write a five-line correction; retest next morning with a new 10-minute problem.

D068 · M03W2D5 · OverTheWire Bandit intermediate independent lab

Phase: Academic month · **Optional date:** 2027-01-02 · **Domain resources:** L02 / L03

1. Day / sequence	Global Day 68 of 365; calendar-free M03W2D5; optional date 2027-01-02.
2. Exact topic	OverTheWire Bandit intermediate independent lab
3. Exact subtopics	Pipes, redirection, permissions, hidden files, encoding, compression and shell escapes
4. Learning objective	Solve unfamiliar command-line puzzles by reading man pages rather than following a walkthrough
5. Primary lecture / reading + direct URL	OverTheWire Bandit `https://overthewire.org/wargames/bandit/` **Type:** CTF / command-line lab · **Access:** Free
6. Exact section / lesson locator	OverTheWire Bandit Levels 11-20; use official level descriptions and local man pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Solve at least six levels, document the concept and exact reasoning for each, and write one alternate command that would also work. **2.** Standard (~2 h): Solve at least six levels, document the concept and exact reasoning for each, and write one alternate command that would also work. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Solve at least six levels, document the concept and exact reasoning for each, and write one alternate command that would also work. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	bandit-intermediate.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What assumption did each level exploit, and how could you verify it locally? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Six levels solved without copied passwords or step-by-step tutorials Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OverTheWire Bandit Levels 11-20; use official level descriptions and local man pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Linux Journey; write a five-line correction; retest next morning with a new 10-minute problem.

D069 · M03W2D6 · Weekly Linux practical and troubleshooting

Phase: Academic month · **Optional date:** 2027-01-03 · **Domain resources:** L01 / L03

1. Day / sequence	Global Day 69 of 365; calendar-free M03W2D6; optional date 2027-01-03.
2. Exact topic	Weekly Linux practical and troubleshooting
3. Exact subtopics	Filesystem, users, processes, Bash, networking, logs, hardening, systemd and independent puzzles
4. Learning objective	Demonstrate Linux administration under a timed scenario
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey review sections selected from your error log. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 60-minute practical: create a user, harden a service, parse a log, fix one network issue and produce evidence. **2.** Standard (~2 h): Complete a 60-minute practical: create a user, harden a service, parse a log, fix one network issue and produce evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 60-minute practical: create a user, harden a service, parse a log, fix one network issue and produce evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M3W2-practical.md and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would you check if a service starts manually but fails at boot? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and every failing step has a diagnostic note Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey review sections selected from your error log"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D070 · M03W2D7 · Windows architecture, accounts and permissions

Phase: Academic month · **Optional date:** 2027-01-04 · **Domain resources:** F14 / L07 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 70 of 365; calendar-free M03W2D7; optional date 2027-01-04.
2. Exact topic	Windows architecture, accounts and permissions
3. Exact subtopics	NTFS permissions, UAC, SID, access token, local accounts, groups, registry basics and process integrity
4. Learning objective	Compare Windows identity and access control with Linux equivalents
5. Primary lecture / reading + direct URL	Microsoft Learn Windows fundamentals `https://learn.microsoft.com/en-us/training/modules/explore-windows-client/` **Type:** Official learning hub · **Access:** Free learning content; exams separate
6. Exact section / lesson locator	Microsoft Learn, Explore the Windows client module and Windows architecture module. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Windows Server documentation `https://learn.microsoft.com/en-us/windows-server/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — In a Windows VM, inspect a local user, group, file ACL and process token; build a comparison table of Linux and Windows concepts — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — In a Windows VM, inspect a local user, group, file ACL and process token; build a comparison table of Linux and Windows concepts — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — In a Windows VM, inspect a local user, group, file ACL and process token; build a comparison table of Linux and Windows concepts — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	windows-identity.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Why is a SID more stable than a username for access control? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Table is accurate and observations come from the lab VM Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft Learn, Explore the Windows client module and Windows architecture module"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PowerShell documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 03 · Week 3 · daily cards

D071 · M03W3D1 · Windows processes, services and Sysinternals

Phase: Academic month · **Optional date:** 2027-01-05 · **Domain resources:** L10 / L08

1. Day / sequence	Global Day 71 of 365; calendar-free M03W3D1; optional date 2027-01-05.
2. Exact topic	Windows processes, services and Sysinternals
3. Exact subtopics	Process tree, handles, DLLs, services, scheduled tasks, autoruns and Process Explorer/Process Monitor
4. Learning objective	Investigate a benign process launch and identify its parent, modules and file activity
5. Primary lecture / reading + direct URL	Microsoft Sysinternals `https://learn.microsoft.com/en-us/sysinternals/` **Type:** Official tools/docs · **Access:** Free
6. Exact section / lesson locator	Microsoft Sysinternals documentation for Process Explorer, Process Monitor and Autoruns. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	PowerShell documentation `https://learn.microsoft.com/en-us/powershell/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Launch notepad or another benign app, inspect process properties, capture a short ProcMon trace, and explain three event classes. **2.** Standard (~2 h): Launch notepad or another benign app, inspect process properties, capture a short ProcMon trace, and explain three event classes. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Launch notepad or another benign app, inspect process properties, capture a short ProcMon trace, and explain three event classes. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	windows-process-trace.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How can a process with a legitimate name still be suspicious? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Process tree and trace are consistent and no unrelated personal data is captured Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft Sysinternals documentation for Process Explorer, Process Monitor and Autoruns"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PowerShell documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D072 · M03W3D2 · PowerShell fundamentals and safe scripting

Phase: Academic month · **Optional date:** 2027-01-06 · **Domain resources:** L08 / L03

1. Day / sequence	Global Day 72 of 365; calendar-free M03W3D2; optional date 2027-01-06.
2. Exact topic	PowerShell fundamentals and safe scripting
3. Exact subtopics	Cmdlets, pipeline, objects versus text, Get-Help, variables, execution policy, remoting basics and encoding
4. Learning objective	Write a benign Windows inventory script using PowerShell objects
5. Primary lecture / reading + direct URL	PowerShell documentation `https://learn.microsoft.com/en-us/powershell/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	PowerShell documentation, getting started and scripting sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run Get-Process, Get-Service and Get-LocalUser; export structured JSON/CSV; write a script that handles empty results and logs errors. **2.** Standard (~2 h): Run Get-Process, Get-Service and Get-LocalUser; export structured JSON/CSV; write a script that handles empty results and logs errors. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run Get-Process, Get-Service and Get-LocalUser; export structured JSON/CSV; write a script that handles empty results and logs errors. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	windows-inventory.ps1 and sample output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why are PowerShell pipeline objects different from Bash text streams? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Script works, output is structured, and errors are handled without suppressing them Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PowerShell documentation, getting started and scripting sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Greg's Wiki BashGuide; write a five-line correction; retest next morning with a new 10-minute problem.

D073 · M03W3D3 · Windows Event Viewer and event IDs

Phase: Academic month · **Optional date:** 2027-01-07 · **Domain resources:** L09 / L11

1. Day / sequence	Global Day 73 of 365; calendar-free M03W3D3; optional date 2027-01-07.
2. Exact topic	Windows Event Viewer and event IDs
3. Exact subtopics	Channels, providers, levels, EventID, RecordId, Security log, System log, XPath queries and EVTX export
4. Learning objective	Find and interpret security-relevant Windows events from your own lab
5. Primary lecture / reading + direct URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	Windows Event Log documentation, event records and queries. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Sysmon documentation and download `https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Generate logon success/failure and service start events, locate them in Event Viewer, export a sanitized EVTX and record key fields. **2.** Standard (~2 h): Generate logon success/failure and service start events, locate them in Event Viewer, export a sanitized EVTX and record key fields. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Generate logon success/failure and service start events, locate them in Event Viewer, export a sanitized EVTX and record key fields. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	windows-events.csv and sanitization note; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can the same EventID require different interpretation depending on logon type or subject? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Events are found, fields are labelled and private identifiers are sanitized Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Windows Event Log documentation, event records and queries"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Sysmon documentation and download; write a five-line correction; retest next morning with a new 10-minute problem.

D074 · M03W3D4 · Sysmon installation and process telemetry

Phase: Academic month · **Optional date:** 2027-01-08 · **Domain resources:** L11 / L09

1. Day / sequence	Global Day 74 of 365; calendar-free M03W3D4; optional date 2027-01-08.
2. Exact topic	Sysmon installation and process telemetry
3. Exact subtopics	Sysmon event IDs, configuration, process creation, network connection, image loads and hashing in a disposable VM
4. Learning objective	Collect high-value endpoint telemetry and understand its privacy implications
5. Primary lecture / reading + direct URL	Sysmon documentation and download `https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon` **Type:** Official tool/doc · **Access:** Free
6. Exact section / lesson locator	Sysmon documentation and event ID schema. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Install Sysmon with a minimal official/sample config in a disposable VM, generate a benign process and network connection, and export two events. **2.** Standard (~2 h): Install Sysmon with a minimal official/sample config in a disposable VM, generate a benign process and network connection, and export two events. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Install Sysmon with a minimal official/sample config in a disposable VM, generate a benign process and network connection, and export two events. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	sysmon-config.xml and event samples; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which Sysmon field would you use to distinguish a renamed executable from a legitimate one? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Sysmon events appear and the answer includes image, hash, command line or parent process Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sysmon documentation and event ID schema"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Windows Event Log documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D075 · M03W3D5 · Active Directory architecture and identity boundaries

Phase: Academic month · **Optional date:** 2027-01-09 · **Domain resources:** L12 / L07

1. Day / sequence	Global Day 75 of 365; calendar-free M03W3D5; optional date 2027-01-09.
2. Exact topic	Active Directory architecture and identity boundaries
3. Exact subtopics	Forest, domain, OU, domain controller, DNS, LDAP, Kerberos, users, groups, GPO, trusts and tiering concept
4. Learning objective	Diagram an AD environment and explain authentication versus authorization boundaries
5. Primary lecture / reading + direct URL	Active Directory Domain Services overview `https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Active Directory Domain Services overview. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Windows Server documentation `https://learn.microsoft.com/en-us/windows-server/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a detailed diagram of a fictional university domain with OUs, groups, GPOs and trust boundaries; do not deploy AD if hardware is insufficient. **2.** Standard (~2 h): Build a detailed diagram of a fictional university domain with OUs, groups, GPOs and trust boundaries; do not deploy AD if hardware is insufficient. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a detailed diagram of a fictional university domain with OUs, groups, GPOs and trust boundaries; do not deploy AD if hardware is insufficient. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ad-architecture.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should administrative accounts not be used for ordinary email and web browsing? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram correctly separates identity, administration and resource boundaries Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Active Directory Domain Services overview"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Windows Server documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D076 · M03W3D6 · Weekly Windows and AD review

Phase: Academic month · **Optional date:** 2027-01-10 · **Domain resources:** L08 / L12

1. Day / sequence	Global Day 76 of 365; calendar-free M03W3D6; optional date 2027-01-10.
2. Exact topic	Weekly Windows and AD review
3. Exact subtopics	Windows architecture, permissions, processes, PowerShell, events, Sysmon and AD; retrieval practice
4. Learning objective	Consolidate Month 3 Week 3 and prepare endpoint hardening
5. Primary lecture / reading + direct URL	PowerShell documentation `https://learn.microsoft.com/en-us/powershell/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	PowerShell documentation, cmdlet reference and remoting overview. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Active Directory Domain Services overview `https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 20 retrieval questions, explain five Windows event fields from memory, and map one synthetic attack path to AD trust boundaries. **2.** Standard (~2 h): Complete 20 retrieval questions, explain five Windows event fields from memory, and map one synthetic attack path to AD trust boundaries. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 20 retrieval questions, explain five Windows event fields from memory, and map one synthetic attack path to AD trust boundaries. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M3W3-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which single log source would you add first to improve endpoint visibility and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70 percent and AD diagram passes peer-style self-review Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PowerShell documentation, cmdlet reference and remoting overview"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Greg's Wiki BashGuide; write a five-line correction; retest next morning with a new 10-minute problem.

D077 · M03W3D7 · Group Policy and security baselines

Phase: Academic month · **Optional date:** 2027-01-11 · **Domain resources:** L13 / L14 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 77 of 365; calendar-free M03W3D7; optional date 2027-01-11.
2. Exact topic	Group Policy and security baselines
3. Exact subtopics	GPO processing, scope, precedence, security settings, audit policy, baseline packages and policy analyzer
4. Learning objective	Apply a safe lab GPO or, without a domain, interpret a baseline and predict its effect
5. Primary lecture / reading + direct URL	Windows security baselines `https://learn.microsoft.com/en-us/windows/security/operating-system-security/device-management/windows-security-configuration-framework/windows-security-baselines` **Type:** Official configuration guidance · **Access:** Free
6. Exact section / lesson locator	Windows security baselines, deployment and analyzer guidance. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CIS Benchmarks `https://www.cisecurity.org/cis-benchmarks`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a GPO plan for password policy, audit policy and service restriction; test one setting locally if no domain controller exists — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a GPO plan for password policy, audit policy and service restriction; test one setting locally if no domain controller exists — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a GPO plan for password policy, audit policy and service restriction; test one setting locally if no domain controller exists — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	gpo-plan.md and before/after screenshots; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Why can a GPO setting fail to apply even though it is configured at the domain level? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan includes scope, precedence, validation and rollback Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Windows security baselines, deployment and analyzer guidance"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Benchmarks; write a five-line correction; retest next morning with a new 10-minute problem.

Month 03 · Week 4 · daily cards

D078 · M03W4D1 · Endpoint security controls and patch management

Phase: Academic month · **Optional date:** 2027-01-12 · **Domain resources:** C01 / C21

1. Day / sequence	Global Day 78 of 365; calendar-free M03W4D1; optional date 2027-01-12.
2. Exact topic	Endpoint security controls and patch management
3. Exact subtopics	Defender/antivirus concepts, EDR/telemetry, attack surface reduction, application control, patch rings and vulnerability remediation
4. Learning objective	Design a small endpoint security policy with measurable validation tests
5. Primary lecture / reading + direct URL	Professor Messer SY0-701 Security+ Course Index https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/ **Type:** Course index · **Access:** Free videos; paid notes/practice exams optional
6. Exact section / lesson locator	Professor Messer SY0-701 objective 2 and 4 videos on endpoint security and mitigation techniques. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CIS Controls Version 8 https://www.cisecurity.org/controls/v8
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: On a Windows VM, inspect current protections and update state; create a hardening checklist with ten controls and tests without disabling essential protections. **2.** Standard (~2 h): On a Windows VM, inspect current protections and update state; create a hardening checklist with ten controls and tests without disabling essential protections. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): On a Windows VM, inspect current protections and update state; create a hardening checklist with ten controls and tests without disabling essential protections. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	endpoint-baseline.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is disabling security controls as a routine practice unacceptable even in a lab? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Checklist contains safe controls and no instruction to turn off host protection Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer SY0-701 objective 2 and 4 videos on endpoint security and mitigation techniques"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer SY0-701 Security+ official playlist; write a five-line correction; retest next morning with a new 10-minute problem.

D079 · M03W4D2 · Windows image and event-log investigation project

Phase: Academic month · **Optional date:** 2027-01-13 · **Domain resources:** L09 / B05

1. Day / sequence	Global Day 79 of 365; calendar-free M03W4D2; optional date 2027-01-13.
2. Exact topic	Windows image and event-log investigation project
3. Exact subtopics	Evidence acquisition, timeline, logon events, service events, Sysmon process creation and report writing
4. Learning objective	Investigate a benign simulated incident from provided or self-generated logs
5. Primary lecture / reading + direct URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	Windows Event Log documentation and B05 EVTX sample repository for optional extra samples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SBousseaden EVTX-ATTACK-SAMPLES `https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a timeline of five synthetic events, identify the root cause injected in the dataset, and write an incident report with confidence levels. **2.** Standard (~2 h): Create a timeline of five synthetic events, identify the root cause injected in the dataset, and write an incident report with confidence levels. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a timeline of five synthetic events, identify the root cause injected in the dataset, and write an incident report with confidence levels. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-06 evidence package and incident-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which evidence corroborates your conclusion and what alternative explanation remains? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Timeline is consistent, report states limitations, and all data is sanitized Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Windows Event Log documentation and B05 EVTX sample repository for optional extra samples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Sysmon documentation and download; write a five-line correction; retest next morning with a new 10-minute problem.

D080 · M03W4D3 · Linux and Windows audit utility project

Phase: Academic month · **Optional date:** 2027-01-14 · **Domain resources:** L03 / P05

1. Day / sequence	Global Day 80 of 365; calendar-free M03W4D3; optional date 2027-01-14.
2. Exact topic	Linux and Windows audit utility project
3. Exact subtopics	Inventory, permission checks, service state, patch state, listening ports, log collection and JSON output
4. Learning objective	Build a useful defensive automation tool that works in the student's own lab
5. Primary lecture / reading + direct URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide` **Type:** Free scripting guide · **Access:** Free
6. Exact section / lesson locator	BashGuide functions, tests and quoting sections; L08 for PowerShell objects. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write separate Bash and PowerShell scripts that output a common JSON schema; add tests for empty values and malformed input. **2.** Standard (~2 h): Write separate Bash and PowerShell scripts that output a common JSON schema; add tests for empty values and malformed input. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write separate Bash and PowerShell scripts that output a common JSON schema; add tests for empty values and malformed input. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	audit-tool repository with README and test results; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why must an audit tool avoid executing untrusted file contents? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Both scripts run, tests pass, and no secrets or personal data are in sample output Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "BashGuide functions, tests and quoting sections; L08 for PowerShell objects"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RegexOne; write a five-line correction; retest next morning with a new 10-minute problem.

D081 · M03W4D4 · Month 3 practical examination and milestone gate

Phase: Academic month · **Optional date:** 2027-01-15 · **Domain resources:** L01 / L09

1. Day / sequence	Global Day 81 of 365; calendar-free M03W4D4; optional date 2027-01-15.
2. Exact topic	Month 3 practical examination and milestone gate
3. Exact subtopics	Linux administration, Windows administration, PowerShell, logs, AD architecture, hardening and scripting
4. Learning objective	Demonstrate system-administration competence with independent troubleshooting
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey review sections chosen from the error log. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 120-minute practical: harden Linux and Windows artifacts, interpret logs, write a PowerShell query and explain an AD diagram. **2.** Standard (~2 h): Complete a 120-minute practical: harden Linux and Windows artifacts, interpret logs, write a PowerShell query and explain an AD diagram. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 120-minute practical: harden Linux and Windows artifacts, interpret logs, write a PowerShell query and explain an AD diagram. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M3-assessment.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How would you determine whether an account change was authorized using only available logs? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, milestone gate passed, practical at least 18/25 and safety at least 9/10 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey review sections chosen from the error log"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D082 · M03W4D5 · Monthly review, cleanup and Month 4 preview

Phase: Academic month · **Optional date:** 2027-01-16 · **Domain resources:** L03 / F08

1. Day / sequence	Global Day 82 of 365; calendar-free M03W4D5; optional date 2027-01-16.
2. Exact topic	Monthly review, cleanup and Month 4 preview
3. Exact subtopics	Command notebook, error log, script bugs, AD misconceptions and project limitations
4. Learning objective	Repair Month 3 gaps and prepare programming automation prerequisites
5. Primary lecture / reading + direct URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide` **Type:** Free scripting guide · **Access:** Free
6. Exact section / lesson locator	BashGuide review sections and P04 RegexOne review lessons. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Revisit failed assessment items, refactor two scripts, produce ten flashcards, and install Python/testing tools in a clean virtual environment. **2.** Standard (~2 h): Revisit failed assessment items, refactor two scripts, produce ten flashcards, and install Python/testing tools in a clean virtual environment. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Revisit failed assessment items, refactor two scripts, produce ten flashcards, and install Python/testing tools in a clean virtual environment. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M3-review.md and tools-check.txt; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which programming concept is the biggest bottleneck for your security automation goal? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest passes, lab snapshots are clean, and Python environment is ready Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "BashGuide review sections and P04 RegexOne review lessons"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RegexOne; write a five-line correction; retest next morning with a new 10-minute problem.

D083 · M03W4D6 · Kerberos, NTLM and AD authentication flows

Phase: Academic month · **Optional date:** 2027-01-17 · **Domain resources:** L12 / C18

1. Day / sequence	Global Day 83 of 365; calendar-free M03W4D6; optional date 2027-01-17.
2. Exact topic	Kerberos, NTLM and AD authentication flows
3. Exact subtopics	Ticket-granting ticket, service ticket, KDC, realm, SPN, NTLM challenge-response, golden/silver ticket concept and pass-the-hash concept
4. Learning objective	Explain how AD authentication differs from simply sending a password to each service
5. Primary lecture / reading + direct URL	Active Directory Domain Services overview `https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Active Directory Domain Services overview and Microsoft authentication documentation linked from that page. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a Kerberos login and service-access sequence; identify where a credential or ticket could be stolen and one detection data source. **2.** Standard (~2 h): Draw a Kerberos login and service-access sequence; identify where a credential or ticket could be stolen and one detection data source. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a Kerberos login and service-access sequence; identify where a credential or ticket could be stolen and one detection data source. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	kerberos-flow.svg and ATT&CK mapping; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a stolen ticket useful even if the attacker does not know the plaintext password? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Sequence is correct and detection data sources are specific rather than tool-name lists Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Active Directory Domain Services overview and Microsoft authentication documentation linked from that page"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Windows Server documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D084 · M03W4D7 · Spaced revision, recovery and portfolio update

Phase: Academic month · **Optional date:** 2027-01-18 · **Domain resources:** F09 / L01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 84 of 365; calendar-free M03W4D7; optional date 2027-01-18.
2. Exact topic	Spaced revision, recovery and portfolio update
3. Exact subtopics	Linux and Windows command retrieval, event IDs, permissions and identity boundaries
4. Learning objective	Preserve Month 3 gains and publish sanitized project evidence
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapter 2 for clean commits and tags. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Update PRJ-05 and PRJ-06 READMEs, add limitations and validation tests, tag releases, and do a light 30-minute retrieval session — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Update PRJ-05 and PRJ-06 READMEs, add limitations and validation tests, tag releases, and do a light 30-minute retrieval session — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Update PRJ-05 and PRJ-06 READMEs, add limitations and validation tests, tag releases, and do a light 30-minute retrieval session — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	updated portfolio links and M3-consolidation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which three commands or event fields do you still confuse, and when will you review them? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Repositories contain no secrets and each project includes reproducible instructions Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapter 2 for clean commits and tags"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

Month 04 — Python, scripting, Git and security automation

Days D085-D112. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Python setup, execution model and development workflow D2: Variables, primitive data types and expressions D3: Collections: lists, tuples, sets and dictionaries D4: Control flow, conditions and loops D5: Functions, scope, parameters and return values D6: Files, paths, exceptions and context managers D7: Weekly Python fundamentals review and mini project
Week 2	D1: Regular expressions and defensive parsing D2: Modules, packages, imports and virtual environments D3: CSV, JSON and data normalization D4: HTTP requests, APIs and safe client behavior D5: SQL and parameterized database access D6: Logging, observability and defensive error handling D7: Weekly automation review and debugging practical
Week 3	D1: Bash for security automation D2: PowerShell objects and structured output D3: Git workflows, code review and documentation D4: Testing security utilities and adversarial inputs D5: Weekly automation practical and code quality gate D6: Security audit utility design and requirements D7: Log parser engineering sprint: parser and normalization
Week 4	D1: Log parser engineering sprint: analysis and reporting D2: Secure coding review of the automation tool D3: Bash and PowerShell automation comparison lab D4: Project publication and professional README D5: Month 4 practical assessment and milestone review D6: Monthly review and prerequisite gate for security principles D7: Spaced revision, recovery and Month 5 preview

> **Gate:** score the Month 4 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 04 · Week 1 · daily cards

D085 · M04W1D1 · Python setup, execution model and development workflow

Phase: Academic month · **Optional date:** 2027-01-19 · **Domain resources:** F08 / P01

1. Day / sequence	Global Day 85 of 365; calendar-free M04W1D1; optional date 2027-01-19.
2. Exact topic	Python setup, execution model and development workflow
3. Exact subtopics	Interpreter, REPL, scripts, virtual environments, pip, modules, indentation and syntax errors
4. Learning objective	Set up a clean Python environment and explain how a script runs
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Sections 1-2: Whetting Your Appetite and Using the Python Interpreter. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python for Everybody `https://www.py4e.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a venv, install one harmless library, run three scripts that fail in different ways, and document the tracebacks. **2.** Standard (~2 h): Create a venv, install one harmless library, run three scripts that fail in different ways, and document the tracebacks. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a venv, install one harmless library, run three scripts that fail in different ways, and document the tracebacks. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	python-setup.md and error-log; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What does a traceback tell you about the call stack and the first real error? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Virtual environment is reproducible and all three errors are classified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Sections 1-2: Whetting Your Appetite and Using the Python Interpreter"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D086 · M04W1D2 · Variables, primitive data types and expressions

Phase: Academic month · **Optional date:** 2027-01-20 · **Domain resources:** F08 / P01

1. Day / sequence	Global Day 86 of 365; calendar-free M04W1D2; optional date 2027-01-20.
2. Exact topic	Variables, primitive data types and expressions
3. Exact subtopics	Integers, floats, strings, booleans, None, operators, type conversion, immutability and f-strings
4. Learning objective	Predict types and values of expressions without running code
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Section 3: An Informal Introduction to Python. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python for Everybody `https://www.py4e.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 30 expression-prediction exercises; verify with the interpreter and correct wrong predictions in the error log. **2.** Standard (~2 h): Complete 30 expression-prediction exercises; verify with the interpreter and correct wrong predictions in the error log. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 30 expression-prediction exercises; verify with the interpreter and correct wrong predictions in the error log. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	expressions.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does string concatenation with an integer raise TypeError and how can that prevent silent data corruption? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	At least 85 percent predictions correct and all mistakes explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Section 3: An Informal Introduction to Python"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D087 · M04W1D3 · Collections: lists, tuples, sets and dictionaries

Phase: Academic month · **Optional date:** 2027-01-21 · **Domain resources:** F08 / P02

1. Day / sequence	Global Day 87 of 365; calendar-free M04W1D3; optional date 2027-01-21.
2. Exact topic	Collections: lists, tuples, sets and dictionaries
3. Exact subtopics	Indexing, slicing, mutability, comprehensions, dictionaries, sets, nested structures and copying
4. Learning objective	Choose the appropriate collection for a security-data task
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Sections 3-5: informal introduction, control flow and data structures. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Learn Python - Full Course for Beginners (freeCodeCamp) `https://www.youtube.com/watch?v=eWRfhZUzrAc`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Store ten synthetic events in each collection type, write queries to count unique IPs, and explain shallow versus deep copies. **2.** Standard (~2 h): Store ten synthetic events in each collection type, write queries to count unique IPs, and explain shallow versus deep copies. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Store ten synthetic events in each collection type, write queries to count unique IPs, and explain shallow versus deep copies. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	collections-lab.py; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: When is a set better than a list for IOC lookup, and what information is lost? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All programs run and the collection choices are justified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Sections 3-5: informal introduction, control flow and data structures"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D088 · M04W1D4 · Control flow, conditions and loops

Phase: Academic month · **Optional date:** 2027-01-22 · **Domain resources:** F08 / P02

1. Day / sequence	Global Day 88 of 365; calendar-free M04W1D4; optional date 2027-01-22.
2. Exact topic	Control flow, conditions and loops
3. Exact subtopics	Boolean logic, comparison, if/elif/else, for/while, break/continue, range, enumerate, zip and off-by-one errors
4. Learning objective	Implement a robust loop with explicit termination and edge-case handling
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Section 4: More Control Flow Tools. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Learn Python - Full Course for Beginners (freeCodeCamp) `https://www.youtube.com/watch?v=eWRfhZUzrAc`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a parser that classifies 100 synthetic log lines by severity; include empty lines, unknown severity and very long lines. **2.** Standard (~2 h): Write a parser that classifies 100 synthetic log lines by severity; include empty lines, unknown severity and very long lines. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a parser that classifies 100 synthetic log lines by severity; include empty lines, unknown severity and very long lines. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	classifier.py and test-results.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which loop invariant guarantees that every line is processed exactly once? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All edge cases are handled and output counts match a manually verified sample Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Section 4: More Control Flow Tools"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D089 · M04W1D5 · Functions, scope, parameters and return values

Phase: Academic month · **Optional date:** 2027-01-23 · **Domain resources:** F08 / P05

1. Day / sequence	Global Day 89 of 365; calendar-free M04W1D5; optional date 2027-01-23.
2. Exact topic	Functions, scope, parameters and return values
3. Exact subtopics	Function definitions, arguments, defaults, keyword arguments, return values, local/global scope and docstrings
4. Learning objective	Design reusable functions with clear contracts and no hidden side effects
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Section 4.6-4.8: defining functions, default arguments and coding style. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Refactor the classifier into five small functions; write docstrings and unit tests for normal, malformed and boundary inputs. **2.** Standard (~2 h): Refactor the classifier into five small functions; write docstrings and unit tests for normal, malformed and boundary inputs. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Refactor the classifier into five small functions; write docstrings and unit tests for normal, malformed and boundary inputs. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	functions.py and pytest results; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a mutable default argument cause surprising state persistence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Tests pass and no hidden global state remains Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Section 4.6-4.8: defining functions, default arguments and coding style"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D090 · M04W1D6 · Files, paths, exceptions and context managers

Phase: Academic month · **Optional date:** 2027-01-24 · **Domain resources:** F08 / P03

1. Day / sequence	Global Day 90 of 365; calendar-free M04W1D6; optional date 2027-01-24.
2. Exact topic	Files, paths, exceptions and context managers
3. Exact subtopics	open modes, pathlib, with-statement, encoding, CSV, JSON, exceptions, try/except/else/finally and atomic writes
4. Learning objective	Read and write structured security data without corrupting files
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Sections 7 and 8: input/output and errors/exceptions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Automate the Boring Stuff with Python `https://automatetheboringstuff.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Convert a synthetic auth.log to JSON and CSV, force a UnicodeDecodeError and a JSONDecodeError, then handle both safely. **2.** Standard (~2 h): Convert a synthetic auth.log to JSON and CSV, force a UnicodeDecodeError and a JSONDecodeError, then handle both safely. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Convert a synthetic auth.log to JSON and CSV, force a UnicodeDecodeError and a JSONDecodeError, then handle both safely. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	file-conversion.py and error tests; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should you write to a temporary file and rename it atomically for important outputs? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Conversion works, exceptions are explicit, and no partial output is accepted Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Sections 7 and 8: input/output and errors/exceptions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D091 · M04W1D7 · Weekly Python fundamentals review and mini project

Phase: Academic month · **Optional date:** 2027-01-25 · **Domain resources:** F08 / P05 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 91 of 365; calendar-free M04W1D7; optional date 2027-01-25.
2. Exact topic	Weekly Python fundamentals review and mini project
3. Exact subtopics	Types, collections, conditions, loops, functions, files, exceptions and debugging
4. Learning objective	Build a small independent tool from specifications rather than a tutorial
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Sections 1-8 review pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a CLI tool that reads a synthetic log and prints a summary; include tests, a README and one deliberate bug fixed through debugging — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a CLI tool that reads a synthetic log and prints a summary; include tests, a README and one deliberate bug fixed through debugging — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Create a CLI tool that reads a synthetic log and prints a summary; include tests, a README and one deliberate bug fixed through debugging — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	mini-log-summary repository; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which test caught the most important bug and what did it assume? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 on the monthly rubric and all tests pass Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Sections 1-8 review pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

Month 04 · Week 2 · daily cards

D092 · M04W2D1 · Regular expressions and defensive parsing

Phase: Academic month · **Optional date:** 2027-01-26 · **Domain resources:** P04 / F08

1. Day / sequence	Global Day 92 of 365; calendar-free M04W2D1; optional date 2027-01-26.
2. Exact topic	Regular expressions and defensive parsing
3. Exact subtopics	Character classes, quantifiers, anchors, groups, named groups, lookarounds, greedy/lazy matching and catastrophic backtracking risk
4. Learning objective	Parse untrusted logs with bounded, readable patterns and avoid regex denial of service
5. Primary lecture / reading + direct URL	RegexOne `https://regexone.com/` **Type:** Interactive lessons · **Access:** Free
6. Exact section / lesson locator	RegexOne Lessons 1-15, especially practical examples and backreferences. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write regexes for Apache, SSH and JSON-ish log lines; test against 50 valid and 10 malformed lines; measure one pathological pattern safely. **2.** Standard (~2 h): Write regexes for Apache, SSH and JSON-ish log lines; test against 50 valid and 10 malformed lines; measure one pathological pattern safely. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write regexes for Apache, SSH and JSON-ish log lines; test against 50 valid and 10 malformed lines; measure one pathological pattern safely. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	regex-lab.py and timing notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should a regex for untrusted input avoid nested ambiguous quantifiers? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All valid lines parse, malformed lines fail safely, and regex limitations are documented Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "RegexOne Lessons 1-15, especially practical examples and backreferences"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D093 · M04W2D2 · Modules, packages, imports and virtual environments

Phase: Academic month · **Optional date:** 2027-01-27 · **Domain resources:** F08 / W13

1. Day / sequence	Global Day 93 of 365; calendar-free M04W2D2; optional date 2027-01-27.
2. Exact topic	Modules, packages, imports and virtual environments
3. Exact subtopics	Module search path, __name__, packages, pip, requirements.txt, pyproject.toml, dependency pinning and malicious package risk
4. Learning objective	Structure a small project and explain supply-chain trust boundaries
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Section 6: Modules. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Split the log summary into parser, analyzer and CLI modules; pin dependencies; inspect package metadata and write a dependency-risk note. **2.** Standard (~2 h): Split the log summary into parser, analyzer and CLI modules; pin dependencies; inspect package metadata and write a dependency-risk note. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Split the log summary into parser, analyzer and CLI modules; pin dependencies; inspect package metadata and write a dependency-risk note. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	project-structure.md and requirements.txt; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a package name typo a security issue rather than just a spelling mistake? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Package imports work offline and dependency decisions are documented Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Section 6: Modules"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D094 · M04W2D3 · CSV, JSON and data normalization

Phase: Academic month · **Optional date:** 2027-01-28 · **Domain resources:** P03 / P07

1. Day / sequence	Global Day 94 of 365; calendar-free M04W2D3; optional date 2027-01-28.
2. Exact topic	CSV, JSON and data normalization
3. Exact subtopics	csv module, json module, schemas, nested JSON, data types, timestamp formats, missing fields and canonicalization
4. Learning objective	Normalize three log formats into one common event schema
5. Primary lecture / reading + direct URL	Automate the Boring Stuff with Python `https://automatetheboringstuff.com/` **Type:** Free online book · **Access:** Free online; paid print/course optional
6. Exact section / lesson locator	Automate the Boring Stuff, CSV/JSON chapters. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SQLite documentation `https://www.sqlite.org/docs.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create three synthetic input files with inconsistent fields, normalize them into common JSON, and reject records without required fields. **2.** Standard (~2 h): Create three synthetic input files with inconsistent fields, normalize them into common JSON, and reject records without required fields. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create three synthetic input files with inconsistent fields, normalize them into common JSON, and reject records without required fields. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	normalized-events.json and schema.json; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why must timestamps include a time zone before correlating events? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every output record satisfies the schema and all rejected records have a reason Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Automate the Boring Stuff, CSV/JSON chapters"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D095 · M04W2D4 · HTTP requests, APIs and safe client behavior

Phase: Academic month · **Optional date:** 2027-01-29 · **Domain resources:** P06 / F10

1. Day / sequence	Global Day 95 of 365; calendar-free M04W2D4; optional date 2027-01-29.
2. Exact topic	HTTP requests, APIs and safe client behavior
3. Exact subtopics	requests, GET/POST, status codes, headers, query parameters, timeouts, retries, TLS verification, sessions and rate limits
4. Learning objective	Query an API safely and handle network failures without disabling security controls
5. Primary lecture / reading + direct URL	Requests: HTTP for Humans `https://requests.readthedocs.io/en/latest/` **Type:** Official library docs · **Access:** Free
6. Exact section / lesson locator	Requests Quickstart and Advanced Usage, especially timeouts, SSL verification and sessions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MDN HTTP Overview `https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Send requests to a local mock API; test timeout, 4xx, 5xx, malformed JSON and rate-limit response; implement bounded retry logic. **2.** Standard (~2 h): Send requests to a local mock API; test timeout, 4xx, 5xx, malformed JSON and rate-limit response; implement bounded retry logic. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Send requests to a local mock API; test timeout, 4xx, 5xx, malformed JSON and rate-limit response; implement bounded retry logic. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	api-client.py and test log; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is verify=False dangerous and what is the correct local-lab alternative? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All failure modes are handled and TLS verification remains enabled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Requests Quickstart and Advanced Usage, especially timeouts, SSL verification and sessions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MDN HTTP Overview; write a five-line correction; retest next morning with a new 10-minute problem.

D096 · M04W2D5 · SQL and parameterized database access

Phase: Academic month · **Optional date:** 2027-01-30 · **Domain resources:** P07 / F11

1. Day / sequence	Global Day 96 of 365; calendar-free M04W2D5; optional date 2027-01-30.
2. Exact topic	SQL and parameterized database access
3. Exact subtopics	sqlite3, parameterized queries, transactions, indexes, joins, GROUP BY, injection risk and least-privilege database accounts
4. Learning objective	Store and query security events safely with parameterized SQL
5. Primary lecture / reading + direct URL	SQLite documentation `https://www.sqlite.org/docs.html` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	SQLite documentation, SQL syntax, transactions and security considerations. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SQLBolt `https://sqlbolt.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a database of normalized events, write five analytical queries, and demonstrate that a malicious string cannot alter query structure. **2.** Standard (~2 h): Build a database of normalized events, write five analytical queries, and demonstrate that a malicious string cannot alter query structure. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a database of normalized events, write five analytical queries, and demonstrate that a malicious string cannot alter query structure. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	events.db, queries.sql and injection-test.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does parameterization preserve the SQL parse tree while string concatenation does not? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Queries are correct and injection tests are safely performed on synthetic data Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SQLite documentation, SQL syntax, transactions and security considerations"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SQLBolt; write a five-line correction; retest next morning with a new 10-minute problem.

D097 · M04W2D6 · Logging, observability and defensive error handling

Phase: Academic month · **Optional date:** 2027-01-31 · **Domain resources:** F08 / W17

1. Day / sequence	Global Day 97 of 365; calendar-free M04W2D6; optional date 2027-01-31.
2. Exact topic	Logging, observability and defensive error handling
3. Exact subtopics	logging module, levels, structured logs, correlation IDs, exception logging, secret redaction and avoiding noisy logs
4. Learning objective	Instrument a security utility so failures are diagnosable without leaking secrets
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Standard Library logging HOWTO linked from Python Tutorial Section 8. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Add structured logging and redaction to the log parser; generate debug/info/warning/error events and verify no sample secrets in logs. **2.** Standard (~2 h): Add structured logging and redaction to the log parser; generate debug/info/warning/error events and verify no sample secrets in logs. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Add structured logging and redaction to the log parser; generate debug/info/warning/error events and verify no sample secrets in logs. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	structured-logs.jsonl; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which fields are useful for incident response but unsafe to log without redaction? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Logs are useful for debugging and all synthetic secrets are redacted Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Standard Library logging HOWTO linked from Python Tutorial Section 8"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D098 · M04W2D7 · Weekly automation review and debugging practical

Phase: Academic month · **Optional date:** 2027-02-01 · **Domain resources:** P05 / P03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 98 of 365; calendar-free M04W2D7; optional date 2027-02-01.
2. Exact topic	Weekly automation review and debugging practical
3. Exact subtopics	Modules, files, regex, APIs, SQL, logging and error handling; independent debugging
4. Learning objective	Repair an unfamiliar broken script using a disciplined debugging workflow
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest documentation, getting started and assertion rewriting sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Automate the Boring Stuff with Python `https://automatetheboringstuff.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Instructor/self-created broken parser with five bugs; reproduce, isolate, fix, add regression tests, and write a postmortem — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Instructor/self-created broken parser with five bugs; reproduce, isolate, fix, add regression tests, and write a postmortem — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Instructor/self-created broken parser with five bugs; reproduce, isolate, fix, add regression tests, and write a postmortem — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	debugging-postmortem.md and tests; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which two pieces of evidence narrowed each bug to a single function? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All five bugs are fixed and regression tests pass Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest documentation, getting started and assertion rewriting sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

Month 04 · Week 3 · daily cards

D099 · M04W3D1 · Bash for security automation

Phase: Academic month · **Optional date:** 2027-02-02 · **Domain resources:** L03 / P03

1. Day / sequence	Global Day 99 of 365; calendar-free M04W3D1; optional date 2027-02-02.
2. Exact topic	Bash for security automation
3. Exact subtopics	set -euo pipefail, functions, arrays, traps, getopts, quoting, exit codes, process substitution and safe temporary files
4. Learning objective	Write a robust CLI audit wrapper around the Python tool
5. Primary lecture / reading + direct URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide` **Type:** Free scripting guide · **Access:** Free
6. Exact section / lesson locator	Greg's Wiki BashGuide, quoting, arrays, functions and traps sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Automate the Boring Stuff with Python `https://automatetheboringstuff.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a Bash wrapper that validates arguments, runs the parser in the venv, handles signals and cleans temporary files. **2.** Standard (~2 h): Create a Bash wrapper that validates arguments, runs the parser in the venv, handles signals and cleans temporary files. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a Bash wrapper that validates arguments, runs the parser in the venv, handles signals and cleans temporary files. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	run-audit.sh and test cases; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can an unquoted glob expression delete unintended files? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Wrapper passes normal, empty and malformed argument tests Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Greg's Wiki BashGuide, quoting, arrays, functions and traps sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RegexOne; write a five-line correction; retest next morning with a new 10-minute problem.

D100 · M04W3D2 · PowerShell objects and structured output

Phase: Academic month · **Optional date:** 2027-02-03 · **Domain resources:** L08 / F09

1. Day / sequence	Global Day 100 of 365; calendar-free M04W3D2; optional date 2027-02-03.
2. Exact topic	PowerShell objects and structured output
3. Exact subtopics	Cmdlets, objects, formatting, ConvertTo-Json, error handling, execution policy and safe invocation
4. Learning objective	Build a Windows counterpart and compare its output schema with Bash/Python
5. Primary lecture / reading + direct URL	PowerShell documentation `https://learn.microsoft.com/en-us/powershell/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	PowerShell documentation, scripting, modules and error handling sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a PowerShell inventory script that outputs the same common JSON schema as the Linux audit tool; test on a Windows VM. **2.** Standard (~2 h): Write a PowerShell inventory script that outputs the same common JSON schema as the Linux audit tool; test on a Windows VM. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a PowerShell inventory script that outputs the same common JSON schema as the Linux audit tool; test on a Windows VM. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	windows-inventory.ps1 and sample JSON; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is Export-Csv with objects safer than parsing formatted table text? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Output parses as JSON and field names match the shared schema Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PowerShell documentation, scripting, modules and error handling sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Greg's Wiki BashGuide; write a five-line correction; retest next morning with a new 10-minute problem.

D101 · M04W3D3 · Git workflows, code review and documentation

Phase: Academic month · **Optional date:** 2027-02-04 · **Domain resources:** F09 / W13

1. Day / sequence	Global Day 101 of 365; calendar-free M04W3D3; optional date 2027-02-04.
2. Exact topic	Git workflows, code review and documentation
3. Exact subtopics	Feature branches, diff review, semantic commit messages, issues, pull requests, tags, CHANGELOG and reproducible setup
4. Learning objective	Use version control as professional engineering evidence rather than a file dump
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 2-3 and Chapter 5 distributed Git basics. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a feature branch for the parser, write a pull request to yourself, review the diff, fix two issues, tag v0.1 and write a CHANGELOG. **2.** Standard (~2 h): Create a feature branch for the parser, write a pull request to yourself, review the diff, fix two issues, tag v0.1 and write a CHANGELOG. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a feature branch for the parser, write a pull request to yourself, review the diff, fix two issues, tag v0.1 and write a CHANGELOG. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PR screenshot/diff and CHANGELOG.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What evidence in a commit history shows independent engineering rather than copied code? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Clean history, meaningful commits and reproducible setup instructions Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 2-3 and Chapter 5 distributed Git basics"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D102 · M04W3D4 · Testing security utilities and adversarial inputs

Phase: Academic month · **Optional date:** 2027-02-05 · **Domain resources:** P05 / P04

1. Day / sequence	Global Day 102 of 365; calendar-free M04W3D4; optional date 2027-02-05.
2. Exact topic	Testing security utilities and adversarial inputs
3. Exact subtopics	Unit, integration and property-style tests; fixtures; parametrization; malformed inputs; boundary values; fuzzing concepts and failure injection
4. Learning objective	Prove that a defensive tool fails safely on hostile input
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest documentation, fixtures, parametrization and marks sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RegexOne `https://regexone.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Add tests for empty files, huge lines, Unicode, null bytes, invalid JSON, regex backtracking and missing permissions; use a simple random input generator. **2.** Standard (~2 h): Add tests for empty files, huge lines, Unicode, null bytes, invalid JSON, regex backtracking and missing permissions; use a simple random input generator. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Add tests for empty files, huge lines, Unicode, null bytes, invalid JSON, regex backtracking and missing permissions; use a simple random input generator. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	test-results.xml and fuzz-corpus/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which input caused the worst runtime and how did you bound it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All tests pass, runtime is bounded, and no crashes expose tracebacks to users Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest documentation, fixtures, parametrization and marks sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D103 · M04W3D5 · Weekly automation practical and code quality gate

Phase: Academic month · **Optional date:** 2027-02-06 · **Domain resources:** P05 / F09

1. Day / sequence	Global Day 103 of 365; calendar-free M04W3D5; optional date 2027-02-06.
2. Exact topic	Weekly automation practical and code quality gate
3. Exact subtopics	Python, Bash, PowerShell, Git, APIs, SQL, regex, testing and documentation
4. Learning objective	Deliver a small independent automation tool with tests and code review
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest documentation and P03 Automate the Boring Stuff project chapters. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a CLI that ingests two log formats, normalizes them, stores events, queries top talkers and outputs JSON/CSV; include a README and threat model. **2.** Standard (~2 h): Build a CLI that ingests two log formats, normalizes them, stores events, queries top talkers and outputs JSON/CSV; include a README and threat model. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a CLI that ingests two log formats, normalizes them, stores events, queries top talkers and outputs JSON/CSV; include a README and threat model. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	automation-tool repository and code-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Where could an attacker influence input, and how does your design constrain it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, tests pass, and documentation is sufficient for another student to run it Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest documentation and P03 Automate the Boring Stuff project chapters"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D104 · M04W3D6 · Security audit utility design and requirements

Phase: Academic month · **Optional date:** 2027-02-07 · **Domain resources:** P08 / C21

1. Day / sequence	Global Day 104 of 365; calendar-free M04W3D6; optional date 2027-02-07.
2. Exact topic	Security audit utility design and requirements
3. Exact subtopics	Asset inventory, secure defaults, consent, scope, data minimization, read-only operations, exit codes and reporting
4. Learning objective	Design a safe, useful audit tool before adding features
5. Primary lecture / reading + direct URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/` **Type:** Checklist/reference · **Access:** Free
6. Exact section / lesson locator	OWASP Secure Coding Practices Quick Reference Guide, input validation, error handling and data protection sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CIS Controls Version 8 `https://www.cisecurity.org/controls/v8`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a one-page requirements and threat model for a Linux/Windows audit utility; define what it must never do. **2.** Standard (~2 h): Write a one-page requirements and threat model for a Linux/Windows audit utility; define what it must never do. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a one-page requirements and threat model for a Linux/Windows audit utility; define what it must never do. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	audit-tool-requirements.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which feature would create unacceptable risk and how would you redesign it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Requirements are testable and explicitly forbid credential collection and destructive changes Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Secure Coding Practices Quick Reference Guide, input validation, error handling and data protection sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D105 · M04W3D7 · Log parser engineering sprint: parser and normalization

Phase: Academic month · **Optional date:** 2027-02-08 · **Domain resources:** P03 / P05 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 105 of 365; calendar-free M04W3D7; optional date 2027-02-08.
2. Exact topic	Log parser engineering sprint: parser and normalization
3. Exact subtopics	Streaming input, multiple formats, schema validation, timestamp parsing, field extraction and error budgets
4. Learning objective	Implement an efficient parser that never crashes on malformed data
5. Primary lecture / reading + direct URL	Automate the Boring Stuff with Python `https://automatetheboringstuff.com/` **Type:** Free online book · **Access:** Free online; paid print/course optional
6. Exact section / lesson locator	Automate the Boring Stuff chapters on regular expressions, files and debugging. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Implement streaming parsing for Apache, SSH and JSON lines; cap line length and errors; write 20 unit tests — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Implement streaming parsing for Apache, SSH and JSON lines; cap line length and errors; write 20 unit tests — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Implement streaming parsing for Apache, SSH and JSON lines; cap line length and errors; write 20 unit tests — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	parser.py and test_parser.py; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: How does streaming change memory usage and error recovery? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All tests pass and a 100 MB synthetic file can be processed within a documented memory limit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Automate the Boring Stuff chapters on regular expressions, files and debugging"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

Month 04 · Week 4 · daily cards

D106 · M04W4D1 · Log parser engineering sprint: analysis and reporting

Phase: Academic month · **Optional date:** 2027-02-09 · **Domain resources:** P07 / P06

1. Day / sequence	Global Day 106 of 365; calendar-free M04W4D1; optional date 2027-02-09.
2. Exact topic	Log parser engineering sprint: analysis and reporting
3. Exact subtopics	Counting, top-N, time bucketing, correlation, output formats, plotting optional and statistical false positives
4. Learning objective	Generate a report that supports a conclusion rather than a data dump
5. Primary lecture / reading + direct URL	SQLite documentation `https://www.sqlite.org/docs.html` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	SQLite aggregation and indexing documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Requests: HTTP for Humans `https://requests.readthedocs.io/en/latest/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Query normalized events for failed logins by source, hour and username; create a Markdown report with confidence and limitations. **2.** Standard (~2 h): Query normalized events for failed logins by source, hour and username; create a Markdown report with confidence and limitations. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Query normalized events for failed logins by source, hour and username; create a Markdown report with confidence and limitations. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	security-log-report.md and SQL queries; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a high failed-login count be normal behaviour in some environments? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Report includes baseline, context, limitations and at least one alternative explanation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SQLite aggregation and indexing documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SQLBolt; write a five-line correction; retest next morning with a new 10-minute problem.

D107 · M04W4D2 · Secure coding review of the automation tool

Phase: Academic month · **Optional date:** 2027-02-10 · **Domain resources:** P08 / W14

1. Day / sequence	Global Day 107 of 365; calendar-free M04W4D2; optional date 2027-02-10.
2. Exact topic	Secure coding review of the automation tool
3. Exact subtopics	Input validation, command injection, path traversal, secrets, dependency risk, logging, permissions and least privilege
4. Learning objective	Find and fix security flaws in your own code
5. Primary lecture / reading + direct URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/` **Type:** Checklist/reference · **Access:** Free
6. Exact section / lesson locator	OWASP Secure Coding Practices and OWASP Cheat Sheet Series via W17. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Semgrep documentation `https://semgrep.dev/docs/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run Semgrep locally on the tool, triage findings, fix three real issues, and document one false positive. **2.** Standard (~2 h): Run Semgrep locally on the tool, triage findings, fix three real issues, and document one false positive. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run Semgrep locally on the tool, triage findings, fix three real issues, and document one false positive. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	secure-code-review.md and diff; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which bug would have allowed user-controlled input to reach a shell command? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Fixes are tested and no dangerous shell interpolation remains Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Secure Coding Practices and OWASP Cheat Sheet Series via W17"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D108 · M04W4D3 · Bash and PowerShell automation comparison lab

Phase: Academic month · **Optional date:** 2027-02-11 · **Domain resources:** L03 / P05

1. Day / sequence	Global Day 108 of 365; calendar-free M04W4D3; optional date 2027-02-11.
2. Exact topic	Bash and PowerShell automation comparison lab
3. Exact subtopics	Cross-platform telemetry schema, process inventory, listening ports, users, services and error handling
4. Learning objective	Produce equivalent evidence from Linux and Windows without leaking secrets
5. Primary lecture / reading + direct URL	Greg's Wiki BashGuide `https://mywiki.wooledge.org/BashGuide` **Type:** Free scripting guide · **Access:** Free
6. Exact section / lesson locator	BashGuide review of arrays and quoting; L08 PowerShell error handling docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run both audit scripts, compare schemas, normalize differences, and explain two platform-specific limitations. **2.** Standard (~2 h): Run both audit scripts, compare schemas, normalize differences, and explain two platform-specific limitations. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run both audit scripts, compare schemas, normalize differences, and explain two platform-specific limitations. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cross-platform-report.md and sample outputs; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should both tools expose the same field names even when the underlying APIs differ? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Both tools produce valid common-schema JSON and limitations are explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "BashGuide review of arrays and quoting; L08 PowerShell error handling docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RegexOne; write a five-line correction; retest next morning with a new 10-minute problem.

D109 · M04W4D4 · Project publication and professional README

Phase: Academic month · **Optional date:** 2027-02-12 · **Domain resources:** F09 / W13

1. Day / sequence	Global Day 109 of 365; calendar-free M04W4D4; optional date 2027-02-12.
2. Exact topic	Project publication and professional README
3. Exact subtopics	Repository structure, setup, usage, examples, tests, screenshots, threat model, limitations, licence and changelog
4. Learning objective	Publish the security-log-parser and audit utility as portfolio evidence
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 1-3 and GitHub Docs README conventions linked from F09 repository navigation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Finalize PRJ-07, sanitize sample logs, add verification commands, tag a release and write a succinct professional README. **2.** Standard (~2 h): Finalize PRJ-07, sanitize sample logs, add verification commands, tag a release and write a succinct professional README. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Finalize PRJ-07, sanitize sample logs, add verification commands, tag a release and write a succinct professional README. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	published PRJ-07 repository; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which three commands prove the project works on a clean machine? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Repository passes the portfolio checklist and contains no credentials or personal data Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 1-3 and GitHub Docs README conventions linked from F09 repository navigation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D110 · M04W4D5 · Month 4 practical assessment and milestone review

Phase: Academic month · **Optional date:** 2027-02-13 · **Domain resources:** F08 / P05

1. Day / sequence	Global Day 110 of 365; calendar-free M04W4D5; optional date 2027-02-13.
2. Exact topic	Month 4 practical assessment and milestone review
3. Exact subtopics	Python, Bash, PowerShell, Git, regex, APIs, SQL, testing, logging and secure coding
4. Learning objective	Demonstrate intermediate scripting competence through an unfamiliar task
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial Sections 3-10 review pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: In 120 minutes, receive an unseen synthetic log format, write a parser, tests, normalization schema and short report without tutorial guidance. **2.** Standard (~2 h): In 120 minutes, receive an unseen synthetic log format, write a parser, tests, normalization schema and short report without tutorial guidance. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): In 120 minutes, receive an unseen synthetic log format, write a parser, tests, normalization schema and short report without tutorial guidance. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M4-assessment.pdf and code repository; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which requirement was ambiguous and how did you resolve it safely? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, independent problem-solving at least 11/15, and code runs cleanly Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial Sections 3-10 review pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D111 · M04W4D6 · Monthly review and prerequisite gate for security principles

Phase: Academic month · **Optional date:** 2027-02-14 · **Domain resources:** P01 / F09

1. Day / sequence	Global Day 111 of 365; calendar-free M04W4D6; optional date 2027-02-14.
2. Exact topic	Monthly review and prerequisite gate for security principles
3. Exact subtopics	Error-log analysis, debugging habits, test gaps, command notebook and spaced repetition
4. Learning objective	Close Month 4 gaps and verify readiness for core security theory
5. Primary lecture / reading + direct URL	Python for Everybody `https://www.py4e.com/` **Type:** University course · **Access:** Free site; Coursera specialization freemium
6. Exact section / lesson locator	Python for Everybody review chapters on files, dictionaries and regular expressions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework all failed assessment items, add missing tests, create ten flashcards, and map each Month 5 topic to a Month 1-4 prerequisite. **2.** Standard (~2 h): Rework all failed assessment items, add missing tests, create ten flashcards, and map each Month 5 topic to a Month 1-4 prerequisite. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework all failed assessment items, add missing tests, create ten flashcards, and map each Month 5 topic to a Month 1-4 prerequisite. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M4-review.md and prerequisite-map.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which programming weakness would block secure coding in Month 6? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and prerequisite map has no unexplained critical gap Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python for Everybody review chapters on files, dictionaries and regular expressions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D112 · M04W4D7 · Spaced revision, recovery and Month 5 preview

Phase: Academic month · **Optional date:** 2027-02-15 · **Domain resources:** P04 / F04 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 112 of 365; calendar-free M04W4D7; optional date 2027-02-15.
2. Exact topic	Spaced revision, recovery and Month 5 preview
3. Exact subtopics	Active recall of commands, APIs, regex, SQL and testing; portfolio cleanup; lighter workload
4. Learning objective	Preserve automation skill and prepare CIA/AAA/risk vocabulary
5. Primary lecture / reading + direct URL	RegexOne `https://regexone.com/` **Type:** Interactive lessons · **Access:** Free
6. Exact section / lesson locator	RegexOne review lessons 10-15. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute mixed retrieval session, fix one README issue, review the CS50 Cybersecurity syllabus, and rest or take a light walkthrough of the Month 5 lab — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute mixed retrieval session, fix one README issue, review the CS50 Cybersecurity syllabus, and rest or take a light walkthrough of the Month 5 lab — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute mixed retrieval session, fix one README issue, review the CS50 Cybersecurity syllabus, and rest or take a light walkthrough of the Month 5 lab — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M4-consolidation.md and updated portfolio; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Explain how input validation, logging and least privilege appear in your own tool without notes Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All scheduled cards are reviewed and the three concepts are explained correctly Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "RegexOne review lessons 10-15"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.



ULTIMATE CYBERSECURITY MASTER BLUEPRINT 2026-2027

Volume 3 — Daily Curriculum: Months 5-8

112 daily cards for core security, web/API, ethical testing and blue team

Edition: Volume 3
Resource verification date: 2026-10-27
Plan window: 2026-10-27 to 2027-10-26 (shift dates if you start later)
System: 365 days, 12 academic months, 52 weeks + 1 graduation day
Profile: BS CS student in Pakistan; free/high-value resources; safe authorized labs

Academic-style self-study manual. No employment, salary, visa, certification or top-1% guarantee.

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Month 05 — Core security, cryptography, identity, risk and controls

Days D113-D140. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: CIA triad, non-repudiation and security trade-offs D2: AAA, least privilege and defense in depth D3: Threats, vulnerabilities, exploits, risk and impact D4: Attack surfaces and data-flow threat modelling D5: Security architecture and control categories D6: Security policies, standards, procedures and baselines D7: Weekly core-security review and risk practical
Week 2	D1: Hash functions and password storage D2: Symmetric encryption, block modes and key management D3: Public-key cryptography, key exchange and digital signatures D4: TLS, certificates and chain of trust D5: Identity, federation, SSO and MFA D6: IAM lifecycle and privilege management D7: Weekly cryptography and identity review
Week 3	D1: Endpoint, network, application and data security controls D2: Patch and vulnerability management lifecycle D3: Phishing defense and security awareness D4: Incident response lifecycle and tabletop exercise D5: Business continuity, disaster recovery and backup validation D6: GRC frameworks and legal-obligation distinctions D7: Weekly controls, GRC and incident response review
Week 4	D1: Privacy by design and data classification D2: Security architecture review of the capstone D3: Certification readiness without exam obsession D4: Month 5 capstone: security control baseline D5: Month 5 practical examination and milestone gate D6: Monthly review, gap repair and Month 6 preview D7: Spaced revision, recovery and readiness check

> **Gate:** score the Month 5 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 05 · Week 1 · daily cards

D113 · M05W1D1 · CIA triad, non-repudiation and security trade-offs

Phase: Academic month · **Optional date:** 2027-02-16 · **Domain resources:** F04 / C01

1. Day / sequence	Global Day 113 of 365; calendar-free M05W1D1; optional date 2027-02-16.
2. Exact topic	CIA triad, non-repudiation and security trade-offs
3. Exact subtopics	Confidentiality, integrity, availability, authenticity, non-repudiation, usability and cost/benefit
4. Learning objective	Classify controls by the property they protect and identify trade-offs
5. Primary lecture / reading + direct URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/` **Type:** University course · **Access:** Free OCW; paid edX verified certificate optional
6. Exact section / lesson locator	CS50 Cybersecurity Week 0-1 notes; F04 course home links to Week 0 and Week 1 pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write ten control examples from your lab and label primary/secondary CIA effects plus one usability cost. **2.** Standard (~2 h): Write ten control examples from your lab and label primary/secondary CIA effects plus one usability cost. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write ten control examples from your lab and label primary/secondary CIA effects plus one usability cost. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cia-matrix.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can encrypting backups improve confidentiality but harm availability if keys are lost? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every label is justified and at least three trade-offs are concrete Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50 Cybersecurity Week 0-1 notes; F04 course home links to Week 0 and Week 1 pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50's Introduction to Cybersecurity (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D114 · M05W1D2 · AAA, least privilege and defense in depth

Phase: Academic month · **Optional date:** 2027-02-17 · **Domain resources:** C01 / C06

1. Day / sequence	Global Day 114 of 365; calendar-free M05W1D2; optional date 2027-02-17.
2. Exact topic	AAA, least privilege and defense in depth
3. Exact subtopics	Identification, authentication, authorization, accounting, least privilege, separation of duties, layered controls and fail-safe defaults
4. Learning objective	Design layered access control for a small lab service
5. Primary lecture / reading + direct URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/` **Type:** Course index · **Access:** Free videos; paid notes/practice exams optional
6. Exact section / lesson locator	Professor Messer SY0-701 course index, objectives 1.2 and 3.3; use exact objective videos. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Model three layers of controls for a local web app, remove one layer in a thought experiment and describe the impact. **2.** Standard (~2 h): Model three layers of controls for a local web app, remove one layer in a thought experiment and describe the impact. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Model three layers of controls for a local web app, remove one layer in a thought experiment and describe the impact. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	aaa-design.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is logging part of accountability rather than authentication? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram clearly separates identity, permission and evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer SY0-701 course index, objectives 1.2 and 3.3; use exact objective videos"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer SY0-701 Security+ official playlist; write a five-line correction; retest next morning with a new 10-minute problem.

D115 · M05W1D3 · Threats, vulnerabilities, exploits, risk and impact

Phase: Academic month · **Optional date:** 2027-02-18 · **Domain resources:** C26 / C06

1. Day / sequence	Global Day 115 of 365; calendar-free M05W1D3; optional date 2027-02-18.
2. Exact topic	Threats, vulnerabilities, exploits, risk and impact
3. Exact subtopics	Threat actor, capability, intent, vulnerability, exploit, asset, likelihood, impact, risk and residual risk
4. Learning objective	Decompose a news-style incident into precise risk language
5. Primary lecture / reading + direct URL	NIST SP 800-30 Revision 1 `https://csrc.nist.gov/pubs/sp/800/30/r1/final` **Type:** Official risk guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-30, risk model, threat sources and risk determination sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Analyze one synthetic scenario, fill a risk register with five entries, and explain why exploitability alone is not business risk. **2.** Standard (~2 h): Analyze one synthetic scenario, fill a risk register with five entries, and explain why exploitability alone is not business risk. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Analyze one synthetic scenario, fill a risk register with five entries, and explain why exploitability alone is not business risk. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	risk-register.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How can a low-severity bug be high risk in a specific environment? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All terms are used correctly and risk ranking includes context Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-30, risk model, threat sources and risk determination sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D116 · M05W1D4 · Attack surfaces and data-flow threat modelling

Phase: Academic month · **Optional date:** 2027-02-19 · **Domain resources:** C17 / A08

1. Day / sequence	Global Day 116 of 365; calendar-free M05W1D4; optional date 2027-02-19.
2. Exact topic	Attack surfaces and data-flow threat modelling
3. Exact subtopics	Assets, trust boundaries, data flows, entry points, privilege levels, STRIDE and misuse cases
4. Learning objective	Draw a defensible threat model for a system you own
5. Primary lecture / reading + direct URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling` **Type:** Reference/process · **Access:** Free
6. Exact section / lesson locator	OWASP Threat Modeling community page, process and data-flow diagram sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a DFD for the Month 4 log parser, identify six threats with STRIDE and propose mitigations. **2.** Standard (~2 h): Draw a DFD for the Month 4 log parser, identify six threats with STRIDE and propose mitigations. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a DFD for the Month 4 log parser, identify six threats with STRIDE and propose mitigations. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	threat-model.svg and STRIDE table; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which trust boundary was most often forgotten and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	DFD has labelled boundaries and each threat maps to a testable mitigation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Threat Modeling community page, process and data-flow diagram sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D117 · M05W1D5 · Security architecture and control categories

Phase: Academic month · **Optional date:** 2027-02-20 · **Domain resources:** C01 / C21

1. Day / sequence	Global Day 117 of 365; calendar-free M05W1D5; optional date 2027-02-20.
2. Exact topic	Security architecture and control categories
3. Exact subtopics	Preventive, detective, corrective, deterrent, compensating, directive, administrative, technical and physical controls
4. Learning objective	Select controls and explain where each sits in an attack chain
5. Primary lecture / reading + direct URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/` **Type:** Course index · **Access:** Free videos; paid notes/practice exams optional
6. Exact section / lesson locator	Professor Messer SY0-701 course index, objective 1.1 security controls video. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CIS Controls Version 8 `https://www.cisecurity.org/controls/v8`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: For one lab attack path, place nine control categories on a timeline and identify missing feedback loops. **2.** Standard (~2 h): For one lab attack path, place nine control categories on a timeline and identify missing feedback loops. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): For one lab attack path, place nine control categories on a timeline and identify missing feedback loops. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	control-timeline.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a detective control insufficient without a response process? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Timeline is internally consistent and includes escalation/response Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer SY0-701 course index, objective 1.1 security controls video"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer SY0-701 Security+ official playlist; write a five-line correction; retest next morning with a new 10-minute problem.

D118 · M05W1D6 · Security policies, standards, procedures and baselines

Phase: Academic month · **Optional date:** 2027-02-21 · **Domain resources:** C22 / C06

1. Day / sequence	Global Day 118 of 365; calendar-free M05W1D6; optional date 2027-02-21.
2. Exact topic	Security policies, standards, procedures and baselines
3. Exact subtopics	Governance hierarchy, policy lifecycle, exceptions, evidence, acceptable use and change management
4. Learning objective	Distinguish a policy from a standard and a procedure
5. Primary lecture / reading + direct URL	NIST SP 800-53 Revision 5 `https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final` **Type:** Official controls catalog · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-53 control families AC, AU, CM, IR and SI; use the control catalog search. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write one mini-policy, one standard and one procedure for lab VM patching; include roles and exception handling. **2.** Standard (~2 h): Write one mini-policy, one standard and one procedure for lab VM patching; include roles and exception handling. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write one mini-policy, one standard and one procedure for lab VM patching; include roles and exception handling. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	governance-pack.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a policy without an owner and exception path unenforceable? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Documents reference one another correctly and include measurable compliance evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-53 control families AC, AU, CM, IR and SI; use the control catalog search"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D119 · M05W1D7 · Weekly core-security review and risk practical

Phase: Academic month · **Optional date:** 2027-02-22 · **Domain resources:** C26 / C01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 119 of 365; calendar-free M05W1D7; optional date 2027-02-22.
2. Exact topic	Weekly core-security review and risk practical
3. Exact subtopics	CIA, AAA, least privilege, threat models, risk, controls and governance; active recall
4. Learning objective	Demonstrate Month 5 Week 1 concepts through an unfamiliar scenario
5. Primary lecture / reading + direct URL	NIST SP 800-30 Revision 1 `https://csrc.nist.gov/pubs/sp/800/30/r1/final` **Type:** Official risk guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-30 risk assessment process and risk register tables. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute risk assessment for a fictional web service; identify assets, threats, controls, likelihood, impact and treatment — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute risk assessment for a fictional web service; identify assets, threats, controls, likelihood, impact and treatment — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute risk assessment for a fictional web service; identify assets, threats, controls, likelihood, impact and treatment — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M5W1-practical.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which assumption has the greatest influence on your risk ranking? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and every high risk has a named owner and treatment Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-30 risk assessment process and risk register tables"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Month 05 · Week 2 · daily cards

D120 · M05W2D1 · Hash functions and password storage

Phase: Academic month · **Optional date:** 2027-02-23 · **Domain resources:** F04 / C09

1. Day / sequence	Global Day 120 of 365; calendar-free M05W2D1; optional date 2027-02-23.
2. Exact topic	Hash functions and password storage
3. Exact subtopics	One-way functions, collision/preimage resistance, salts, pepper, slow KDFs, password hashing and cracking economics
4. Learning objective	Explain why a general-purpose hash is unsuitable for passwords
5. Primary lecture / reading + direct URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/` **Type:** University course · **Access:** Free OCW; paid edX verified certificate optional
6. Exact section / lesson locator	CS50 Cybersecurity Week 1 Securing Data notes, hashing/salting/rainbow tables sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Crypto 101 `https://www.crypto101.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Hash synthetic passwords with a fast hash and a KDF in Python; compare timing and explain offline attack economics safely on your own inputs. **2.** Standard (~2 h): Hash synthetic passwords with a fast hash and a KDF in Python; compare timing and explain offline attack economics safely on your own inputs. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Hash synthetic passwords with a fast hash and a KDF in Python; compare timing and explain offline attack economics safely on your own inputs. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	password-hashing.md and timing results; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does salting not make a weak password strong? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Results are measured and the explanation distinguishes salt from work factor Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50 Cybersecurity Week 1 Securing Data notes, hashing/salting/rainbow tables sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50's Introduction to Cybersecurity (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D121 · M05W2D2 · Symmetric encryption, block modes and key management

Phase: Academic month · **Optional date:** 2027-02-24 · **Domain resources:** C09 / C10

1. Day / sequence	Global Day 121 of 365; calendar-free M05W2D2; optional date 2027-02-24.
2. Exact topic	Symmetric encryption, block modes and key management
3. Exact subtopics	AES concepts, block/stream ciphers, modes, IV/nonce, padding, authenticated encryption and key lifecycle
4. Learning objective	Identify misuse patterns without implementing production cryptography
5. Primary lecture / reading + direct URL	Crypto 101 `https://www.crypto101.io/` **Type:** Free course/book · **Access:** Free
6. Exact section / lesson locator	Crypto 101 block ciphers, stream ciphers and authenticated encryption sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Stanford Cryptography I `https://www.coursera.org/learn/crypto`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Use a vetted library only to encrypt/decrypt synthetic data; compare ECB and an authenticated mode visually; document never-to-use patterns. **2.** Standard (~2 h): Use a vetted library only to encrypt/decrypt synthetic data; compare ECB and an authenticated mode visually; document never-to-use patterns. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Use a vetted library only to encrypt/decrypt synthetic data; compare ECB and an authenticated mode visually; document never-to-use patterns. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	crypto-lab.py and notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does encryption without integrity allow tampering? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All examples use vetted libraries and the answer explains malleability Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Crypto 101 block ciphers, stream ciphers and authenticated encryption sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Stanford Cryptography I; write a five-line correction; retest next morning with a new 10-minute problem.

D122 · M05W2D3 · Public-key cryptography, key exchange and digital signatures

Phase: Academic month · **Optional date:** 2027-02-25 · **Domain resources:** C10 / C11

1. Day / sequence	Global Day 122 of 365; calendar-free M05W2D3; optional date 2027-02-25.
2. Exact topic	Public-key cryptography, key exchange and digital signatures
3. Exact subtopics	RSA/Diffie-Hellman concepts, public/private keys, signatures, certificates and non-repudiation limits
4. Learning objective	Explain which key is used for encryption versus signing
5. Primary lecture / reading + direct URL	Stanford Cryptography I https://www.coursera.org/learn/crypto **Type:** University course · **Access:** Free audit/lectures may be available; verified certificate and graded homework paid
6. Exact section / lesson locator	Stanford Cryptography I, public-key encryption and digital signatures lecture pages accessible via the Coursera audit/preview. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	A Graduate Course in Applied Cryptography https://toc.cryptobook.us/
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Generate a local key pair, sign a benign file and verify it; then corrupt one byte and observe verification failure. **2.** Standard (~2 h): Generate a local key pair, sign a benign file and verify it; then corrupt one byte and observe verification failure. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Generate a local key pair, sign a benign file and verify it; then corrupt one byte and observe verification failure. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	signature-demo.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does a valid signature prove key possession, not necessarily a person's intent? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Sign/verify workflow works and the legal/technical distinction is explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Stanford Cryptography I, public-key encryption and digital signatures lecture pages accessible via the Coursera audit/preview"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Crypto 101; write a five-line correction; retest next morning with a new 10-minute problem.

D123 · M05W2D4 · TLS, certificates and chain of trust

Phase: Academic month · **Optional date:** 2027-02-26 · **Domain resources:** C12 / C13

1. Day / sequence	Global Day 123 of 365; calendar-free M05W2D4; optional date 2027-02-26.
2. Exact topic	TLS, certificates and chain of trust
3. Exact subtopics	X.509, root/intermediate/leaf certificates, CA, SAN, expiry, revocation, chain validation and hostname verification
4. Learning objective	Inspect a certificate chain and explain each validation step
5. Primary lecture / reading + direct URL	Let's Encrypt documentation `https://letsencrypt.org/docs/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Let's Encrypt documentation, certificate chain, validation and renewal pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	RFC 8446: TLS 1.3 `https://www.rfc-editor.org/rfc/rfc8446`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Use browser or openssl to inspect a certificate chain for an authorized site or local test certificate; draw the chain and list validation checks. **2.** Standard (~2 h): Use browser or openssl to inspect a certificate chain for an authorized site or local test certificate; draw the chain and list validation checks. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Use browser or openssl to inspect a certificate chain for an authorized site or local test certificate; draw the chain and list validation checks. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	tls-chain.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What does certificate transparency add and what does it not prevent? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Chain is correctly labelled and validation is not reduced to encryption Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Let's Encrypt documentation, certificate chain, validation and renewal pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RFC 8446: TLS 1.3; write a five-line correction; retest next morning with a new 10-minute problem.

D124 · M05W2D5 · Identity, federation, SSO and MFA

Phase: Academic month · **Optional date:** 2027-02-27 · **Domain resources:** C08 / A20

1. Day / sequence	Global Day 124 of 365; calendar-free M05W2D5; optional date 2027-02-27.
2. Exact topic	Identity, federation, SSO and MFA
3. Exact subtopics	OAuth 2.0, OpenID Connect, SAML concepts, IdP, SP, tokens, MFA factors, session lifetime and phishing-resistant MFA
4. Learning objective	Design an authentication flow and identify token/session failure modes
5. Primary lecture / reading + direct URL	NIST SP 800-63B `https://pages.nist.gov/800-63-4/sp800-63b.html` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-63B, authentication and authenticator management sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-900 official certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-compliance-and-identity-fundamentals/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw an OIDC authorization-code flow with PKCE for a fictional app; annotate where tokens live and where phishing-resistant MFA applies. **2.** Standard (~2 h): Draw an OIDC authorization-code flow with PKCE for a fictional app; annotate where tokens live and where phishing-resistant MFA applies. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw an OIDC authorization-code flow with PKCE for a fictional app; annotate where tokens live and where phishing-resistant MFA applies. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	oidc-flow.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is an access token bearer evidence rather than proof of user identity? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Flow is technically correct and token storage risks are identified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-63B, authentication and authenticator management sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Crypto 101; write a five-line correction; retest next morning with a new 10-minute problem.

D125 · M05W2D6 · IAM lifecycle and privilege management

Phase: Academic month · **Optional date:** 2027-02-28 · **Domain resources:** C01 / C20

1. Day / sequence	Global Day 125 of 365; calendar-free M05W2D6; optional date 2027-02-28.
2. Exact topic	IAM lifecycle and privilege management
3. Exact subtopics	Provisioning, deprovisioning, joiner/mover/leaver, RBAC, ABAC, PAM, access reviews, entitlements and privilege creep
4. Learning objective	Design an access review that detects privilege creep
5. Primary lecture / reading + direct URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/` **Type:** Course index · **Access:** Free videos; paid notes/practice exams optional
6. Exact section / lesson locator	Professor Messer SY0-701 course index, IAM objectives 1.5 and 4.6. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a synthetic employee lifecycle with three role changes, simulate accumulated entitlements and write an access-review decision log. **2.** Standard (~2 h): Create a synthetic employee lifecycle with three role changes, simulate accumulated entitlements and write an access-review decision log. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a synthetic employee lifecycle with three role changes, simulate accumulated entitlements and write an access-review decision log. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	iam-review.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can RBAC fail when users accumulate exceptions? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Scenario is complete and review removes excess rights with documented approval Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer SY0-701 course index, IAM objectives 1.5 and 4.6"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer SY0-701 Security+ official playlist; write a five-line correction; retest next morning with a new 10-minute problem.

D126 · M05W2D7 · Weekly cryptography and identity review

Phase: Academic month · **Optional date:** 2027-03-01 · **Domain resources:** C09 / C08 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 126 of 365; calendar-free M05W2D7; optional date 2027-03-01.
2. Exact topic	Weekly cryptography and identity review
3. Exact subtopics	Hash/KDF, symmetric crypto, AEAD, public-key crypto, TLS, federation, MFA and IAM; active recall
4. Learning objective	Demonstrate correct use of cryptographic primitives and identity terminology
5. Primary lecture / reading + direct URL	Crypto 101 https://www.crypto101.io/ **Type:** Free course/book · **Access:** Free
6. Exact section / lesson locator	Crypto 101 review sections and C10 selected lectures. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-63B https://pages.nist.gov/800-63-4/sp800-63b.html
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete 20 closed-book questions, one certificate-chain analysis and one IAM design exercise; correct all mistakes with new examples — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete 20 closed-book questions, one certificate-chain analysis and one IAM design exercise; correct all mistakes with new examples — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete 20 closed-book questions, one certificate-chain analysis and one IAM design exercise; correct all mistakes with new examples — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M5W2-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which cryptographic primitive would you use for each of confidentiality, integrity and authentication? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 75 percent and no primitive is confused with another Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Crypto 101 review sections and C10 selected lectures"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Stanford Cryptography I; write a five-line correction; retest next morning with a new 10-minute problem.

Month 05 · Week 3 · daily cards

D127 · M05W3D1 · Endpoint, network, application and data security controls

Phase: Academic month · **Optional date:** 2027-03-02 · **Domain resources:** C01 / C21

1. Day / sequence	Global Day 127 of 365; calendar-free M05W3D1; optional date 2027-03-02.
2. Exact topic	Endpoint, network, application and data security controls
3. Exact subtopics	EDR, hardening, segmentation, WAF, secure coding, DLP, encryption and backups
4. Learning objective	Map controls to attack paths and identify residual risk
5. Primary lecture / reading + direct URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/` **Type:** Course index · **Access:** Free videos; paid notes/practice exams optional
6. Exact section / lesson locator	Professor Messer SY0-701 objectives 2, 3 and 4; use exact video titles in the course index. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CIS Controls Version 8 `https://www.cisecurity.org/controls/v8`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a control map for a fictional three-tier app, with five attack paths and a detective control for each. **2.** Standard (~2 h): Build a control map for a fictional three-tier app, with five attack paths and a detective control for each. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a control map for a fictional three-tier app, with five attack paths and a detective control for each. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	controls-map.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which two controls share a single failure domain? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Map includes prevention, detection, response and shared-failure analysis Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer SY0-701 objectives 2, 3 and 4; use exact video titles in the course index"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer SY0-701 Security+ official playlist; write a five-line correction; retest next morning with a new 10-minute problem.

D128 · M05W3D2 · Patch and vulnerability management lifecycle

Phase: Academic month · **Optional date:** 2027-03-03 · **Domain resources:** C18 / E08

1. Day / sequence	Global Day 128 of 365; calendar-free M05W3D2; optional date 2027-03-03.
2. Exact topic	Patch and vulnerability management lifecycle
3. Exact subtopics	Asset inventory, discovery, prioritization, testing, deployment, verification, exception and metrics
4. Learning objective	Build a repeatable patch workflow with evidence and rollback
5. Primary lecture / reading + direct URL	MITRE ATT&CK `https://attack.mitre.org/` **Type:** Knowledge base · **Access:** Free
6. Exact section / lesson locator	CISA KEV catalog and MITRE ATT&CK pages for one relevant technique. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NVD `https://nvd.nist.gov/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Select five synthetic vulnerabilities with CVSS/KEV-like metadata, prioritize them by business context, and define test/rollback for each. **2.** Standard (~2 h): Select five synthetic vulnerabilities with CVSS/KEV-like metadata, prioritize them by business context, and define test/rollback for each. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Select five synthetic vulnerabilities with CVSS/KEV-like metadata, prioritize them by business context, and define test/rollback for each. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	vuln-prioritization.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a lower CVSS score be patched first in a real environment? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Priorities are justified with context and each item has verification evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CISA KEV catalog and MITRE ATT&CK pages for one relevant technique"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE D3FEND; write a five-line correction; retest next morning with a new 10-minute problem.

D129 · M05W3D3 · Phishing defense and security awareness

Phase: Academic month · **Optional date:** 2027-03-04 · **Domain resources:** C01 / B17

1. Day / sequence	Global Day 129 of 365; calendar-free M05W3D3; optional date 2027-03-04.
2. Exact topic	Phishing defense and security awareness
3. Exact subtopics	Social engineering, phishing indicators, URL parsing, attachments, reporting culture and measurable awareness
4. Learning objective	Analyze a synthetic phishing email without clicking or downloading
5. Primary lecture / reading + direct URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/` **Type:** Course index · **Access:** Free videos; paid notes/practice exams optional
6. Exact section / lesson locator	Professor Messer SY0-701 objective 5.6 user training and phishing videos. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CISA cybersecurity advisories `https://www.cisa.gov/news-events/cybersecurity-advisories`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create or use a synthetic email sample, identify five indicators, write a user-facing report and one technical control. **2.** Standard (~2 h): Create or use a synthetic email sample, identify five indicators, write a user-facing report and one technical control. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create or use a synthetic email sample, identify five indicators, write a user-facing report and one technical control. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	phishing-analysis.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is blaming users a weak control compared with report-driven engineering? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Analysis is safe, indicators are specific, and both human and technical controls are present Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer SY0-701 objective 5.6 user training and phishing videos"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer SY0-701 Security+ official playlist; write a five-line correction; retest next morning with a new 10-minute problem.

D130 · M05W3D4 · Incident response lifecycle and tabletop exercise

Phase: Academic month · **Optional date:** 2027-03-05 · **Domain resources:** C07 / C06

1. Day / sequence	Global Day 130 of 365; calendar-free M05W3D4; optional date 2027-03-05.
2. Exact topic	Incident response lifecycle and tabletop exercise
3. Exact subtopics	Preparation, detection, analysis, containment, eradication, recovery, lessons learned, roles and communications
4. Learning objective	Run a tabletop incident using only synthetic data
5. Primary lecture / reading + direct URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-61r3 incident response life cycle and CSF function alignment. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Conduct a 60-minute tabletop for a synthetic credential-theft incident; record decisions, evidence needs, escalation and communication. **2.** Standard (~2 h): Conduct a 60-minute tabletop for a synthetic credential-theft incident; record decisions, evidence needs, escalation and communication. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Conduct a 60-minute tabletop for a synthetic credential-theft incident; record decisions, evidence needs, escalation and communication. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	tabletop-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which decision required information you did not have and what detection would provide it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Report includes timeline, decisions, evidence gaps and lessons learned Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-61r3 incident response life cycle and CSF function alignment"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D131 · M05W3D5 · Business continuity, disaster recovery and backup validation

Phase: Academic month · **Optional date:** 2027-03-06 · **Domain resources:** C27 / F12

1. Day / sequence	Global Day 131 of 365; calendar-free M05W3D5; optional date 2027-03-06.
2. Exact topic	Business continuity, disaster recovery and backup validation
3. Exact subtopics	BIA, RTO, RPO, backup types, restore testing, failover, dependencies and crisis communication
4. Learning objective	Prove that a backup can be restored and quantify the test result
5. Primary lecture / reading + direct URL	NIST SP 800-34 Revision 1 https://csrc.nist.gov/pubs/sp/800/34/r1/final **Type:** Official continuity guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-34r1, BIA and contingency-planning sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Oracle VirtualBox User Manual https://www.virtualbox.org/manual/
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Restore the Month 1 VM snapshot or file backup, measure elapsed time, compare hashes and state observed RTO/RPO. **2.** Standard (~2 h): Restore the Month 1 VM snapshot or file backup, measure elapsed time, compare hashes and state observed RTO/RPO. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Restore the Month 1 VM snapshot or file backup, measure elapsed time, compare hashes and state observed RTO/RPO. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	dr-test.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does an untested backup fail the business objective even if the job reports success? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Restore is proven by hashes and test has measured RTO/RPO Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-34r1, BIA and contingency-planning sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D132 · M05W3D6 · GRC frameworks and legal-obligation distinctions

Phase: Academic month · **Optional date:** 2027-03-07 · **Domain resources:** C06 / C23

1. Day / sequence	Global Day 132 of 365; calendar-free M05W3D6; optional date 2027-03-07.
2. Exact topic	GRC frameworks and legal-obligation distinctions
3. Exact subtopics	Regulations, laws, contracts, standards, frameworks, best practices, controls, audits and compliance evidence
4. Learning objective	Correctly classify common cybersecurity documents and obligations
5. Primary lecture / reading + direct URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST CSF 2.0, Govern function and implementation examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	ISO/IEC 27001 overview `https://www.iso.org/standard/27001`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a comparison table for GDPR, ISO/IEC 27001, NIST CSF, CIS Controls and PCI DSS-like contract obligations; label each as law, regulation, standard, framework or best practice. **2.** Standard (~2 h): Build a comparison table for GDPR, ISO/IEC 27001, NIST CSF, CIS Controls and PCI DSS-like contract obligations; label each as law, regulation, standard, framework or best practice. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a comparison table for GDPR, ISO/IEC 27001, NIST CSF, CIS Controls and PCI DSS-like contract obligations; label each as law, regulation, standard, framework or best practice. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	grc-classification.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does following a framework not automatically make an organization legally compliant? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All classifications are correct in a defined jurisdiction context Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST CSF 2.0, Govern function and implementation examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Controls Version 8; write a five-line correction; retest next morning with a new 10-minute problem.

D133 · M05W3D7 · Weekly controls, GRC and incident response review

Phase: Academic month · **Optional date:** 2027-03-08 · **Domain resources:** C07 / C26 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 133 of 365; calendar-free M05W3D7; optional date 2027-03-08.
2. Exact topic	Weekly controls, GRC and incident response review
3. Exact subtopics	Endpoint/network/app/data controls, vuln management, awareness, IR, BCDR and frameworks; retrieval
4. Learning objective	Demonstrate Month 5 Week 3 competence through a scenario-based assessment
5. Primary lecture / reading + direct URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-61r3 and C06 CSF 2.0 crosswalk. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-30 Revision 1 `https://csrc.nist.gov/pubs/sp/800/30/r1/final`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete an integrated scenario: triage an incident, classify a vulnerability, choose controls and identify legal/framework implications — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete an integrated scenario: triage an incident, classify a vulnerability, choose controls and identify legal/framework implications — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete an integrated scenario: triage an incident, classify a vulnerability, choose controls and identify legal/framework implications — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M5W3-practical.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which evidence would support both an incident report and an audit? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and all classifications have evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-61r3 and C06 CSF 2.0 crosswalk"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Month 05 · Week 4 · daily cards

D134 · M05W4D1 · Privacy by design and data classification

Phase: Academic month · **Optional date:** 2027-03-09 · **Domain resources:** C24 / C25

1. Day / sequence	Global Day 134 of 365; calendar-free M05W4D1; optional date 2027-03-09.
2. Exact topic	Privacy by design and data classification
3. Exact subtopics	Personal data, special categories, purpose limitation, minimization, retention, anonymization, pseudonymization and data flow
4. Learning objective	Add privacy controls to a software design and risk register
5. Primary lecture / reading + direct URL	GDPR full text (EUR-Lex) `https://eur-lex.europa.eu/eli/reg/2016/679/oj` **Type:** Regulation/legal text · **Access:** Free
6. Exact section / lesson locator	GDPR Articles 5, 25, 32, 33 and 35; NIST Privacy Framework via C25 for supporting structure. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Privacy Framework `https://www.nist.gov/privacy-framework`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Classify ten synthetic data fields, draw a data flow, define retention and propose three privacy controls. **2.** Standard (~2 h): Classify ten synthetic data fields, draw a data flow, define retention and propose three privacy controls. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Classify ten synthetic data fields, draw a data flow, define retention and propose three privacy controls. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	privacy-design.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which data field is most sensitive and what technical control reduces exposure while preserving function? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Classifications are justified and data flow includes retention and deletion points Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GDPR Articles 5, 25, 32, 33 and 35; NIST Privacy Framework via C25 for supporting structure"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Privacy Framework; write a five-line correction; retest next morning with a new 10-minute problem.

D135 · M05W4D2 · Security architecture review of the capstone

Phase: Academic month · **Optional date:** 2027-03-10 · **Domain resources:** C15 / C17

1. Day / sequence	Global Day 135 of 365; calendar-free M05W4D2; optional date 2027-03-10.
2. Exact topic	Security architecture review of the capstone
3. Exact subtopics	Trust boundaries, identity, crypto, logging, input validation, failure modes and control dependencies
4. Learning objective	Review the Month 4 automation tool as a security architect
5. Primary lecture / reading + direct URL	OWASP Application Security Verification Standard `https://owasp.org/www-project-application-security-verification-standard/` **Type:** Standard/reference · **Access:** Free
6. Exact section / lesson locator	OWASP ASVS selected Level 1 requirements for authentication, logging, data protection and validation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Perform a design review of the log parser, create a findings register and fix the highest architectural risk. **2.** Standard (~2 h): Perform a design review of the log parser, create a findings register and fix the highest architectural risk. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Perform a design review of the log parser, create a findings register and fix the highest architectural risk. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	architecture-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control failure would invalidate multiple other controls? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Findings are prioritized by impact and at least one design fix is implemented Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP ASVS selected Level 1 requirements for authentication, logging, data protection and validation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Secure Coding Practices Quick Reference Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D136 · M05W4D3 · Certification readiness without exam obsession

Phase: Academic month · **Optional date:** 2027-03-11 · **Domain resources:** C03 / C01

1. Day / sequence	Global Day 136 of 365; calendar-free M05W4D3; optional date 2027-03-11.
2. Exact topic	Certification readiness without exam obsession
3. Exact subtopics	ISC2 CC and Security+ objective mapping, practice-exam strategy, evidence of competence and cost-benefit analysis
4. Learning objective	Decide rationally whether to pursue an entry-level certification at Month 12
5. Primary lecture / reading + direct URL	ISC2 Certified in Cybersecurity Exam Outline `https://www.isc2.org/certifications/cc/cc-certification-exam-outline` **Type:** Official exam outline · **Access:** Free
6. Exact section / lesson locator	ISC2 CC Exam Outline effective 2026-09-01; A18 CompTIA Security+ V7/V8 official pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Map your portfolio evidence to five CC domains and five Security+ domains; take only free official practice questions and write a decision memo. **2.** Standard (~2 h): Map your portfolio evidence to five CC domains and five Security+ domains; take only free official practice questions and write a decision memo. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Map your portfolio evidence to five CC domains and five Security+ domains; take only free official practice questions and write a decision memo. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	certification-decision.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What does the certification prove that your projects do not, and vice versa? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Memo is evidence-based, budget-aware and does not promise employment Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "ISC2 CC Exam Outline effective 2026-09-01; A18 CompTIA Security+ V7/V8 official pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup ISC2 CC self-study resources; write a five-line correction; retest next morning with a new 10-minute problem.

D137 · M05W4D4 · Month 5 capstone: security control baseline

Phase: Academic month · **Optional date:** 2027-03-12 · **Domain resources:** C06 / C21

1. Day / sequence	Global Day 137 of 365; calendar-free M05W4D4; optional date 2027-03-12.
2. Exact topic	Month 5 capstone: security control baseline
3. Exact subtopics	Risk register, control selection, implementation, testing, exceptions and residual risk
4. Learning objective	Produce a complete security baseline for the student's own lab
5. Primary lecture / reading + direct URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST CSF 2.0 quick-start guides and implementation examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CIS Controls Version 8 `https://www.cisecurity.org/controls/v8`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Select ten controls from your lab, test each with evidence, document exceptions and produce a 90-day improvement plan. **2.** Standard (~2 h): Select ten controls from your lab, test each with evidence, document exceptions and produce a 90-day improvement plan. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Select ten controls from your lab, test each with evidence, document exceptions and produce a 90-day improvement plan. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-02 security-baseline repository; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control failed its test and what was the root cause? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All ten controls have objective tests and residual risk is stated Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST CSF 2.0 quick-start guides and implementation examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Controls Version 8; write a five-line correction; retest next morning with a new 10-minute problem.

D138 · M05W4D5 · Month 5 practical examination and milestone gate

Phase: Academic month · **Optional date:** 2027-03-13 · **Domain resources:** C03 / C06

1. Day / sequence	Global Day 138 of 365; calendar-free M05W4D5; optional date 2027-03-13.
2. Exact topic	Month 5 practical examination and milestone gate
3. Exact subtopics	Security principles, crypto, identity, controls, risk, IR, privacy and GRC; practical scenario
4. Learning objective	Demonstrate core security competence with independent reasoning under time pressure
5. Primary lecture / reading + direct URL	ISC2 Certified in Cybersecurity Exam Outline `https://www.isc2.org/certifications/cc/cc-certification-exam-outline` **Type:** Official exam outline · **Access:** Free
6. Exact section / lesson locator	ISC2 CC Exam Outline, five domains; use it only as a revision checklist. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 120-minute practical: analyze an incident, select crypto primitives, design IAM, prioritize vulnerabilities, and write a risk register. **2.** Standard (~2 h): Complete a 120-minute practical: analyze an incident, select crypto primitives, design IAM, prioritize vulnerabilities, and write a risk register. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 120-minute practical: analyze an incident, select crypto primitives, design IAM, prioritize vulnerabilities, and write a risk register. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M5-assessment.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How would you justify a control decision to a non-technical manager using evidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, milestone gate passed, practical at least 18/25 and safety at least 9/10 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "ISC2 CC Exam Outline, five domains; use it only as a revision checklist"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup ISC2 CC self-study resources; write a five-line correction; retest next morning with a new 10-minute problem.

D139 · M05W4D6 · Monthly review, gap repair and Month 6 preview

Phase: Academic month · **Optional date:** 2027-03-14 · **Domain resources:** C01 / F04

1. Day / sequence	Global Day 139 of 365; calendar-free M05W4D6; optional date 2027-03-14.
2. Exact topic	Monthly review, gap repair and Month 6 preview
3. Exact subtopics	Error-log analysis, crypto misconceptions, risk ranking errors, framework classifications and project feedback
4. Learning objective	Repair Month 5 gaps and prepare web/API prerequisites
5. Primary lecture / reading + direct URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/` **Type:** Course index · **Access:** Free videos; paid notes/practice exams optional
6. Exact section / lesson locator	Professor Messer SY0-701 course index; choose three weakest objectives. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework every failed assessment item, write ten flashcards, update the security baseline, and review HTTP/cookies from Month 1 and Month 2. **2.** Standard (~2 h): Rework every failed assessment item, write ten flashcards, update the security baseline, and review HTTP/cookies from Month 1 and Month 2. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework every failed assessment item, write ten flashcards, update the security baseline, and review HTTP/cookies from Month 1 and Month 2. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M5-review.md and retest score; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which core security concept is still fuzzy enough to weaken web-security decisions? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and each original gap has a new worked example Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Professor Messer SY0-701 course index; choose three weakest objectives"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer SY0-701 Security+ official playlist; write a five-line correction; retest next morning with a new 10-minute problem.

D140 · M05W4D7 · Spaced revision, recovery and readiness check

Phase: Academic month · **Optional date:** 2027-03-15 · **Domain resources:** F04 / C09 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 140 of 365; calendar-free M05W4D7; optional date 2027-03-15.
2. Exact topic	Spaced revision, recovery and readiness check
3. Exact subtopics	CIA/AAA, risk, crypto, TLS, IAM, IR and GRC; active recall and light lab maintenance
4. Learning objective	Preserve core security knowledge and prepare a safe web lab
5. Primary lecture / reading + direct URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/` **Type:** University course · **Access:** Free OCW; paid edX verified certificate optional
6. Exact section / lesson locator	CS50 Cybersecurity Weeks 0-4 review notes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Crypto 101 `https://www.crypto101.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, review all flashcards, verify VM snapshots, install Docker or confirm PortSwigger access, and rest — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, review all flashcards, verify VM snapshots, install Docker or confirm PortSwigger access, and rest — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, review all flashcards, verify VM snapshots, install Docker or confirm PortSwigger access, and rest — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M5-consolidation.md and lab-readiness.txt; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Explain the difference between authentication, authorization, accountability and non-repudiation in one minute Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All prerequisite tools are available and the explanation is correct Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50 Cybersecurity Weeks 0-4 review notes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50's Introduction to Cybersecurity (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

Month 06 — Web security, API security and secure coding

Days D141-D168. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Web architecture and HTTP request lifecycle D2: HTTP methods, headers, cookies and security headers D3: Sessions, cookies and authentication flaws D4: Access control and IDOR/BOLA D5: OWASP Top 10 as a coverage index D6: Secure coding: input validation and output encoding D7: Weekly web foundations review and local lab setup
Week 2	D1: SQL injection and parameterized queries D2: Cross-site scripting and DOM security D3: CSRF, SameSite and anti-CSRF tokens D4: SSRF and outbound request controls D5: Path traversal and file upload security D6: Weekly injection and client-side security review D7: API fundamentals and authorization models
Week 3	D1: Rate limiting, input validation and abuse cases D2: Secrets management and configuration security D3: Secure SDLC and code review gates D4: Dependency and supply-chain security D5: Weekly API, SDLC and supply-chain review D6: Secure web application engineering sprint: design D7: Secure web application engineering sprint: implementation
Week 4	D1: Secure web application engineering sprint: testing D2: Local web security assessment of the app D3: Dependency and CI security for the capstone D4: Portfolio publication of secure app and assessment D5: Month 6 practical examination and milestone gate D6: Monthly review and gap repairs D7: Spaced revision, recovery and Month 7 preview

> **Gate:** score the Month 6 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 06 · Week 1 · daily cards

D141 · M06W1D1 · Web architecture and HTTP request lifecycle

Phase: Academic month · **Optional date:** 2027-03-16 · **Domain resources:** W01 / W04

1. Day / sequence	Global Day 141 of 365; calendar-free M06W1D1; optional date 2027-03-16.
2. Exact topic	Web architecture and HTTP request lifecycle
3. Exact subtopics	DNS, TCP, TLS, HTTP request, routing, application, database, response, browser rendering and same-origin policy
4. Learning objective	Trace one request across the full web stack and identify trust boundaries
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy, Getting Started and Essential Skills topics. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MDN HTTP documentation `https://developer.mozilla.org/en-US/docs/Web/HTTP`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw an end-to-end request lifecycle diagram for a local app and capture one request/response with browser DevTools or Burp. **2.** Standard (~2 h): Draw an end-to-end request lifecycle diagram for a local app and capture one request/response with browser DevTools or Burp. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw an end-to-end request lifecycle diagram for a local app and capture one request/response with browser DevTools or Burp. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	web-request-lifecycle.svg and annotated HTTP exchange; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Where can input be transformed or validated and why is client-side validation insufficient? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram includes at least seven stages and all client-side checks are marked untrusted Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy, Getting Started and Essential Skills topics"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

D142 · M06W1D2 · HTTP methods, headers, cookies and security headers

Phase: Academic month · **Optional date:** 2027-03-17 · **Domain resources:** W04 / W17

1. Day / sequence	Global Day 142 of 365; calendar-free M06W1D2; optional date 2027-03-17.
2. Exact topic	HTTP methods, headers, cookies and security headers
3. Exact subtopics	GET/POST/PUT/PATCH/DELETE, idempotency, status codes, Set-Cookie, Secure, HttpOnly, SameSite, CSP, HSTS and CORS basics
4. Learning objective	Interpret security-relevant headers and explain their limitations
5. Primary lecture / reading + direct URL	MDN HTTP documentation `https://developer.mozilla.org/en-US/docs/Web/HTTP` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	MDN HTTP documentation, Methods, Headers, Cookies, CSP and CORS sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Inspect headers from a local app and an authorized site; propose a header policy and predict one functional breakage. **2.** Standard (~2 h): Inspect headers from a local app and an authorized site; propose a header policy and predict one functional breakage. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Inspect headers from a local app and an authorized site; propose a header policy and predict one functional breakage. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	headers-policy.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does SameSite=Lax not solve every CSRF problem? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All assigned headers are classified and limitations are explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MDN HTTP documentation, Methods, Headers, Cookies, CSP and CORS sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MDN HTTP Overview; write a five-line correction; retest next morning with a new 10-minute problem.

D143 · M06W1D3 · Sessions, cookies and authentication flaws

Phase: Academic month · **Optional date:** 2027-03-18 · **Domain resources:** W01 / C08

1. Day / sequence	Global Day 143 of 365; calendar-free M06W1D3; optional date 2027-03-18.
2. Exact topic	Sessions, cookies and authentication flaws
3. Exact subtopics	Session identifiers, fixation, entropy, timeout, logout, rotation, password reset and multi-factor flows
4. Learning objective	Design a secure session lifecycle and identify weak token handling
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy, Authentication topic and Apprentice labs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-63B `https://pages.nist.gov/800-63-4/sp800-63b.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Solve one authentication Apprentice lab without a walkthrough, then write secure session requirements and a remediation test. **2.** Standard (~2 h): Solve one authentication Apprentice lab without a walkthrough, then write secure session requirements and a remediation test. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Solve one authentication Apprentice lab without a walkthrough, then write secure session requirements and a remediation test. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	auth-lab-notes.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why must session identifiers rotate after privilege change? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Lab is solved in the authorized academy and remediation includes server-side invalidation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy, Authentication topic and Apprentice labs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

D144 · M06W1D4 · Access control and IDOR/BOLA

Phase: Academic month · **Optional date:** 2027-03-19 · **Domain resources:** W01 / C16

1. Day / sequence	Global Day 144 of 365; calendar-free M06W1D4; optional date 2027-03-19.
2. Exact topic	Access control and IDOR/BOLA
3. Exact subtopics	Vertical/horizontal privilege escalation, object references, authorization on every request, deny by default and multi-step checks
4. Learning objective	Prove where authorization must occur in a request flow
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy, Access Control topic; C16 OWASP API Top 10 API1 BOLA. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP API Security Top 10 `https://owasp.org/API-Security/editions/2023/en/0x11-t10/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Solve one IDOR/access-control Apprentice lab; implement a small local function that authorizes by user and object owner. **2.** Standard (~2 h): Solve one IDOR/access-control Apprentice lab; implement a small local function that authorizes by user and object owner. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Solve one IDOR/access-control Apprentice lab; implement a small local function that authorizes by user and object owner. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	access-control.md and code snippet; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is hiding an object ID in the UI not an authorization control? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Lab is solved and local code denies access server-side Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy, Access Control topic; C16 OWASP API Top 10 API1 BOLA"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

D145 · M06W1D5 · OWASP Top 10 as a coverage index

Phase: Academic month · **Optional date:** 2027-03-20 · **Domain resources:** C14 / W01

1. Day / sequence	Global Day 145 of 365; calendar-free M06W1D5; optional date 2027-03-20.
2. Exact topic	OWASP Top 10 as a coverage index
3. Exact subtopics	A01-A10 categories, root causes, prevention patterns and limitations of a risk taxonomy
4. Learning objective	Map each category to a lab, a prevention control and an evidence source
5. Primary lecture / reading + direct URL	OWASP Top 10 `https://owasp.org/www-project-top-ten/` **Type:** Reference · **Access:** Free
6. Exact section / lesson locator	OWASP Top 10 project, A01 through A10 pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	PortSwigger Web Security Academy `https://portswigger.net/web-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a personal coverage matrix with ten categories and a PortSwigger lab or local exercise for each; label which are not yet completed. **2.** Standard (~2 h): Build a personal coverage matrix with ten categories and a PortSwigger lab or local exercise for each; label which are not yet completed. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a personal coverage matrix with ten categories and a PortSwigger lab or local exercise for each; label which are not yet completed. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	owasp-matrix.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does completing an OWASP lab not prove an entire application is secure? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Matrix is complete and explicitly states no claim of full coverage Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Top 10 project, A01 through A10 pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D146 · M06W1D6 · Secure coding: input validation and output encoding

Phase: Academic month · **Optional date:** 2027-03-21 · **Domain resources:** W17 / P08

1. Day / sequence	Global Day 146 of 365; calendar-free M06W1D6; optional date 2027-03-21.
2. Exact topic	Secure coding: input validation and output encoding
3. Exact subtopics	Allow-lists, canonicalization, type/length/range checks, context-aware encoding, template escaping and Unicode pitfalls
4. Learning objective	Fix a deliberately vulnerable local code snippet using safe library functions
5. Primary lecture / reading + direct URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/` **Type:** Official checklist library · **Access:** Free
6. Exact section / lesson locator	OWASP Cheat Sheet Series, Input Validation and XSS Prevention cheat sheets. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Implement server-side validation and context-aware output encoding in three synthetic snippets; add tests for Unicode, null bytes and long strings. **2.** Standard (~2 h): Implement server-side validation and context-aware output encoding in three synthetic snippets; add tests for Unicode, null bytes and long strings. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Implement server-side validation and context-aware output encoding in three synthetic snippets; add tests for Unicode, null bytes and long strings. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	secure-validation.py and tests; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can validation fail if canonicalization happens after the security check? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Tests pass and all outputs are escaped in the correct context Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Cheat Sheet Series, Input Validation and XSS Prevention cheat sheets"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Application Security Verification Standard; write a five-line correction; retest next morning with a new 10-minute problem.

D147 · M06W1D7 · Weekly web foundations review and local lab setup

Phase: Academic month · **Optional date:** 2027-03-22 · **Domain resources:** W11 / W08 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 147 of 365; calendar-free M06W1D7; optional date 2027-03-22.
2. Exact topic	Weekly web foundations review and local lab setup
3. Exact subtopics	HTTP, headers, sessions, access control, OWASP Top 10, validation and output encoding; active recall
4. Learning objective	Demonstrate Month 6 Week 1 understanding with a small local app
5. Primary lecture / reading + direct URL	OWASP WebGoat `https://owasp.org/www-project-webgoat/` **Type:** Vulnerable-by-design lessons · **Access:** Free/open-source
6. Exact section / lesson locator	OWASP WebGoat HTTP basics and access-control lessons; use only in an isolated local container/VM. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Juice Shop `https://owasp.org/www-project-juice-shop/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Deploy a local vulnerable app on loopback only, complete three WebGoat or PortSwigger labs, and document the defensive fix for each — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Deploy a local vulnerable app on loopback only, complete three WebGoat or PortSwigger labs, and document the defensive fix for each — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Deploy a local vulnerable app on loopback only, complete three WebGoat or PortSwigger labs, and document the defensive fix for each — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M6W1-review.md and lab evidence; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which vulnerability was caused by missing server-side authorization rather than bad input? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and all services are bound to localhost/internal network Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP WebGoat HTTP basics and access-control lessons; use only in an isolated local container/VM"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

Month 06 · Week 2 · daily cards

D148 · M06W2D1 · SQL injection and parameterized queries

Phase: Academic month · **Optional date:** 2027-03-23 · **Domain resources:** W01 / P08

1. Day / sequence	Global Day 148 of 365; calendar-free M06W2D1; optional date 2027-03-23.
2. Exact topic	SQL injection and parameterized queries
3. Exact subtopics	In-band, blind and second-order injection; UNION concepts; stacked queries; stored procedures; ORMs; parameterization; least privilege
4. Learning objective	Demonstrate a PortSwigger SQLi lab and repair an equivalent local vulnerable query
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy, SQL Injection topic and SQL injection learning path. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete two Apprentice SQLi labs in the Academy, then rewrite a local synthetic vulnerable query with bound parameters and tests. **2.** Standard (~2 h): Complete two Apprentice SQLi labs in the Academy, then rewrite a local synthetic vulnerable query with bound parameters and tests. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete two Apprentice SQLi labs in the Academy, then rewrite a local synthetic vulnerable query with bound parameters and tests. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	sqli-remediation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does parameterization prevent data from changing SQL structure? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Labs are solved only in authorized environments and local tests include malicious strings Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy, SQL Injection topic and SQL injection learning path"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

D149 · M06W2D2 · Cross-site scripting and DOM security

Phase: Academic month · **Optional date:** 2027-03-24 · **Domain resources:** W01 / W17

1. Day / sequence	Global Day 149 of 365; calendar-free M06W2D2; optional date 2027-03-24.
2. Exact topic	Cross-site scripting and DOM security
3. Exact subtopics	Reflected, stored and DOM XSS; contexts; CSP; Trusted Types concepts; sanitization; framework auto-escaping and dangerous sinks
4. Learning objective	Identify sinks and apply context-specific defenses
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy, Cross-site scripting topic and Apprentice labs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete two XSS Apprentice labs, inspect the DOM sink, and write a local safe-rendering example plus CSP policy. **2.** Standard (~2 h): Complete two XSS Apprentice labs, inspect the DOM sink, and write a local safe-rendering example plus CSP policy. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete two XSS Apprentice labs, inspect the DOM sink, and write a local safe-rendering example plus CSP policy. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	xss-notes.md and CSP example; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can HTML entity encoding fail inside a JavaScript string or URL context? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each lab has a root cause and each local output uses the correct encoding Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy, Cross-site scripting topic and Apprentice labs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

D150 · M06W2D3 · CSRF, SameSite and anti-CSRF tokens

Phase: Academic month · **Optional date:** 2027-03-25 · **Domain resources:** W01 / W17

1. Day / sequence	Global Day 150 of 365; calendar-free M06W2D3; optional date 2027-03-25.
2. Exact topic	CSRF, SameSite and anti-CSRF tokens
3. Exact subtopics	State-changing requests, cookie ambient authority, synchronizer tokens, double-submit, Origin/ Referer checks and same-site policy
4. Learning objective	Explain which requests require CSRF defense and how tokens should be validated
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy, CSRF topic and Apprentice labs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Solve one CSRF lab, implement a signed anti-CSRF token in a local app, and test token reuse and mismatch cases. **2.** Standard (~2 h): Solve one CSRF lab, implement a signed anti-CSRF token in a local app, and test token reuse and mismatch cases. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Solve one CSRF lab, implement a signed anti-CSRF token in a local app, and test token reuse and mismatch cases. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	csrf-defense.py and lab notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does a CSRF token need to be tied to a user session? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Lab is solved and local token validation rejects all assigned negative tests Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy, CSRF topic and Apprentice labs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

D151 · M06W2D4 · SSRF and outbound request controls

Phase: Academic month · **Optional date:** 2027-03-26 · **Domain resources:** W01 / C17

1. Day / sequence	Global Day 151 of 365; calendar-free M06W2D4; optional date 2027-03-26.
2. Exact topic	SSRF and outbound request controls
3. Exact subtopics	Internal service access, URL parsing, DNS rebinding concepts, metadata services, allow-lists, network egress and response handling
4. Learning objective	Design server-side controls for a feature that fetches a URL
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy, SSRF topic and Apprentice labs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Solve one basic SSRF lab, then write a local fetcher that enforces scheme, host allow-list, DNS resolution checks and egress restrictions. **2.** Standard (~2 h): Solve one basic SSRF lab, then write a local fetcher that enforces scheme, host allow-list, DNS resolution checks and egress restrictions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Solve one basic SSRF lab, then write a local fetcher that enforces scheme, host allow-list, DNS resolution checks and egress restrictions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ssrf-lab.md and fetcher.py; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a string blocklist of localhost insufficient against SSRF? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Authorized lab is solved and local implementation includes multiple independent checks Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy, SSRF topic and Apprentice labs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

D152 · M06W2D5 · Path traversal and file upload security

Phase: Academic month · **Optional date:** 2027-03-27 · **Domain resources:** W01 / W17

1. Day / sequence	Global Day 152 of 365; calendar-free M06W2D5; optional date 2027-03-27.
2. Exact topic	Path traversal and file upload security
3. Exact subtopics	Canonicalization, traversal sequences, zip slip, file type confusion, content disposition, storage outside web root, size limits and malware scanning
4. Learning objective	Build a safe file-upload workflow from untrusted input
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy, Path traversal and File upload topics; choose one Apprentice lab each. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete one lab, then implement safe filename generation, extension/content checks, size cap and storage outside the web root in a local test app. **2.** Standard (~2 h): Complete one lab, then implement safe filename generation, extension/content checks, size cap and storage outside the web root in a local test app. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete one lab, then implement safe filename generation, extension/content checks, size cap and storage outside the web root in a local test app. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	secure-upload.py and tests; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a MIME type supplied by the client not trustworthy? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Tests reject all malicious samples and no uploaded file is executable Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy, Path traversal and File upload topics; choose one Apprentice lab each"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

D153 · M06W2D6 · Weekly injection and client-side security review

Phase: Academic month · **Optional date:** 2027-03-28 · **Domain resources:** W02 / W17

1. Day / sequence	Global Day 153 of 365; calendar-free M06W2D6; optional date 2027-03-28.
2. Exact topic	Weekly injection and client-side security review
3. Exact subtopics	SQLi, XSS, CSRF, SSRF, path traversal and file upload; defensive mapping and retrieval
4. Learning objective	Demonstrate that each exploit has a concrete preventive control and validation test
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy Learning Paths `https://portswigger.net/web-security/learning-paths` **Type:** Structured learning paths · **Access:** Free
6. Exact section / lesson locator	PortSwigger Web Security Academy Learning Paths, server-side vulnerabilities path; complete assigned labs only. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a table mapping six vulnerabilities to root cause, control, test and residual risk; retake one failed lab from memory. **2.** Standard (~2 h): Create a table mapping six vulnerabilities to root cause, control, test and residual risk; retake one failed lab from memory. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a table mapping six vulnerabilities to root cause, control, test and residual risk; retake one failed lab from memory. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M6W2-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which two vulnerability classes share canonicalization as a root cause? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and every row has a working validation test Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy Learning Paths, server-side vulnerabilities path; complete assigned labs only"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D154 · M06W2D7 · API fundamentals and authorization models

Phase: Academic month · **Optional date:** 2027-03-29 · **Domain resources:** C16 / W01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 154 of 365; calendar-free M06W2D7; optional date 2027-03-29.
2. Exact topic	API fundamentals and authorization models
3. Exact subtopics	REST/GraphQL basics, resource identifiers, verbs, JSON schemas, pagination, BOLA, BFLA, object property authorization and mass assignment
4. Learning objective	Test an API for object-level authorization using synthetic data
5. Primary lecture / reading + direct URL	OWASP API Security Top 10 `https://owasp.org/API-Security/editions/2023/en/0x11-t10/` **Type:** Reference · **Access:** Free
6. Exact section / lesson locator	OWASP API Security Top 10, API1 BOLA, API3 property authorization and API5 BFLA. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	PortSwigger Web Security Academy `https://portswigger.net/web-security`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Build a small local API with two users and five resources; write tests that prove one user cannot read or modify another's objects — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Build a small local API with two users and five resources; write tests that prove one user cannot read or modify another's objects — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Build a small local API with two users and five resources; write tests that prove one user cannot read or modify another's objects — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	api-authz-tests.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Why must authorization use the authenticated server-side identity rather than a user-supplied ID? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All cross-user tests fail closed and legitimate tests pass Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP API Security Top 10, API1 BOLA, API3 property authorization and API5 BFLA"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Top 10; write a five-line correction; retest next morning with a new 10-minute problem.

Month 06 · Week 3 · daily cards

D155 · M06W3D1 · Rate limiting, input validation and abuse cases

Phase: Academic month · **Optional date:** 2027-03-30 · **Domain resources:** W01 / C15

1. Day / sequence	Global Day 155 of 365; calendar-free M06W3D1; optional date 2027-03-30.
2. Exact topic	Rate limiting, input validation and abuse cases
3. Exact subtopics	Authentication throttling, resource limits, pagination bounds, idempotency, request size, enumeration and business-logic abuse
4. Learning objective	Design abuse controls that preserve legitimate function
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy, Rate limiting and Business logic vulnerabilities topics. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Application Security Verification Standard `https://owasp.org/www-project-application-security-verification-standard/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Add rate limits and pagination bounds to the local API; test against a synthetic burst and one business-logic bypass. **2.** Standard (~2 h): Add rate limits and pagination bounds to the local API; test against a synthetic burst and one business-logic bypass. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Add rate limits and pagination bounds to the local API; test against a synthetic burst and one business-logic bypass. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	rate-limit-tests.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a global rate limit be both too strict and too weak for an API? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Controls are tested and limitations are explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy, Rate limiting and Business logic vulnerabilities topics"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

D156 · M06W3D2 · Secrets management and configuration security

Phase: Academic month · **Optional date:** 2027-03-31 · **Domain resources:** W13 / W17

1. Day / sequence	Global Day 156 of 365; calendar-free M06W3D2; optional date 2027-03-31.
2. Exact topic	Secrets management and configuration security
3. Exact subtopics	Environment variables, secret managers, rotation, .gitignore, entropy, logging redaction, config drift and default credentials
4. Learning objective	Remove secrets from a deliberately unsafe local repository and prevent recurrence
5. Primary lecture / reading + direct URL	GitHub code security documentation `https://docs.github.com/en/code-security` **Type:** Official documentation · **Access:** Free features vary by repository visibility and plan
6. Exact section / lesson locator	GitHub code security documentation, secret scanning and repository security sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a synthetic repository with dummy secrets, remove them from working code/config, add .gitignore, pre-commit-style checks and a rotation plan. **2.** Standard (~2 h): Create a synthetic repository with dummy secrets, remove them from working code/config, add .gitignore, pre-commit-style checks and a rotation plan. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a synthetic repository with dummy secrets, remove them from working code/config, add .gitignore, pre-commit-style checks and a rotation plan. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	secrets-remediation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does deleting a secret from the latest commit not remove it from Git history? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No secret remains in current files and the plan addresses history/rotation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub code security documentation, secret scanning and repository security sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Semgrep documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D157 · M06W3D3 · Secure SDLC and code review gates

Phase: Academic month · **Optional date:** 2027-04-01 · **Domain resources:** P08 / W13

1. Day / sequence	Global Day 157 of 365; calendar-free M06W3D3; optional date 2027-04-01.
2. Exact topic	Secure SDLC and code review gates
3. Exact subtopics	Requirements, threat modelling, secure design, dependencies, code review, SAST/DAST, testing, release and incident feedback
4. Learning objective	Design a small secure development pipeline with measurable gates
5. Primary lecture / reading + direct URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/` **Type:** Checklist/reference · **Access:** Free
6. Exact section / lesson locator	OWASP Secure Coding Practices Quick Reference Guide and OWASP ASVS Level 1 selected requirements. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a CI pipeline with at least five security gates; define pass/fail evidence and a risk-acceptance path for each. **2.** Standard (~2 h): Draw a CI pipeline with at least five security gates; define pass/fail evidence and a risk-acceptance path for each. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a CI pipeline with at least five security gates; define pass/fail evidence and a risk-acceptance path for each. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	sdlc-pipeline.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which gate would have detected the most bugs in your Month 4 tool? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Pipeline is realistic for a solo student and each gate has objective criteria Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Secure Coding Practices Quick Reference Guide and OWASP ASVS Level 1 selected requirements"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D158 · M06W3D4 · Dependency and supply-chain security

Phase: Academic month · **Optional date:** 2027-04-02 · **Domain resources:** W12 / W16

1. Day / sequence	Global Day 158 of 365; calendar-free M06W3D4; optional date 2027-04-02.
2. Exact topic	Dependency and supply-chain security
3. Exact subtopics	SBOMs, lockfiles, provenance, pinning, typosquatting, vulnerable dependencies, transitive risk and update policy
4. Learning objective	Triage a synthetic vulnerable dependency and improve repository hygiene
5. Primary lecture / reading + direct URL	OWASP Dependency-Check `https://owasp.org/www-project-dependency-check/` **Type:** Open-source SCA tool · **Access:** Free/open-source
6. Exact section / lesson locator	OWASP Dependency-Check documentation; W15 OpenSSF Scorecard checks. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SLSA specification `https://slsa.dev/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Scan a synthetic project with a deliberately old dependency, triage findings, pin versions, add an SBOM and improve two Scorecard checks. **2.** Standard (~2 h): Scan a synthetic project with a deliberately old dependency, triage findings, pin versions, add an SBOM and improve two Scorecard checks. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Scan a synthetic project with a deliberately old dependency, triage findings, pin versions, add an SBOM and improve two Scorecard checks. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	supply-chain-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a high/critical CVE be unexploitable in your specific usage? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Findings are triaged with context and no blind upgrade without testing Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Dependency-Check documentation; W15 OpenSSF Scorecard checks"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OpenSSF Scorecard; write a five-line correction; retest next morning with a new 10-minute problem.

D159 · M06W3D5 · Weekly API, SDLC and supply-chain review

Phase: Academic month · **Optional date:** 2027-04-03 · **Domain resources:** W13 / C15

1. Day / sequence	Global Day 159 of 365; calendar-free M06W3D5; optional date 2027-04-03.
2. Exact topic	Weekly API, SDLC and supply-chain review
3. Exact subtopics	API authorization, rate limits, validation, secrets, secure SDLC, SAST, SCA and provenance; active recall
4. Learning objective	Demonstrate Month 6 Week 3 competence through a mini secure-release review
5. Primary lecture / reading + direct URL	GitHub code security documentation `https://docs.github.com/en/code-security` **Type:** Official documentation · **Access:** Free features vary by repository visibility and plan
6. Exact section / lesson locator	GitHub code security docs, code scanning, secret scanning and Dependabot sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Application Security Verification Standard `https://owasp.org/www-project-application-security-verification-standard/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review a synthetic release candidate, decide pass/fail for five gates, and justify one temporary risk acceptance with compensating controls. **2.** Standard (~2 h): Review a synthetic release candidate, decide pass/fail for five gates, and justify one temporary risk acceptance with compensating controls. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review a synthetic release candidate, decide pass/fail for five gates, and justify one temporary risk acceptance with compensating controls. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M6W3-practical.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which supply-chain risk is not detectable by a vulnerability scanner alone? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and every gate decision has evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub code security docs, code scanning, secret scanning and Dependabot sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Semgrep documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D160 · M06W3D6 · Secure web application engineering sprint: design

Phase: Academic month · **Optional date:** 2027-04-04 · **Domain resources:** C15 / C17

1. Day / sequence	Global Day 160 of 365; calendar-free M06W3D6; optional date 2027-04-04.
2. Exact topic	Secure web application engineering sprint: design
3. Exact subtopics	Functional requirements, data model, auth model, roles, threat model, trust boundaries, abuse cases and test plan
4. Learning objective	Design a small secure notes API with explicit security requirements
5. Primary lecture / reading + direct URL	OWASP Application Security Verification Standard `https://owasp.org/www-project-application-security-verification-standard/` **Type:** Standard/reference · **Access:** Free
6. Exact section / lesson locator	OWASP ASVS Level 1 requirements V1, V2, V3, V4, V5 and V7. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a specification for a local notes API, including user roles, session policy, object ownership, validation rules and security test cases. **2.** Standard (~2 h): Write a specification for a local notes API, including user roles, session policy, object ownership, validation rules and security test cases. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a specification for a local notes API, including user roles, session policy, object ownership, validation rules and security test cases. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-08 design.md and threat-model.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which requirement prevents the most likely authorization bug? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Design is complete and every security requirement has a planned test Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP ASVS Level 1 requirements V1, V2, V3, V4, V5 and V7"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Secure Coding Practices Quick Reference Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D161 · M06W3D7 · Secure web application engineering sprint: implementation

Phase: Academic month · **Optional date:** 2027-04-05 · **Domain resources:** W17 / P08 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 161 of 365; calendar-free M06W3D7; optional date 2027-04-05.
2. Exact topic	Secure web application engineering sprint: implementation
3. Exact subtopics	Parameterized queries, password hashing/KDF, session cookies, anti-CSRF, output encoding, rate limiting and structured errors
4. Learning objective	Implement the core API with secure defaults and no secrets in code
5. Primary lecture / reading + direct URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/` **Type:** Official checklist library · **Access:** Free
6. Exact section / lesson locator	OWASP Cheat Sheet Series, Authentication, Session Management, CSRF and SQL Injection Prevention cheat sheets. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Implement registration, login, logout, create/read/update/delete notes, object ownership checks and input validation; use a local database only — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Implement registration, login, logout, create/read/update/delete notes, object ownership checks and input validation; use a local database only — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Implement registration, login, logout, create/read/update/delete notes, object ownership checks and input validation; use a local database only — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-08 source code and migrations; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Why is error handling a security control rather than just a user-experience concern? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All core functions work locally and no sensitive data appears in errors or logs Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Cheat Sheet Series, Authentication, Session Management, CSRF and SQL Injection Prevention cheat sheets"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Application Security Verification Standard; write a five-line correction; retest next morning with a new 10-minute problem.

Month 06 · Week 4 · daily cards

D162 · M06W4D1 · Secure web application engineering sprint: testing

Phase: Academic month · **Optional date:** 2027-04-06 · **Domain resources:** P05 / W14

1. Day / sequence	Global Day 162 of 365; calendar-free M06W4D1; optional date 2027-04-06.
2. Exact topic	Secure web application engineering sprint: testing
3. Exact subtopics	Unit, integration, authorization, injection, XSS, CSRF, SSRF-like fetch controls, dependency scan and negative tests
4. Learning objective	Prove the app resists assigned abuse cases
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest documentation and fixtures/parametrization sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Semgrep documentation `https://semgrep.dev/docs/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write at least 20 automated tests, including cross-user access, malicious input, expired session and invalid token; run Semgrep and triage findings. **2.** Standard (~2 h): Write at least 20 automated tests, including cross-user access, malicious input, expired session and invalid token; run Semgrep and triage findings. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write at least 20 automated tests, including cross-user access, malicious input, expired session and invalid token; run Semgrep and triage findings. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-08 test report and scanner output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which test was most valuable and what bug did it expose? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All tests pass and scanner findings are triaged with evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest documentation and fixtures/parametrization sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D163 · M06W4D2 · Local web security assessment of the app

Phase: Academic month · **Optional date:** 2027-04-07 · **Domain resources:** W06 / E01

1. Day / sequence	Global Day 163 of 365; calendar-free M06W4D2; optional date 2027-04-07.
2. Exact topic	Local web security assessment of the app
3. Exact subtopics	Written scope, permitted discovery, enumeration, validation, impact, evidence, remediation and retest
4. Learning objective	Perform an authorized assessment against your own loopback app and fix the findings
5. Primary lecture / reading + direct URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/` **Type:** Testing methodology · **Access:** Free
6. Exact section / lesson locator	OWASP Web Security Testing Guide, relevant sections for authentication, authorization, session and input validation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a rules-of-engagement document, test five areas of your own local app, record three findings with sanitized evidence, then remediate and retest. **2.** Standard (~2 h): Create a rules-of-engagement document, test five areas of your own local app, record three findings with sanitized evidence, then remediate and retest. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a rules-of-engagement document, test five areas of your own local app, record three findings with sanitized evidence, then remediate and retest. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-09 assessment report; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding was a false positive and why did you conclude that? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Scope is explicit, targets are local/owned, and all findings have retest evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Web Security Testing Guide, relevant sections for authentication, authorization, session and input validation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D164 · M06W4D3 · Dependency and CI security for the capstone

Phase: Academic month · **Optional date:** 2027-04-08 · **Domain resources:** W13 / W14

1. Day / sequence	Global Day 164 of 365; calendar-free M06W4D3; optional date 2027-04-08.
2. Exact topic	Dependency and CI security for the capstone
3. Exact subtopics	Debug mode, default accounts, security headers, CORS, pinned actions, least permissions, secret scanning, code scanning, dependency scanning, SBOM and reproducible builds
4. Learning objective	Harden a GitHub Actions workflow for the secure API repository
5. Primary lecture / reading + direct URL	GitHub code security documentation `https://docs.github.com/en/code-security` **Type:** Official documentation · **Access:** Free features vary by repository visibility and plan
6. Exact section / lesson locator	GitHub Actions security hardening, GITHUB_TOKEN permissions, secrets and third-party action risks. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Semgrep documentation `https://semgrep.dev/docs/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a CI workflow with pinned action SHAs, least privileges, tests, Semgrep and dependency scanning; verify no secrets in logs. **2.** Standard (~2 h): Create a CI workflow with pinned action SHAs, least privileges, tests, Semgrep and dependency scanning; verify no secrets in logs. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a CI workflow with pinned action SHAs, least privileges, tests, Semgrep and dependency scanning; verify no secrets in logs. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	.github/workflows/security.yml and run evidence; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should third-party actions be pinned to a commit SHA rather than a mutable tag? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Workflow runs and all assigned controls are demonstrably present Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub Actions security hardening, GITHUB_TOKEN permissions, secrets and third-party action risks"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Semgrep documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D165 · M06W4D4 · Portfolio publication of secure app and assessment

Phase: Academic month · **Optional date:** 2027-04-09 · **Domain resources:** F09 / W13

1. Day / sequence	Global Day 165 of 365; calendar-free M06W4D4; optional date 2027-04-09.
2. Exact topic	Portfolio publication of secure app and assessment
3. Exact subtopics	README, architecture, threat model, secure-design trade-offs, sanitized evidence, tests, limitations and reproduction steps
4. Learning objective	Publish both attacker and defender evidence without leaking secrets
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 2-3 and GitHub Docs README conventions linked from F09. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Finalize PRJ-08 and PRJ-09, add setup scripts, sanitized screenshots, test instructions and a lessons-learned section. **2.** Standard (~2 h): Finalize PRJ-08 and PRJ-09, add setup scripts, sanitized screenshots, test instructions and a lessons-learned section. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Finalize PRJ-08 and PRJ-09, add setup scripts, sanitized screenshots, test instructions and a lessons-learned section. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	published repositories and release tags; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which README section proves independent reasoning rather than tutorial copying? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Portfolio checklist passes and no credentials, real personal data or active exploit endpoints are published Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 2-3 and GitHub Docs README conventions linked from F09"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D166 · M06W4D5 · Month 6 practical examination and milestone gate

Phase: Academic month · **Optional date:** 2027-04-10 · **Domain resources:** W01 / C15

1. Day / sequence	Global Day 166 of 365; calendar-free M06W4D5; optional date 2027-04-10.
2. Exact topic	Month 6 practical examination and milestone gate
3. Exact subtopics	Web/API security, secure coding, OWASP Top 10, defensive fixes, SDLC and supply chain; practical scenario
4. Learning objective	Demonstrate secure web engineering and authorized assessment competence
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy all-labs index; select only Apprentice labs for revision. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Application Security Verification Standard `https://owasp.org/www-project-application-security-verification-standard/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 120-minute practical: find and fix two vulnerabilities in an unseen local snippet, design API authorization controls, and write a remediation report. **2.** Standard (~2 h): Complete a 120-minute practical: find and fix two vulnerabilities in an unseen local snippet, design API authorization controls, and write a remediation report. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 120-minute practical: find and fix two vulnerabilities in an unseen local snippet, design API authorization controls, and write a remediation report. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M6-assessment.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which fix addresses the root cause and which only blocks one payload? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, milestone gate passed, practical at least 18/25 and safety at least 9/10 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy all-labs index; select only Apprentice labs for revision"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

D167 · M06W4D6 · Monthly review and gap repairs

Phase: Academic month · **Optional date:** 2027-04-11 · **Domain resources:** W17 / W06

1. Day / sequence	Global Day 167 of 365; calendar-free M06W4D6; optional date 2027-04-11.
2. Exact topic	Monthly review and gap repairs
3. Exact subtopics	Injection/XSS/CSRF/SSRF root causes, authorization, session controls, headers, secrets and CI gates; active recall
4. Learning objective	Repair Month 6 gaps and prepare ethical testing methodology
5. Primary lecture / reading + direct URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/` **Type:** Official checklist library · **Access:** Free
6. Exact section / lesson locator	OWASP Cheat Sheet Series selected cheat sheets for your three weakest topics. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework failed assessment items with new code examples, add missing tests, and review the OWASP WSTG and NIST SP 800-115 structure. **2.** Standard (~2 h): Rework failed assessment items with new code examples, add missing tests, and review the OWASP WSTG and NIST SP 800-115 structure. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework failed assessment items with new code examples, add missing tests, and review the OWASP WSTG and NIST SP 800-115 structure. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M6-review.md and retest score; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which web vulnerability class is still easy to identify but hard to remediate correctly? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and each gap has a new defensive test Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Cheat Sheet Series selected cheat sheets for your three weakest topics"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Application Security Verification Standard; write a five-line correction; retest next morning with a new 10-minute problem.

D168 · M06W4D7 · Spaced revision, recovery and Month 7 preview

Phase: Academic month · **Optional date:** 2027-04-12 · **Domain resources:** W01 / E01 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 168 of 365; calendar-free M06W4D7; optional date 2027-04-12.
2. Exact topic	Spaced revision, recovery and Month 7 preview
3. Exact subtopics	Secure headers, auth/authz, injection, XSS, CSRF, SSRF, file upload, API and supply-chain retrieval; lighter workload
4. Learning objective	Preserve web-security knowledge and prepare authorized assessment rules
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy dashboard for spaced lab review, one old Apprentice lab only. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, solve one old lab without notes, read the first sections of NIST SP 800-115, and rest — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, solve one old lab without notes, read the first sections of NIST SP 800-115, and rest — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, solve one old lab without notes, read the first sections of NIST SP 800-115, and rest — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M6-consolidation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: How would you explain the difference between a vulnerability, an exploit and a finding to a client? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retrieval score is recorded and explanation is accurate Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy dashboard for spaced lab review, one old Apprentice lab only"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

Month 07 — Ethical testing, vulnerability assessment and authorized labs

Days D169-D196. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Ethics, law and written authorization D2: Rules of engagement and assessment planning D3: Asset inventory and permitted discovery D4: Threat modelling for an authorized assessment D5: Discovery and enumeration in an isolated lab D6: Vulnerability scanning concepts and validation D7: Weekly ethical testing review and scope gate
Week 2	D1: Web vulnerability validation in PortSwigger labs D2: Network service validation in a local VM lab D3: Risk prioritization beyond CVSS D4: Evidence collection and chain of custody D5: Remediation recommendations that engineers can implement D6: Weekly validation and remediation review D7: Retesting and closure criteria
Week 3	D1: Technical report writing: executive summary and findings D2: Professional communication and defensive debrief D3: Responsible disclosure and coordinated vulnerability handling D4: Common penetration-testing pitfalls and professional limits D5: Weekly report-writing and ethics review D6: Authorized assessment capstone: planning and discovery D7: Authorized assessment capstone: execution and evidence
Week 4	D1: Authorized assessment capstone: analysis and report D2: Authorized assessment capstone: remediation and retest D3: Portfolio publication and independent reflection D4: Certification and specialization decision point D5: Month 7 practical examination and mastery gate D6: Monthly review and gap repairs D7: Spaced revision, recovery and Month 8 preview

> **Gate:** score the Month 7 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 07 · Week 1 · daily cards

D169 · M07W1D1 · Ethics, law and written authorization

Phase: Academic month · **Optional date:** 2027-04-13 · **Domain resources:** E01 / E05

1. Day / sequence	Global Day 169 of 365; calendar-free M07W1D1; optional date 2027-04-13.
2. Exact topic	Ethics, law and written authorization
3. Exact subtopics	Ownership, consent, written scope, jurisdiction, computer-misuse concepts, privacy, professional codes and consequences
4. Learning objective	Explain why authorization is a technical prerequisite, not a formality
5. Primary lecture / reading + direct URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final` **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115, planning and legal/ethical considerations sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	HackerOne Disclosure Guidelines `https://www.hackerone.com/disclosure-guidelines`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a one-page ethics statement and a written authorization template for a fictional lab owned by the student. **2.** Standard (~2 h): Write a one-page ethics statement and a written authorization template for a fictional lab owned by the student. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a one-page ethics statement and a written authorization template for a fictional lab owned by the student. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ethics-statement.md and authorization-template.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which three facts must be present before any active scanning begins? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Documents define owner, assets, time, permitted actions, emergency contact and stop conditions Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115, planning and legal/ethical considerations sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D170 · M07W1D2 · Rules of engagement and assessment planning

Phase: Academic month · **Optional date:** 2027-04-14 · **Domain resources:** E01 / C26

1. Day / sequence	Global Day 170 of 365; calendar-free M07W1D2; optional date 2027-04-14.
2. Exact topic	Rules of engagement and assessment planning
3. Exact subtopics	Objectives, in-scope/out-of-scope assets, testing windows, allowed techniques, prohibited actions, data handling, escalation and abort criteria
4. Learning objective	Build a complete RoE for an isolated student lab
5. Primary lecture / reading + direct URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final` **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115, assessment planning and rules of behaviour sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-30 Revision 1 `https://csrc.nist.gov/pubs/sp/800/30/r1/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a detailed RoE for a two-VM local lab, including evidence handling, rate limits and destruction/cleanup steps. **2.** Standard (~2 h): Create a detailed RoE for a two-VM local lab, including evidence handling, rate limits and destruction/cleanup steps. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a detailed RoE for a two-VM local lab, including evidence handling, rate limits and destruction/cleanup steps. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	rules-of-engagement.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which clause protects availability if a scanner causes service failure? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	RoE is unambiguous and includes a specific stop condition and contact path Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115, assessment planning and rules of behaviour sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D171 · M07W1D3 · Asset inventory and permitted discovery

Phase: Academic month · **Optional date:** 2027-04-15 · **Domain resources:** C21 / N09

1. Day / sequence	Global Day 171 of 365; calendar-free M07W1D3; optional date 2027-04-15.
2. Exact topic	Asset inventory and permitted discovery
3. Exact subtopics	Asset register, IP ranges, domains, owners, services, data sensitivity, passive discovery and active confirmation
4. Learning objective	Build an asset inventory before any testing activity
5. Primary lecture / reading + direct URL	CIS Controls Version 8 `https://www.cisecurity.org/controls/v8` **Type:** Best-practice controls · **Access:** Free overview; some tools paid
6. Exact section / lesson locator	CIS Controls Version 8, controls on enterprise/data asset inventory. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Nmap Reference Guide `https://nmap.org/book/man.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Inventory your own lab VMs and containers, record IPs, ports, services, owner and authorization status, then verify with one passive and one active method. **2.** Standard (~2 h): Inventory your own lab VMs and containers, record IPs, ports, services, owner and authorization status, then verify with one passive and one active method. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Inventory your own lab VMs and containers, record IPs, ports, services, owner and authorization status, then verify with one passive and one active method. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	asset-inventory.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is an unknown asset outside the inventory unsafe to scan even if it is nearby? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All active discovery targets appear in the signed scope/inventory Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CIS Controls Version 8, controls on enterprise/data asset inventory"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D172 · M07W1D4 · Threat modelling for an authorized assessment

Phase: Academic month · **Optional date:** 2027-04-16 · **Domain resources:** C17 / C18

1. Day / sequence	Global Day 172 of 365; calendar-free M07W1D4; optional date 2027-04-16.
2. Exact topic	Threat modelling for an authorized assessment
3. Exact subtopics	Attack trees, trust boundaries, likely threat actors, abuse cases, assumptions and test hypotheses
4. Learning objective	Turn scope into prioritized test hypotheses
5. Primary lecture / reading + direct URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling` **Type:** Reference/process · **Access:** Free
6. Exact section / lesson locator	OWASP Threat Modeling community page and W06 OWASP WSTG information-gathering sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Produce an attack tree for a fictional local web app and derive five testable hypotheses with required evidence. **2.** Standard (~2 h): Produce an attack tree for a fictional local web app and derive five testable hypotheses with required evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Produce an attack tree for a fictional local web app and derive five testable hypotheses with required evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	attack-tree.svg and hypotheses.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which hypothesis can be disproved with a single benign request? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each hypothesis has a prediction and a safe falsification test Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Threat Modeling community page and W06 OWASP WSTG information-gathering sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D173 · M07W1D5 · Discovery and enumeration in an isolated lab

Phase: Academic month · **Optional date:** 2027-04-17 · **Domain resources:** N09 / N08

1. Day / sequence	Global Day 173 of 365; calendar-free M07W1D5; optional date 2027-04-17.
2. Exact topic	Discovery and enumeration in an isolated lab
3. Exact subtopics	Host discovery, port/service enumeration, banner grabbing, version mapping, OS fingerprinting, DNS and web-content discovery
4. Learning objective	Perform authorized discovery without exploiting anything yet
5. Primary lecture / reading + direct URL	Nmap Reference Guide `https://nmap.org/book/man.html` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Nmap Reference Guide, host discovery, port scanning, service/version detection and output sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	tcpdump manual page `https://www.tcpdump.org/manpages/tcpdump.1.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Scan only an isolated lab VM; save raw output and a normalized service table; capture packets for one scan type. **2.** Standard (~2 h): Scan only an isolated lab VM; save raw output and a normalized service table; capture packets for one scan type. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Scan only an isolated lab VM; save raw output and a normalized service table; capture packets for one scan type. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	discovery-evidence/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which result is ambiguous and what second observation would resolve it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Written scope is attached, output is preserved, and ambiguous results are labelled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Nmap Reference Guide, host discovery, port scanning, service/version detection and output sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RFC 9293: Transmission Control Protocol; write a five-line correction; retest next morning with a new 10-minute problem.

D174 · M07W1D6 · Vulnerability scanning concepts and validation

Phase: Academic month · **Optional date:** 2027-04-18 · **Domain resources:** W07 / E08

1. Day / sequence	Global Day 174 of 365; calendar-free M07W1D6; optional date 2027-04-18.
2. Exact topic	Vulnerability scanning concepts and validation
3. Exact subtopics	Scanner architecture, signatures, authenticated/unauthenticated scans, false positives/negatives, CVE/CVSS and safe validation
4. Learning objective	Distinguish a scanner alert from a validated vulnerability
5. Primary lecture / reading + direct URL	OWASP ZAP documentation `https://www.zaproxy.org/docs/` **Type:** Official tool docs · **Access:** Free/open-source
6. Exact section / lesson locator	OWASP ZAP documentation, passive/active scanning and alert interpretation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NVD `https://nvd.nist.gov/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run a passive/local scan on an intentionally vulnerable local app, choose three alerts, and write benign validation steps that prove or disprove each. **2.** Standard (~2 h): Run a passive/local scan on an intentionally vulnerable local app, choose three alerts, and write benign validation steps that prove or disprove each. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run a passive/local scan on an intentionally vulnerable local app, choose three alerts, and write benign validation steps that prove or disprove each. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	vulnerability-validation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is banner version alone insufficient proof of exploitability? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each alert has evidence, confidence and a safe validation decision Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP ZAP documentation, passive/active scanning and alert interpretation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Burp Suite documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D175 · M07W1D7 · Weekly ethical testing review and scope gate

Phase: Academic month · **Optional date:** 2027-04-19 · **Domain resources:** E01 / C26 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 175 of 365; calendar-free M07W1D7; optional date 2027-04-19.
2. Exact topic	Weekly ethical testing review and scope gate
3. Exact subtopics	Ethics, law, RoE, asset inventory, threat modelling, discovery, enumeration and scanner interpretation
4. Learning objective	Pass a scope-and-safety gate before any exploitation labs
5. Primary lecture / reading + direct URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final` **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115, technical assessment planning and validation sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-30 Revision 1 `https://csrc.nist.gov/pubs/sp/800/30/r1/final`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute case with five scope traps; write decisions and justify each stop/continue choice — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute case with five scope traps; write decisions and justify each stop/continue choice — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute case with five scope traps; write decisions and justify each stop/continue choice — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M7W1-scope-gate.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which action is legal/authorized but still unsafe for availability? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 80/100 on safety and no unauthorized target appears in any evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115, technical assessment planning and validation sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

Month 07 · Week 2 · daily cards

D176 · M07W2D1 · Web vulnerability validation in PortSwigger labs

Phase: Academic month · **Optional date:** 2027-04-20 · **Domain resources:** W01 / W06

1. Day / sequence	Global Day 176 of 365; calendar-free M07W2D1; optional date 2027-04-20.
2. Exact topic	Web vulnerability validation in PortSwigger labs
3. Exact subtopics	Reproducible request/response evidence, impact, root cause, minimal proof of concept and avoiding collateral actions
4. Learning objective	Validate one vulnerability class without using walkthroughs
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy, SQL injection or XSS Apprentice labs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Choose one unseen Apprentice lab, solve it with a documented hypothesis chain, and save sanitized request/response evidence. **2.** Standard (~2 h): Choose one unseen Apprentice lab, solve it with a documented hypothesis chain, and save sanitized request/response evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Choose one unseen Apprentice lab, solve it with a documented hypothesis chain, and save sanitized request/response evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	lab-validation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What observation changed your hypothesis from possible to confirmed? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Lab is solved within the authorized Academy and evidence is reproducible Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy, SQL injection or XSS Apprentice labs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

D177 · M07W2D2 · Network service validation in a local VM lab

Phase: Academic month · **Optional date:** 2027-04-21 · **Domain resources:** N09 / C18

1. Day / sequence	Global Day 177 of 365; calendar-free M07W2D2; optional date 2027-04-21.
2. Exact topic	Network service validation in a local VM lab
3. Exact subtopics	Service version, configuration weakness, default credentials, unnecessary services and controlled proof of impact
4. Learning objective	Validate a synthetic service finding with minimal privileged actions
5. Primary lecture / reading + direct URL	Nmap Reference Guide `https://nmap.org/book/man.html` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Nmap Reference Guide service/version detection and NSE safety sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: In an isolated local VM, identify one intentionally weak service, confirm the issue with benign evidence, and stop before persistence or lateral movement. **2.** Standard (~2 h): In an isolated local VM, identify one intentionally weak service, confirm the issue with benign evidence, and stop before persistence or lateral movement. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): In an isolated local VM, identify one intentionally weak service, confirm the issue with benign evidence, and stop before persistence or lateral movement. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	service-finding.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What is the least invasive evidence that proves the weakness? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Evidence is sufficient, no destructive action occurs, and findings are reproducible Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Nmap Reference Guide service/version detection and NSE safety sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RFC 9293: Transmission Control Protocol; write a five-line correction; retest next morning with a new 10-minute problem.

D178 · M07W2D3 · Risk prioritization beyond CVSS

Phase: Academic month · **Optional date:** 2027-04-22 · **Domain resources:** E06 / B18

1. Day / sequence	Global Day 178 of 365; calendar-free M07W2D3; optional date 2027-04-22.
2. Exact topic	Risk prioritization beyond CVSS
3. Exact subtopics	CVSS base/temporal/environmental, EPSS-like concepts, KEV, asset criticality, exposure, compensating controls and business context
4. Learning objective	Rank five findings with explicit reasons and uncertainty
5. Primary lecture / reading + direct URL	FIRST CVSS specification `https://www.first.org/cvss/` **Type:** Scoring standard · **Access:** Free
6. Exact section / lesson locator	FIRST CVSS specification, calculator and metric groups. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CISA Known Exploited Vulnerabilities Catalog `https://www.cisa.gov/known-exploited-vulnerabilities-catalog`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Score five synthetic findings, then re-rank them after learning asset criticality and compensating controls; explain every change. **2.** Standard (~2 h): Score five synthetic findings, then re-rank them after learning asset criticality and compensating controls; explain every change. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Score five synthetic findings, then re-rank them after learning asset criticality and compensating controls; explain every change. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	finding-prioritization.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a critical CVSS score be lower priority than a medium exploited-in-the-wild bug? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Rankings are consistent with the stated scenario and uncertainty is documented Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "FIRST CVSS specification, calculator and metric groups"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NVD; write a five-line correction; retest next morning with a new 10-minute problem.

D179 · M07W2D4 · Evidence collection and chain of custody

Phase: Academic month · **Optional date:** 2027-04-23 · **Domain resources:** E01 / F09

1. Day / sequence	Global Day 179 of 365; calendar-free M07W2D4; optional date 2027-04-23.
2. Exact topic	Evidence collection and chain of custody
3. Exact subtopics	Screenshots, raw logs, timestamps, hashes, tool versions, command line, provenance, sanitization and evidence manifest
4. Learning objective	Build a defensible evidence package for a synthetic finding
5. Primary lecture / reading + direct URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final` **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115 post-test activities and reporting sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create an evidence bundle for one local lab finding with hashes, timestamps, tool versions, raw output and a README manifest. **2.** Standard (~2 h): Create an evidence bundle for one local lab finding with hashes, timestamps, tool versions, raw output and a README manifest. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create an evidence bundle for one local lab finding with hashes, timestamps, tool versions, raw output and a README manifest. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	evidence-bundle/ with SHA-256 manifest; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which detail would make the evidence inadmissible or unconvincing to a reviewer? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Manifest verifies all files and no personal/unauthorized data is present Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115 post-test activities and reporting sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D180 · M07W2D5 · Remediation recommendations that engineers can implement

Phase: Academic month · **Optional date:** 2027-04-24 · **Domain resources:** W17 / P08

1. Day / sequence	Global Day 180 of 365; calendar-free M07W2D5; optional date 2027-04-24.
2. Exact topic	Remediation recommendations that engineers can implement
3. Exact subtopics	Root cause, configuration/code fix, compensating control, priority, owner, test, rollback and residual risk
4. Learning objective	Convert an offensive finding into an actionable defensive fix
5. Primary lecture / reading + direct URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/` **Type:** Official checklist library · **Access:** Free
6. Exact section / lesson locator	OWASP Cheat Sheet Series matching the assigned vulnerability. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write three remediation tickets for one finding at code, configuration and architecture levels; include acceptance tests. **2.** Standard (~2 h): Write three remediation tickets for one finding at code, configuration and architecture levels; include acceptance tests. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write three remediation tickets for one finding at code, configuration and architecture levels; include acceptance tests. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	remediation-tickets.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is patch the software an incomplete recommendation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each ticket has owner, test, rollback and residual risk Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Cheat Sheet Series matching the assigned vulnerability"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Application Security Verification Standard; write a five-line correction; retest next morning with a new 10-minute problem.

D181 · M07W2D6 · Weekly validation and remediation review

Phase: Academic month · **Optional date:** 2027-04-25 · **Domain resources:** E06 / W17

1. Day / sequence	Global Day 181 of 365; calendar-free M07W2D6; optional date 2027-04-25.
2. Exact topic	Weekly validation and remediation review
3. Exact subtopics	Discovery, scanner interpretation, web/service validation, CVSS, evidence and remediation; active recall
4. Learning objective	Demonstrate Month 7 Week 2 competence with a synthetic finding pack
5. Primary lecture / reading + direct URL	FIRST CVSS specification `https://www.first.org/cvss/` **Type:** Scoring standard · **Access:** Free
6. Exact section / lesson locator	FIRST CVSS and E08 NVD/CVE pages for one sample finding. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Receive three synthetic findings, validate evidence, prioritize, write remediations and identify one false positive. **2.** Standard (~2 h): Receive three synthetic findings, validate evidence, prioritize, write remediations and identify one false positive. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Receive three synthetic findings, validate evidence, prioritize, write remediations and identify one false positive. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M7W2-practical.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding lacks enough evidence to justify a high-confidence conclusion? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and every conclusion has evidence and confidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "FIRST CVSS and E08 NVD/CVE pages for one sample finding"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NVD; write a five-line correction; retest next morning with a new 10-minute problem.

D182 · M07W2D7 · Retesting and closure criteria

Phase: Academic month · **Optional date:** 2027-04-26 · **Domain resources:** W06 / P05 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 182 of 365; calendar-free M07W2D7; optional date 2027-04-26.
2. Exact topic	Retesting and closure criteria
3. Exact subtopics	Reproduce original evidence, verify fix, regression-test surrounding functions, detect workaround bypass and update status
4. Learning objective	Design a retest that could fail even after a patch appears applied
5. Primary lecture / reading + direct URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/` **Type:** Testing methodology · **Access:** Free
6. Exact section / lesson locator	OWASP WSTG reporting and verification sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — For one local remediated bug, write three retest cases: original payload, variant and functional regression; record pass/fail and residual risk — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — For one local remediated bug, write three retest cases: original payload, variant and functional regression; record pass/fail and residual risk — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — For one local remediated bug, write three retest cases: original payload, variant and functional regression; record pass/fail and residual risk — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	retest-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Why can a WAF block one payload while the underlying vulnerability remains? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest distinguishes blocked symptom from fixed root cause Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP WSTG reporting and verification sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

Month 07 · Week 3 · daily cards

D183 · M07W3D1 · Technical report writing: executive summary and findings

Phase: Academic month · **Optional date:** 2027-04-27 · **Domain resources:** W06 / E01

1. Day / sequence	Global Day 183 of 365; calendar-free M07W3D1; optional date 2027-04-27.
2. Exact topic	Technical report writing: executive summary and findings
3. Exact subtopics	Audience, scope, methodology, risk rating, evidence, impact, likelihood, recommendation and appendices
4. Learning objective	Write a professional report that a developer and manager can both use
5. Primary lecture / reading + direct URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/` **Type:** Testing methodology · **Access:** Free
6. Exact section / lesson locator	OWASP Web Security Testing Guide, reporting section and report structure. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draft a five-page report for a fictional local assessment with three findings and one false positive. **2.** Standard (~2 h): Draft a five-page report for a fictional local assessment with three findings and one false positive. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draft a five-page report for a fictional local assessment with three findings and one false positive. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	assessment-report-draft.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which sentence confuses observation with interpretation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Report separates facts, interpretation and recommendation and scores at least 70/100 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Web Security Testing Guide, reporting section and report structure"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D184 · M07W3D2 · Professional communication and defensive debrief

Phase: Academic month · **Optional date:** 2027-04-28 · **Domain resources:** E01 / C06

1. Day / sequence	Global Day 184 of 365; calendar-free M07W3D2; optional date 2027-04-28.
2. Exact topic	Professional communication and defensive debrief
3. Exact subtopics	Findings walkthrough, uncertainty, disagreement, remediation negotiation, non-blaming language and follow-up
4. Learning objective	Defend conclusions using evidence without exaggeration
5. Primary lecture / reading + direct URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final` **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115 reporting and post-test activities. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Conduct a 20-minute recorded self-presentation or written Q&A that answers five skeptical stakeholder questions. **2.** Standard (~2 h): Conduct a 20-minute recorded self-presentation or written Q&A that answers five skeptical stakeholder questions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Conduct a 20-minute recorded self-presentation or written Q&A that answers five skeptical stakeholder questions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	debrief-notes.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which stakeholder concern reveals a missing business-context fact? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Answers are concise, honest and cite specific evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115 reporting and post-test activities"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D185 · M07W3D3 · Responsible disclosure and coordinated vulnerability handling

Phase: Academic month · **Optional date:** 2027-04-29 · **Domain resources:** E05 / E08

1. Day / sequence	Global Day 185 of 365; calendar-free M07W3D3; optional date 2027-04-29.
2. Exact topic	Responsible disclosure and coordinated vulnerability handling
3. Exact subtopics	Disclosure policy, safe harbour, private reporting, timelines, embargo, CVE request, vendor communication and public write-up ethics
4. Learning objective	Draft a disclosure timeline for a synthetic finding
5. Primary lecture / reading + direct URL	HackerOne Disclosure Guidelines `https://www.hackerone.com/disclosure-guidelines` **Type:** Responsible disclosure guidance · **Access:** Free
6. Exact section / lesson locator	HackerOne Disclosure Guidelines and E07 CVE Program information. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NVD `https://nvd.nist.gov/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a synthetic disclosure email, timeline and post-fix write-up; explicitly state that no real target was tested. **2.** Standard (~2 h): Create a synthetic disclosure email, timeline and post-fix write-up; explicitly state that no real target was tested. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a synthetic disclosure email, timeline and post-fix write-up; explicitly state that no real target was tested. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	disclosure-pack.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should a public write-up wait for a fix or agreed disclosure date? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All documents are synthetic and safe harbour/authorization boundaries are clear Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "HackerOne Disclosure Guidelines and E07 CVE Program information"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup FIRST CVSS specification; write a five-line correction; retest next morning with a new 10-minute problem.

D186 · M07W3D4 · Common penetration-testing pitfalls and professional limits

Phase: Academic month · **Optional date:** 2027-04-30 · **Domain resources:** E01 / E05

1. Day / sequence	Global Day 186 of 365; calendar-free M07W3D4; optional date 2027-04-30.
2. Exact topic	Common penetration-testing pitfalls and professional limits
3. Exact subtopics	Scanner dependence, scope creep, stale objectives, destructive proof, credential exposure, unsupported claims and report inflation
4. Learning objective	Audit an intentionally bad assessment report and correct it
5. Primary lecture / reading + direct URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final` **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115 and W06 WSTG methodology sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	HackerOne Disclosure Guidelines `https://www.hackerone.com/disclosure-guidelines`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review a synthetic report containing ten errors; correct each with evidence, ethical reasoning or clearer language. **2.** Standard (~2 h): Review a synthetic report containing ten errors; correct each with evidence, ethical reasoning or clearer language. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review a synthetic report containing ten errors; correct each with evidence, ethical reasoning or clearer language. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	report-audit.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which error could cause legal or operational harm? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All ten errors are found and corrections are defensible Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115 and W06 WSTG methodology sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D187 · M07W3D5 · Weekly report-writing and ethics review

Phase: Academic month · **Optional date:** 2027-05-01 · **Domain resources:** W06 / E05

1. Day / sequence	Global Day 187 of 365; calendar-free M07W3D5; optional date 2027-05-01.
2. Exact topic	Weekly report-writing and ethics review
3. Exact subtopics	Retesting, reporting, communication, disclosure, CVE/CVSS and professional limits; peer-style review
4. Learning objective	Demonstrate Month 7 Week 3 competence through an integrated reporting scenario
5. Primary lecture / reading + direct URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/` **Type:** Testing methodology · **Access:** Free
6. Exact section / lesson locator	OWASP WSTG reporting appendix and E06 FIRST CVSS examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	HackerOne Disclosure Guidelines `https://www.hackerone.com/disclosure-guidelines`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 90-minute scenario: validate, score, report and disclose two synthetic findings under time limits. **2.** Standard (~2 h): Complete a 90-minute scenario: validate, score, report and disclose two synthetic findings under time limits. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 90-minute scenario: validate, score, report and disclose two synthetic findings under time limits. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M7W3-practical.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How would you phrase uncertainty without weakening a real risk? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and safety/ethics section at least 9/10 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP WSTG reporting appendix and E06 FIRST CVSS examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D188 · M07W3D6 · Authorized assessment capstone: planning and discovery

Phase: Academic month · **Optional date:** 2027-05-02 · **Domain resources:** E01 / C21

1. Day / sequence	Global Day 188 of 365; calendar-free M07W3D6; optional date 2027-05-02.
2. Exact topic	Authorized assessment capstone: planning and discovery
3. Exact subtopics	Scoping, RoE, asset inventory, threat model, discovery plan and evidence manifest
4. Learning objective	Plan an end-to-end assessment of your own isolated lab
5. Primary lecture / reading + direct URL	NIST SP 800-115 https://csrc.nist.gov/pubs/sp/800/115/final **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115 full assessment workflow. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CIS Controls Version 8 https://www.cisecurity.org/controls/v8
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a complete assessment plan for two local VMs and one vulnerable web app, with signed scope, timeline and success criteria. **2.** Standard (~2 h): Create a complete assessment plan for two local VMs and one vulnerable web app, with signed scope, timeline and success criteria. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a complete assessment plan for two local VMs and one vulnerable web app, with signed scope, timeline and success criteria. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-10/plan/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which asset or technique is explicitly out of scope and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan passes the scope gate and all targets are owned/isolated Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115 full assessment workflow"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D189 · M07W3D7 · Authorized assessment capstone: execution and evidence

Phase: Academic month · **Optional date:** 2027-05-03 · **Domain resources:** N09 / W06 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 189 of 365; calendar-free M07W3D7; optional date 2027-05-03.
2. Exact topic	Authorized assessment capstone: execution and evidence
3. Exact subtopics	Discovery, enumeration, vulnerability validation, evidence preservation and minimal impact testing
4. Learning objective	Execute the planned assessment under controlled conditions
5. Primary lecture / reading + direct URL	Nmap Reference Guide `https://nmap.org/book/man.html` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Nmap Reference Guide for authorized scanning; W05 Burp Suite scope documentation for local web testing. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Perform only in-scope tests, collect raw evidence, stop at proof of vulnerability, and record every command/request in a timeline — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Perform only in-scope tests, collect raw evidence, stop at proof of vulnerability, and record every command/request in a timeline — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Perform only in-scope tests, collect raw evidence, stop at proof of vulnerability, and record every command/request in a timeline — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-10/evidence/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which test caused an unexpected service impact and how did you respond? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Timeline is complete and no out-of-scope or destructive action is present Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Nmap Reference Guide for authorized scanning; W05 Burp Suite scope documentation for local web testing"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup RFC 9293: Transmission Control Protocol; write a five-line correction; retest next morning with a new 10-minute problem.

Month 07 · Week 4 · daily cards

D190 · M07W4D1 · Authorized assessment capstone: analysis and report

Phase: Academic month · **Optional date:** 2027-05-04 · **Domain resources:** W06 / W17

1. Day / sequence	Global Day 190 of 365; calendar-free M07W4D1; optional date 2027-05-04.
2. Exact topic	Authorized assessment capstone: analysis and report
3. Exact subtopics	Root cause, risk, impact, evidence correlation, false positives, remediation and retest
4. Learning objective	Produce a defensible assessment report with prioritized recommendations
5. Primary lecture / reading + direct URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/` **Type:** Testing methodology · **Access:** Free
6. Exact section / lesson locator	OWASP WSTG reporting structure and E06 FIRST CVSS specification. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write an eight-page report, include an executive summary, three findings with evidence, one rejected false positive and a remediation roadmap. **2.** Standard (~2 h): Write an eight-page report, include an executive summary, three findings with evidence, one rejected false positive and a remediation roadmap. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write an eight-page report, include an executive summary, three findings with evidence, one rejected false positive and a remediation roadmap. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-10/report.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding's evidence is strongest and which is most uncertain? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Report passes the 100-point rubric and no proprietary or real personal data is exposed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP WSTG reporting structure and E06 FIRST CVSS specification"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D191 · M07W4D2 · Authorized assessment capstone: remediation and retest

Phase: Academic month · **Optional date:** 2027-05-05 · **Domain resources:** W17 / P05

1. Day / sequence	Global Day 191 of 365; calendar-free M07W4D2; optional date 2027-05-05.
2. Exact topic	Authorized assessment capstone: remediation and retest
3. Exact subtopics	Fixes, verification, regression testing, residual risk and client-style handover
4. Learning objective	Prove that remediation closed the root cause without breaking functionality
5. Primary lecture / reading + direct URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/` **Type:** Official checklist library · **Access:** Free
6. Exact section / lesson locator	OWASP Cheat Sheet Series for each selected finding. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Implement fixes in the local lab, rerun the original and variant tests, check functional regression, and write a closure memo. **2.** Standard (~2 h): Implement fixes in the local lab, rerun the original and variant tests, check functional regression, and write a closure memo. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Implement fixes in the local lab, rerun the original and variant tests, check functional regression, and write a closure memo. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-10/retest-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What evidence proves root-cause closure rather than payload filtering? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All assigned findings are retested and closure criteria are explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Cheat Sheet Series for each selected finding"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Application Security Verification Standard; write a five-line correction; retest next morning with a new 10-minute problem.

D192 · M07W4D3 · Portfolio publication and independent reflection

Phase: Academic month · **Optional date:** 2027-05-06 · **Domain resources:** F09 / W13

1. Day / sequence	Global Day 192 of 365; calendar-free M07W4D3; optional date 2027-05-06.
2. Exact topic	Portfolio publication and independent reflection
3. Exact subtopics	Sanitization, scope statement, limitations, ethical boundaries, evidence redaction and independent contribution
4. Learning objective	Publish a professional portfolio case study from the assessment
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 2-3 and GitHub security docs via W13. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a sanitized public repository with report, sample evidence, README, scope statement, limitations and lessons learned; remove IPs/secrets/personal data. **2.** Standard (~2 h): Create a sanitized public repository with report, sample evidence, README, scope statement, limitations and lessons learned; remove IPs/secrets/personal data. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a sanitized public repository with report, sample evidence, README, scope statement, limitations and lessons learned; remove IPs/secrets/personal data. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	published PRJ-10 repository; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which part of the project demonstrates independent problem-solving rather than tool output? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Portfolio checklist passes and no sensitive evidence is published Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 2-3 and GitHub security docs via W13"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D193 · M07W4D4 · Certification and specialization decision point

Phase: Academic month · **Optional date:** 2027-05-07 · **Domain resources:** A22 / E01

1. Day / sequence	Global Day 193 of 365; calendar-free M07W4D4; optional date 2027-05-07.
2. Exact topic	Certification and specialization decision point
3. Exact subtopics	eJPT, OSCP, Security+, CCST and free alternatives; skills demonstrated versus credentials
4. Learning objective	Decide whether a paid pentesting certification is appropriate after Year 1
5. Primary lecture / reading + direct URL	INE eJPT official certification page `https://ine.com/security/certifications/ejpt-certification` **Type:** Official certification page · **Access:** Paid; official page states an INE subscription and exam voucher are needed; checkout lists voucher/bundle prices
6. Exact section / lesson locator	INE eJPT official certification page; A21 OffSec PEN-200 OSCP+ page; E01 free NIST methodology. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Compare three certifications with official objectives, costs, prerequisites and free alternatives; write a budget-aware recommendation for Month 12 and Year 2. **2.** Standard (~2 h): Compare three certifications with official objectives, costs, prerequisites and free alternatives; write a budget-aware recommendation for Month 12 and Year 2. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Compare three certifications with official objectives, costs, prerequisites and free alternatives; write a budget-aware recommendation for Month 12 and Year 2. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	pentest-cert-decision.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What skill would the exam validate that your capstone does not? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Current official facts and prices are cited with verification date; no purchase is treated as mandatory Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "INE eJPT official certification page; A21 OffSec PEN-200 OSCP+ page; E01 free NIST methodology"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST SP 800-115; write a five-line correction; retest next morning with a new 10-minute problem.

D194 · M07W4D5 · Month 7 practical examination and mastery gate

Phase: Academic month · **Optional date:** 2027-05-08 · **Domain resources:** E01 / W06

1. Day / sequence	Global Day 194 of 365; calendar-free M07W4D5; optional date 2027-05-08.
2. Exact topic	Month 7 practical examination and mastery gate
3. Exact subtopics	Authorization, scope, asset inventory, discovery, validation, risk, evidence, remediation, retest, reporting and disclosure
4. Learning objective	Demonstrate ethical testing competence in a time-boxed authorized assessment
5. Primary lecture / reading + direct URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final` **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115 full workflow. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 150-minute practical on a new isolated lab with an unseen scope document and two synthetic vulnerabilities; produce evidence and a report. **2.** Standard (~2 h): Complete a 150-minute practical on a new isolated lab with an unseen scope document and two synthetic vulnerabilities; produce evidence and a report. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 150-minute practical on a new isolated lab with an unseen scope document and two synthetic vulnerabilities; produce evidence and a report. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M7-assessment.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which action would you refuse to perform even if the client requested it informally? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, safety at least 9/10 and no scope violation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115 full workflow"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D195 · M07W4D6 · Monthly review and gap repairs

Phase: Academic month · **Optional date:** 2027-05-09 · **Domain resources:** E01 / C01

1. Day / sequence	Global Day 195 of 365; calendar-free M07W4D6; optional date 2027-05-09.
2. Exact topic	Monthly review and gap repairs
3. Exact subtopics	Scope errors, scan interpretation, validation logic, CVSS misuse, evidence quality and report clarity
4. Learning objective	Repair Month 7 gaps and prepare blue-team log analysis
5. Primary lecture / reading + direct URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final` **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115 and W06 WSTG reporting sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework failed assessment items, rewrite one weak remediation, add ten flashcards, and map each Month 8 topic to skills from Months 1-7. **2.** Standard (~2 h): Rework failed assessment items, rewrite one weak remediation, add ten flashcards, and map each Month 8 topic to skills from Months 1-7. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework failed assessment items, rewrite one weak remediation, add ten flashcards, and map each Month 8 topic to skills from Months 1-7. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M7-review.md and retest score; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which offensive validation skill will be most useful for detection engineering? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and no unresolved safety gaps Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115 and W06 WSTG reporting sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D196 · M07W4D7 · Spaced revision, recovery and Month 8 preview

Phase: Academic month · **Optional date:** 2027-05-10 · **Domain resources:** E05 / B02 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 196 of 365; calendar-free M07W4D7; optional date 2027-05-10.
2. Exact topic	Spaced revision, recovery and Month 8 preview
3. Exact subtopics	Ethics, RoE, evidence, validation, risk and reporting; active recall and lab cleanup
4. Learning objective	Preserve ethical-testing knowledge and shift to defensive investigation mindset
5. Primary lecture / reading + direct URL	HackerOne Disclosure Guidelines `https://www.hackerone.com/disclosure-guidelines` **Type:** Responsible disclosure guidance · **Access:** Free
6. Exact section / lesson locator	HackerOne Disclosure Guidelines and E06 FIRST CVSS calculator. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, destroy/reset assessment VMs from clean snapshots, organize evidence backups, and preview KQL basics — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, destroy/reset assessment VMs from clean snapshots, organize evidence backups, and preview KQL basics — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, destroy/reset assessment VMs from clean snapshots, organize evidence backups, and preview KQL basics — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M7-consolidation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: How does an attacker's evidence chain become a defender's detection hypothesis? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Lab is clean, evidence is archived, and the explanation is coherent Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "HackerOne Disclosure Guidelines and E06 FIRST CVSS calculator"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup FIRST CVSS specification; write a five-line correction; retest next morning with a new 10-minute problem.

Month 08 — Blue team, SOC, logs, SIEM and incident response

Days D197-D224. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: SOC roles, workflows and alert lifecycle D2: SIEM concepts and log pipeline D3: Windows and Linux security logs D4: Alert triage and hypotheses D5: Indicators of compromise and confidence D6: MITRE ATT&CK and detection coverage D7: Weekly blue-team foundations review
Week 2	D1: Incident response lifecycle and evidence handling D2: KQL fundamentals for security logs D3: Detection rule anatomy and Sigma D4: Detection engineering lifecycle D5: Endpoint detection and response concepts D6: Weekly detection engineering review D7: Threat hunting methodology
Week 3	D1: Threat intelligence and advisory reading D2: False-positive analysis and tuning D3: Digital forensics fundamentals and evidence integrity D4: Weekly hunting, DFIR and tuning review D5: Incident report writing and stakeholder communication D6: SIEM alert investigation capstone: data preparation D7: SIEM alert investigation capstone: investigation
Week 4	D1: SIEM alert investigation capstone: response and lessons D2: Blue-team portfolio publication D3: Blue-team specialization taster and decision evidence D4: Month 8 practical examination and milestone gate D5: Monthly review and gap repairs D6: Network forensics and malware-traffic safety D7: Spaced revision, recovery and Month 9 preview

> **Gate:** score the Month 8 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 08 · Week 1 · daily cards

D197 · M08W1D1 · SOC roles, workflows and alert lifecycle

Phase: Academic month · **Optional date:** 2027-05-11 · **Domain resources:** B01 / C20

1. Day / sequence	Global Day 197 of 365; calendar-free M08W1D1; optional date 2027-05-11.
2. Exact topic	SOC roles, workflows and alert lifecycle
3. Exact subtopics	Tier 1/2/3, monitoring, intake, triage, escalation, investigation, containment recommendation, closure and handoff
4. Learning objective	Explain an alert's path from telemetry to decision and evidence
5. Primary lecture / reading + direct URL	Microsoft Sentinel documentation `https://learn.microsoft.com/en-us/azure/sentinel/` **Type:** Official documentation · **Access:** Docs free; Azure/Sentinel usage may incur costs
6. Exact section / lesson locator	Microsoft Sentinel documentation, overview, incidents and SOC workflow sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a SOC workflow for a synthetic failed-login spike and label decision points, data sources and handoff notes. **2.** Standard (~2 h): Draw a SOC workflow for a synthetic failed-login spike and label decision points, data sources and handoff notes. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a SOC workflow for a synthetic failed-login spike and label decision points, data sources and handoff notes. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	soc-workflow.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which step is missing if alerts are closed without a documented rationale? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Workflow includes feedback, false-positive learning and escalation criteria Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft Sentinel documentation, overview, incidents and SOC workflow sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kusto Query Language overview; write a five-line correction; retest next morning with a new 10-minute problem.

D198 · M08W1D2 · SIEM concepts and log pipeline

Phase: Academic month · **Optional date:** 2027-05-12 · **Domain resources:** B01 / B02

1. Day / sequence	Global Day 198 of 365; calendar-free M08W1D2; optional date 2027-05-12.
2. Exact topic	SIEM concepts and log pipeline
3. Exact subtopics	Collection, parsing, normalization, enrichment, indexing, retention, correlation, dashboards, case management and cost drivers
4. Learning objective	Design a log pipeline for a small lab with synthetic data
5. Primary lecture / reading + direct URL	Microsoft Sentinel documentation `https://learn.microsoft.com/en-us/azure/sentinel/` **Type:** Official documentation · **Access:** Docs free; Azure/Sentinel usage may incur costs
6. Exact section / lesson locator	Microsoft Sentinel documentation, data connectors, workspace, analytics and hunting sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Ingest a small synthetic CSV/JSON log set into a local notebook or free-tier lab only if budgets are configured; otherwise simulate with SQLite/KQL-style queries. **2.** Standard (~2 h): Ingest a small synthetic CSV/JSON log set into a local notebook or free-tier lab only if budgets are configured; otherwise simulate with SQLite/KQL-style queries. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Ingest a small synthetic CSV/JSON log set into a local notebook or free-tier lab only if budgets are configured; otherwise simulate with SQLite/KQL-style queries. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	log-pipeline-design.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does normalization matter more than collecting raw logs without schema? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Design includes parsing, timestamp/time zone, retention and privacy controls Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft Sentinel documentation, data connectors, workspace, analytics and hunting sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kusto Query Language overview; write a five-line correction; retest next morning with a new 10-minute problem.

D199 · M08W1D3 · Windows and Linux security logs

Phase: Academic month · **Optional date:** 2027-05-13 · **Domain resources:** L09 / B05

1. Day / sequence	Global Day 199 of 365; calendar-free M08W1D3; optional date 2027-05-13.
2. Exact topic	Windows and Linux security logs
3. Exact subtopics	Security/System/Application channels, Sysmon, auth.log, journald, process creation, logon, service, network and file events
4. Learning objective	Map an adversary behaviour to the log fields that reveal it
5. Primary lecture / reading + direct URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	Windows Event Log documentation and L05 systemd/journald manual. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SBousseaden EVTX-ATTACK-SAMPLES `https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a cross-platform log-source matrix for process creation, authentication, network connection, service change and file access. **2.** Standard (~2 h): Create a cross-platform log-source matrix for process creation, authentication, network connection, service change and file access. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a cross-platform log-source matrix for process creation, authentication, network connection, service change and file access. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	log-source-matrix.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which behaviour has the weakest visibility in your current lab and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Matrix includes event IDs/field names and explicit visibility gaps Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Windows Event Log documentation and L05 systemd/ journald manual"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Sysmon documentation and download; write a five-line correction; retest next morning with a new 10-minute problem.

D200 · M08W1D4 · Alert triage and hypotheses

Phase: Academic month · **Optional date:** 2027-05-14 · **Domain resources:** B15 / C18

1. Day / sequence	Global Day 200 of 365; calendar-free M08W1D4; optional date 2027-05-14.
2. Exact topic	Alert triage and hypotheses
3. Exact subtopics	Alert context, affected assets, severity, confidence, prevalence, first/last seen, known baseline and competing hypotheses
4. Learning objective	Triage a synthetic alert without jumping to a conclusion
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 certification page and official study guide, threat response/hunting sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: For one synthetic alert, write three competing hypotheses and the exact query or evidence that would discriminate among them. **2.** Standard (~2 h): For one synthetic alert, write three competing hypotheses and the exact query or evidence that would discriminate among them. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): For one synthetic alert, write three competing hypotheses and the exact query or evidence that would discriminate among them. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	alert-triage.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What observation would prove the alert is a false positive? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Each hypothesis has a discriminating test and outcome criteria Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 certification page and official study guide, threat response/hunting sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D201 · M08W1D5 · Indicators of compromise and confidence

Phase: Academic month · **Optional date:** 2027-05-15 · **Domain resources:** C18 / B17

1. Day / sequence	Global Day 201 of 365; calendar-free M08W1D5; optional date 2027-05-15.
2. Exact topic	Indicators of compromise and confidence
3. Exact subtopics	Atomic, computed and behavioural indicators; hashes, domains, IPs, user agents, mutexes, TTPs; confidence and expiration
4. Learning objective	Explain why an IOC is not equivalent to an incident
5. Primary lecture / reading + direct URL	MITRE ATT&CK `https://attack.mitre.org/` **Type:** Knowledge base · **Access:** Free
6. Exact section / lesson locator	MITRE ATT&CK, tactics/techniques/data sources and detection notes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CISA cybersecurity advisories `https://www.cisa.gov/news-events/cybersecurity-advisories`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build an IOC table for a synthetic scenario with type, source, confidence, expiration, false-positive risk and recommended action. **2.** Standard (~2 h): Build an IOC table for a synthetic scenario with type, source, confidence, expiration, false-positive risk and recommended action. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build an IOC table for a synthetic scenario with type, source, confidence, expiration, false-positive risk and recommended action. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ioc-table.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a behavioural TTP often more durable than an IP address? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Table includes decay/expiration and no claim that an IOC alone confirms compromise Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MITRE ATT&CK, tactics/techniques/data sources and detection notes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE D3FEND; write a five-line correction; retest next morning with a new 10-minute problem.

D202 · M08W1D6 · MITRE ATT&CK and detection coverage

Phase: Academic month · **Optional date:** 2027-05-16 · **Domain resources:** C18 / C19

1. Day / sequence	Global Day 202 of 365; calendar-free M08W1D6; optional date 2027-05-16.
2. Exact topic	MITRE ATT&CK and detection coverage
3. Exact subtopics	Tactics, techniques, sub-techniques, procedures, data sources, mitigations, Navigator and coverage caveats
4. Learning objective	Map one synthetic incident to techniques and required telemetry
5. Primary lecture / reading + direct URL	MITRE ATT&CK `https://attack.mitre.org/` **Type:** Knowledge base · **Access:** Free
6. Exact section / lesson locator	MITRE ATT&CK, Enterprise matrix and technique pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE D3FEND `https://d3fend.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create an ATT&CK Navigator-style layer for five techniques in your own lab; mark data required and current detection confidence. **2.** Standard (~2 h): Create an ATT&CK Navigator-style layer for five techniques in your own lab; mark data required and current detection confidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create an ATT&CK Navigator-style layer for five techniques in your own lab; mark data required and current detection confidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	attack-layer.json and explanation; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does a claimed technique coverage not prove a useful detection? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Layer is correctly formatted and each technique has explicit data-source evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MITRE ATT&CK, Enterprise matrix and technique pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE D3FEND; write a five-line correction; retest next morning with a new 10-minute problem.

D203 · M08W1D7 · Weekly blue-team foundations review

Phase: Academic month · **Optional date:** 2027-05-17 · **Domain resources:** B02 / B03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 203 of 365; calendar-free M08W1D7; optional date 2027-05-17.
2. Exact topic	Weekly blue-team foundations review
3. Exact subtopics	SOC workflow, SIEM pipeline, logs, triage, IOCs and ATT&CK; active recall and scenario
4. Learning objective	Demonstrate Month 8 Week 1 competence with an unfamiliar synthetic alert
5. Primary lecture / reading + direct URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric` **Type:** Official language docs · **Access:** Free
6. Exact section / lesson locator	KQL overview, where/tabular operators and summarize sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute scenario: normalize logs, query an alert, map ATT&CK and recommend escalation or closure with confidence — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute scenario: normalize logs, query an alert, map ATT&CK and recommend escalation or closure with confidence — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute scenario: normalize logs, query an alert, map ATT&CK and recommend escalation or closure with confidence — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M8W1-practical.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which evidence would change your escalation decision? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and all conclusions cite specific fields Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "KQL overview, where/tabular operators and summarize sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

Month 08 · Week 2 · daily cards

D204 · M08W2D1 · Incident response lifecycle and evidence handling

Phase: Academic month · **Optional date:** 2027-05-18 · **Domain resources:** C07 / B11

1. Day / sequence	Global Day 204 of 365; calendar-free M08W2D1; optional date 2027-05-18.
2. Exact topic	Incident response lifecycle and evidence handling
3. Exact subtopics	Preparation, detection, analysis, containment, eradication, recovery, lessons learned, legal hold and chain of custody
4. Learning objective	Build an IR runbook for a synthetic credential-theft incident
5. Primary lecture / reading + direct URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-61r3, incident response life cycle and CSF alignment. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Autopsy documentation `https://www.autopsy.com/support/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a runbook with roles, decision tree, evidence preservation, containment options, communication and recovery verification. **2.** Standard (~2 h): Write a runbook with roles, decision tree, evidence preservation, containment options, communication and recovery verification. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a runbook with roles, decision tree, evidence preservation, containment options, communication and recovery verification. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ir-runbook.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which containment action could destroy volatile evidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Runbook balances containment with evidence preservation and includes rollback Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-61r3, incident response life cycle and CSF alignment"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D205 · M08W2D2 · KQL fundamentals for security logs

Phase: Academic month · **Optional date:** 2027-05-19 · **Domain resources:** B02 / A23

1. Day / sequence	Global Day 205 of 365; calendar-free M08W2D2; optional date 2027-05-19.
2. Exact topic	KQL fundamentals for security logs
3. Exact subtopics	Tables, where, project, extend, summarize, count, join, let, time filters, dynamic fields and render basics
4. Learning objective	Query synthetic security events to answer investigative questions
5. Primary lecture / reading + direct URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric` **Type:** Official language docs · **Access:** Free
6. Exact section / lesson locator	KQL overview and tutorial, learn common operators sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	KQL Essentials official Azure Data Explorer playlist `https://www.youtube.com/playlist?list=PLKHIhZJiQ5AkOMEC54mQQcAkfg4UXT8uK`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write ten KQL-style queries against a free sample dataset or a local equivalent; explain each operator and output. **2.** Standard (~2 h): Write ten KQL-style queries against a free sample dataset or a local equivalent; explain each operator and output. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write ten KQL-style queries against a free sample dataset or a local equivalent; explain each operator and output. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	kql-queries.kql and results; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can a join on a high-cardinality field create misleading or expensive results? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All queries are syntactically valid in the chosen environment and results are interpreted Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "KQL overview and tutorial, learn common operators sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D206 · M08W2D3 · Detection rule anatomy and Sigma

Phase: Academic month · **Optional date:** 2027-05-20 · **Domain resources:** B04 / B03

1. Day / sequence	Global Day 206 of 365; calendar-free M08W2D3; optional date 2027-05-20.
2. Exact topic	Detection rule anatomy and Sigma
3. Exact subtopics	logsource, detection, selection, condition, falsepositives, level, tags, metadata and conversion to SIEM queries
4. Learning objective	Write a readable, vendor-neutral detection rule
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started documentation, rule structure and conversion sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Author one original Sigma rule for a benign lab behaviour and convert it to a local query; include true/false-positive test cases. **2.** Standard (~2 h): Author one original Sigma rule for a benign lab behaviour and convert it to a local query; include true/false-positive test cases. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Author one original Sigma rule for a benign lab behaviour and convert it to a local query; include true/false-positive test cases. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	detection-rule.yml and test-data.json; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which field in your rule is most likely to cause a false positive in a real environment? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Rule validates, converts, and test results are documented Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started documentation, rule structure and conversion sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D207 · M08W2D4 · Detection engineering lifecycle

Phase: Academic month · **Optional date:** 2027-05-21 · **Domain resources:** B14 / B03

1. Day / sequence	Global Day 207 of 365; calendar-free M08W2D4; optional date 2027-05-21.
2. Exact topic	Detection engineering lifecycle
3. Exact subtopics	Hypothesis, data requirements, rule design, backtesting, tuning, deployment, version control, measurement and retirement
4. Learning objective	Treat detections as software with tests and feedback loops
5. Primary lecture / reading + direct URL	Elastic detection rules documentation `https://www.elastic.co/guide/en/security/current/detection-engine-overview.html` **Type:** Official docs / open-source rules · **Access:** Free docs/open-source rules; Elastic Stack may have paid features
6. Exact section / lesson locator	Elastic detection rules documentation, rule types, exceptions and alerting. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a detection-as-code repository with one rule, unit tests, sample data, README and version history. **2.** Standard (~2 h): Create a detection-as-code repository with one rule, unit tests, sample data, README and version history. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a detection-as-code repository with one rule, unit tests, sample data, README and version history. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-12 repository draft; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How would you measure the detection's value beyond alert count? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Repository includes tests, owners, tuning notes and false-positive metrics Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Elastic detection rules documentation, rule types, exceptions and alerting"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D208 · M08W2D5 · Endpoint detection and response concepts

Phase: Academic month · **Optional date:** 2027-05-22 · **Domain resources:** B15 / C18

1. Day / sequence	Global Day 208 of 365; calendar-free M08W2D5; optional date 2027-05-22.
2. Exact topic	Endpoint detection and response concepts
3. Exact subtopics	EDR telemetry, process trees, behavioural detection, isolation, response actions, tamper protection and privacy
4. Learning objective	Explain EDR's strengths and blind spots without relying on vendor marketing
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide and B01 Defender/Sentinel documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Compare EDR telemetry with Windows Security and Sysmon logs for one synthetic process chain; list two blind spots. **2.** Standard (~2 h): Compare EDR telemetry with Windows Security and Sysmon logs for one synthetic process chain; list two blind spots. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Compare EDR telemetry with Windows Security and Sysmon logs for one synthetic process chain; list two blind spots. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	edr-notes.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can endpoint isolation be a containment control and a business-risk decision at the same time? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Comparison is grounded in data fields and includes privacy/operational limits Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide and B01 Defender/Sentinel documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D209 · M08W2D6 · Weekly detection engineering review

Phase: Academic month · **Optional date:** 2027-05-23 · **Domain resources:** B04 / C18

1. Day / sequence	Global Day 209 of 365; calendar-free M08W2D6; optional date 2027-05-23.
2. Exact topic	Weekly detection engineering review
3. Exact subtopics	IR lifecycle, KQL, Sigma, detection-as-code, EDR and ATT&CK; active recall and code review
4. Learning objective	Demonstrate Month 8 Week 2 competence through a detection challenge
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started rule conversion examples and B14 detection rule repository. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Receive one synthetic behaviour, write a detection hypothesis, create a rule, test true/false positives and explain residual risk. **2.** Standard (~2 h): Receive one synthetic behaviour, write a detection hypothesis, create a rule, test true/false positives and explain residual risk. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Receive one synthetic behaviour, write a detection hypothesis, create a rule, test true/false positives and explain residual risk. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M8W2-practical.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which test data would make your rule fail silently? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and the rule has both passing and failing test cases Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started rule conversion examples and B14 detection rule repository"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D210 · M08W2D7 · Threat hunting methodology

Phase: Academic month · **Optional date:** 2027-05-24 · **Domain resources:** B15 / C18 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 210 of 365; calendar-free M08W2D7; optional date 2027-05-24.
2. Exact topic	Threat hunting methodology
3. Exact subtopics	Hypothesis-driven hunting, baseline, anomaly, known-behhaviour search, data quality, pivoting and documenting negative results
4. Learning objective	Design a hunt from a technique rather than a tool
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide, threat hunting section and B01 Sentinel hunting docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Choose one ATT&CK technique, list required fields, write three hunt queries and define what a negative result means — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Choose one ATT&CK technique, list required fields, write three hunt queries and define what a negative result means — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Choose one ATT&CK technique, list required fields, write three hunt queries and define what a negative result means — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	hunt-plan.md and query output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Why is a negative hunt result only meaningful if data coverage is verified? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Hunt plan includes data-quality checks and explicit limitations Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide, threat hunting section and B01 Sentinel hunting docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 08 · Week 3 · daily cards

D211 · M08W3D1 · Threat intelligence and advisory reading

Phase: Academic month · **Optional date:** 2027-05-25 · **Domain resources:** B17 / C18

1. Day / sequence	Global Day 211 of 365; calendar-free M08W3D1; optional date 2027-05-25.
2. Exact topic	Threat intelligence and advisory reading
3. Exact subtopics	Strategic/operational/tactical/technical intelligence, source reliability, confidence, indicator decay, STIX/TAXII concepts and analytic bias
4. Learning objective	Convert an advisory into a prioritized local detection hypothesis
5. Primary lecture / reading + direct URL	CISA cybersecurity advisories `https://www.cisa.gov/news-events/cybersecurity-advisories` **Type:** Official advisories · **Access:** Free
6. Exact section / lesson locator	CISA cybersecurity advisories and B18 KEV catalog; choose one current advisory. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Read one official advisory, extract two TTPs and two mitigations, map to ATT&CK and write one detection hypothesis. **2.** Standard (~2 h): Read one official advisory, extract two TTPs and two mitigations, map to ATT&CK and write one detection hypothesis. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Read one official advisory, extract two TTPs and two mitigations, map to ATT&CK and write one detection hypothesis. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	advisory-analysis.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which part of the advisory is an observation and which is an assessment? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Sources are cited, confidence is stated, and no IOC is treated as proof Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CISA cybersecurity advisories and B18 KEV catalog; choose one current advisory"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D212 · M08W3D2 · False-positive analysis and tuning

Phase: Academic month · **Optional date:** 2027-05-26 · **Domain resources:** B14 / B03

1. Day / sequence	Global Day 212 of 365; calendar-free M08W3D2; optional date 2027-05-26.
2. Exact topic	False-positive analysis and tuning
3. Exact subtopics	Benign triggers, prevalence, suppression, exceptions, risk-based scoring, rule decay and feedback from analysts
4. Learning objective	Tune a noisy rule without hiding true positives
5. Primary lecture / reading + direct URL	Elastic detection rules documentation `https://www.elastic.co/guide/en/security/current/detection-engine-overview.html` **Type:** Official docs / open-source rules · **Access:** Free docs/open-source rules; Elastic Stack may have paid features
6. Exact section / lesson locator	Elastic detection rules documentation, exceptions and alert investigation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a synthetic noisy dataset, run a rule, classify ten alerts, tune with a narrow exception and document what was excluded. **2.** Standard (~2 h): Create a synthetic noisy dataset, run a rule, classify ten alerts, tune with a narrow exception and document what was excluded. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a synthetic noisy dataset, run a rule, classify ten alerts, tune with a narrow exception and document what was excluded. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	false-positive-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is blanket suppression more dangerous than a narrow, reviewed exception? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Tuning reduces benign alerts and preserves all assigned malicious test cases Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Elastic detection rules documentation, exceptions and alert investigation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D213 · M08W3D3 · Digital forensics fundamentals and evidence integrity

Phase: Academic month · **Optional date:** 2027-05-27 · **Domain resources:** B11 / C07

1. Day / sequence	Global Day 213 of 365; calendar-free M08W3D3; optional date 2027-05-27.
2. Exact topic	Digital forensics fundamentals and evidence integrity
3. Exact subtopics	Order of volatility, acquisition, write blocking, hashing, timelines, file system artifacts, memory concepts and legal authorization
4. Learning objective	Explain how to preserve evidence before analysis
5. Primary lecture / reading + direct URL	Autopsy documentation `https://www.autopsy.com/support/` **Type:** Official tool docs · **Access:** Free/open-source
6. Exact section / lesson locator	Autopsy documentation, case creation, disk images and reports. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a benign test disk image or use a provided synthetic image, hash it, acquire it, and build a simple timeline without modifying the original. **2.** Standard (~2 h): Create a benign test disk image or use a provided synthetic image, hash it, acquire it, and build a simple timeline without modifying the original. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a benign test disk image or use a provided synthetic image, hash it, acquire it, and build a simple timeline without modifying the original. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	forensic-case.md and hash manifest; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why must analysis happen on a copy rather than the original evidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Hashes match, chain of custody is documented and original remains unchanged Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Autopsy documentation, case creation, disk images and reports"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SANS SIFT Workstation; write a five-line correction; retest next morning with a new 10-minute problem.

D214 · M08W3D4 · Weekly hunting, DFIR and tuning review

Phase: Academic month · **Optional date:** 2027-05-28 · **Domain resources:** B17 / B03

1. Day / sequence	Global Day 214 of 365; calendar-free M08W3D4; optional date 2027-05-28.
2. Exact topic	Weekly hunting, DFIR and tuning review
3. Exact subtopics	Hypothesis hunting, threat intel, false positives, forensics and evidence integrity; active recall
4. Learning objective	Demonstrate Month 8 Week 3 competence with a mixed incident pack
5. Primary lecture / reading + direct URL	CISA cybersecurity advisories `https://www.cisa.gov/news-events/cybersecurity-advisories` **Type:** Official advisories · **Access:** Free
6. Exact section / lesson locator	CISA advisories and B11 Autopsy documentation case workflow. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Analyze a synthetic incident pack containing logs, one IOC and one disk artifact; decide true/false positive and document evidence limits. **2.** Standard (~2 h): Analyze a synthetic incident pack containing logs, one IOC and one disk artifact; decide true/false positive and document evidence limits. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Analyze a synthetic incident pack containing logs, one IOC and one disk artifact; decide true/false positive and document evidence limits. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M8W3-practical.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which artifact corroborates the log timeline and which only suggests a possibility? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and evidence integrity rules are followed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CISA advisories and B11 Autopsy documentation case workflow"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D215 · M08W3D5 · Incident report writing and stakeholder communication

Phase: Academic month · **Optional date:** 2027-05-29 · **Domain resources:** C07 / B17

1. Day / sequence	Global Day 215 of 365; calendar-free M08W3D5; optional date 2027-05-29.
2. Exact topic	Incident report writing and stakeholder communication
3. Exact subtopics	Executive summary, timeline, scope, impact, root cause, evidence, confidence, containment, recovery, recommendations and lessons learned
4. Learning objective	Write an incident report that separates confirmed facts from hypotheses
5. Primary lecture / reading + direct URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-61r3, improvement and communications sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CISA cybersecurity advisories `https://www.cisa.gov/news-events/cybersecurity-advisories`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a six-page incident report for a synthetic credential-theft case with timestamps, evidence references and uncertainty labels. **2.** Standard (~2 h): Write a six-page incident report for a synthetic credential-theft case with timestamps, evidence references and uncertainty labels. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a six-page incident report for a synthetic credential-theft case with timestamps, evidence references and uncertainty labels. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-11 incident-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which sentence would be inappropriate because it overstates the evidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Report scores at least 70/100 and all claims have confidence labels Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-61r3, improvement and communications sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D216 · M08W3D6 · SIEM alert investigation capstone: data preparation

Phase: Academic month · **Optional date:** 2027-05-30 · **Domain resources:** B02 / B05

1. Day / sequence	Global Day 216 of 365; calendar-free M08W3D6; optional date 2027-05-30.
2. Exact topic	SIEM alert investigation capstone: data preparation
3. Exact subtopics	Synthetic log generation, normalization, time synchronization, entity mapping and case metadata
4. Learning objective	Build a safe dataset that can reproduce an alert investigation
5. Primary lecture / reading + direct URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric` **Type:** Official language docs · **Access:** Free
6. Exact section / lesson locator	KQL overview and B01 Sentinel data collection overview. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SBousseaden EVTX-ATTACK-SAMPLES `https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Generate or download synthetic Windows/ Linux logs, normalize fields, document schema and hash all input files. **2.** Standard (~2 h): Generate or download synthetic Windows/Linux logs, normalize fields, document schema and hash all input files. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Generate or download synthetic Windows/Linux logs, normalize fields, document schema and hash all input files. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-11/data/ with README and manifest; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why must all event times use one time zone before correlation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Dataset is reproducible, synthetic and schema-documented Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "KQL overview and B01 Sentinel data collection overview"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D217 · M08W3D7 · SIEM alert investigation capstone: investigation

Phase: Academic month · **Optional date:** 2027-05-31 · **Domain resources:** B02 / B15 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 217 of 365; calendar-free M08W3D7; optional date 2027-05-31.
2. Exact topic	SIEM alert investigation capstone: investigation
3. Exact subtopics	Query, pivot, correlate, enrich, assess scope, dismiss benign explanations and build a timeline
4. Learning objective	Investigate an alert end-to-end using only authorized synthetic data
5. Primary lecture / reading + direct URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric` **Type:** Official language docs · **Access:** Free
6. Exact section / lesson locator	KQL operators, joins, summarize and let statements. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run at least eight investigative queries, pivot across user/host/IP, produce a timeline and state what is known, likely and unknown — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run at least eight investigative queries, pivot across user/host/IP, produce a timeline and state what is known, likely and unknown — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run at least eight investigative queries, pivot across user/host/IP, produce a timeline and state what is known, likely and unknown — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-11/investigation.md and query log; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which alternative explanation remains plausible after your investigation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Timeline is evidence-backed and uncertainty is explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "KQL operators, joins, summarize and let statements"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

Month 08 · Week 4 · daily cards

D218 · M08W4D1 · SIEM alert investigation capstone: response and lessons

Phase: Academic month · **Optional date:** 2027-06-01 · **Domain resources:** C07 / B03

1. Day / sequence	Global Day 218 of 365; calendar-free M08W4D1; optional date 2027-06-01.
2. Exact topic	SIEM alert investigation capstone: response and lessons
3. Exact subtopics	Containment recommendation, eradication, recovery validation, detection gap, rule improvement and stakeholder summary
4. Learning objective	Convert investigation findings into measurable improvements
5. Primary lecture / reading + direct URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-61r3 post-incident activity and lessons learned. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a response plan, improve one detection rule, add three regression tests and produce a one-page executive summary. **2.** Standard (~2 h): Write a response plan, improve one detection rule, add three regression tests and produce a one-page executive summary. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a response plan, improve one detection rule, add three regression tests and produce a one-page executive summary. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-11/response.md and updated rule; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: How would you prove recovery rather than assume it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Response plan includes validation and the new rule passes assigned tests Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-61r3 post-incident activity and lessons learned"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D219 · M08W4D2 · Blue-team portfolio publication

Phase: Academic month · **Optional date:** 2027-06-02 · **Domain resources:** F09 / B01

1. Day / sequence	Global Day 219 of 365; calendar-free M08W4D2; optional date 2027-06-02.
2. Exact topic	Blue-team portfolio publication
3. Exact subtopics	Sanitization, architecture, dataset provenance, queries, report, detection rule, test results and limitations
4. Learning objective	Publish a professional defensive case study
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 2-3 and W13 GitHub security docs for public repository hygiene. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Sentinel documentation `https://learn.microsoft.com/en-us/azure/sentinel/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Finalize PRJ-11 and PRJ-12 with READMEs, screenshots, query files, synthetic data, threat model and limitations. **2.** Standard (~2 h): Finalize PRJ-11 and PRJ-12 with READMEs, screenshots, query files, synthetic data, threat model and limitations. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Finalize PRJ-11 and PRJ-12 with READMEs, screenshots, query files, synthetic data, threat model and limitations. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	published blue-team repositories; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which file proves the detection was tested on both true and false positives? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Portfolio checklist passes and no sensitive log data or credentials are public Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 2-3 and W13 GitHub security docs for public repository hygiene"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D220 · M08W4D3 · Blue-team specialization taster and decision evidence

Phase: Academic month · **Optional date:** 2027-06-03 · **Domain resources:** C20 / B15

1. Day / sequence	Global Day 220 of 365; calendar-free M08W4D3; optional date 2027-06-03.
2. Exact topic	Blue-team specialization taster and decision evidence
3. Exact subtopics	SOC analyst, detection engineering, DFIR, cloud security and security engineering roles; evidence from Month 8 projects
4. Learning objective	Compare specialization fit using demonstrated results rather than popularity
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST NICE Workforce Framework, work roles and task statements. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Score five roles against your project results, interests, programming comfort and local opportunities; choose a top two for Month 10 tasting. **2.** Standard (~2 h): Score five roles against your project results, interests, programming comfort and local opportunities; choose a top two for Month 10 tasting. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Score five roles against your project results, interests, programming comfort and local opportunities; choose a top two for Month 10 tasting. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	specialization-scorecard.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which project result is strongest evidence for your top role? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Scorecard uses evidence and no role is selected only because of prestige or salary assumptions Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST NICE Workforce Framework, work roles and task statements"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D221 · M08W4D4 · Month 8 practical examination and milestone gate

Phase: Academic month · **Optional date:** 2027-06-04 · **Domain resources:** B15 / B02

1. Day / sequence	Global Day 221 of 365; calendar-free M08W4D4; optional date 2027-06-04.
2. Exact topic	Month 8 practical examination and milestone gate
3. Exact subtopics	SOC workflow, logs, SIEM, triage, ATT&CK, IR, KQL, Sigma, hunting, DFIR and reporting
4. Learning objective	Demonstrate blue-team competence through an unfamiliar incident simulation
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 official study guide, skills measured as of 2026; use free materials only. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 150-minute practical with a synthetic alert, logs, IOC and detection rule; investigate, report and propose a tested detection. **2.** Standard (~2 h): Complete a 150-minute practical with a synthetic alert, logs, IOC and detection rule; investigate, report and propose a tested detection. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 150-minute practical with a synthetic alert, logs, IOC and detection rule; investigate, report and propose a tested detection. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M8-assessment.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which data source was decisive and what was its limitation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, milestone gate passed, practical at least 18/25 and safety at least 9/10 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 official study guide, skills measured as of 2026; use free materials only"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D222 · M08W4D5 · Monthly review and gap repairs

Phase: Academic month · **Optional date:** 2027-06-05 · **Domain resources:** B02 / C18

1. Day / sequence	Global Day 222 of 365; calendar-free M08W4D5; optional date 2027-06-05.
2. Exact topic	Monthly review and gap repairs
3. Exact subtopics	SIEM concepts, KQL, detection logic, ATT&CK mapping, incident reporting and evidence handling; active recall
4. Learning objective	Repair Month 8 gaps and prepare cloud/container foundations
5. Primary lecture / reading + direct URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric` **Type:** Official language docs · **Access:** Free
6. Exact section / lesson locator	KQL overview and B04 Sigma getting-started review sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework failed assessment items with new queries and test data, update PRJ-11/12, and review basic Linux/Docker prerequisites. **2.** Standard (~2 h): Rework failed assessment items with new queries and test data, update PRJ-11/12, and review basic Linux/Docker prerequisites. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework failed assessment items with new queries and test data, update PRJ-11/12, and review basic Linux/Docker prerequisites. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M8-review.md and retest score; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which detection still lacks a false-positive test and how will you add one? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and every project gap has an issue ticket Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "KQL overview and B04 Sigma getting-started review sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D223 · M08W4D6 · Network forensics and malware-traffic safety

Phase: Academic month · **Optional date:** 2027-06-06 · **Domain resources:** N07 / B08

1. Day / sequence	Global Day 223 of 365; calendar-free M08W4D6; optional date 2027-06-06.
2. Exact topic	Network forensics and malware-traffic safety
3. Exact subtopics	PCAP triage, DNS/HTTP/TLS metadata, beaconing, file extraction, IOC confidence, isolated analysis and no-sample execution
4. Learning objective	Analyze a legal sample capture without executing any malware
5. Primary lecture / reading + direct URL	Wireshark SampleCaptures `https://wiki.wireshark.org/SampleCaptures` **Type:** Packet capture repository · **Access:** Free
6. Exact section / lesson locator	Wireshark SampleCaptures and N06 Wireshark User's Guide statistics/export sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Malware Traffic Analysis `https://malware-traffic-analysis.net/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: In a disposable VM, analyze one Malware Traffic Analysis or Wireshark sample capture; identify endpoints, protocols, suspicious patterns and confidence limits. **2.** Standard (~2 h): In a disposable VM, analyze one Malware Traffic Analysis or Wireshark sample capture; identify endpoints, protocols, suspicious patterns and confidence limits. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): In a disposable VM, analyze one Malware Traffic Analysis or Wireshark sample capture; identify endpoints, protocols, suspicious patterns and confidence limits. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	network-forensics.md and sanitized packet evidence; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which metadata supports your hypothesis and what payload data is unavailable because of encryption? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No sample is executed, findings are packet-backed, and uncertainty is explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Wireshark SampleCaptures and N06 Wireshark User's Guide statistics/export sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Wireshark User's Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D224 · M08W4D7 · Spaced revision, recovery and Month 9 preview

Phase: Academic month · **Optional date:** 2027-06-07 · **Domain resources:** B03 / CL04 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 224 of 365; calendar-free M08W4D7; optional date 2027-06-07.
2. Exact topic	Spaced revision, recovery and Month 9 preview
3. Exact subtopics	Log fields, KQL operators, detection lifecycle, ATT&CK and IR phases; active recall and lab cleanup
4. Learning objective	Preserve blue-team knowledge and prepare shared-responsibility thinking
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ main documentation and B01 Sentinel overview. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Free Tier `https://aws.amazon.com/free/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, archive synthetic logs, delete unneeded cloud resources if any were created, and read AWS Free Tier terms without creating an account — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, archive synthetic logs, delete unneeded cloud resources if any were created, and read AWS Free Tier terms without creating an account — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, archive synthetic logs, delete unneeded cloud resources if any were created, and read AWS Free Tier terms without creating an account — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M8-consolidation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which cloud responsibility belongs to the customer even when the provider secures the physical infrastructure? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No active billable resources exist and the explanation is correct Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ main documentation and B01 Sentinel overview"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.



ULTIMATE CYBERSECURITY MASTER BLUEPRINT 2026-2027

Volume 4 — Daily Curriculum: Months 9-12 + Final Consolidation

141 daily cards covering cloud, specialization, capstone, final exams and Week 49-52 integration

Edition: Volume 4
Resource verification date: 2026-10-27
Plan window: 2026-10-27 to 2027-10-26 (shift dates if you start later)
System: 365 days, 12 academic months, 52 weeks + 1 graduation day
Profile: BS CS student in Pakistan; free/high-value resources; safe authorized labs

Academic-style self-study manual. No employment, salary, visa, certification or top-1% guarantee.

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Month 09 — Cloud, containers, Kubernetes and infrastructure security

Days D225-D252. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Cloud service models and shared responsibility D2: Cloud account structure and IAM foundations D3: Cloud networking and segmentation D4: Cloud storage, encryption and data lifecycle D5: Cloud logging, monitoring and incident response D6: Free-tier safety and cloud cost controls D7: Weekly cloud foundations review and paper design
Week 2	D1: Docker architecture and local setup D2: Dockerfile security and image hygiene D3: Container networking, volumes and secrets D4: Image scanning, SBOM and vulnerability triage D5: Container runtime security and isolation D6: Weekly container security review D7: Kubernetes architecture and control plane
Week 3	D1: Kubernetes RBAC and service accounts D2: Network policies and workload isolation D3: Pod security, secrets and admission D4: Kubernetes supply chain and audit logging D5: Local Kubernetes security lab and review D6: CI/CD security and GitHub Actions hardening D7: Infrastructure as Code security
Week 4	D1: Dependency scanning, provenance and release gates D2: Cloud configuration security review project D3: Docker image and container security review project D4: Month 9 practical examination and milestone gate D5: Monthly review and gap repairs D6: Multi-cloud comparison and vendor lock-in controls D7: Spaced revision, recovery and Month 10 preview

> **Gate:** score the Month 9 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 09 · Week 1 · daily cards

D225 · M09W1D1 · Cloud service models and shared responsibility

Phase: Academic month · **Optional date:** 2027-06-08 · **Domain resources:** CL03 / CL05

1. Day / sequence	Global Day 225 of 365; calendar-free M09W1D1; optional date 2027-06-08.
2. Exact topic	Cloud service models and shared responsibility
3. Exact subtopics	IaaS, PaaS, SaaS, physical/infrastructure/platform/application/data layers, customer/provider duties and control planes
4. Learning objective	Explain which security tasks remain the customer's responsibility in each service model
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar, Security Foundations and shared responsibility sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Azure security documentation `https://learn.microsoft.com/en-us/azure/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a responsibility matrix for a VM, managed database and serverless function; identify three common misunderstanding risks. **2.** Standard (~2 h): Build a responsibility matrix for a VM, managed database and serverless function; identify three common misunderstanding risks. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a responsibility matrix for a VM, managed database and serverless function; identify three common misunderstanding risks. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	shared-responsibility.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does a managed service not remove the need for identity and data classification? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every row correctly separates provider and customer duties Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar, Security Foundations and shared responsibility sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D226 · M09W1D2 · Cloud account structure and IAM foundations

Phase: Academic month · **Optional date:** 2027-06-09 · **Domain resources:** CL02 / CL01

1. Day / sequence	Global Day 226 of 365; calendar-free M09W1D2; optional date 2027-06-09.
2. Exact topic	Cloud account structure and IAM foundations
3. Exact subtopics	Accounts/subscriptions, organizations, root user, principals, policies, roles, groups, temporary credentials and least privilege
4. Learning objective	Design an IAM policy from deny-by-default principles
5. Primary lecture / reading + direct URL	AWS security documentation `https://docs.aws.amazon.com/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	AWS security documentation, IAM best practices and identity sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Skill Builder `https://skillbuilder.aws/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write three synthetic IAM policies: read-only analyst, deployment role and break-glass admin; explain why one is too broad. **2.** Standard (~2 h): Write three synthetic IAM policies: read-only analyst, deployment role and break-glass admin; explain why one is too broad. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write three synthetic IAM policies: read-only analyst, deployment role and break-glass admin; explain why one is too broad. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	iam-policies.json and review notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should daily work avoid long-lived root or administrator credentials? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Policies are syntactically plausible and least-privilege analysis is explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS security documentation, IAM best practices and identity sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS Well-Architected Security Pillar; write a five-line correction; retest next morning with a new 10-minute problem.

D227 · M09W1D3 · Cloud networking and segmentation

Phase: Academic month · **Optional date:** 2027-06-10 · **Domain resources:** CL03 / CL05

1. Day / sequence	Global Day 227 of 365; calendar-free M09W1D3; optional date 2027-06-10.
2. Exact topic	Cloud networking and segmentation
3. Exact subtopics	VPC/VNet, subnets, route tables, security groups, NACLs, private endpoints, load balancers and egress control
4. Learning objective	Draw a segmented cloud network with explicit allow/deny paths
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar, Infrastructure Protection section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Azure security documentation `https://learn.microsoft.com/en-us/azure/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a paper architecture with public/private subnets, one database and one admin path; write a rule table and failure-domain analysis. **2.** Standard (~2 h): Create a paper architecture with public/private subnets, one database and one admin path; write a rule table and failure-domain analysis. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a paper architecture with public/private subnets, one database and one admin path; write a rule table and failure-domain analysis. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-network.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which rule unintentionally allows broad lateral movement? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram and rule table agree and all high-risk paths have compensating controls Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar, Infrastructure Protection section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D228 · M09W1D4 · Cloud storage, encryption and data lifecycle

Phase: Academic month · **Optional date:** 2027-06-11 · **Domain resources:** CL02 / C25

1. Day / sequence	Global Day 228 of 365; calendar-free M09W1D4; optional date 2027-06-11.
2. Exact topic	Cloud storage, encryption and data lifecycle
3. Exact subtopics	Object storage, block storage, buckets, ACLs/policies, public access blocks, encryption at rest/in transit, KMS keys, versioning, logging and retention
4. Learning objective	Review a synthetic storage configuration and fix public-data exposure
5. Primary lecture / reading + direct URL	AWS security documentation `https://docs.aws.amazon.com/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	AWS security documentation, data protection and encryption sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Privacy Framework `https://www.nist.gov/privacy-framework`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create ten synthetic storage resources with five misconfigurations; classify data and write remediation with verification commands. **2.** Standard (~2 h): Create ten synthetic storage resources with five misconfigurations; classify data and write remediation with verification commands. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create ten synthetic storage resources with five misconfigurations; classify data and write remediation with verification commands. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-storage-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can encryption at rest fail to protect an object that is publicly readable? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All five findings are fixed in the design and encryption versus access control is distinguished Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS security documentation, data protection and encryption sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS Well-Architected Security Pillar; write a five-line correction; retest next morning with a new 10-minute problem.

D229 · M09W1D5 · Cloud logging, monitoring and incident response

Phase: Academic month · **Optional date:** 2027-06-12 · **Domain resources:** B01 / CL02

1. Day / sequence	Global Day 229 of 365; calendar-free M09W1D5; optional date 2027-06-12.
2. Exact topic	Cloud logging, monitoring and incident response
3. Exact subtopics	Control-plane logs, data-plane logs, flow logs, audit trails, metric alerts, retention, tamper resistance and incident evidence
4. Learning objective	Design a minimum viable cloud detection pipeline without enabling costly services
5. Primary lecture / reading + direct URL	Microsoft Sentinel documentation `https://learn.microsoft.com/en-us/azure/sentinel/` **Type:** Official documentation · **Access:** Docs free; Azure/Sentinel usage may incur costs
6. Exact section / lesson locator	Microsoft Sentinel documentation for comparative cloud SIEM concepts; CL03 AWS Security Pillar Detection section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS security documentation `https://docs.aws.amazon.com/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a log-source matrix and two detection hypotheses for suspicious IAM changes and public bucket creation. **2.** Standard (~2 h): Create a log-source matrix and two detection hypotheses for suspicious IAM changes and public bucket creation. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a log-source matrix and two detection hypotheses for suspicious IAM changes and public bucket creation. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-logging-design.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which log is necessary to prove who changed a security policy? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Matrix includes source, owner, retention, integrity and required fields Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft Sentinel documentation for comparative cloud SIEM concepts; CL03 AWS Security Pillar Detection section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kusto Query Language overview; write a five-line correction; retest next morning with a new 10-minute problem.

D230 · M09W1D6 · Free-tier safety and cloud cost controls

Phase: Academic month · **Optional date:** 2027-06-13 · **Domain resources:** CL04 / CL01

1. Day / sequence	Global Day 230 of 365; calendar-free M09W1D6; optional date 2027-06-13.
2. Exact topic	Free-tier safety and cloud cost controls
3. Exact subtopics	Free Tier types, budgets, billing alarms, account aliases, MFA, resource tags, cleanup, region selection and IaC teardown
4. Learning objective	Prepare a zero-surprise cloud lab policy before creating any resources
5. Primary lecture / reading + direct URL	AWS Free Tier `https://aws.amazon.com/free/` **Type:** Official program page · **Access:** Program terms; some services free temporarily, others always free
6. Exact section / lesson locator	AWS Free Tier official page and CL03 Well-Architected Security Pillar operational practices. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Skill Builder `https://skillbuilder.aws/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a pre-flight checklist including MFA, budget alert, resource tagging, daily cleanup and a do-not-deploy list; do not create billable resources yet. **2.** Standard (~2 h): Write a pre-flight checklist including MFA, budget alert, resource tagging, daily cleanup and a do-not-deploy list; do not create billable resources yet. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a pre-flight checklist including MFA, budget alert, resource tagging, daily cleanup and a do-not-deploy list; do not create billable resources yet. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-preflight.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which free-tier offer expires and which is always free, and why does that matter? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Checklist is complete and no cloud account has active paid resources Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Free Tier official page and CL03 Well-Architected Security Pillar operational practices"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS Skill Builder; write a five-line correction; retest next morning with a new 10-minute problem.

D231 · M09W1D7 · Weekly cloud foundations review and paper design

Phase: Academic month · **Optional date:** 2027-06-14 · **Domain resources:** CL03 / CL16 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 231 of 365; calendar-free M09W1D7; optional date 2027-06-14.
2. Exact topic	Weekly cloud foundations review and paper design
3. Exact subtopics	Shared responsibility, IAM, network, storage, logging and cost controls; active recall
4. Learning objective	Demonstrate Month 9 Week 1 competence without paid infrastructure
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar review questions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Cloud Security Alliance Cloud Controls Matrix `https://cloudsecurityalliance.org/research/cloud-controls-matrix`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute architecture review of a fictional small cloud app; identify ten risks, map to CCM/CSF and propose prioritized fixes — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute architecture review of a fictional small cloud app; identify ten risks, map to CCM/CSF and propose prioritized fixes — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute architecture review of a fictional small cloud app; identify ten risks, map to CCM/CSF and propose prioritized fixes — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M9W1-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which risk is caused by identity design rather than network design? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and all recommendations are verifiable Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar review questions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 09 · Week 2 · daily cards

D232 · M09W2D1 · Docker architecture and local setup

Phase: Academic month · **Optional date:** 2027-06-15 · **Domain resources:** CL06 / CL07

1. Day / sequence	Global Day 232 of 365; calendar-free M09W2D1; optional date 2027-06-15.
2. Exact topic	Docker architecture and local setup
3. Exact subtopics	Engine, daemon, client, images, layers, containers, registries, Dockerfile, Compose, namespaces, cgroups and root risk
4. Learning objective	Build and inspect a tiny local image without exposing the daemon publicly
5. Primary lecture / reading + direct URL	Docker documentation `https://docs.docker.com/` **Type:** Official documentation · **Access:** Free; Docker Desktop licensing may apply in some organizations
6. Exact section / lesson locator	Docker documentation, Get Started and Docker Engine overview sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Docker Engine security documentation `https://docs.docker.com/engine/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Install Docker Engine or Podman in an isolated Linux VM; build a hello-world image, inspect layers and list daemon attack-surface concerns. **2.** Standard (~2 h): Install Docker Engine or Podman in an isolated Linux VM; build a hello-world image, inspect layers and list daemon attack-surface concerns. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Install Docker Engine or Podman in an isolated Linux VM; build a hello-world image, inspect layers and list daemon attack-surface concerns. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	docker-basics.md and image metadata; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is access to the Docker daemon effectively root-level access on many hosts? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Image builds and runs locally and answer explains the daemon trust boundary Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Docker documentation, Get Started and Docker Engine overview sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Docker Engine security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D233 · M09W2D2 · Dockerfile security and image hygiene

Phase: Academic month · **Optional date:** 2027-06-16 · **Domain resources:** CL07 / W13

1. Day / sequence	Global Day 233 of 365; calendar-free M09W2D2; optional date 2027-06-16.
2. Exact topic	Dockerfile security and image hygiene
3. Exact subtopics	Minimal base images, pinned versions, non-root user, package cleanup, multi-stage builds, COPY versus ADD, secrets, labels and reproducibility
4. Learning objective	Harden a deliberately poor Dockerfile and compare before/after
5. Primary lecture / reading + direct URL	Docker Engine security documentation `https://docs.docker.com/engine/security/` **Type:** Official security docs · **Access:** Free
6. Exact section / lesson locator	Docker Engine security documentation and CL12 Trivy getting-started documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rewrite a sample Dockerfile with five security improvements, build it, and scan both versions with Trivy in offline/local mode if available. **2.** Standard (~2 h): Rewrite a sample Dockerfile with five security improvements, build it, and scan both versions with Trivy in offline/local mode if available. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rewrite a sample Dockerfile with five security improvements, build it, and scan both versions with Trivy in offline/local mode if available. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	Dockerfile.before/after and scan report; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which flaw remains even after vulnerability patching and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Hardened image runs as non-root and has no secret or mutable build tag Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Docker Engine security documentation and CL12 Trivy getting-started documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Docker documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D234 · M09W2D3 · Container networking, volumes and secrets

Phase: Academic month · **Optional date:** 2027-06-17 · **Domain resources:** CL06 / CL07

1. Day / sequence	Global Day 234 of 365; calendar-free M09W2D3; optional date 2027-06-17.
2. Exact topic	Container networking, volumes and secrets
3. Exact subtopics	Bridge/host/none networks, port publishing, inter-container communication, volumes, bind mounts, read-only root filesystem and secret injection
4. Learning objective	Explain how data and secrets can leak through container configuration
5. Primary lecture / reading + direct URL	Docker documentation `https://docs.docker.com/` **Type:** Official documentation · **Access:** Free; Docker Desktop licensing may apply in some organizations
6. Exact section / lesson locator	Docker documentation, networking, volumes and Compose sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Docker Engine security documentation `https://docs.docker.com/engine/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create two local containers on an isolated network, intentionally misconfigure one and then fix it; demonstrate secrets not present in image history. **2.** Standard (~2 h): Create two local containers on an isolated network, intentionally misconfigure one and then fix it; demonstrate secrets not present in image history. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create two local containers on an isolated network, intentionally misconfigure one and then fix it; demonstrate secrets not present in image history. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	container-network.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is an environment variable sometimes a weak secret-delivery mechanism? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Final configuration is isolated and image history contains no secret Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Docker documentation, networking, volumes and Compose sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Docker Engine security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D235 · M09W2D4 · Image scanning, SBOM and vulnerability triage

Phase: Academic month · **Optional date:** 2027-06-18 · **Domain resources:** W12 / E08

1. Day / sequence	Global Day 235 of 365; calendar-free M09W2D4; optional date 2027-06-18.
2. Exact topic	Image scanning, SBOM and vulnerability triage
3. Exact subtopics	CVEs, OS/package layers, SBOM, scanner limitations, false positives, base-image provenance and patch SLAs
4. Learning objective	Triage a container scan and decide what to patch, accept or investigate
5. Primary lecture / reading + direct URL	OWASP Dependency-Check `https://owasp.org/www-project-dependency-check/` **Type:** Open-source SCA tool · **Access:** Free/open-source
6. Exact section / lesson locator	OWASP Dependency-Check and CL12 Trivy documentation, scanning and output interpretation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NVD `https://nvd.nist.gov/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Scan two local images, produce SBOM-like inventory, triage three findings by reachability and context, and write remediation actions. **2.** Standard (~2 h): Scan two local images, produce SBOM-like inventory, triage three findings by reachability and context, and write remediation actions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Scan two local images, produce SBOM-like inventory, triage three findings by reachability and context, and write remediation actions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	image-scan-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does a vulnerable package in an unused code path still require a documented decision? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every finding has context, owner, action and retest plan Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Dependency-Check and CL12 Trivy documentation, scanning and output interpretation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OpenSSF Scorecard; write a five-line correction; retest next morning with a new 10-minute problem.

D236 · M09W2D5 · Container runtime security and isolation

Phase: Academic month · **Optional date:** 2027-06-19 · **Domain resources:** CL07 / L04

1. Day / sequence	Global Day 236 of 365; calendar-free M09W2D5; optional date 2027-06-19.
2. Exact topic	Container runtime security and isolation
3. Exact subtopics	Capabilities, seccomp, AppArmor/SELinux, rootless mode, privileged containers, resource limits, no-new-privileges and read-only filesystem
4. Learning objective	Run a container with reduced privileges and explain each control's failure mode
5. Primary lecture / reading + direct URL	Docker Engine security documentation `https://docs.docker.com/engine/security/` **Type:** Official security docs · **Access:** Free
6. Exact section / lesson locator	Docker Engine security documentation, namespaces, capabilities, seccomp and rootless sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux man-pages project `https://man7.org/linux/man-pages/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Compare default and hardened docker run configurations; test one blocked operation and one allowed operation. **2.** Standard (~2 h): Compare default and hardened docker run configurations; test one blocked operation and one allowed operation. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Compare default and hardened docker run configurations; test one blocked operation and one allowed operation. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	runtime-hardening.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is privileged mode especially dangerous even in a lab? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Hardened container works and privileged option is not used Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Docker Engine security documentation, namespaces, capabilities, seccomp and rootless sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Docker documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D237 · M09W2D6 · Weekly container security review

Phase: Academic month · **Optional date:** 2027-06-20 · **Domain resources:** CL12 / W15

1. Day / sequence	Global Day 237 of 365; calendar-free M09W2D6; optional date 2027-06-20.
2. Exact topic	Weekly container security review
3. Exact subtopics	Docker architecture, Dockerfile, networking, volumes, secrets, scanning and runtime isolation; active recall
4. Learning objective	Demonstrate Month 9 Week 2 competence with a local image review
5. Primary lecture / reading + direct URL	Trivy documentation `https://trivy.dev/docs/` **Type:** Official scanner docs · **Access:** Free/open-source
6. Exact section / lesson locator	Trivy documentation and CL10 OWASP Kubernetes Top 10 for image/supply-chain context. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OpenSSF Scorecard `https://securityscorecards.dev/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review an unseen Dockerfile and runtime configuration, find ten issues and implement five fixes with tests. **2.** Standard (~2 h): Review an unseen Dockerfile and runtime configuration, find ten issues and implement five fixes with tests. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review an unseen Dockerfile and runtime configuration, find ten issues and implement five fixes with tests. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M9W2-practical.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding would allow host compromise rather than just container compromise? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and all fixes are tested Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Trivy documentation and CL10 OWASP Kubernetes Top 10 for image/supply-chain context"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Dependency-Check; write a five-line correction; retest next morning with a new 10-minute problem.

D238 · M09W2D7 · Kubernetes architecture and control plane

Phase: Academic month · **Optional date:** 2027-06-21 · **Domain resources:** CL08 / CL09 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 238 of 365; calendar-free M09W2D7; optional date 2027-06-21.
2. Exact topic	Kubernetes architecture and control plane
3. Exact subtopics	API server, etcd, scheduler, controller manager, kubelet, kube-proxy, pods, nodes, namespaces and reconciliation loop
4. Learning objective	Explain how a desired-state request becomes a running workload
5. Primary lecture / reading + direct URL	Kubernetes official documentation `https://kubernetes.io/docs/home/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Kubernetes official documentation, Concepts, Cluster Architecture and Workloads sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Install kind/minikube only if hardware permits; otherwise use the official interactive tutorial or draw the architecture and annotate trust boundaries — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Install kind/minikube only if hardware permits; otherwise use the official interactive tutorial or draw the architecture and annotate trust boundaries — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Install kind/minikube only if hardware permits; otherwise use the official interactive tutorial or draw the architecture and annotate trust boundaries — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	k8s-architecture.svg or tutorial evidence; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which component stores cluster state and why must access be tightly controlled? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Architecture is complete and control-plane/data-plane boundaries are labelled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Kubernetes official documentation, Concepts, Cluster Architecture and Workloads sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 09 · Week 3 · daily cards

D239 · M09W3D1 · Kubernetes RBAC and service accounts

Phase: Academic month · **Optional date:** 2027-06-22 · **Domain resources:** CL09 / CL08

1. Day / sequence	Global Day 239 of 365; calendar-free M09W3D1; optional date 2027-06-22.
2. Exact topic	Kubernetes RBAC and service accounts
3. Exact subtopics	Subjects, roles, bindings, ClusterRole, ServiceAccount, verbs, resources, default permissions and privilege escalation
4. Learning objective	Write least-privilege RBAC and test denial explicitly
5. Primary lecture / reading + direct URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Kubernetes security documentation, RBAC and service accounts sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes official documentation `https://kubernetes.io/docs/home/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a namespace, service account, role and binding that can only list pods; test one allowed and three denied actions. **2.** Standard (~2 h): Create a namespace, service account, role and binding that can only list pods; test one allowed and three denied actions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a namespace, service account, role and binding that can only list pods; test one allowed and three denied actions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	rbac-manifests/ and kubectl output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can wildcard verbs and resources silently violate least privilege? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Tests show expected allow/deny and wildcards are avoided Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Kubernetes security documentation, RBAC and service accounts sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes official documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D240 · M09W3D2 · Network policies and workload isolation

Phase: Academic month · **Optional date:** 2027-06-23 · **Domain resources:** CL09 / CL08

1. Day / sequence	Global Day 240 of 365; calendar-free M09W3D2; optional date 2027-06-23.
2. Exact topic	Network policies and workload isolation
3. Exact subtopics	Default allow/deny, pod selectors, namespace selectors, ingress/egress and DNS/egress exceptions
4. Learning objective	Design a policy that blocks lateral movement without breaking DNS
5. Primary lecture / reading + direct URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Kubernetes security documentation, Network Policies and Pod Security sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes official documentation `https://kubernetes.io/docs/home/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Apply a default-deny ingress policy in a local cluster, add a minimal allow rule, and verify connectivity with two test pods. **2.** Standard (~2 h): Apply a default-deny ingress policy in a local cluster, add a minimal allow rule, and verify connectivity with two test pods. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Apply a default-deny ingress policy in a local cluster, add a minimal allow rule, and verify connectivity with two test pods. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	networkpolicy.yaml and test results; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does a default-deny ingress policy not automatically block egress? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Tests demonstrate allow and deny paths and DNS behaviour is explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Kubernetes security documentation, Network Policies and Pod Security sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes official documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D241 · M09W3D3 · Pod security, secrets and admission

Phase: Academic month · **Optional date:** 2027-06-24 · **Domain resources:** CL10 / CL08

1. Day / sequence	Global Day 241 of 365; calendar-free M09W3D3; optional date 2027-06-24.
2. Exact topic	Pod security, secrets and admission
3. Exact subtopics	Security contexts, runAsNonRoot, allowPrivilegeEscalation, capabilities, seccomp, Pod Security Admission, Secrets and encryption at rest
4. Learning objective	Harden a pod manifest and explain which controls are enforced at admission
5. Primary lecture / reading + direct URL	OWASP Kubernetes Top 10 `https://owasp.org/www-project-kubernetes-top-ten/` **Type:** Risk reference · **Access:** Free
6. Exact section / lesson locator	OWASP Kubernetes Top 10, K01 insecure workloads and K07 secrets; CL09 Kubernetes security docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes official documentation `https://kubernetes.io/docs/home/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Convert a deliberately privileged pod manifest into a hardened manifest; include Secret handling caveats and test admission policy. **2.** Standard (~2 h): Convert a deliberately privileged pod manifest into a hardened manifest; include Secret handling caveats and test admission policy. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Convert a deliberately privileged pod manifest into a hardened manifest; include Secret handling caveats and test admission policy. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	pod-hardened.yaml; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a Kubernetes Secret not automatically end-to-end secure? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Cluster rejects privileged settings or paper review identifies exact admission behaviour Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Kubernetes Top 10, K01 insecure workloads and K07 secrets; CL09 Kubernetes security docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D242 · M09W3D4 · Kubernetes supply chain and audit logging

Phase: Academic month · **Optional date:** 2027-06-25 · **Domain resources:** CL10 / W16

1. Day / sequence	Global Day 242 of 365; calendar-free M09W3D4; optional date 2027-06-25.
2. Exact topic	Kubernetes supply chain and audit logging
3. Exact subtopics	Image provenance, registries, tags/digests, admission policy, API audit events, runtime detection and cluster upgrade responsibility
4. Learning objective	Build a cluster supply-chain trust map with evidence sources
5. Primary lecture / reading + direct URL	OWASP Kubernetes Top 10 `https://owasp.org/www-project-kubernetes-top-ten/` **Type:** Risk reference · **Access:** Free
6. Exact section / lesson locator	OWASP Kubernetes Top 10 K02 supply chain and K09 misconfigured components. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SLSA specification `https://slsa.dev/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw an image-to-pod trust chain, list five controls and specify which audit event would reveal a suspicious deployment. **2.** Standard (~2 h): Draw an image-to-pod trust chain, list five controls and specify which audit event would reveal a suspicious deployment. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw an image-to-pod trust chain, list five controls and specify which audit event would reveal a suspicious deployment. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	k8s-supply-chain.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is an image tag weaker evidence than a digest? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Map includes builder, registry, admission, runtime and audit evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Kubernetes Top 10 K02 supply chain and K09 misconfigured components"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D243 · M09W3D5 · Local Kubernetes security lab and review

Phase: Academic month · **Optional date:** 2027-06-26 · **Domain resources:** CL11 / CL10

1. Day / sequence	Global Day 243 of 365; calendar-free M09W3D5; optional date 2027-06-26.
2. Exact topic	Local Kubernetes security lab and review
3. Exact subtopics	kind/minikube setup, namespaces, RBAC, NetworkPolicy, pod security, image scanning and audit
4. Learning objective	Demonstrate Month 9 Week 3 competence with a practical cluster review
5. Primary lecture / reading + direct URL	Kubernetes Goat `https://madhuakula.com/kubernetes-goat/` **Type:** Vulnerable-by-design K8s lab · **Access:** Free/open-source
6. Exact section / lesson locator	Kubernetes Goat beginner scenarios; use only in a disposable local cluster and only if hardware permits. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Kubernetes Top 10 `https://owasp.org/www-project-kubernetes-top-ten/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete two beginner Kubernetes Goat scenarios or an equivalent manifest review; remediate each and document resource limits. **2.** Standard (~2 h): Complete two beginner Kubernetes Goat scenarios or an equivalent manifest review; remediate each and document resource limits. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete two beginner Kubernetes Goat scenarios or an equivalent manifest review; remediate each and document resource limits. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	k8s-lab-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control failed because of cluster configuration rather than manifest syntax? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and no cluster is publicly exposed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Kubernetes Goat beginner scenarios; use only in a disposable local cluster and only if hardware permits"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Kubernetes Top 10; write a five-line correction; retest next morning with a new 10-minute problem.

D244 · M09W3D6 · CI/CD security and GitHub Actions hardening

Phase: Academic month · **Optional date:** 2027-06-27 · **Domain resources:** CL14 / W13

1. Day / sequence	Global Day 244 of 365; calendar-free M09W3D6; optional date 2027-06-27.
2. Exact topic	CI/CD security and GitHub Actions hardening
3. Exact subtopics	Pipeline trust boundaries, secrets, GITHUB_TOKEN permissions, fork pull requests, action pinning, runners, artifacts and environments
4. Learning objective	Harden a deliberately unsafe workflow and explain each change
5. Primary lecture / reading + direct URL	GitHub Actions security hardening `https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	GitHub Actions security hardening guide, permissions, secrets and third-party actions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a safe workflow for PRJ-08 with least permissions, pinned action SHAs, tests and dependency scanning; verify no secret in logs. **2.** Standard (~2 h): Create a safe workflow for PRJ-08 with least permissions, pinned action SHAs, tests and dependency scanning; verify no secret in logs. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a safe workflow for PRJ-08 with least permissions, pinned action SHAs, tests and dependency scanning; verify no secret in logs. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	security.yml and CI run evidence; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should untrusted PR code not receive repository secrets? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Workflow runs and all assigned hardening controls are present Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub Actions security hardening guide, permissions, secrets and third-party actions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup GitHub code security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D245 · M09W3D7 · Infrastructure as Code security

Phase: Academic month · **Optional date:** 2027-06-28 · **Domain resources:** CL15 / CL13 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 245 of 365; calendar-free M09W3D7; optional date 2027-06-28.
2. Exact topic	Infrastructure as Code security
3. Exact subtopics	Declarative resources, state, providers, drift, modules, secrets in state, plan/apply, Checkov policies and immutable infrastructure
4. Learning objective	Scan and remediate a synthetic IaC configuration without deploying it
5. Primary lecture / reading + direct URL	Terraform documentation `https://developer.hashicorp.com/terraform/docs` **Type:** Official documentation · **Access:** Free/open-source; HCP Terraform has paid tiers
6. Exact section / lesson locator	Terraform documentation, configuration, state and plan sections; CL13 Checkov documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Checkov documentation `https://www.checkov.io/1.Welcome/What%20is%20Checkov.html`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a synthetic Terraform file with five misconfigurations, run Checkov or an offline checklist, remediate and compare plans — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a synthetic Terraform file with five misconfigurations, run Checkov or an offline checklist, remediate and compare plans — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a synthetic Terraform file with five misconfigurations, run Checkov or an offline checklist, remediate and compare plans — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	iac-review.tf and scan report; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which secret could be committed if state files are mishandled? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No apply to cloud account occurs and all five issues are remediated Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Terraform documentation, configuration, state and plan sections; CL13 Checkov documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Checkov documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 09 · Week 4 · daily cards

D246 · M09W4D1 · Dependency scanning, provenance and release gates

Phase: Academic month · **Optional date:** 2027-06-29 · **Domain resources:** W16 / W13

1. Day / sequence	Global Day 246 of 365; calendar-free M09W4D1; optional date 2027-06-29.
2. Exact topic	Dependency scanning, provenance and release gates
3. Exact subtopics	SBOM, lockfiles, signed builds, SLSA levels, artifacts, attestations, release approval and incident rollback
4. Learning objective	Define a release gate that blocks unverifiable artifacts
5. Primary lecture / reading + direct URL	SLSA specification `https://slsa.dev/` **Type:** Supply-chain framework · **Access:** Free
6. Exact section / lesson locator	SLSA specification, levels and provenance; W15 OpenSSF Scorecard checks. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a release checklist with artifact hashes, provenance evidence, vulnerability thresholds and rollback owner. **2.** Standard (~2 h): Create a release checklist with artifact hashes, provenance evidence, vulnerability thresholds and rollback owner. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a release checklist with artifact hashes, provenance evidence, vulnerability thresholds and rollback owner. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	release-gate.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a green unit-test job insufficient to trust a released artifact? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Checklist includes build identity, integrity, vulnerability and rollback evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SLSA specification, levels and provenance; W15 OpenSSF Scorecard checks"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup GitHub code security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D247 · M09W4D2 · Cloud configuration security review project

Phase: Academic month · **Optional date:** 2027-06-30 · **Domain resources:** CL03 / CL02

1. Day / sequence	Global Day 247 of 365; calendar-free M09W4D2; optional date 2027-06-30.
2. Exact topic	Cloud configuration security review project
3. Exact subtopics	Asset inventory, IAM, network, storage, logging, encryption, detection and cost boundaries
4. Learning objective	Produce a professional cloud configuration review using synthetic architecture/data
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar review questions and CL16 Cloud Controls Matrix selected controls. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS security documentation `https://docs.aws.amazon.com/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review a fictional ten-resource cloud architecture, identify twelve misconfigurations, prioritize and provide verification steps; do not deploy paid resources. **2.** Standard (~2 h): Review a fictional ten-resource cloud architecture, identify twelve misconfigurations, prioritize and provide verification steps; do not deploy paid resources. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review a fictional ten-resource cloud architecture, identify twelve misconfigurations, prioritize and provide verification steps; do not deploy paid resources. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-13 report and risk register; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding has the highest blast radius and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Report passes the 100-point rubric and no live cloud resources were required Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar review questions and CL16 Cloud Controls Matrix selected controls"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D248 · M09W4D3 · Docker image and container security review project

Phase: Academic month · **Optional date:** 2027-07-01 · **Domain resources:** CL12 / W12

1. Day / sequence	Global Day 248 of 365; calendar-free M09W4D3; optional date 2027-07-01.
2. Exact topic	Docker image and container security review project
3. Exact subtopics	Image layers, Dockerfile, secrets, runtime privileges, networking, volumes, scanning and SBOM
4. Learning objective	Produce an image/container security review with reproducible local evidence
5. Primary lecture / reading + direct URL	Trivy documentation `https://trivy.dev/docs/` **Type:** Official scanner docs · **Access:** Free/open-source
6. Exact section / lesson locator	Trivy documentation and CL07 Docker Engine security documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Dependency-Check `https://owasp.org/www-project-dependency-check/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review one intentionally weak local image/container, record ten findings, fix five, rescan and document residual risk. **2.** Standard (~2 h): Review one intentionally weak local image/container, record ten findings, fix five, rescan and document residual risk. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review one intentionally weak local image/container, record ten findings, fix five, rescan and document residual risk. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-14 report and before/after scan; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding was a false positive or not exploitable and how did you decide? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Containers remain isolated and all fixes are verified by retest Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Trivy documentation and CL07 Docker Engine security documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Dependency-Check; write a five-line correction; retest next morning with a new 10-minute problem.

D249 · M09W4D4 · Month 9 practical examination and milestone gate

Phase: Academic month · **Optional date:** 2027-07-02 · **Domain resources:** CL03 / CL09

1. Day / sequence	Global Day 249 of 365; calendar-free M09W4D4; optional date 2027-07-02.
2. Exact topic	Month 9 practical examination and milestone gate
3. Exact subtopics	Cloud shared responsibility, IAM, storage, logging, Docker, Kubernetes, CI/CD and IaC; practical scenario
4. Learning objective	Demonstrate infrastructure-security competence under time pressure
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar and CL08 Kubernetes concepts review pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 150-minute practical with an unseen cloud architecture, Dockerfile, Kubernetes manifest and workflow; identify and fix prioritized risks. **2.** Standard (~2 h): Complete a 150-minute practical with an unseen cloud architecture, Dockerfile, Kubernetes manifest and workflow; identify and fix prioritized risks. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 150-minute practical with an unseen cloud architecture, Dockerfile, Kubernetes manifest and workflow; identify and fix prioritized risks. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M9-assessment.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which risk crosses from workload compromise to control-plane compromise? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, milestone gate passed, practical at least 18/25 and safety at least 9/10 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar and CL08 Kubernetes concepts review pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D250 · M09W4D5 · Monthly review and gap repairs

Phase: Academic month · **Optional date:** 2027-07-03 · **Domain resources:** CL09 / C20

1. Day / sequence	Global Day 250 of 365; calendar-free M09W4D5; optional date 2027-07-03.
2. Exact topic	Monthly review and gap repairs
3. Exact subtopics	IAM reasoning, Docker privileges, Kubernetes RBAC/NetworkPolicy, CI secrets and IaC state; active recall
4. Learning objective	Repair Month 9 gaps and choose specialization evidence for Month 10
5. Primary lecture / reading + direct URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Kubernetes security documentation and CL14 GitHub Actions hardening guide. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework failed assessment items, update PRJ-13/14, write ten flashcards and score your interest/results in cloud security versus detection engineering. **2.** Standard (~2 h): Rework failed assessment items, update PRJ-13/14, write ten flashcards and score your interest/results in cloud security versus detection engineering. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework failed assessment items, update PRJ-13/14, write ten flashcards and score your interest/results in cloud security versus detection engineering. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M9-review.md and specialization-prep.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which infrastructure topic would be most valuable as a secondary skill? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and scorecard uses actual project evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Kubernetes security documentation and CL14 GitHub Actions hardening guide"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes official documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D251 · M09W4D6 · Multi-cloud comparison and vendor lock-in controls

Phase: Academic month · **Optional date:** 2027-07-04 · **Domain resources:** CL05 / CL02

1. Day / sequence	Global Day 251 of 365; calendar-free M09W4D6; optional date 2027-07-04.
2. Exact topic	Multi-cloud comparison and vendor lock-in controls
3. Exact subtopics	AWS/Azure equivalents for identity, network, storage, logging, KMS and incident response; portable controls and service-specific gaps
4. Learning objective	Translate one AWS control design into Azure terminology without assuming identical defaults
5. Primary lecture / reading + direct URL	Microsoft Azure security documentation `https://learn.microsoft.com/en-us/azure/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Microsoft Azure security documentation, identity, network, data protection and monitoring sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS security documentation `https://docs.aws.amazon.com/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a side-by-side control table for ten AWS/Azure concepts and identify three behaviours that must be verified in current docs. **2.** Standard (~2 h): Build a side-by-side control table for ten AWS/Azure concepts and identify three behaviours that must be verified in current docs. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a side-by-side control table for ten AWS/Azure concepts and identify three behaviours that must be verified in current docs. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	multicloud-controls.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control name sounds equivalent but has different default behaviour? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Table uses current official docs and explicitly marks unverified assumptions Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft Azure security documentation, identity, network, data protection and monitoring sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D252 · M09W4D7 · Spaced revision, recovery and Month 10 preview

Phase: Academic month · **Optional date:** 2027-07-05 · **Domain resources:** CL04 / C20 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 252 of 365; calendar-free M09W4D7; optional date 2027-07-05.
2. Exact topic	Spaced revision, recovery and Month 10 preview
3. Exact subtopics	Cloud responsibility, IAM, Docker/Kubernetes controls, CI/CD and IaC retrieval; lab cleanup
4. Learning objective	Preserve infrastructure-security knowledge and prepare specialization selection
5. Primary lecture / reading + direct URL	AWS Free Tier `https://aws.amazon.com/free/` **Type:** Official program page · **Access:** Program terms; some services free temporarily, others always free
6. Exact section / lesson locator	AWS Free Tier and CL01 AWS Skill Builder free course catalogue. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, remove all local test containers/clusters or save clean snapshots, and read the specialization comparison in Volume 7 — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, remove all local test containers/clusters or save clean snapshots, and read the specialization comparison in Volume 7 — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, remove all local test containers/clusters or save clean snapshots, and read the specialization comparison in Volume 7 — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M9-consolidation.md and cleanup log; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which specialization best matches your strongest project evidence so far? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No active billable resources remain and decision notes cite at least three projects Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Free Tier and CL01 AWS Skill Builder free course catalogue"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS Skill Builder; write a five-line correction; retest next morning with a new 10-minute problem.

Month 10 — Specialization selection and focused technical development

Days D253-D280. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Specialization decision framework and evidence audit D2: Ten specialization maps and prerequisite reality D3: Primary specialization recommendation: detection engineering and security automation D4: Complementary secondary skill: cloud and container security D5: Designing a four-week specialization tasting sprint D6: Baseline assessment for specialization sprint D7: Weekly decision review and rest
Week 2	D1: Detection engineering sprint: telemetry requirements D2: Detection engineering sprint: rule design and unit tests D3: Detection engineering sprint: backtesting and tuning D4: Detection engineering sprint: automation and CI D5: Detection engineering sprint: threat hunting with the rule D6: Detection engineering sprint: documentation and defence D7: Weekly detection sprint review and score
Week 3	D1: Cloud security sprint: IAM threat modelling D2: Cloud security sprint: network and storage detections D3: Cloud security sprint: Kubernetes RBAC and NetworkPolicy review D4: Cloud security sprint: IaC and CI hardening lab D5: Cloud security sprint: incident in a containerized app D6: Cloud security sprint: cost, resilience and recovery D7: Weekly cloud sprint review and score
Week 4	D1: Decision review with a technical peer or self-defence D2: Specialization project requirements and architecture D3: Specialization literature and documentation reading D4: Independent problem-solving baseline with no walkthrough D5: Month 10 practical examination and gate D6: Monthly review and gap repairs D7: Spaced revision, recovery and Month 11 preview

> **Gate:** score the Month 10 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 10 · Week 1 · daily cards

D253 · M10W1D1 · Specialization decision framework and evidence audit

Phase: Academic month · **Optional date:** 2027-07-06 · **Domain resources:** C20 / A08

1. Day / sequence	Global Day 253 of 365; calendar-free M10W1D1; optional date 2027-07-06.
2. Exact topic	Specialization decision framework and evidence audit
3. Exact subtopics	Interest, demonstrated results, technical depth, programming comfort, local opportunity, feedback and opportunity cost
4. Learning objective	Choose a primary and secondary specialization from evidence rather than trends
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST NICE Workforce Framework, work roles, tasks, knowledge and skills. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a weighted decision matrix for ten paths using scores from M1-M9 assessments, project quality and genuine interest. **2.** Standard (~2 h): Build a weighted decision matrix for ten paths using scores from M1-M9 assessments, project quality and genuine interest. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a weighted decision matrix for ten paths using scores from M1-M9 assessments, project quality and genuine interest. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	specialization-decision.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which score is based on evidence and which is based on assumption? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Matrix is transparent, weighted and identifies a primary plus a complementary secondary Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST NICE Workforce Framework, work roles, tasks, knowledge and skills"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D254 · M10W1D2 · Ten specialization maps and prerequisite reality

Phase: Academic month · **Optional date:** 2027-07-07 · **Domain resources:** B15 / C20

1. Day / sequence	Global Day 254 of 365; calendar-free M10W1D2; optional date 2027-07-07.
2. Exact topic	Ten specialization maps and prerequisite reality
3. Exact subtopics	SOC, detection engineering, DFIR, web/API security, pentesting, cloud, security engineering, GRC, vulnerability research and malware/RE
4. Learning objective	Compare daily work, tools, programming, portfolio and realistic Year 1 limits
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide and A21 PEN-200 page for two contrasting role examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a one-page comparison of ten paths with required prerequisites, first job evidence and what 12 months does not prove. **2.** Standard (~2 h): Write a one-page comparison of ten paths with required prerequisites, first job evidence and what 12 months does not prove. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a one-page comparison of ten paths with required prerequisites, first job evidence and what 12 months does not prove. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	specialization-map.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which path would require years of low-level experience before credible expertise? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No path is described as mastered and prerequisites are explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide and A21 PEN-200 page for two contrasting role examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D255 · M10W1D3 · Primary specialization recommendation: detection engineering and security automation

Phase: Academic month · **Optional date:** 2027-07-08 · **Domain resources:** B03 / B14

1. Day / sequence	Global Day 255 of 365; calendar-free M10W1D3; optional date 2027-07-08.
2. Exact topic	Primary specialization recommendation: detection engineering and security automation
3. Exact subtopics	Detection lifecycle, telemetry, KQL/Sigma, Python automation, cloud logs, measurement and communication
4. Learning objective	Validate whether this recommended primary fits your evidence and interest
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ and B04 Sigma getting-started documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Elastic detection rules documentation `https://www.elastic.co/guide/en/security/current/detection-engine-overview.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a personal fit memo for detection engineering with three supportive and three contradictory pieces of project evidence. **2.** Standard (~2 h): Write a personal fit memo for detection engineering with three supportive and three contradictory pieces of project evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a personal fit memo for detection engineering with three supportive and three contradictory pieces of project evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	primary-fit-memo.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which project result would convince a working detection engineer of your potential? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Recommendation is justified but treated as revisable after the Month 10 sprint Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ and B04 Sigma getting-started documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

D256 · M10W1D4 · Complementary secondary skill: cloud and container security

Phase: Academic month · **Optional date:** 2027-07-09 · **Domain resources:** CL03 / B01

1. Day / sequence	Global Day 256 of 365; calendar-free M10W1D4; optional date 2027-07-09.
2. Exact topic	Complementary secondary skill: cloud and container security
3. Exact subtopics	Cloud IAM, Kubernetes, Docker, CI/CD, IaC, logging and detection overlap
4. Learning objective	Show how the secondary skill strengthens the primary specialization
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar and CL09 Kubernetes security docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Sentinel documentation `https://learn.microsoft.com/en-us/azure/sentinel/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a skill-integration map connecting cloud logs, containers, detections and automation projects. **2.** Standard (~2 h): Draw a skill-integration map connecting cloud logs, containers, detections and automation projects. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a skill-integration map connecting cloud logs, containers, detections and automation projects. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	skill-integration.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which cloud telemetry source would be most useful for your detection work? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Map has at least six concrete connections and no vague tool names Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar and CL09 Kubernetes security docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D257 · M10W1D5 · Designing a four-week specialization tasting sprint

Phase: Academic month · **Optional date:** 2027-07-10 · **Domain resources:** C06 / F09

1. Day / sequence	Global Day 257 of 365; calendar-free M10W1D5; optional date 2027-07-10.
2. Exact topic	Designing a four-week specialization tasting sprint
3. Exact subtopics	SMART objectives, hypotheses, timeboxing, deliverables, evidence, feedback and stopping criteria
4. Learning objective	Create a fair experiment to compare two specializations without switching randomly
5. Primary lecture / reading + direct URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST CSF 2.0, Govern/Identify/Protect/Detect/Respond/Recover functions as outcome categories. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Define two two-week sprints with measurable deliverables, rubrics, daily logs and a final decision review. **2.** Standard (~2 h): Define two two-week sprints with measurable deliverables, rubrics, daily logs and a final decision review. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Define two two-week sprints with measurable deliverables, rubrics, daily logs and a final decision review. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	sprint-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What evidence would cause you to reject your preferred path? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan includes falsifiable criteria and a fixed decision date Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST CSF 2.0, Govern/Identify/Protect/Detect/Respond/Recover functions as outcome categories"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Controls Version 8; write a five-line correction; retest next morning with a new 10-minute problem.

D258 · M10W1D6 · Baseline assessment for specialization sprint

Phase: Academic month · **Optional date:** 2027-07-11 · **Domain resources:** B04 / B02

1. Day / sequence	Global Day 258 of 365; calendar-free M10W1D6; optional date 2027-07-11.
2. Exact topic	Baseline assessment for specialization sprint
3. Exact subtopics	Detection rule quality, query fluency, cloud misconfiguration reasoning, report clarity and independent debugging
4. Learning objective	Establish a baseline so improvement can be measured
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started rule structure and CL03 Security Pillar review questions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 90-minute baseline: one detection rule with test data, one cloud config review and one written explanation; score with the standard rubric. **2.** Standard (~2 h): Complete a 90-minute baseline: one detection rule with test data, one cloud config review and one written explanation; score with the standard rubric. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 90-minute baseline: one detection rule with test data, one cloud config review and one written explanation; score with the standard rubric. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10-baseline.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which task took longest and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Baseline is scored with evidence and no tutorial assistance Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started rule structure and CL03 Security Pillar review questions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D259 · M10W1D7 · Weekly decision review and rest

Phase: Academic month · **Optional date:** 2027-07-12 · **Domain resources:** C20 / B03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 259 of 365; calendar-free M10W1D7; optional date 2027-07-12.
2. Exact topic	Weekly decision review and rest
3. Exact subtopics	Specialization map, evidence, decision framework, primary/secondary choice, sprint plan and baseline; active recall
4. Learning objective	Consolidate Month 10 Week 1 and commit to a fair experiment
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role pages and C18 ATT&CK matrix for role-to-skill mapping. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do 15 retrieval questions, review baseline errors, adjust sprint difficulty and rest or take a light walkthrough of Week 2 tasks — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do 15 retrieval questions, review baseline errors, adjust sprint difficulty and rest or take a light walkthrough of Week 2 tasks — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do 15 retrieval questions, review baseline errors, adjust sprint difficulty and rest or take a light walkthrough of Week 2 tasks — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10W1-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which baseline weakness is most important for your chosen primary? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Sprint plan is realistic and no decision changes without documented evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role pages and C18 ATT&CK matrix for role-to-skill mapping"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Month 10 · Week 2 · daily cards

D260 · M10W2D1 · Detection engineering sprint: telemetry requirements

Phase: Academic month · **Optional date:** 2027-07-13 · **Domain resources:** C18 / B05

1. Day / sequence	Global Day 260 of 365; calendar-free M10W2D1; optional date 2027-07-13.
2. Exact topic	Detection engineering sprint: telemetry requirements
3. Exact subtopics	Data source, event schema, field coverage, time synchronization, entity mapping and blind spots
4. Learning objective	Define telemetry needed for a specific adversary behaviour
5. Primary lecture / reading + direct URL	MITRE ATT&CK `https://attack.mitre.org/` **Type:** Knowledge base · **Access:** Free
6. Exact section / lesson locator	MITRE ATT&CK technique pages and data sources. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SBousseaden EVTX-ATTACK-SAMPLES `https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Pick one technique, create a telemetry requirements table and identify which fields are missing from your synthetic logs. **2.** Standard (~2 h): Pick one technique, create a telemetry requirements table and identify which fields are missing from your synthetic logs. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Pick one technique, create a telemetry requirements table and identify which fields are missing from your synthetic logs. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	telemetry-requirements.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a technique description insufficient for writing a detection? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Table maps each detection field to a source and reliability Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MITRE ATT&CK technique pages and data sources"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE D3FEND; write a five-line correction; retest next morning with a new 10-minute problem.

D261 · M10W2D2 · Detection engineering sprint: rule design and unit tests

Phase: Academic month · **Optional date:** 2027-07-14 · **Domain resources:** B04 / B03

1. Day / sequence	Global Day 261 of 365; calendar-free M10W2D2; optional date 2027-07-14.
2. Exact topic	Detection engineering sprint: rule design and unit tests
3. Exact subtopics	Modifiers, conditions, correlations, thresholds, exceptions, metadata and test cases
4. Learning objective	Write a detection with explicit true/false-positive tests
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started guide and B14 Elastic detection rules docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Implement one advanced Sigma rule for a synthetic lab behaviour, create ten positive/negative events and validate conversion to a query. **2.** Standard (~2 h): Implement one advanced Sigma rule for a synthetic lab behaviour, create ten positive/negative events and validate conversion to a query. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Implement one advanced Sigma rule for a synthetic lab behaviour, create ten positive/negative events and validate conversion to a query. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	advanced-rule.yml and tests; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which test event represents a realistic benign lookalike? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Rule passes all tests and conversion output is explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started guide and B14 Elastic detection rules docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D262 · M10W2D3 · Detection engineering sprint: backtesting and tuning

Phase: Academic month · **Optional date:** 2027-07-15 · **Domain resources:** B14 / B02

1. Day / sequence	Global Day 262 of 365; calendar-free M10W2D3; optional date 2027-07-15.
2. Exact topic	Detection engineering sprint: backtesting and tuning
3. Exact subtopics	Historical data, prevalence, benign baselines, alert volume, precision proxy, suppression and exception design
4. Learning objective	Measure rule behaviour on a larger synthetic dataset without hiding true positives
5. Primary lecture / reading + direct URL	Elastic detection rules documentation `https://www.elastic.co/guide/en/security/current/detection-engine-overview.html` **Type:** Official docs / open-source rules · **Access:** Free docs/open-source rules; Elastic Stack may have paid features
6. Exact section / lesson locator	Elastic detection rules, exceptions and alert investigation docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Generate 1,000 synthetic events with known labels, run the rule, calculate precision/recall-like measures and tune with narrow exceptions. **2.** Standard (~2 h): Generate 1,000 synthetic events with known labels, run the rule, calculate precision/recall-like measures and tune with narrow exceptions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Generate 1,000 synthetic events with known labels, run the rule, calculate precision/recall-like measures and tune with narrow exceptions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	backtest-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which exception could hide a true positive variant? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Metrics are computed from labelled data and tuning is justified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Elastic detection rules, exceptions and alert investigation docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D263 · M10W2D4 · Detection engineering sprint: automation and CI

Phase: Academic month · **Optional date:** 2027-07-16 · **Domain resources:** P05 / B03

1. Day / sequence	Global Day 263 of 365; calendar-free M10W2D4; optional date 2027-07-16.
2. Exact topic	Detection engineering sprint: automation and CI
3. Exact subtopics	Detection-as-code repository, linting, schema validation, tests, versioning, review and deployment simulation
4. Learning objective	Build a repeatable workflow for rule quality
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest documentation and F09 Pro Git branching sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create CI-like local scripts that validate Sigma YAML, run test events, produce a report and fail on regression. **2.** Standard (~2 h): Create CI-like local scripts that validate Sigma YAML, run test events, produce a report and fail on regression. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create CI-like local scripts that validate Sigma YAML, run test events, produce a report and fail on regression. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	detection-ci/ repository; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which test should block a rule deployment? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Workflow fails on intentionally broken rule and passes on fixed version Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest documentation and F09 Pro Git branching sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D264 · M10W2D5 · Detection engineering sprint: threat hunting with the rule

Phase: Academic month · **Optional date:** 2027-07-17 · **Domain resources:** B15 / C18

1. Day / sequence	Global Day 264 of 365; calendar-free M10W2D5; optional date 2027-07-17.
2. Exact topic	Detection engineering sprint: threat hunting with the rule
3. Exact subtopics	Hypothesis, query, pivots, temporal correlation, entity risk and negative results
4. Learning objective	Use the new rule to answer an investigative question rather than just generate an alert
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide threat hunting section and B02 KQL operators. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a synthetic hunt dataset, run three KQL-style queries, document pivots and state what remains unknown. **2.** Standard (~2 h): Create a synthetic hunt dataset, run three KQL-style queries, document pivots and state what remains unknown. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a synthetic hunt dataset, run three KQL-style queries, document pivots and state what remains unknown. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	hunt-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which data gap prevents a confident conclusion? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Hunt is reproducible and includes negative-result interpretation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide threat hunting section and B02 KQL operators"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D265 · M10W2D6 · Detection engineering sprint: documentation and defence

Phase: Academic month · **Optional date:** 2027-07-18 · **Domain resources:** B03 / B14

1. Day / sequence	Global Day 265 of 365; calendar-free M10W2D6; optional date 2027-07-18.
2. Exact topic	Detection engineering sprint: documentation and defence
3. Exact subtopics	Rule rationale, ATT&CK mapping, assumptions, test evidence, known bypasses and reviewer questions
4. Learning objective	Defend the detection to a skeptical peer in writing
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ rule metadata guidance and C18 ATT&CK technique page. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Elastic detection rules documentation `https://www.elastic.co/guide/en/security/current/detection-engine-overview.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a two-page detection design document, list three bypasses and answer five reviewer questions from a self-critique checklist. **2.** Standard (~2 h): Write a two-page detection design document, list three bypasses and answer five reviewer questions from a self-critique checklist. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a two-page detection design document, list three bypasses and answer five reviewer questions from a self-critique checklist. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	detection-design.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which bypass is most realistic and what additional data would detect it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Document is technically specific and includes a mitigation or accepted limitation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ rule metadata guidance and C18 ATT&CK technique page"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

D266 · M10W2D7 · Weekly detection sprint review and score

Phase: Academic month · **Optional date:** 2027-07-19 · **Domain resources:** B04 / C18 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 266 of 365; calendar-free M10W2D7; optional date 2027-07-19.
2. Exact topic	Weekly detection sprint review and score
3. Exact subtopics	Telemetry, rule design, backtesting, CI, hunting and defence; independent evaluation
4. Learning objective	Measure improvement against the Week 1 baseline and compare specialization fit
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started and B14 Elastic detection rule docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score the week's artifacts with the 100-point rubric, repair one major gap, and write a one-page reaction on interest and difficulty — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score the week's artifacts with the 100-point rubric, repair one major gap, and write a one-page reaction on interest and difficulty — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score the week's artifacts with the 100-point rubric, repair one major gap, and write a one-page reaction on interest and difficulty — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10W2-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which part of detection engineering felt effortful but rewarding? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score is evidence-based and a follow-up task is scheduled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started and B14 Elastic detection rule docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

Month 10 · Week 3 · daily cards

D267 · M10W3D1 · Cloud security sprint: IAM threat modelling

Phase: Academic month · **Optional date:** 2027-07-20 · **Domain resources:** CL02 / C17

1. Day / sequence	Global Day 267 of 365; calendar-free M10W3D1; optional date 2027-07-20.
2. Exact topic	Cloud security sprint: IAM threat modelling
3. Exact subtopics	Principal, action, resource, condition, privilege escalation, confused deputy, cross-account access and logging
4. Learning objective	Threat-model an IAM design before writing policies
5. Primary lecture / reading + direct URL	AWS security documentation `https://docs.aws.amazon.com/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	AWS security documentation IAM best practices and CL03 Security Pillar identity section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw an IAM attack graph for a fictional app, identify five escalation paths and propose least-privilege fixes. **2.** Standard (~2 h): Draw an IAM attack graph for a fictional app, identify five escalation paths and propose least-privilege fixes. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw an IAM attack graph for a fictional app, identify five escalation paths and propose least-privilege fixes. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	iam-threat-model.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which permission is dangerous only in combination with another? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Graph is plausible and mitigations break specific edges Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS security documentation IAM best practices and CL03 Security Pillar identity section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS Well-Architected Security Pillar; write a five-line correction; retest next morning with a new 10-minute problem.

D268 · M10W3D2 · Cloud security sprint: network and storage detections

Phase: Academic month · **Optional date:** 2027-07-21 · **Domain resources:** CL03 / B01

1. Day / sequence	Global Day 268 of 365; calendar-free M10W3D2; optional date 2027-07-21.
2. Exact topic	Cloud security sprint: network and storage detections
3. Exact subtopics	Public exposure, policy changes, unusual data access, egress anomalies and log sources
4. Learning objective	Design cloud detections that connect infrastructure misconfiguration to monitoring
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar, Detection and Data Protection sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Sentinel documentation `https://learn.microsoft.com/en-us/azure/sentinel/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write two detection hypotheses with required cloud log fields, one query-like pseudo-rule and a validation plan. **2.** Standard (~2 h): Write two detection hypotheses with required cloud log fields, one query-like pseudo-rule and a validation plan. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write two detection hypotheses with required cloud log fields, one query-like pseudo-rule and a validation plan. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-detections.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which detection depends on a log source that is not enabled by default? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Hypotheses specify fields, owner and enabling steps Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar, Detection and Data Protection sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D269 · M10W3D3 · Cloud security sprint: Kubernetes RBAC and NetworkPolicy review

Phase: Academic month · **Optional date:** 2027-07-22 · **Domain resources:** CL09 / CL10

1. Day / sequence	Global Day 269 of 365; calendar-free M10W3D3; optional date 2027-07-22.
2. Exact topic	Cloud security sprint: Kubernetes RBAC and NetworkPolicy review
3. Exact subtopics	Role bindings, service accounts, default-deny, egress, namespace isolation and admission controls
4. Learning objective	Review a cluster design for both prevention and detection of privilege misuse
5. Primary lecture / reading + direct URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Kubernetes security documentation, RBAC, NetworkPolicy and audit sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Kubernetes Top 10 `https://owasp.org/www-project-kubernetes-top-ten/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Analyze ten synthetic manifests, find five privilege or segmentation issues, and write one audit-event detection hypothesis. **2.** Standard (~2 h): Analyze ten synthetic manifests, find five privilege or segmentation issues, and write one audit-event detection hypothesis. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Analyze ten synthetic manifests, find five privilege or segmentation issues, and write one audit-event detection hypothesis. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	k8s-skill-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which RBAC binding creates cross-namespace access? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Findings are technically correct and no live cluster is exposed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Kubernetes security documentation, RBAC, NetworkPolicy and audit sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes official documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D270 · M10W3D4 · Cloud security sprint: IaC and CI hardening lab

Phase: Academic month · **Optional date:** 2027-07-23 · **Domain resources:** CL14 / W16

1. Day / sequence	Global Day 270 of 365; calendar-free M10W3D4; optional date 2027-07-23.
2. Exact topic	Cloud security sprint: IaC and CI hardening lab
3. Exact subtopics	Static IaC scanning, workflow permissions, action pinning, secret handling and plan review
4. Learning objective	Combine preventive and detective controls in a software delivery pipeline
5. Primary lecture / reading + direct URL	GitHub Actions security hardening `https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	GitHub Actions security hardening and CL13 Checkov documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SLSA specification `https://slsa.dev/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a local pipeline that scans a synthetic Terraform/Kubernetes repo, fails on three issues and produces a remediation report. **2.** Standard (~2 h): Build a local pipeline that scans a synthetic Terraform/Kubernetes repo, fails on three issues and produces a remediation report. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a local pipeline that scans a synthetic Terraform/Kubernetes repo, fails on three issues and produces a remediation report. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	iac-ci-demo/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control prevents the issue and which detects recurrence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Pipeline passes after fixes and report distinguishes prevention from detection Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub Actions security hardening and CL13 Checkov documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup GitHub code security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D271 · M10W3D5 · Cloud security sprint: incident in a containerized app

Phase: Academic month · **Optional date:** 2027-07-24 · **Domain resources:** C07 / B02

1. Day / sequence	Global Day 271 of 365; calendar-free M10W3D5; optional date 2027-07-24.
2. Exact topic	Cloud security sprint: incident in a containerized app
3. Exact subtopics	Container logs, process metadata, IAM audit events, network evidence and timeline correlation
4. Learning objective	Investigate a synthetic cloud/container incident with incomplete data
5. Primary lecture / reading + direct URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-61r3 incident response lifecycle and CL03 Security Pillar incident response. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Analyze a provided synthetic log bundle, build a timeline, identify the likely root cause and state evidence gaps. **2.** Standard (~2 h): Analyze a provided synthetic log bundle, build a timeline, identify the likely root cause and state evidence gaps. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Analyze a provided synthetic log bundle, build a timeline, identify the likely root cause and state evidence gaps. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-incident-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which claim is probable but not proven by the available telemetry? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Timeline is consistent and uncertainty is explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-61r3 incident response lifecycle and CL03 Security Pillar incident response"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D272 · M10W3D6 · Cloud security sprint: cost, resilience and recovery

Phase: Academic month · **Optional date:** 2027-07-25 · **Domain resources:** C27 / CL01

1. Day / sequence	Global Day 272 of 365; calendar-free M10W3D6; optional date 2027-07-25.
2. Exact topic	Cloud security sprint: cost, resilience and recovery
3. Exact subtopics	Resource cleanup, backups, multi-AZ concepts, recovery objectives, budgets, quotas and blast-radius reduction
4. Learning objective	Add operational realism to a cloud security design
5. Primary lecture / reading + direct URL	NIST SP 800-34 Revision 1 `https://csrc.nist.gov/pubs/sp/800/34/r1/final` **Type:** Official continuity guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-34r1 and CL04 AWS Free Tier/current cost-control documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Skill Builder `https://skillbuilder.aws/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Revise the fictional architecture for resilience and cost limits; write one restore test and one budget alert plan. **2.** Standard (~2 h): Revise the fictional architecture for resilience and cost limits; write one restore test and one budget alert plan. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Revise the fictional architecture for resilience and cost limits; write one restore test and one budget alert plan. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-resilience.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which resilience control adds a new security dependency? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Design includes RTO/RPO, validation and teardown evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-34r1 and CL04 AWS Free Tier/current cost-control documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D273 · M10W3D7 · Weekly cloud sprint review and score

Phase: Academic month · **Optional date:** 2027-07-26 · **Domain resources:** CL03 / B15 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 273 of 365; calendar-free M10W3D7; optional date 2027-07-26.
2. Exact topic	Weekly cloud sprint review and score
3. Exact subtopics	IAM, network/storage, Kubernetes, IaC/CI, incident response and resilience; independent evaluation
4. Learning objective	Compare cloud sprint results with detection sprint results using the same rubric
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar review questions and CL14 GitHub Actions hardening. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score all artifacts, repair one major gap and write a one-page comparison of the two sprints — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score all artifacts, repair one major gap and write a one-page comparison of the two sprints — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score all artifacts, repair one major gap and write a one-page comparison of the two sprints — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10W3-review.md and comparison.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which sprint produced stronger evidence of independent problem-solving? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Both sprints are scored with the same 100-point rubric Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar review questions and CL14 GitHub Actions hardening"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 10 · Week 4 · daily cards

D274 · M10W4D1 · Decision review with a technical peer or self-defence

Phase: Academic month · **Optional date:** 2027-07-27 · **Domain resources:** C20 / B15

1. Day / sequence	Global Day 274 of 365; calendar-free M10W4D1; optional date 2027-07-27.
2. Exact topic	Decision review with a technical peer or self-defence
3. Exact subtopics	Evidence, assumptions, opportunity cost, local job market constraints and 12-month limits
4. Learning objective	Make a defensible final specialization decision for Months 11-12
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Workforce Framework and official role-skill resources for selected roles. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Present your two sprint portfolios to a peer/ teacher or record a self-defence; answer five challenge questions and update the decision matrix. **2.** Standard (~2 h): Present your two sprint portfolios to a peer/teacher or record a self-defence; answer five challenge questions and update the decision matrix. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Present your two sprint portfolios to a peer/teacher or record a self-defence; answer five challenge questions and update the decision matrix. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	specialization-final-decision.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which evidence was strongest against your preferred choice? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Final decision is documented with primary, secondary and review date Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Workforce Framework and official role-skill resources for selected roles"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D275 · M10W4D2 · Specialization project requirements and architecture

Phase: Academic month · **Optional date:** 2027-07-28 · **Domain resources:** C17 / B03

1. Day / sequence	Global Day 275 of 365; calendar-free M10W4D2; optional date 2027-07-28.
2. Exact topic	Specialization project requirements and architecture
3. Exact subtopics	Problem statement, users, assets, threat model, data sources, architecture, metrics, constraints and ethics
4. Learning objective	Define a Month 11 capstone aligned with the chosen specialization
5. Primary lecture / reading + direct URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling` **Type:** Reference/process · **Access:** Free
6. Exact section / lesson locator	OWASP Threat Modeling and C06 NIST CSF 2.0 outcome categories. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a two-page capstone specification with testable objectives, security boundaries, synthetic data plan and success metrics. **2.** Standard (~2 h): Write a two-page capstone specification with testable objectives, security boundaries, synthetic data plan and success metrics. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a two-page capstone specification with testable objectives, security boundaries, synthetic data plan and success metrics. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	capstone-spec.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which requirement is both measurable and genuinely difficult? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Specification is approved by the self-review checklist and has no illegal targets Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Threat Modeling and C06 NIST CSF 2.0 outcome categories"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D276 · M10W4D3 · Specialization literature and documentation reading

Phase: Academic month · **Optional date:** 2027-07-29 · **Domain resources:** B03 / B17

1. Day / sequence	Global Day 276 of 365; calendar-free M10W4D3; optional date 2027-07-29.
2. Exact topic	Specialization literature and documentation reading
3. Exact subtopics	Primary docs, standards, papers, advisories, known bypasses and implementation details
4. Learning objective	Learn to read primary sources instead of only tutorials
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ documentation and B14 Elastic detection rules repository for primary detection content. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CISA cybersecurity advisories `https://www.cisa.gov/news-events/cybersecurity-advisories`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Read two primary technical sources, write an annotated bibliography and extract three design decisions for your capstone. **2.** Standard (~2 h): Read two primary technical sources, write an annotated bibliography and extract three design decisions for your capstone. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Read two primary technical sources, write an annotated bibliography and extract three design decisions for your capstone. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	annotated-bibliography.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which source is authoritative and which is only commentary? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Notes distinguish primary evidence from opinion and cite exact sections Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ documentation and B14 Elastic detection rules repository for primary detection content"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

D277 · M10W4D4 · Independent problem-solving baseline with no walkthrough

Phase: Academic month · **Optional date:** 2027-07-30 · **Domain resources:** B04 / B02

1. Day / sequence	Global Day 277 of 365; calendar-free M10W4D4; optional date 2027-07-30.
2. Exact topic	Independent problem-solving baseline with no walkthrough
3. Exact subtopics	Unfamiliar rule/schema/configuration bug, hypothesis testing, documentation search and debugging
4. Learning objective	Measure ability to work without step-by-step guidance
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started and CL09 Kubernetes security documentation, chosen according to specialization. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Solve one unfamiliar technical task in 90 minutes using only official docs and your error log; record every failed hypothesis. **2.** Standard (~2 h): Solve one unfamiliar technical task in 90 minutes using only official docs and your error log; record every failed hypothesis. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Solve one unfamiliar technical task in 90 minutes using only official docs and your error log; record every failed hypothesis. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	independent-baseline.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which documentation section resolved the ambiguity? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Task is completed or failure is analyzed with a concrete next experiment Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started and CL09 Kubernetes security documentation, chosen according to specialization"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D278 · M10W4D5 · Month 10 practical examination and gate

Phase: Academic month · **Optional date:** 2027-07-31 · **Domain resources:** B04 / C20

1. Day / sequence	Global Day 278 of 365; calendar-free M10W4D5; optional date 2027-07-31.
2. Exact topic	Month 10 practical examination and gate
3. Exact subtopics	Specialization decision, detection engineering, cloud security, independent research, documentation and portfolio quality
4. Learning objective	Demonstrate intermediate specialization competence and decision maturity
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started and CL03 Well-Architected Security Pillar as primary references. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 150-minute practical in the chosen primary: solve an unfamiliar technical problem, write documentation and defend the specialization decision. **2.** Standard (~2 h): Complete a 150-minute practical in the chosen primary: solve an unfamiliar technical problem, write documentation and defend the specialization decision. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 150-minute practical in the chosen primary: solve an unfamiliar technical problem, write documentation and defend the specialization decision. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10-assessment.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would change your decision in the next six months? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and independent problem-solving at least 11/15 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started and CL03 Well-Architected Security Pillar as primary references"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D279 · M10W4D6 · Monthly review and gap repairs

Phase: Academic month · **Optional date:** 2027-08-01 · **Domain resources:** B03 / C17

1. Day / sequence	Global Day 279 of 365; calendar-free M10W4D6; optional date 2027-08-01.
2. Exact topic	Monthly review and gap repairs
3. Exact subtopics	Decision bias, telemetry gaps, query/detection quality, cloud IAM, Kubernetes controls and reading notes
4. Learning objective	Close Month 10 gaps and finalize the Month 11 capstone plan
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ and B14 detection rule docs; CL09 Kubernetes security docs for secondary gaps. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework failed assessment items, update capstone spec, add ten flashcards and define a weekly milestone schedule for Month 11. **2.** Standard (~2 h): Rework failed assessment items, update capstone spec, add ten flashcards and define a weekly milestone schedule for Month 11. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework failed assessment items, update capstone spec, add ten flashcards and define a weekly milestone schedule for Month 11. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10-review.md and capstone-milestones.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which prerequisite is still too weak for the capstone and how will you fix it first? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and capstone plan has clear weekly deliverables Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ and B14 detection rule docs; CL09 Kubernetes security docs for secondary gaps"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

D280 · M10W4D7 · Spaced revision, recovery and Month 11 preview

Phase: Academic month · **Optional date:** 2027-08-02 · **Domain resources:** A08 / B03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 280 of 365; calendar-free M10W4D7; optional date 2027-08-02.
2. Exact topic	Spaced revision, recovery and Month 11 preview
3. Exact subtopics	Detection lifecycle, cloud controls, research reading and decision rationale; active recall and rest
4. Learning objective	Preserve specialization gains before a month of independent engineering
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering selected chapter for long-term perspective, read only one section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, clean repositories, create the Month 11 project board, and rest or take a light walkthrough of the capstone architecture — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, clean repositories, create the Month 11 project board, and rest or take a light walkthrough of the capstone architecture — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, clean repositories, create the Month 11 project board, and rest or take a light walkthrough of the capstone architecture — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10-consolidation.md and project-board.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: What is the single riskiest assumption in your capstone plan? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Board is ready, snapshots are clean, and the risky assumption has a validation experiment Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering selected chapter for long-term perspective, read only one section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Month 11 — Capstone engineering, independent investigation and portfolio

Days D281-D308. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Capstone kickoff and problem definition D2: Capstone architecture and data-flow design D3: Threat model and abuse-case tests D4: Development environment and reproducibility D5: Build sprint 1: core application and data model D6: Build sprint 1: telemetry and evidence generation D7: Weekly capstone review 1 and design defence
Week 2	D1: Build sprint 2: automation and detection rules D2: Build sprint 2: containerization and CI/CD D3: Build sprint 2: cloud/IaC security alternative D4: Build sprint 2: failure injection and resilience D5: Build sprint 2: independent investigation exercise D6: Build sprint 2: remediation and regression testing D7: Weekly capstone review 2 and score
Week 3	D1: Independent technical challenge 1: unfamiliar log format D2: Independent technical challenge 2: unknown detection gap D3: Independent technical challenge 4: ethical decision scenario D4: Capstone security review and red-team self-test D5: Capstone remediation sprint D6: Weekly independent work review D7: Capstone technical report: architecture and methods
Week 4	D1: Capstone technical report: results and evaluation D2: Capstone report: discussion, ethics and future work D3: Capstone demo and oral defence script D4: Final portfolio integration and README polish D5: Month 11 practical examination and mastery gate D6: Monthly review and gap repairs D7: Spaced revision, recovery and Month 12 preview

> **Gate:** score the Month 11 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 11 · Week 1 · daily cards

D281 · M11W1D1 · Capstone kickoff and problem definition

Phase: Academic month · **Optional date:** 2027-08-03 · **Domain resources:** C17 / F09

1. Day / sequence	Global Day 281 of 365; calendar-free M11W1D1; optional date 2027-08-03.
2. Exact topic	Capstone kickoff and problem definition
3. Exact subtopics	Stakeholder, assets, problem, scope, constraints, success metrics, non-goals and security boundaries
4. Learning objective	Convert the Month 10 specification into a buildable engineering plan
5. Primary lecture / reading + direct URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling` **Type:** Reference/process · **Access:** Free
6. Exact section / lesson locator	OWASP Threat Modeling community page and C06 NIST CSF 2.0 outcome categories. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a project charter, milestone board, risk register and evidence plan; define what the capstone will not do. **2.** Standard (~2 h): Create a project charter, milestone board, risk register and evidence plan; define what the capstone will not do. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a project charter, milestone board, risk register and evidence plan; define what the capstone will not do. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/charter.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which non-goal prevents scope creep without making the project trivial? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Charter has measurable outcomes and explicit ethical boundaries Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Threat Modeling community page and C06 NIST CSF 2.0 outcome categories"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D282 · M11W1D2 · Capstone architecture and data-flow design

Phase: Academic month · **Optional date:** 2027-08-04 · **Domain resources:** C15 / CL03

1. Day / sequence	Global Day 282 of 365; calendar-free M11W1D2; optional date 2027-08-04.
2. Exact topic	Capstone architecture and data-flow design
3. Exact subtopics	Components, trust boundaries, identities, data stores, APIs, logs, CI/CD, containers and recovery
4. Learning objective	Produce a security architecture that can be implemented on student hardware
5. Primary lecture / reading + direct URL	OWASP Application Security Verification Standard `https://owasp.org/www-project-application-security-verification-standard/` **Type:** Standard/reference · **Access:** Free
6. Exact section / lesson locator	OWASP ASVS Level 1 selected architecture and logging requirements. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a three-tier architecture with data flows, trust boundaries, failure domains and a resource budget under 8 GB RAM where possible. **2.** Standard (~2 h): Draw a three-tier architecture with data flows, trust boundaries, failure domains and a resource budget under 8 GB RAM where possible. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a three-tier architecture with data flows, trust boundaries, failure domains and a resource budget under 8 GB RAM where possible. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/architecture.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which component has the highest blast radius if compromised? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Architecture is feasible, labelled and has a specific isolation plan Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP ASVS Level 1 selected architecture and logging requirements"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Secure Coding Practices Quick Reference Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D283 · M11W1D3 · Threat model and abuse-case tests

Phase: Academic month · **Optional date:** 2027-08-05 · **Domain resources:** C17 / C26

1. Day / sequence	Global Day 283 of 365; calendar-free M11W1D3; optional date 2027-08-05.
2. Exact topic	Threat model and abuse-case tests
3. Exact subtopics	STRIDE, attack trees, misuse cases, telemetry mapping, risk ranking and mitigations
4. Learning objective	Derive testable security requirements from threats rather than a generic checklist
5. Primary lecture / reading + direct URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling` **Type:** Reference/process · **Access:** Free
6. Exact section / lesson locator	OWASP Threat Modeling and C18 MITRE ATT&CK technique pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-30 Revision 1 `https://csrc.nist.gov/pubs/sp/800/30/r1/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a threat model with at least twelve threats; map each to a mitigation, test and detection data source. **2.** Standard (~2 h): Build a threat model with at least twelve threats; map each to a mitigation, test and detection data source. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a threat model with at least twelve threats; map each to a mitigation, test and detection data source. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/threat-model.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which threat has no current telemetry and how will you add it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every high/critical threat has a mitigation and validation test Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Threat Modeling and C18 MITRE ATT&CK technique pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D284 · M11W1D4 · Development environment and reproducibility

Phase: Academic month · **Optional date:** 2027-08-06 · **Domain resources:** F09 / CL14

1. Day / sequence	Global Day 284 of 365; calendar-free M11W1D4; optional date 2027-08-06.
2. Exact topic	Development environment and reproducibility
3. Exact subtopics	Repository structure, containers, dependencies, virtual environments, configuration, secrets placeholders, Makefile/scripts and clean setup
4. Learning objective	Make the capstone reproducible on a clean machine with no cloud spending
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 1-3 and CL06 Docker Get Started documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub Actions security hardening `https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a repository skeleton, setup script, .gitignore, sample configuration and a clean-install test from scratch. **2.** Standard (~2 h): Create a repository skeleton, setup script, .gitignore, sample configuration and a clean-install test from scratch. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a repository skeleton, setup script, .gitignore, sample configuration and a clean-install test from scratch. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15 repository skeleton; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which setup step still depends on undocumented manual action? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	A second user or a fresh VM can reproduce the environment from README Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 1-3 and CL06 Docker Get Started documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D285 · M11W1D5 · Build sprint 1: core application and data model

Phase: Academic month · **Optional date:** 2027-08-07 · **Domain resources:** W17 / P08

1. Day / sequence	Global Day 285 of 365; calendar-free M11W1D5; optional date 2027-08-07.
2. Exact topic	Build sprint 1: core application and data model
3. Exact subtopics	Minimal API/service, authentication, authorization, object ownership, validation, database schema and errors
4. Learning objective	Implement the smallest trustworthy core rather than an oversized feature set
5. Primary lecture / reading + direct URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/` **Type:** Official checklist library · **Access:** Free
6. Exact section / lesson locator	OWASP Cheat Sheet Series for Authentication, Session Management and Input Validation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Implement core endpoints, server-side authorization, parameterized queries, password hashing/KDF and structured error handling. **2.** Standard (~2 h): Implement core endpoints, server-side authorization, parameterized queries, password hashing/KDF and structured error handling. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Implement core endpoints, server-side authorization, parameterized queries, password hashing/KDF and structured error handling. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/src/ and migration files; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which input path is not yet validated and how could that fail? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Core functional tests pass and all assigned negative tests fail closed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Cheat Sheet Series for Authentication, Session Management and Input Validation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Application Security Verification Standard; write a five-line correction; retest next morning with a new 10-minute problem.

D286 · M11W1D6 · Build sprint 1: telemetry and evidence generation

Phase: Academic month · **Optional date:** 2027-08-08 · **Domain resources:** B02 / W17

1. Day / sequence	Global Day 286 of 365; calendar-free M11W1D6; optional date 2027-08-08.
2. Exact topic	Build sprint 1: telemetry and evidence generation
3. Exact subtopics	Structured logs, correlation IDs, audit events, time sync, redaction, synthetic malicious/benign events and data retention
4. Learning objective	Ensure the capstone produces evidence useful for investigation
5. Primary lecture / reading + direct URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric` **Type:** Official language docs · **Access:** Free
6. Exact section / lesson locator	KQL overview for schema design and B01 Sentinel data collection concepts. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Add structured logging for authentication, authorization failures, privileged actions and errors; generate a labelled synthetic event dataset. **2.** Standard (~2 h): Add structured logging for authentication, authorization failures, privileged actions and errors; generate a labelled synthetic event dataset. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Add structured logging for authentication, authorization failures, privileged actions and errors; generate a labelled synthetic event dataset. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/data/ and logging schema; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which privacy-sensitive field must never appear in logs? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Logs are structured, redacted and include reproducible benign/malicious examples Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "KQL overview for schema design and B01 Sentinel data collection concepts"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D287 · M11W1D7 · Weekly capstone review 1 and design defence

Phase: Academic month · **Optional date:** 2027-08-09 · **Domain resources:** C15 / F09 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 287 of 365; calendar-free M11W1D7; optional date 2027-08-09.
2. Exact topic	Weekly capstone review 1 and design defence
3. Exact subtopics	Charter, architecture, threat model, reproducibility, core implementation and telemetry; peer-style review
4. Learning objective	Catch design errors before deeper implementation
5. Primary lecture / reading + direct URL	OWASP Application Security Verification Standard `https://owasp.org/www-project-application-security-verification-standard/` **Type:** Standard/reference · **Access:** Free
6. Exact section / lesson locator	OWASP ASVS Level 1 and C17 threat model review. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Conduct a 45-minute self-design review, create ten issues, fix the highest three, and record a short architecture walkthrough — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Conduct a 45-minute self-design review, create ten issues, fix the highest three, and record a short architecture walkthrough — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Conduct a 45-minute self-design review, create ten issues, fix the highest three, and record a short architecture walkthrough — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M11W1-review.md and issue list; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which design change improved security without adding complexity? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All high-priority issues have owners and tests Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP ASVS Level 1 and C17 threat model review"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Secure Coding Practices Quick Reference Guide; write a five-line correction; retest next morning with a new 10-minute problem.

Month 11 · Week 2 · daily cards

D288 · M11W2D1 · Build sprint 2: automation and detection rules

Phase: Academic month · **Optional date:** 2027-08-10 · **Domain resources:** B04 / B02

1. Day / sequence	Global Day 288 of 365; calendar-free M11W2D1; optional date 2027-08-10.
2. Exact topic	Build sprint 2: automation and detection rules
3. Exact subtopics	Log normalization, Sigma/KQL-style rules, true/false-positive tests, alert metadata and measurement
4. Learning objective	Integrate the primary specialization capability into the capstone
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started guide and B14 Elastic detection rules documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Implement three detections with labelled test data, CI validation and a short design note for each. **2.** Standard (~2 h): Implement three detections with labelled test data, CI validation and a short design note for each. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Implement three detections with labelled test data, CI validation and a short design note for each. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/detections/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which detection has the highest false-positive risk and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Rules validate, tests pass and all false-positive cases are labelled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started guide and B14 Elastic detection rules documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D289 · M11W2D2 · Build sprint 2: containerization and CI/CD

Phase: Academic month · **Optional date:** 2027-08-11 · **Domain resources:** CL14 / W13

1. Day / sequence	Global Day 289 of 365; calendar-free M11W2D2; optional date 2027-08-11.
2. Exact topic	Build sprint 2: containerization and CI/CD
3. Exact subtopics	Dockerfile, non-root user, image scanning, GitHub Actions, least permissions, pinned actions, tests and SBOM
4. Learning objective	Build a reproducible, secure local delivery pipeline
5. Primary lecture / reading + direct URL	GitHub Actions security hardening `https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	GitHub Actions security hardening and CL12 Trivy documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Containerize the app, harden the Dockerfile, scan the image, add CI tests and generate an SBOM-like artifact inventory. **2.** Standard (~2 h): Containerize the app, harden the Dockerfile, scan the image, add CI tests and generate an SBOM-like artifact inventory. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Containerize the app, harden the Dockerfile, scan the image, add CI tests and generate an SBOM-like artifact inventory. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/Dockerfile and workflow evidence; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which CI secret could leak through a fork pull request? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Pipeline runs locally or in CI and no secret appears in logs Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub Actions security hardening and CL12 Trivy documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup GitHub code security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D290 · M11W2D3 · Build sprint 2: cloud/IaC security alternative

Phase: Academic month · **Optional date:** 2027-08-12 · **Domain resources:** CL15 / CL09

1. Day / sequence	Global Day 290 of 365; calendar-free M11W2D3; optional date 2027-08-12.
2. Exact topic	Build sprint 2: cloud/IaC security alternative
3. Exact subtopics	Synthetic Terraform/Kubernetes manifests, IAM/network policies, Checkov checks and no live deployment
4. Learning objective	Integrate infrastructure security without creating cloud costs
5. Primary lecture / reading + direct URL	Terraform documentation `https://developer.hashicorp.com/terraform/docs` **Type:** Official documentation · **Access:** Free/open-source; HCP Terraform has paid tiers
6. Exact section / lesson locator	Terraform documentation and CL13 Checkov documentation; CL03 Security Pillar for controls. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Add a local IaC representation of the capstone, scan it, fix five findings and document the shared-responsibility boundary. **2.** Standard (~2 h): Add a local IaC representation of the capstone, scan it, fix five findings and document the shared-responsibility boundary. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Add a local IaC representation of the capstone, scan it, fix five findings and document the shared-responsibility boundary. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/iac/ and scan report; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control would still be the customer's responsibility in a managed service? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No billable resource is deployed and scan findings are triaged Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Terraform documentation and CL13 Checkov documentation; CL03 Security Pillar for controls"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Checkov documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D291 · M11W2D4 · Build sprint 2: failure injection and resilience

Phase: Academic month · **Optional date:** 2027-08-13 · **Domain resources:** C27 / P05

1. Day / sequence	Global Day 291 of 365; calendar-free M11W2D4; optional date 2027-08-13.
2. Exact topic	Build sprint 2: failure injection and resilience
3. Exact subtopics	Service failure, malformed log, database lock, missing permission, expired credential, network partition and restore
4. Learning objective	Prove the capstone fails safely and can be recovered
5. Primary lecture / reading + direct URL	NIST SP 800-34 Revision 1 `https://csrc.nist.gov/pubs/sp/800/34/r1/final` **Type:** Official continuity guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-34r1 backup, recovery and testing sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create six failure-injection tests, measure detection and recovery, fix two failures and document residual risk. **2.** Standard (~2 h): Create six failure-injection tests, measure detection and recovery, fix two failures and document residual risk. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create six failure-injection tests, measure detection and recovery, fix two failures and document residual risk. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/tests/failure-injection.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which failure was silent and how did you add an observable signal? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All tests are reproducible and at least two fixes are verified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-34r1 backup, recovery and testing sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D292 · M11W2D5 · Build sprint 2: independent investigation exercise

Phase: Academic month · **Optional date:** 2027-08-14 · **Domain resources:** B15 / B02

1. Day / sequence	Global Day 292 of 365; calendar-free M11W2D5; optional date 2027-08-14.
2. Exact topic	Build sprint 2: independent investigation exercise
3. Exact subtopics	Unknown alert, incomplete logs, competing hypotheses, pivoting, confidence and evidence gaps
4. Learning objective	Investigate your own system without using a known answer
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide, hunt/investigation sections; C07 NIST SP 800-61r3. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Ask another person or use a script to inject one unknown benign/malicious event; investigate from logs only and write a case file. **2.** Standard (~2 h): Ask another person or use a script to inject one unknown benign/malicious event; investigate from logs only and write a case file. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Ask another person or use a script to inject one unknown benign/malicious event; investigate from logs only and write a case file. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/investigation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which hypothesis was rejected by a specific log field? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Case file includes evidence, timeline, uncertainty and a conclusion Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide, hunt/investigation sections; C07 NIST SP 800-61r3"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D293 · M11W2D6 · Build sprint 2: remediation and regression testing

Phase: Academic month · **Optional date:** 2027-08-15 · **Domain resources:** W17 / B03

1. Day / sequence	Global Day 293 of 365; calendar-free M11W2D6; optional date 2027-08-15.
2. Exact topic	Build sprint 2: remediation and regression testing
3. Exact subtopics	Root-cause fixes, rule tuning, new tests, rollback, functional regression and documentation
4. Learning objective	Close the investigation loop with engineering changes rather than a report alone
5. Primary lecture / reading + direct URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/` **Type:** Official checklist library · **Access:** Free
6. Exact section / lesson locator	OWASP Cheat Sheet Series relevant to the finding and P05 pytest docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Implement fixes for the highest finding, add regression and detection tests, rerun full suite and record before/after metrics. **2.** Standard (~2 h): Implement fixes for the highest finding, add regression and detection tests, rerun full suite and record before/after metrics. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Implement fixes for the highest finding, add regression and detection tests, rerun full suite and record before/after metrics. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/remediation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which test proves the fix did not only block one payload? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All tests pass and the report distinguishes root cause from symptom Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Cheat Sheet Series relevant to the finding and P05 pytest docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Application Security Verification Standard; write a five-line correction; retest next morning with a new 10-minute problem.

D294 · M11W2D7 · Weekly capstone review 2 and score

Phase: Academic month · **Optional date:** 2027-08-16 · **Domain resources:** B03 / CL14 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 294 of 365; calendar-free M11W2D7; optional date 2027-08-16.
2. Exact topic	Weekly capstone review 2 and score
3. Exact subtopics	Automation, detection, containers, CI/CD, IaC, resilience, investigation and remediation; active recall
4. Learning objective	Evaluate progress against the capstone rubric and adjust Month 11 Week 3 scope
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ and B14 detection rule documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub Actions security hardening `https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score the current capstone with the 100-point rubric, identify three largest gaps and replan the remaining work with explicit cut lines — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score the current capstone with the 100-point rubric, identify three largest gaps and replan the remaining work with explicit cut lines — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score the current capstone with the 100-point rubric, identify three largest gaps and replan the remaining work with explicit cut lines — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M11W2-review.md and updated board; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which feature should be cut to protect quality? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Scope is reduced where necessary and evidence gaps have owners Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ and B14 detection rule documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

Month 11 · Week 3 · daily cards

D295 · M11W3D1 · Independent technical challenge 1: unfamiliar log format

Phase: Academic month · **Optional date:** 2027-08-17 · **Domain resources:** F08 / P03

1. Day / sequence	Global Day 295 of 365; calendar-free M11W3D1; optional date 2027-08-17.
2. Exact topic	Independent technical challenge 1: unfamiliar log format
3. Exact subtopics	Schema discovery, parsing, normalization, edge cases and validation without a walkthrough
4. Learning objective	Demonstrate independent problem-solving on an unknown but safe dataset
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial files/exceptions sections and P04 RegexOne review lessons. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Automate the Boring Stuff with Python `https://automatetheboringstuff.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Receive or generate an unseen synthetic log format; write a parser and tests in 90 minutes using only docs and your notes. **2.** Standard (~2 h): Receive or generate an unseen synthetic log format; write a parser and tests in 90 minutes using only docs and your notes. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Receive or generate an unseen synthetic log format; write a parser and tests in 90 minutes using only docs and your notes. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	challenge1/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which assumption about time or encoding was wrong and how did you find it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Parser works on all test cases without copied code Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial files/exceptions sections and P04 RegexOne review lessons"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D296 · M11W3D2 · Independent technical challenge 2: unknown detection gap

Phase: Academic month · **Optional date:** 2027-08-18 · **Domain resources:** C18 / B04

1. Day / sequence	Global Day 296 of 365; calendar-free M11W3D2; optional date 2027-08-18.
2. Exact topic	Independent technical challenge 2: unknown detection gap
3. Exact subtopics	Behaviour hypothesis, telemetry requirements, rule, test data, false positives and measurement
4. Learning objective	Design a detection for a behaviour not covered by your existing rules
5. Primary lecture / reading + direct URL	MITRE ATT&CK `https://attack.mitre.org/` **Type:** Knowledge base · **Access:** Free
6. Exact section / lesson locator	MITRE ATT&CK technique page chosen from your threat model. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Pick one uncovered technique, implement a detection, create ten labelled events and measure results; explain one bypass. **2.** Standard (~2 h): Pick one uncovered technique, implement a detection, create ten labelled events and measure results; explain one bypass. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Pick one uncovered technique, implement a detection, create ten labelled events and measure results; explain one bypass. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	challenge2/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which missing field would make the detection impossible? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Rule is original, tested and has explicit limitations Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MITRE ATT&CK technique page chosen from your threat model"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE D3FEND; write a five-line correction; retest next morning with a new 10-minute problem.

D297 · M11W3D3 · Independent technical challenge 4: ethical decision scenario

Phase: Academic month · **Optional date:** 2027-08-19 · **Domain resources:** E01 / C24

1. Day / sequence	Global Day 297 of 365; calendar-free M11W3D3; optional date 2027-08-19.
2. Exact topic	Independent technical challenge 4: ethical decision scenario
3. Exact subtopics	Scope ambiguity, privacy, legal constraints, business impact, safety and communication
4. Learning objective	Make and defend a decision when technical ability conflicts with authorization
5. Primary lecture / reading + direct URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final` **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115 planning and ethical considerations; E05 HackerOne Disclosure Guidelines. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GDPR full text (EUR-Lex) `https://eur-lex.europa.eu/eli/reg/2016/679/oj`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Analyze five grey-area scenarios, choose stop/continue/escalate for each and write the exact message to the stakeholder. **2.** Standard (~2 h): Analyze five grey-area scenarios, choose stop/continue/escalate for each and write the exact message to the stakeholder. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Analyze five grey-area scenarios, choose stop/continue/escalate for each and write the exact message to the stakeholder. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ethics-scenarios.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which action is technically easy but professionally unacceptable? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All decisions prioritize authorization, privacy and availability Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115 planning and ethical considerations; E05 HackerOne Disclosure Guidelines"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D298 · M11W3D4 · Capstone security review and red-team self-test

Phase: Academic month · **Optional date:** 2027-08-20 · **Domain resources:** W06 / C17

1. Day / sequence	Global Day 298 of 365; calendar-free M11W3D4; optional date 2027-08-20.
2. Exact topic	Capstone security review and red-team self-test
3. Exact subtopics	Authorized local assessment, abuse cases, finding validation, evidence, impact and false positives
4. Learning objective	Test your own capstone as an ethical assessor
5. Primary lecture / reading + direct URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/` **Type:** Testing methodology · **Access:** Free
6. Exact section / lesson locator	OWASP Web Security Testing Guide and E01 NIST SP 800-115 validation sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run a scoped assessment against the local capstone, record five findings or rejected hypotheses, and preserve evidence. **2.** Standard (~2 h): Run a scoped assessment against the local capstone, record five findings or rejected hypotheses, and preserve evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run a scoped assessment against the local capstone, record five findings or rejected hypotheses, and preserve evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/assessment/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding surprised you and what design assumption caused it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All tests remain local/authorized and findings are reproducible Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Web Security Testing Guide and E01 NIST SP 800-115 validation sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D299 · M11W3D5 · Capstone remediation sprint

Phase: Academic month · **Optional date:** 2027-08-21 · **Domain resources:** W17 / P05

1. Day / sequence	Global Day 299 of 365; calendar-free M11W3D5; optional date 2027-08-21.
2. Exact topic	Capstone remediation sprint
3. Exact subtopics	Root-cause fixes, compensating controls, tests, rollback, residual risk and retest
4. Learning objective	Close high-priority findings before final reporting
5. Primary lecture / reading + direct URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/` **Type:** Official checklist library · **Access:** Free
6. Exact section / lesson locator	OWASP Cheat Sheet Series relevant to findings. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Fix the three highest findings, write regression tests, rerun the assessment and document any accepted residual risk. **2.** Standard (~2 h): Fix the three highest findings, write regression tests, rerun the assessment and document any accepted residual risk. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Fix the three highest findings, write regression tests, rerun the assessment and document any accepted residual risk. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/remediation-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What evidence proves the root cause was fixed? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All high-priority findings are retested and accepted risks have owner and review date Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Cheat Sheet Series relevant to findings"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Application Security Verification Standard; write a five-line correction; retest next morning with a new 10-minute problem.

D300 · M11W3D6 · Weekly independent work review

Phase: Academic month · **Optional date:** 2027-08-22 · **Domain resources:** C20 / B03

1. Day / sequence	Global Day 300 of 365; calendar-free M11W3D6; optional date 2027-08-22.
2. Exact topic	Weekly independent work review
3. Exact subtopics	Four challenges, security review and remediation; peer-style scoring
4. Learning objective	Demonstrate Month 11 Week 3 competence and identify remaining report gaps
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role tasks for the chosen specialization. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Score all four challenges with the 100-point rubric, choose the strongest as portfolio evidence and list three final improvements. **2.** Standard (~2 h): Score all four challenges with the 100-point rubric, choose the strongest as portfolio evidence and list three final improvements. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Score all four challenges with the 100-point rubric, choose the strongest as portfolio evidence and list three final improvements. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M11W3-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which challenge best demonstrates independent capability? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Scores are evidence-based and the chosen artifact is linked in the portfolio plan Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role tasks for the chosen specialization"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D301 · M11W3D7 · Capstone technical report: architecture and methods

Phase: Academic month · **Optional date:** 2027-08-23 · **Domain resources:** A08 / F09 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 301 of 365; calendar-free M11W3D7; optional date 2027-08-23.
2. Exact topic	Capstone technical report: architecture and methods
3. Exact subtopics	Problem, related work/primary sources, architecture, data, methods, assumptions, threat model and reproducibility
4. Learning objective	Write a technical report that another student could replicate
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering selected chapter on technical communication and A12 USENIX Security paper structure examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Draft the first six sections of the final capstone report with diagrams, exact setup instructions and citations to primary docs — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Draft the first six sections of the final capstone report with diagrams, exact setup instructions and citations to primary docs — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Draft the first six sections of the final capstone report with diagrams, exact setup instructions and citations to primary docs — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/report-draft.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which implementation decision needs a clear rationale? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Report is reproducible and distinguishes your work from cited work Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering selected chapter on technical communication and A12 USENIX Security paper structure examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Month 11 · Week 4 · daily cards

D302 · M11W4D1 · Capstone technical report: results and evaluation

Phase: Academic month · **Optional date:** 2027-08-24 · **Domain resources:** B14 / C26

1. Day / sequence	Global Day 302 of 365; calendar-free M11W4D1; optional date 2027-08-24.
2. Exact topic	Capstone technical report: results and evaluation
3. Exact subtopics	Functional results, security tests, detection metrics, failure injection, performance, limitations and negative results
4. Learning objective	Present evidence honestly, including failures
5. Primary lecture / reading + direct URL	Elastic detection rules documentation `https://www.elastic.co/guide/en/security/current/detection-engine-overview.html` **Type:** Official docs / open-source rules · **Access:** Free docs/open-source rules; Elastic Stack may have paid features
6. Exact section / lesson locator	Elastic detection rules docs for measurement concepts and P05 pytest reporting sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-30 Revision 1 `https://csrc.nist.gov/pubs/sp/800/30/r1/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build tables/charts from real capstone test outputs; report false positives, missed cases, resource use and limitations honestly. **2.** Standard (~2 h): Build tables/charts from real capstone test outputs; report false positives, missed cases, resource use and limitations honestly. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build tables/charts from real capstone test outputs; report false positives, missed cases, resource use and limitations honestly. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/results.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which result did not match your hypothesis and what did you learn? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All numbers come from reproducible tests and no success is exaggerated Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Elastic detection rules docs for measurement concepts and P05 pytest reporting sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D303 · M11W4D2 · Capstone report: discussion, ethics and future work

Phase: Academic month · **Optional date:** 2027-08-25 · **Domain resources:** C24 / C07

1. Day / sequence	Global Day 303 of 365; calendar-free M11W4D2; optional date 2027-08-25.
2. Exact topic	Capstone report: discussion, ethics and future work
3. Exact subtopics	Impact, generalizability, ethical boundaries, privacy, operational constraints, cost and future experiments
4. Learning objective	Defend the capstone's value without claiming more than it proves
5. Primary lecture / reading + direct URL	GDPR full text (EUR-Lex) `https://eur-lex.europa.eu/eli/reg/2016/679/oj` **Type:** Regulation/legal text · **Access:** Free
6. Exact section / lesson locator	GDPR privacy-by-design principles and E05 disclosure guidance. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write discussion and future-work sections, explicitly state authorization, synthetic data use and limitations of a student lab. **2.** Standard (~2 h): Write discussion and future-work sections, explicitly state authorization, synthetic data use and limitations of a student lab. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write discussion and future-work sections, explicitly state authorization, synthetic data use and limitations of a student lab. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/discussion.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would a production environment reveal that your lab cannot? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Section is candid and avoids unrealistic claims Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GDPR privacy-by-design principles and E05 disclosure guidance"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Privacy Framework; write a five-line correction; retest next morning with a new 10-minute problem.

D304 · M11W4D3 · Capstone demo and oral defence script

Phase: Academic month · **Optional date:** 2027-08-26 · **Domain resources:** C17 / B03

1. Day / sequence	Global Day 304 of 365; calendar-free M11W4D3; optional date 2027-08-26.
2. Exact topic	Capstone demo and oral defence script
3. Exact subtopics	Ten-minute demo, architecture walkthrough, threat model, live test, failure, detection, remediation and Q&A
4. Learning objective	Prepare to explain and defend technical choices under questioning
5. Primary lecture / reading + direct URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling` **Type:** Reference/process · **Access:** Free
6. Exact section / lesson locator	Threat model and C15 ASVS requirements as review checklist. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Record or rehearse a ten-minute demo; prepare answers to ten probable reviewer questions with evidence links. **2.** Standard (~2 h): Record or rehearse a ten-minute demo; prepare answers to ten probable reviewer questions with evidence links. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Record or rehearse a ten-minute demo; prepare answers to ten probable reviewer questions with evidence links. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/demo-script.md and recording notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which question exposes the weakest evidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Demo works from a clean setup and all answers cite specific files/tests Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Threat model and C15 ASVS requirements as review checklist"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D305 · M11W4D4 · Final portfolio integration and README polish

Phase: Academic month · **Optional date:** 2027-08-27 · **Domain resources:** F09 / C20

1. Day / sequence	Global Day 305 of 365; calendar-free M11W4D4; optional date 2027-08-27.
2. Exact topic	Final portfolio integration and README polish
3. Exact subtopics	Repository structure, setup, usage, examples, screenshots, reports, limitations, licence, changelog and release tag
4. Learning objective	Make every Month 11 artifact discoverable and reproducible
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 2-3 and W13 GitHub security docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Update the portfolio index, link PRJ-01 through PRJ-15, sanitize all evidence and tag a release candidate. **2.** Standard (~2 h): Update the portfolio index, link PRJ-01 through PRJ-15, sanitize all evidence and tag a release candidate. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Update the portfolio index, link PRJ-01 through PRJ-15, sanitize all evidence and tag a release candidate. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	portfolio index and release tag; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which public file still contains personal data or a secret-like string? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Automated/manual secret scan passes and all links work Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 2-3 and W13 GitHub security docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D306 · M11W4D5 · Month 11 practical examination and mastery gate

Phase: Academic month · **Optional date:** 2027-08-28 · **Domain resources:** C17 / B03

1. Day / sequence	Global Day 306 of 365; calendar-free M11W4D5; optional date 2027-08-28.
2. Exact topic	Month 11 practical examination and mastery gate
3. Exact subtopics	Capstone design, implementation, security, detection, incident response, independent challenges and technical writing
4. Learning objective	Demonstrate advanced foundational competence and independent engineering
5. Primary lecture / reading + direct URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling` **Type:** Reference/process · **Access:** Free
6. Exact section / lesson locator	Threat model and C15 ASVS Level 1 review checklist. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 180-minute capstone defence: implement or fix one unseen requirement, investigate a new event, update the report and answer oral questions. **2.** Standard (~2 h): Complete a 180-minute capstone defence: implement or fix one unseen requirement, investigate a new event, update the report and answer oral questions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 180-minute capstone defence: implement or fix one unseen requirement, investigate a new event, update the report and answer oral questions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M11-assessment.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which part of your capstone is original rather than assembled from tutorials? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 75/100 and independent problem-solving at least 12/15 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Threat model and C15 ASVS Level 1 review checklist"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D307 · M11W4D6 · Monthly review and gap repairs

Phase: Academic month · **Optional date:** 2027-08-29 · **Domain resources:** P05 / F09

1. Day / sequence	Global Day 307 of 365; calendar-free M11W4D6; optional date 2027-08-29.
2. Exact topic	Monthly review and gap repairs
3. Exact subtopics	Capstone architecture, evidence, tests, reports, demo and portfolio; active recall and issue triage
4. Learning objective	Close Month 11 gaps before final Month 12 examinations
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest docs and B04 Sigma guide for weak tests/rules. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework failed assessment items, fix top five portfolio issues, update the capstone report and create a final revision checklist. **2.** Standard (~2 h): Rework failed assessment items, fix top five portfolio issues, update the capstone report and create a final revision checklist. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework failed assessment items, fix top five portfolio issues, update the capstone report and create a final revision checklist. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M11-review.md and final-issues.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which gap could cause failure in the Month 12 practical? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and all critical issues are closed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest docs and B04 Sigma guide for weak tests/rules"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D308 · M11W4D7 · Spaced revision, recovery and Month 12 preview

Phase: Academic month · **Optional date:** 2027-08-30 · **Domain resources:** B03 / C20 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 308 of 365; calendar-free M11W4D7; optional date 2027-08-30.
2. Exact topic	Spaced revision, recovery and Month 12 preview
3. Exact subtopics	Capstone decisions, detection metrics, cloud controls, secure coding and report claims; light retrieval
4. Learning objective	Preserve Month 11 gains and prepare final diagnostic exams
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ and CL03 Security Pillar review pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, take a full clean snapshot of the capstone, archive evidence, review Month 12 exam rules and rest — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, take a full clean snapshot of the capstone, archive evidence, review Month 12 exam rules and rest — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, take a full clean snapshot of the capstone, archive evidence, review Month 12 exam rules and rest — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M11-consolidation.md and clean snapshot log; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which topic from Months 1-9 has decayed most and needs first revision? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Capstone is backed up, no active lab is exposed, and revision order is set Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ and CL03 Security Pillar review pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

Month 12 — Practical examinations, revision, portfolio and long-term planning

Days D309-D336. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Final diagnostic: foundations and networking D2: Gap repair 1: computer systems and OS D3: Gap repair 2: networking and packet analysis D4: Gap repair 3: Linux, Windows and scripting D5: Weekly diagnostic review and revision plan D6: Gap repair 4: core security and GRC D7: Gap repair 5: web and API security
Week 2	D1: Gap repair 6: ethical assessment and blue team D2: Gap repair 7: cloud and infrastructure security D3: Mock practical examination 1: guided recall D4: Mock practical examination 2: independent conditions D5: Weekly review of mocks and targeted repair D6: Final practical examination part 1: systems and network D7: Final practical examination part 2: web and ethical assessment
Week 3	D1: Final practical examination part 3: blue team and cloud D2: Final capstone defence and oral examination D3: Weekly final examination review and grade calculation D4: Final portfolio audit and sanitization D5: CV, LinkedIn and technical story preparation D6: Technical interview practice and scenario answers D7: Certification readiness and final decision
Week 4	D1: Year 2-5 long-term plan and 90-day sprint D2: Monthly review, celebration and responsible handover D3: Master resource directory update and link re-verification D4: Practice-exam strategy and question analysis D5: Internship and entry-level application strategy D6: Sustainable study system and burnout prevention D7: Spaced revision and handover to final consolidation

> **Gate:** score the Month 12 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 12 · Week 1 · daily cards

D309 · M12W1D1 · Final diagnostic: foundations and networking

Phase: Academic month · **Optional date:** 2027-08-31 · **Domain resources:** F01 / N16

1. Day / sequence	Global Day 309 of 365; calendar-free M12W1D1; optional date 2027-08-31.
2. Exact topic	Final diagnostic: foundations and networking
3. Exact subtopics	Computer architecture, OS, CLI, HTTP, SQL, Git, OSI/TCPIP, subnetting, TCP/UDP, DNS, TLS and packet analysis
4. Learning objective	Measure current competence and identify the highest-risk knowledge gaps
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x 2026 Weeks 0-9 review pages; N01 Jeremy's IT Lab CCNA Course Index Days 1-38. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Subnetting Practice `https://subnettingpractice.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 90-minute closed-book diagnostic with 60 questions, 20 subnetting problems and one short packet analysis; classify errors by prerequisite. **2.** Standard (~2 h): Complete a 90-minute closed-book diagnostic with 60 questions, 20 subnetting problems and one short packet analysis; classify errors by prerequisite. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 90-minute closed-book diagnostic with 60 questions, 20 subnetting problems and one short packet analysis; classify errors by prerequisite. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-diagnostic-foundations.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which wrong answer reveals a conceptual error rather than a memory lapse? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All errors are classified and a revision queue is created Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x 2026 Weeks 0-9 review pages; N01 Jeremy's IT Lab CCNA Course Index Days 1-38"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D310 · M12W1D2 · Gap repair 1: computer systems and OS

Phase: Academic month · **Optional date:** 2027-09-01 · **Domain resources:** F06 / L01

1. Day / sequence	Global Day 310 of 365; calendar-free M12W1D2; optional date 2027-09-01.
2. Exact topic	Gap repair 1: computer systems and OS
3. Exact subtopics	Process/thread, memory, permissions, filesystems, services, virtualization and troubleshooting
4. Learning objective	Restore Month 1 competence to at least the passing threshold
5. Primary lecture / reading + direct URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/` **Type:** Free textbook · **Access:** Free online chapters; paid print optional
6. Exact section / lesson locator	OSTEP selected chapters on processes, memory and persistence. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Revisit failed diagnostic topics, perform three practical repairs on a clean VM and retest with new questions. **2.** Standard (~2 h): Revisit failed diagnostic topics, perform three practical repairs on a clean VM and retest with new questions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Revisit failed diagnostic topics, perform three practical repairs on a clean VM and retest with new questions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-OS.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which OS mechanism explains two seemingly unrelated errors? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent with practical evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OSTEP selected chapters on processes, memory and persistence"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x 2026 OpenCourseWare; write a five-line correction; retest next morning with a new 10-minute problem.

D311 · M12W1D3 · Gap repair 2: networking and packet analysis

Phase: Academic month · **Optional date:** 2027-09-02 · **Domain resources:** N01 / N07

1. Day / sequence	Global Day 311 of 365; calendar-free M12W1D3; optional date 2027-09-02.
2. Exact topic	Gap repair 2: networking and packet analysis
3. Exact subtopics	Encapsulation, ARP, VLANs, IPv4/IPv6, CIDR, routing, NAT, TCP, UDP, DNS, DHCP, TLS and Wireshark
4. Learning objective	Restore Month 2 competence and daily subnetting fluency
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Course Index selected Day titles from error log; N06 Wireshark User's Guide. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Wireshark SampleCaptures `https://wiki.wireshark.org/SampleCaptures`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rebuild one small Packet Tracer topology, analyze one unknown sample capture, and complete 30 subnetting questions. **2.** Standard (~2 h): Rebuild one small Packet Tracer topology, analyze one unknown sample capture, and complete 30 subnetting questions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rebuild one small Packet Tracer topology, analyze one unknown sample capture, and complete 30 subnetting questions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-networking.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which packet field disproves your first hypothesis? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and capture analysis is packet-backed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Course Index selected Day titles from error log; N06 Wireshark User's Guide"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D312 · M12W1D4 · Gap repair 3: Linux, Windows and scripting

Phase: Academic month · **Optional date:** 2027-09-03 · **Domain resources:** L01 / P05

1. Day / sequence	Global Day 312 of 365; calendar-free M12W1D4; optional date 2027-09-03.
2. Exact topic	Gap repair 3: Linux, Windows and scripting
3. Exact subtopics	Linux permissions/services, Windows events/PowerShell/AD, Python, Bash, PowerShell, Git and testing
4. Learning objective	Restore Month 3-4 competence with hands-on evidence
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey review sections; L08 PowerShell documentation; F08 Python Tutorial. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Fix one broken Python script, one Bash script and one PowerShell script; add tests and explain the root cause of each. **2.** Standard (~2 h): Fix one broken Python script, one Bash script and one PowerShell script; add tests and explain the root cause of each. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Fix one broken Python script, one Bash script and one PowerShell script; add tests and explain the root cause of each. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-scripting.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which bug was caused by platform-specific assumptions? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All three scripts pass regression tests and explanations are specific Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey review sections; L08 PowerShell documentation; F08 Python Tutorial"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D313 · M12W1D5 · Weekly diagnostic review and revision plan

Phase: Academic month · **Optional date:** 2027-09-04 · **Domain resources:** C20 / F09

1. Day / sequence	Global Day 313 of 365; calendar-free M12W1D5; optional date 2027-09-04.
2. Exact topic	Weekly diagnostic review and revision plan
3. Exact subtopics	Diagnostic results, gap repairs, error patterns, time budget and final exam strategy
4. Learning objective	Consolidate Month 12 Week 1 and schedule remaining revision
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role tasks for competence mapping. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a heat map of all 11 modules, rank the ten weakest topics and allocate the remaining 21 days with catch-up buffers. **2.** Standard (~2 h): Create a heat map of all 11 modules, rank the ten weakest topics and allocate the remaining 21 days with catch-up buffers. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a heat map of all 11 modules, rank the ten weakest topics and allocate the remaining 21 days with catch-up buffers. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-revision-plan.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which two weak topics could cause cascading failure in the final practical? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan is realistic for the chosen 1/2/3-4 hour track Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role tasks for competence mapping"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D314 · M12W1D6 · Gap repair 4: core security and GRC

Phase: Academic month · **Optional date:** 2027-09-05 · **Domain resources:** C06 / C03

1. Day / sequence	Global Day 314 of 365; calendar-free M12W1D6; optional date 2027-09-05.
2. Exact topic	Gap repair 4: core security and GRC
3. Exact subtopics	CIA/AAA, risk, crypto, IAM, TLS, controls, IR, privacy, NIST CSF, ISO/GDPR distinctions and BCDR
4. Learning objective	Restore Month 5 competence using scenarios rather than definitions
5. Primary lecture / reading + direct URL	NIST Cybersecurity Framework 2.0 https://www.nist.gov/cyberframework **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST CSF 2.0 and C26 NIST SP 800-30 risk assessment sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	ISC2 Certified in Cybersecurity Exam Outline https://www.isc2.org/certifications/cc/cc-certification-exam-outline
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Solve an integrated risk/IR scenario, select cryptographic primitives, classify controls and write a risk register. **2.** Standard (~2 h): Solve an integrated risk/IR scenario, select cryptographic primitives, classify controls and write a risk register. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Solve an integrated risk/IR scenario, select cryptographic primitives, classify controls and write a risk register. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-core-security.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control category is missing from your first answer? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and all framework classifications are correct Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST CSF 2.0 and C26 NIST SP 800-30 risk assessment sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Controls Version 8; write a five-line correction; retest next morning with a new 10-minute problem.

D315 · M12W1D7 · Gap repair 5: web and API security

Phase: Academic month · **Optional date:** 2027-09-06 · **Domain resources:** W01 / W06 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 315 of 365; calendar-free M12W1D7; optional date 2027-09-06.
2. Exact topic	Gap repair 5: web and API security
3. Exact subtopics	HTTP, sessions, access control, SQLi, XSS, CSRF, SSRF, file upload, API BOLA, rate limits, secrets, SDLC and supply chain
4. Learning objective	Restore Month 6 competence with defensive fixes
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy all-labs index; W17 OWASP Cheat Sheet Series. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Solve two unseen Apprentice labs and fix three vulnerable local code snippets with regression tests — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Solve two unseen Apprentice labs and fix three vulnerable local code snippets with regression tests — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Solve two unseen Apprentice labs and fix three vulnerable local code snippets with regression tests — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-web.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which fix addresses the root cause and which only blocks one payload? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and all fixes have tests Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy all-labs index; W17 OWASP Cheat Sheet Series"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

Month 12 · Week 2 · daily cards

D316 · M12W2D1 · Gap repair 6: ethical assessment and blue team

Phase: Academic month · **Optional date:** 2027-09-07 · **Domain resources:** E01 / B03

1. Day / sequence	Global Day 316 of 365; calendar-free M12W2D1; optional date 2027-09-07.
2. Exact topic	Gap repair 6: ethical assessment and blue team
3. Exact subtopics	Authorization, scope, discovery, validation, evidence, CVSS, reporting, SOC workflow, KQL, Sigma, ATT&CK, hunting and IR
4. Learning objective	Restore Months 7-8 competence through a mixed investigation
5. Primary lecture / reading + direct URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final` **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115; B04 Sigma getting-started; B02 KQL overview. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: In a synthetic case, write scope boundaries, validate two findings, investigate logs, write one detection rule and an incident summary. **2.** Standard (~2 h): In a synthetic case, write scope boundaries, validate two findings, investigate logs, write one detection rule and an incident summary. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): In a synthetic case, write scope boundaries, validate two findings, investigate logs, write one detection rule and an incident summary. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-offense-defense.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding is unsupported by evidence and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and no scope/safety error remains Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115; B04 Sigma getting-started; B02 KQL overview"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D317 · M12W2D2 · Gap repair 7: cloud and infrastructure security

Phase: Academic month · **Optional date:** 2027-09-08 · **Domain resources:** CL03 / CL09

1. Day / sequence	Global Day 317 of 365; calendar-free M12W2D2; optional date 2027-09-08.
2. Exact topic	Gap repair 7: cloud and infrastructure security
3. Exact subtopics	Shared responsibility, IAM, storage, logging, Docker, Kubernetes, CI/CD, IaC, SBOM and cloud incident response
4. Learning objective	Restore Month 9 competence with local-only evidence
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar; CL14 GitHub Actions hardening. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review an unseen cloud architecture, Dockerfile, Kubernetes manifest and workflow; fix ten prioritized issues without live deployment. **2.** Standard (~2 h): Review an unseen cloud architecture, Dockerfile, Kubernetes manifest and workflow; fix ten prioritized issues without live deployment. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review an unseen cloud architecture, Dockerfile, Kubernetes manifest and workflow; fix ten prioritized issues without live deployment. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-cloud.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding crosses the control-plane trust boundary? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and no billable resource is created Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar; CL14 GitHub Actions hardening"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D318 · M12W2D3 · Mock practical examination 1: guided recall

Phase: Academic month · **Optional date:** 2027-09-09 · **Domain resources:** C20 / B04

1. Day / sequence	Global Day 318 of 365; calendar-free M12W2D3; optional date 2027-09-09.
2. Exact topic	Mock practical examination 1: guided recall
3. Exact subtopics	All modules; time management, question interpretation, evidence and safety
4. Learning objective	Establish a baseline under exam conditions with feedback allowed afterward
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework tasks and C06 CSF 2.0 outcomes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 120-minute mixed practical with networking, scripting, web, detection and cloud tasks; self-score with the 100-point rubric. **2.** Standard (~2 h): Complete a 120-minute mixed practical with networking, scripting, web, detection and cloud tasks; self-score with the 100-point rubric. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 120-minute mixed practical with networking, scripting, web, detection and cloud tasks; self-score with the 100-point rubric. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	mock-exam-1.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which task consumed disproportionate time and what prerequisite caused it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score is recorded with evidence and every failure has a repair action Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework tasks and C06 CSF 2.0 outcomes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D319 · M12W2D4 · Mock practical examination 2: independent conditions

Phase: Academic month · **Optional date:** 2027-09-10 · **Domain resources:** B15 / W06

1. Day / sequence	Global Day 319 of 365; calendar-free M12W2D4; optional date 2027-09-10.
2. Exact topic	Mock practical examination 2: independent conditions
3. Exact subtopics	Unfamiliar incident, ambiguous requirements, incomplete data, independent research and technical writing
4. Learning objective	Simulate the final Month 12 practical without walkthroughs
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide and C07 NIST SP 800-61r3 for reference only during open-book section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 150-minute practical using only primary docs; produce a report, detection, remediation and a short executive summary. **2.** Standard (~2 h): Complete a 150-minute practical using only primary docs; produce a report, detection, remediation and a short executive summary. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 150-minute practical using only primary docs; produce a report, detection, remediation and a short executive summary. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	mock-exam-2.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which assumption was unsafe to make and how did you handle it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and safety at least 9/10 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide and C07 NIST SP 800-61r3 for reference only during open-book section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D320 · M12W2D5 · Weekly review of mocks and targeted repair

Phase: Academic month · **Optional date:** 2027-09-11 · **Domain resources:** P05 / C20

1. Day / sequence	Global Day 320 of 365; calendar-free M12W2D5; optional date 2027-09-11.
2. Exact topic	Weekly review of mocks and targeted repair
3. Exact subtopics	Mock exam errors, time management, documentation gaps, troubleshooting and stress control
4. Learning objective	Convert mock results into a final-week plan
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest documentation and B03 SigmaHQ for weak test/rule areas. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Repair the three lowest rubric categories with new exercises; rewrite one weak report and rehearse one troubleshooting scenario. **2.** Standard (~2 h): Repair the three lowest rubric categories with new exercises; rewrite one weak report and rehearse one troubleshooting scenario. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Repair the three lowest rubric categories with new exercises; rewrite one weak report and rehearse one troubleshooting scenario. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12W2-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which error is most likely to recur under pressure? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All three weak categories have new evidence and a prevention checklist Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest documentation and B03 SigmaHQ for weak test/rule areas"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D321 · M12W2D6 · Final practical examination part 1: systems and network

Phase: Academic month · **Optional date:** 2027-09-12 · **Domain resources:** L01 / F08

1. Day / sequence	Global Day 321 of 365; calendar-free M12W2D6; optional date 2027-09-12.
2. Exact topic	Final practical examination part 1: systems and network
3. Exact subtopics	Unknown VM fault, packet capture, subnetting, network troubleshooting, Linux/Windows logs and scripting
4. Learning objective	Demonstrate practical competence without tutorial support under timed conditions
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey and N06 Wireshark User's Guide as permitted references in open-book section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 120 minutes: fix a network/OS fault, analyze a capture, write a short script and document evidence. **2.** Standard (~2 h): Complete 120 minutes: fix a network/OS fault, analyze a capture, write a short script and document evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 120 minutes: fix a network/OS fault, analyze a capture, write a short script and document evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	final-practical-1/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which evidence proves the fault was fixed rather than masked? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Practical score at least 35/50 and no safety violation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey and N06 Wireshark User's Guide as permitted references in open-book section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D322 · M12W2D7 · Final practical examination part 2: web and ethical assessment

Phase: Academic month · **Optional date:** 2027-09-13 · **Domain resources:** W06 / W17 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 322 of 365; calendar-free M12W2D7; optional date 2027-09-13.
2. Exact topic	Final practical examination part 2: web and ethical assessment
3. Exact subtopics	Authorization, scope, local app assessment, vulnerability validation, secure fix, report and retest
4. Learning objective	Demonstrate authorized testing and secure engineering competence
5. Primary lecture / reading + direct URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/` **Type:** Testing methodology · **Access:** Free
6. Exact section / lesson locator	OWASP WSTG and E01 NIST SP 800-115 as permitted references. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete 120 minutes on a local unseen app: scope, validate, remediate, retest and write a concise report — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete 120 minutes on a local unseen app: scope, validate, remediate, retest and write a concise report — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete 120 minutes on a local unseen app: scope, validate, remediate, retest and write a concise report — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	final-practical-2/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which finding was rejected as a false positive and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Practical score at least 35/50 and scope boundaries are explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP WSTG and E01 NIST SP 800-115 as permitted references"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

Month 12 · Week 3 · daily cards

D323 · M12W3D1 · Final practical examination part 3: blue team and cloud

Phase: Academic month · **Optional date:** 2027-09-14 · **Domain resources:** B02 / CL03

1. Day / sequence	Global Day 323 of 365; calendar-free M12W3D1; optional date 2027-09-14.
2. Exact topic	Final practical examination part 3: blue team and cloud
3. Exact subtopics	Synthetic incident, KQL-style query, detection rule, ATT&CK mapping, cloud/IaC review and executive summary
4. Learning objective	Demonstrate detection, investigation and infrastructure-security competence
5. Primary lecture / reading + direct URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric` **Type:** Official language docs · **Access:** Free
6. Exact section / lesson locator	KQL overview and B04 Sigma getting-started as permitted references. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 120 minutes: investigate an incident, write/test one rule, review a cloud configuration and present a prioritized action plan. **2.** Standard (~2 h): Complete 120 minutes: investigate an incident, write/test one rule, review a cloud configuration and present a prioritized action plan. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 120 minutes: investigate an incident, write/test one rule, review a cloud configuration and present a prioritized action plan. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	final-practical-3/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which data source is missing and how does it affect confidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Practical score at least 35/50 and all conclusions state uncertainty Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "KQL overview and B04 Sigma getting-started as permitted references"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D324 · M12W3D2 · Final capstone defence and oral examination

Phase: Academic month · **Optional date:** 2027-09-15 · **Domain resources:** C15 / A08

1. Day / sequence	Global Day 324 of 365; calendar-free M12W3D2; optional date 2027-09-15.
2. Exact topic	Final capstone defence and oral examination
3. Exact subtopics	Architecture, threat model, original engineering, failures, validation, limitations and ethical boundaries
4. Learning objective	Defend the Month 11 capstone to a technical reviewer using evidence
5. Primary lecture / reading + direct URL	OWASP Application Security Verification Standard `https://owasp.org/www-project-application-security-verification-standard/` **Type:** Standard/reference · **Access:** Free
6. Exact section / lesson locator	OWASP ASVS Level 1 and C17 threat model as review checklist. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Deliver a 20-minute capstone defence and answer ten questions; provide links to tests, reports and clean setup evidence. **2.** Standard (~2 h): Deliver a 20-minute capstone defence and answer ten questions; provide links to tests, reports and clean setup evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Deliver a 20-minute capstone defence and answer ten questions; provide links to tests, reports and clean setup evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	final-defence.pdf and recording notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which claim in your report is weakest under questioning? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 75/100 and all high-risk claims have evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP ASVS Level 1 and C17 threat model as review checklist"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Secure Coding Practices Quick Reference Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D325 · M12W3D3 · Weekly final examination review and grade calculation

Phase: Academic month · **Optional date:** 2027-09-16 · **Domain resources:** C20 / F09

1. Day / sequence	Global Day 325 of 365; calendar-free M12W3D3; optional date 2027-09-16.
2. Exact topic	Weekly final examination review and grade calculation
3. Exact subtopics	Three practicals, capstone defence, monthly scores, portfolio quality and safety
4. Learning objective	Calculate the final 12-month mastery result using consistent weights
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework and the standard 100-point assessment rubric in Volume 6. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete the final scorecard, classify competence as foundational/intermediate/advanced beginner specialization, and write a one-year reflection. **2.** Standard (~2 h): Complete the final scorecard, classify competence as foundational/intermediate/advanced beginner specialization, and write a one-year reflection. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete the final scorecard, classify competence as foundational/intermediate/advanced beginner specialization, and write a one-year reflection. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-final-scorecard.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which measurable competency improved most and which remains weakest? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All scores are evidence-linked and no competency is overstated Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework and the standard 100-point assessment rubric in Volume 6"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D326 · M12W3D4 · Final portfolio audit and sanitization

Phase: Academic month · **Optional date:** 2027-09-17 · **Domain resources:** W13 / F09

1. Day / sequence	Global Day 326 of 365; calendar-free M12W3D4; optional date 2027-09-17.
2. Exact topic	Final portfolio audit and sanitization
3. Exact subtopics	Secrets, personal data, IP addresses, logs, screenshots, authorization statements, reproducibility and licences
4. Learning objective	Ensure every public artifact is safe, professional and reproducible
5. Primary lecture / reading + direct URL	GitHub code security documentation `https://docs.github.com/en/code-security` **Type:** Official documentation · **Access:** Free features vary by repository visibility and plan
6. Exact section / lesson locator	GitHub code security documentation, secret scanning and repository security. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run a secret scan and manual checklist across all repositories; remove or redact sensitive data and verify every direct link. **2.** Standard (~2 h): Run a secret scan and manual checklist across all repositories; remove or redact sensitive data and verify every direct link. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run a secret scan and manual checklist across all repositories; remove or redact sensitive data and verify every direct link. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	portfolio-audit.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which file could still reveal real personal or lab-identifying information? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No secrets or personal data remain and all links open Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub code security documentation, secret scanning and repository security"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Semgrep documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D327 · M12W3D5 · CV, LinkedIn and technical story preparation

Phase: Academic month · **Optional date:** 2027-09-18 · **Domain resources:** C20 / B15

1. Day / sequence	Global Day 327 of 365; calendar-free M12W3D5; optional date 2027-09-18.
2. Exact topic	CV, LinkedIn and technical story preparation
3. Exact subtopics	One-page CV, project bullets, evidence links, technical skills, limitations and a 90-second personal narrative
4. Learning objective	Translate portfolio evidence into employer-readable language without exaggeration
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role vocabulary and official job-description analysis guidance from Volume 6. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draft a CV and LinkedIn summary; rewrite ten project bullets as problem-action-evidence-result statements. **2.** Standard (~2 h): Draft a CV and LinkedIn summary; rewrite ten project bullets as problem-action-evidence-result statements. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draft a CV and LinkedIn summary; rewrite ten project bullets as problem-action-evidence-result statements. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cv-draft.pdf and LinkedIn notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which sentence claims more than your evidence supports? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All claims map to a repository, report or assessment result Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role vocabulary and official job-description analysis guidance from Volume 6"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D328 · M12W3D6 · Technical interview practice and scenario answers

Phase: Academic month · **Optional date:** 2027-09-19 · **Domain resources:** B15 / C18

1. Day / sequence	Global Day 328 of 365; calendar-free M12W3D6; optional date 2027-09-19.
2. Exact topic	Technical interview practice and scenario answers
3. Exact subtopics	OSI/TCP troubleshooting, Linux/Windows logs, SQLi/XSS prevention, detection logic, IAM, cloud shared responsibility and incident response
4. Learning objective	Practice explaining technical reasoning clearly under time pressure
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide and W17 OWASP Cheat Sheet Series for accurate terminology. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Record answers to ten scenario questions, self-score for structure, evidence, uncertainty and safety; repeat the three weakest. **2.** Standard (~2 h): Record answers to ten scenario questions, self-score for structure, evidence, uncertainty and safety; repeat the three weakest. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Record answers to ten scenario questions, self-score for structure, evidence, uncertainty and safety; repeat the three weakest. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	interview-practice.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which answer needs a concrete example rather than a definition? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	At least eight answers are clear, accurate and under three minutes Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide and W17 OWASP Cheat Sheet Series for accurate terminology"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D329 · M12W3D7 · Certification readiness and final decision

Phase: Academic month · **Optional date:** 2027-09-20 · **Domain resources:** A18 / C03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 329 of 365; calendar-free M12W3D7; optional date 2027-09-20.
2. Exact topic	Certification readiness and final decision
3. Exact subtopics	ISC2 CC, Security+ V7/V8, CCST Cybersecurity, SC-900, SC-200, eJPT, OSCP, AWS Security Specialty and BTL1
4. Learning objective	Decide if, when and whether to take a certification based on verified official facts and budget
5. Primary lecture / reading + direct URL	CompTIA Security+ V7 and V8 official pages https://www.comptia.org/en-us/certifications/security/ **Type:** Official certification page · **Access:** Paid exam; V7 US$404 from CompTIA blog dated January 2025; verify current store
6. Exact section / lesson locator	CompTIA Security+ V7/V8 official pages; A19 ISC2 CC; A17 Cisco CCST; A20 Microsoft SC-900. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	ISC2 Certified in Cybersecurity Exam Outline https://www.isc2.org/certifications/cc/cc-certification-exam-outline
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Re-check official costs/objectives, complete free practice assessments, estimate total cost including renewal, and write a go/no-go decision for the next six months — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Re-check official costs/objectives, complete free practice assessments, estimate total cost including renewal, and write a go/no-go decision for the next six months — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Re-check official costs/objectives, complete free practice assessments, estimate total cost including renewal, and write a go/no-go decision for the next six months — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	certification-final-decision.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: What specific job or education requirement would justify paying for the exam now? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Decision cites current official sources and no exam is purchased merely for collection Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CompTIA Security+ V7/V8 official pages; A19 ISC2 CC; A17 Cisco CCST; A20 Microsoft SC-900"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer SY0-701 Security+ Course Index; write a five-line correction; retest next morning with a new 10-minute problem.

Month 12 · Week 4 · daily cards

D330 · M12W4D1 · Year 2-5 long-term plan and 90-day sprint

Phase: Academic month · **Optional date:** 2027-09-21 · **Domain resources:** A08 / C20

1. Day / sequence	Global Day 330 of 365; calendar-free M12W4D1; optional date 2027-09-21.
2. Exact topic	Year 2-5 long-term plan and 90-day sprint
3. Exact subtopics	Specialization, peer group, research reading, open-source contribution, internship/job search, advanced labs and measurable milestones
4. Learning objective	Convert 12 months of foundations into a multi-year development plan
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering free book and A12 USENIX Security proceedings for research pathway examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a Year 2 plan, Year 3-5 milestones, a 90-day sprint, weekly metrics and a review date; define a meaningful peer group. **2.** Standard (~2 h): Write a Year 2 plan, Year 3-5 milestones, a 90-day sprint, weekly metrics and a review date; define a meaningful peer group. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a Year 2 plan, Year 3-5 milestones, a 90-day sprint, weekly metrics and a review date; define a meaningful peer group. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	year2-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which milestone measures technical impact rather than course completion? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan is specific, realistic and does not promise top 1% status Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering free book and A12 USENIX Security proceedings for research pathway examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D331 · M12W4D2 · Monthly review, celebration and responsible handover

Phase: Academic month · **Optional date:** 2027-09-22 · **Domain resources:** C06 / F09

1. Day / sequence	Global Day 331 of 365; calendar-free M12W4D2; optional date 2027-09-22.
2. Exact topic	Monthly review, celebration and responsible handover
3. Exact subtopics	Achievements, evidence, failures, mental health, community contribution and next-year commitments
4. Learning objective	Recognize completion while planning sustainable long-term growth
5. Primary lecture / reading + direct URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST CSF 2.0 and C20 NICE Framework for outcome categories. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a final reflection, thank any mentors, publish a lessons-learned post with sanitized resources, and archive all private evidence. **2.** Standard (~2 h): Write a final reflection, thank any mentors, publish a lessons-learned post with sanitized resources, and archive all private evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a final reflection, thank any mentors, publish a lessons-learned post with sanitized resources, and archive all private evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	final-reflection.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would you change if you repeated the year, and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Reflection is honest and the public post leaks no private data Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST CSF 2.0 and C20 NICE Framework for outcome categories"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Controls Version 8; write a five-line correction; retest next morning with a new 10-minute problem.

D332 · M12W4D3 · Master resource directory update and link re-verification

Phase: Academic month · **Optional date:** 2027-09-23 · **Domain resources:** C03 / A19

1. Day / sequence	Global Day 332 of 365; calendar-free M12W4D3; optional date 2027-09-23.
2. Exact topic	Master resource directory update and link re-verification
3. Exact subtopics	Important URLs, access type, course versions, verification date, backups and outdated material flags
4. Learning objective	Ensure the daily manual still points to current primary sources
5. Primary lecture / reading + direct URL	ISC2 Certified in Cybersecurity Exam Outline `https://www.isc2.org/certifications/cc/cc-certification-exam-outline` **Type:** Official exam outline · **Access:** Free
6. Exact section / lesson locator	ISC2 CC Exam Outline and A18 CompTIA Security+ V8 official page as examples of version-sensitive resources. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	ISC2 Certified in Cybersecurity official page `https://www.isc2.org/Certifications/CC`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Recheck all Tier 1 resource links, record HTTP status/date, replace broken links with verified backups, and update the verification register. **2.** Standard (~2 h): Recheck all Tier 1 resource links, record HTTP status/date, replace broken links with verified backups, and update the verification register. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Recheck all Tier 1 resource links, record HTTP status/date, replace broken links with verified backups, and update the verification register. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-resource-register.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which course or certification link changed version and could mislead future study? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every mandatory resource is verified or has a documented backup and date Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "ISC2 CC Exam Outline and A18 CompTIA Security+ V8 official page as examples of version-sensitive resources"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup ISC2 CC self-study resources; write a five-line correction; retest next morning with a new 10-minute problem.

D333 · M12W4D4 · Practice-exam strategy and question analysis

Phase: Academic month · **Optional date:** 2027-09-24 · **Domain resources:** C03 / C01

1. Day / sequence	Global Day 333 of 365; calendar-free M12W4D4; optional date 2027-09-24.
2. Exact topic	Practice-exam strategy and question analysis
3. Exact subtopics	Objective mapping, retrieval, distractor analysis, confidence marks, timeboxes, error taxonomy and retest intervals
4. Learning objective	Prepare for any future certification without reproducing proprietary exam questions
5. Primary lecture / reading + direct URL	ISC2 Certified in Cybersecurity Exam Outline `https://www.isc2.org/certifications/cc/cc-certification-exam-outline` **Type:** Official exam outline · **Access:** Free
6. Exact section / lesson locator	ISC2 CC official flashcards/quiz and A20 Microsoft SC-900 free practice assessment. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a study plan for one chosen certification using only official free practice materials; take two 30-question self-generated quizzes and analyze errors. **2.** Standard (~2 h): Create a study plan for one chosen certification using only official free practice materials; take two 30-question self-generated quizzes and analyze errors. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a study plan for one chosen certification using only official free practice materials; take two 30-question self-generated quizzes and analyze errors. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-exam-strategy.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which error is due to wording rather than missing technical knowledge? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan includes at least three spaced retests and no brain-dump or pirated content Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "ISC2 CC official flashcards/quiz and A20 Microsoft SC-900 free practice assessment"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup ISC2 CC self-study resources; write a five-line correction; retest next morning with a new 10-minute problem.

D334 · M12W4D5 · Internship and entry-level application strategy

Phase: Academic month · **Optional date:** 2027-09-25 · **Domain resources:** C20 / B15

1. Day / sequence	Global Day 334 of 365; calendar-free M12W4D5; optional date 2027-09-25.
2. Exact topic	Internship and entry-level application strategy
3. Exact subtopics	Role targeting, application tracker, evidence links, cold outreach, referrals, interview prep and work-authorization checks
4. Learning objective	Build a realistic three-month application plan without guarantees
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role vocabulary and official career resources in Volume 6. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Identify 20 role types, draft a tracker, tailor one CV to a real job description, and list work-authorization/remote constraints for target regions. **2.** Standard (~2 h): Identify 20 role types, draft a tracker, tailor one CV to a real job description, and list work-authorization/remote constraints for target regions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Identify 20 role types, draft a tracker, tailor one CV to a real job description, and list work-authorization/remote constraints for target regions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	application-plan.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which role's requirements are closest to your portfolio evidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan is evidence-based and includes follow-up dates but no employment promise Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role vocabulary and official career resources in Volume 6"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D335 · M12W4D6 · Sustainable study system and burnout prevention

Phase: Academic month · **Optional date:** 2027-09-26 · **Domain resources:** C06 / F04

1. Day / sequence	Global Day 335 of 365; calendar-free M12W4D6; optional date 2027-09-26.
2. Exact topic	Sustainable study system and burnout prevention
3. Exact subtopics	Workload variants, recovery, backlog rules, spaced repetition, sleep, exercise, campus workload and exam seasons
4. Learning objective	Design a Year 2 schedule that can survive real life and university deadlines
5. Primary lecture / reading + direct URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST CSF 2.0 Govern function for continuous improvement; C20 NICE for skill planning. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Define monthly/quarterly review rituals, a maximum backlog rule, minimum viable study session and two burnout warning signals with responses. **2.** Standard (~2 h): Define monthly/quarterly review rituals, a maximum backlog rule, minimum viable study session and two burnout warning signals with responses. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Define monthly/quarterly review rituals, a maximum backlog rule, minimum viable study session and two burnout warning signals with responses. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	sustainable-study-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What will you drop when university exams consume two weeks? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan includes explicit catch-up and reduced-intensity weeks Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST CSF 2.0 Govern function for continuous improvement; C20 NICE for skill planning"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Controls Version 8; write a five-line correction; retest next morning with a new 10-minute problem.

D336 · M12W4D7 · Spaced revision and handover to final consolidation

Phase: Academic month · **Optional date:** 2027-09-27 · **Domain resources:** B03 / CL03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 336 of 365; calendar-free M12W4D7; optional date 2027-09-27.
2. Exact topic	Spaced revision and handover to final consolidation
3. Exact subtopics	All 11 modules, capstone, practical exams, portfolio and certification decision; light retrieval
4. Learning objective	Preserve the full year's knowledge and prepare for four weeks of independent consolidation
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ and B02 KQL overview for primary-specialization recall. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do 20 mixed retrieval questions, create a final deck of 30 flashcards, clean snapshots and schedule the Week 49-52 challenges — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do 20 mixed retrieval questions, create a final deck of 30 flashcards, clean snapshots and schedule the Week 49-52 challenges — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do 20 mixed retrieval questions, create a final deck of 30 flashcards, clean snapshots and schedule the Week 49-52 challenges — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-consolidation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which three topics require review on days 7, 30 and 90? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Spaced repetition schedule exists and all lab resources are clean Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ and B02 KQL overview for primary-specialization recall"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

Final consolidation — Weeks 49-52 and Day 365

This is the explicitly scheduled remainder of the 365-day plan: 12 academic months (336 days), four final consolidation weeks (28 days), and one graduation day. It is not vague "continue practising"; every day has a resource, practical task, evidence and assessment.

Week 49 · independent integration

D337 · W49D1 · Final consolidation kickoff: unseen integrated scenario

Phase: Final consolidation · **Optional date:** 2027-09-28 · **Domain resources:** C07 / B02

1. Day / sequence	Global Day 337 of 365; calendar-free W49D1; optional date 2027-09-28.
2. Exact topic	Final consolidation kickoff: unseen integrated scenario
3. Exact subtopics	Incident, network packet, Linux/Windows logs, cloud config, web finding and an ambiguous stakeholder request
4. Learning objective	Establish how you approach an unfamiliar problem before looking up answers
5. Primary lecture / reading + direct URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-61r3 incident response lifecycle and C20 NICE work-role tasks. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Read the scenario once, list known/unknown facts, write five hypotheses and create an evidence plan before any tool use. **2.** Standard (~2 h): Read the scenario once, list known/unknown facts, write five hypotheses and create an evidence plan before any tool use. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Read the scenario once, list known/unknown facts, write five hypotheses and create an evidence plan before any tool use. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-case-brief.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which decision must be made before collecting data and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan is recorded before investigation and assumptions are labelled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-61r3 incident response lifecycle and C20 NICE work-role tasks"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D338 · W49D2 · W49D2: systems and network investigation

Phase: Final consolidation · **Optional date:** 2027-09-29 · **Domain resources:** N06 / N07

1. Day / sequence	Global Day 338 of 365; calendar-free W49D2; optional date 2027-09-29.
2. Exact topic	W49D2: systems and network investigation
3. Exact subtopics	Host inventory, packets, subnetting, DNS, TLS, routing and service reachability
4. Learning objective	Resolve the systems/network portion of the integrated scenario
5. Primary lecture / reading + direct URL	Wireshark User's Guide `https://www.wireshark.org/docs/wsug_html_chunked/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Wireshark User's Guide and N09 Nmap Reference Guide for authorized lab scope. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Wireshark SampleCaptures `https://wiki.wireshark.org/SampleCaptures`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Work only with provided synthetic evidence; build a timeline, endpoint map and three alternative explanations. **2.** Standard (~2 h): Work only with provided synthetic evidence; build a timeline, endpoint map and three alternative explanations. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Work only with provided synthetic evidence; build a timeline, endpoint map and three alternative explanations. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-network.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which frame or log field changes your preferred explanation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Conclusions cite specific evidence and uncertainty Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Wireshark User's Guide and N09 Nmap Reference Guide for authorized lab scope"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Wireshark SampleCaptures; write a five-line correction; retest next morning with a new 10-minute problem.

D339 · W49D3 · W49D3: endpoint and identity investigation

Phase: Final consolidation · **Optional date:** 2027-09-30 · **Domain resources:** L09 / B05

1. Day / sequence	Global Day 339 of 365; calendar-free W49D3; optional date 2027-09-30.
2. Exact topic	W49D3: endpoint and identity investigation
3. Exact subtopics	Windows events, Sysmon, Linux auth logs, process trees, identity changes and privilege boundaries
4. Learning objective	Determine what happened on the endpoint without claiming more than telemetry shows
5. Primary lecture / reading + direct URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	Windows Event Log docs and L11 Sysmon documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SBousseaden EVTX-ATTACK-SAMPLES `https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a process/user timeline, identify privilege changes, and list the telemetry needed to close each gap. **2.** Standard (~2 h): Build a process/user timeline, identify privilege changes, and list the telemetry needed to close each gap. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a process/user timeline, identify privilege changes, and list the telemetry needed to close each gap. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-endpoint.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which event could be benign in one context and malicious in another? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Timeline is consistent and all claims have confidence labels Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Windows Event Log docs and L11 Sysmon documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Sysmon documentation and download; write a five-line correction; retest next morning with a new 10-minute problem.

D340 · W49D4 · W49D4: web and API evidence

Phase: Final consolidation · **Optional date:** 2027-10-01 · **Domain resources:** W06 / W01

1. Day / sequence	Global Day 340 of 365; calendar-free W49D4; optional date 2027-10-01.
2. Exact topic	W49D4: web and API evidence
3. Exact subtopics	HTTP requests, sessions, authorization, input validation, injection/XSS/SSRF indicators and server errors
4. Learning objective	Correlate application events with endpoint and identity evidence
5. Primary lecture / reading + direct URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/` **Type:** Testing methodology · **Access:** Free
6. Exact section / lesson locator	OWASP Web Security Testing Guide and W17 OWASP Cheat Sheet Series. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	PortSwigger Web Security Academy `https://portswigger.net/web-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Analyze a provided synthetic request log, identify two suspicious patterns, propose safe validation tests and reject one false positive. **2.** Standard (~2 h): Analyze a provided synthetic request log, identify two suspicious patterns, propose safe validation tests and reject one false positive. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Analyze a provided synthetic request log, identify two suspicious patterns, propose safe validation tests and reject one false positive. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-web.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which request header or response status is decisive? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Findings are reproducible and one alternative explanation is retained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Web Security Testing Guide and W17 OWASP Cheat Sheet Series"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D341 · W49D5 · W49D5: cloud, container and CI evidence

Phase: Final consolidation · **Optional date:** 2027-10-02 · **Domain resources:** CL03 / CL09

1. Day / sequence	Global Day 341 of 365; calendar-free W49D5; optional date 2027-10-02.
2. Exact topic	W49D5: cloud, container and CI evidence
3. Exact subtopics	IAM audit events, storage changes, Kubernetes RBAC, container logs and workflow artifacts
4. Learning objective	Determine whether the incident crossed into infrastructure or supply chain
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar and CL14 GitHub Actions hardening. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review provided synthetic cloud/container evidence, map blast radius and identify one preventive and one detective control. **2.** Standard (~2 h): Review provided synthetic cloud/container evidence, map blast radius and identify one preventive and one detective control. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review provided synthetic cloud/container evidence, map blast radius and identify one preventive and one detective control. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-cloud.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which identity had the ability to change the build pipeline? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Scope is drawn with evidence and no live cloud resource is touched Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar and CL14 GitHub Actions hardening"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D342 · W49D6 · W49D6: detection engineering and response

Phase: Final consolidation · **Optional date:** 2027-10-03 · **Domain resources:** B04 / B03

1. Day / sequence	Global Day 342 of 365; calendar-free W49D6; optional date 2027-10-03.
2. Exact topic	W49D6: detection engineering and response
3. Exact subtopics	Sigma/KQL rule, test data, true/false positives, containment recommendation and recovery validation
4. Learning objective	Turn the week's findings into an actionable detection and response
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started and C07 NIST SP 800-61r3 response sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write one rule, ten labelled test events, an incident summary, a containment recommendation and a recovery test. **2.** Standard (~2 h): Write one rule, ten labelled test events, an incident summary, a containment recommendation and a recovery test. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write one rule, ten labelled test events, an incident summary, a containment recommendation and a recovery test. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-detection.yml and response.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which benign event could trigger the rule and how is it handled? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Rule passes tests and response preserves evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started and C07 NIST SP 800-61r3 response sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D343 · W49D7 · W49D7: weekly report, scoring and recovery

Phase: Final consolidation · **Optional date:** 2027-10-04 · **Domain resources:** B15 / C06 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 343 of 365; calendar-free W49D7; optional date 2027-10-04.
2. Exact topic	W49D7: weekly report, scoring and recovery
3. Exact subtopics	Integrated scenario, evidence quality, confidence, root cause, residual risk and lessons learned
4. Learning objective	Defend the week's conclusion and recover with lighter revision
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide investigation/reporting sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a five-page report, score it with the 100-point rubric, list three lessons, do 30 minutes of spaced recall and rest — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a five-page report, score it with the 100-point rubric, list three lessons, do 30 minutes of spaced recall and rest — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a five-page report, score it with the 100-point rubric, list three lessons, do 30 minutes of spaced recall and rest — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-report.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which conclusion would a skeptical reviewer challenge first? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Report passes 70/100 and includes a clear post-review action Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide investigation/reporting sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Week 50 · independent integration

D344 · W50D1 · W50D1: research paper selection and reading plan

Phase: Final consolidation · **Optional date:** 2027-10-05 · **Domain resources:** A12 / A13

1. Day / sequence	Global Day 344 of 365; calendar-free W50D1; optional date 2027-10-05.
2. Exact topic	W50D1: research paper selection and reading plan
3. Exact subtopics	Primary source, research question, threat model, method, dataset, evaluation, limitations and reproducibility
4. Learning objective	Select one paper or engineering write-up relevant to the primary specialization
5. Primary lecture / reading + direct URL	USENIX Security conference papers `https://www.usenix.org/conferences/byname/164` **Type:** Conference proceedings · **Access:** Many papers freely available; conference attendance paid
6. Exact section / lesson locator	USENIX Security proceedings or Google Project Zero blog via A11; use one primary source only. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	arXiv cs.CR recent papers `https://arxiv.org/list/cs.CR/recent`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Choose a paper/write-up, record citation, prerequisites, three reading questions and a safe reproduction goal. **2.** Standard (~2 h): Choose a paper/write-up, record citation, prerequisites, three reading questions and a safe reproduction goal. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Choose a paper/write-up, record citation, prerequisites, three reading questions and a safe reproduction goal. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-reading-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would count as successful reproduction without repeating dangerous behaviour? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Source is credible and reproduction is legal and isolated Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "USENIX Security proceedings or Google Project Zero blog via A11; use one primary source only"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Google Project Zero blog; write a five-line correction; retest next morning with a new 10-minute problem.

D345 · W50D2 · W50D2: primary-source close reading

Phase: Final consolidation · **Optional date:** 2027-10-06 · **Domain resources:** A08 / A11

1. Day / sequence	Global Day 345 of 365; calendar-free W50D2; optional date 2027-10-06.
2. Exact topic	W50D2: primary-source close reading
3. Exact subtopics	Terminology, claims, assumptions, experimental setup, baselines and statistical/technical evidence
4. Learning objective	Separate the authors' observations from their interpretations
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering book chapter on evaluating claims; A12 selected paper. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Google Project Zero blog `https://googleprojectzero.blogspot.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Annotate the paper's problem, method, data, results, limitations and five unfamiliar terms with official definitions. **2.** Standard (~2 h): Annotate the paper's problem, method, data, results, limitations and five unfamiliar terms with official definitions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Annotate the paper's problem, method, data, results, limitations and five unfamiliar terms with official definitions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-annotated-notes.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which claim depends on an assumption not tested by the authors? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Notes distinguish evidence, inference and speculation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering book chapter on evaluating claims; A12 selected paper"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D346 · W50D3 · W50D3: reproduction design and lab safety

Phase: Final consolidation · **Optional date:** 2027-10-07 · **Domain resources:** A01 / B03

1. Day / sequence	Global Day 346 of 365; calendar-free W50D3; optional date 2027-10-07.
2. Exact topic	W50D3: reproduction design and lab safety
3. Exact subtopics	Environment, dependencies, synthetic data, metrics, controls, failure criteria and isolation
4. Learning objective	Design a minimal reproduction that tests one claim
5. Primary lecture / reading + direct URL	pwn.college `https://pwn.college/` **Type:** Hands-on university platform · **Access:** Free
6. Exact section / lesson locator	pwn.college dojo index for safe challenge infrastructure; A12 paper artifact if available. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a reproduction plan with exact inputs, expected outputs, abort conditions and a disposable VM/container boundary. **2.** Standard (~2 h): Write a reproduction plan with exact inputs, expected outputs, abort conditions and a disposable VM/container boundary. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a reproduction plan with exact inputs, expected outputs, abort conditions and a disposable VM/container boundary. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-reproduction-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which part of the original experiment cannot be reproduced safely or affordably? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan has a narrow claim, objective metrics and no unauthorized target Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pwn.college dojo index for safe challenge infrastructure; A12 paper artifact if available"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup pwn.college Dojo index; write a five-line correction; retest next morning with a new 10-minute problem.

D347 · W50D4 · W50D4: reproduction execution and evidence

Phase: Final consolidation · **Optional date:** 2027-10-08 · **Domain resources:** B02 / B04

1. Day / sequence	Global Day 347 of 365; calendar-free W50D4; optional date 2027-10-08.
2. Exact topic	W50D4: reproduction execution and evidence
3. Exact subtopics	Setup logs, commands, configuration, raw output, version numbers, errors and deviations
4. Learning objective	Reproduce one narrow result with complete evidence
5. Primary lecture / reading + direct URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric` **Type:** Official language docs · **Access:** Free
6. Exact section / lesson locator	KQL or primary-specialization official docs relevant to the paper's method. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run the minimal reproduction in an isolated lab, record all commands and outputs, and document every deviation from the paper. **2.** Standard (~2 h): Run the minimal reproduction in an isolated lab, record all commands and outputs, and document every deviation from the paper. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run the minimal reproduction in an isolated lab, record all commands and outputs, and document every deviation from the paper. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-reproduction/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which deviation could explain a different result? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All evidence is captured and hashes/versions are recorded Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "KQL or primary-specialization official docs relevant to the paper's method"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D348 · W50D5 · W50D5: result comparison and negative findings

Phase: Final consolidation · **Optional date:** 2027-10-09 · **Domain resources:** A13 / A12

1. Day / sequence	Global Day 348 of 365; calendar-free W50D5; optional date 2027-10-09.
2. Exact topic	W50D5: result comparison and negative findings
3. Exact subtopics	Expected versus observed results, effect size, confidence, alternative explanations and limitations
4. Learning objective	Interpret reproduction results honestly even when they disagree with the paper
5. Primary lecture / reading + direct URL	arXiv cs.CR recent papers `https://arxiv.org/list/cs.CR/recent` **Type:** Preprint archive · **Access:** Free
6. Exact section / lesson locator	arXiv cs.CR for related work context only, not as authoritative evidence. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	USENIX Security conference papers `https://www.usenix.org/conferences/byname/164`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Compare results, calculate at least one simple metric, list three alternative explanations and decide what can be concluded. **2.** Standard (~2 h): Compare results, calculate at least one simple metric, list three alternative explanations and decide what can be concluded. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Compare results, calculate at least one simple metric, list three alternative explanations and decide what can be concluded. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-results.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which result is a negative finding and why is it still valuable? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Conclusion is supported by data and does not overclaim Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "arXiv cs.CR for related work context only, not as authoritative evidence"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup USENIX Security conference papers; write a five-line correction; retest next morning with a new 10-minute problem.

D349 · W50D6 · W50D6: open-source contribution or technical artifact

Phase: Final consolidation · **Optional date:** 2027-10-10 · **Domain resources:** F09 / B03

1. Day / sequence	Global Day 349 of 365; calendar-free W50D6; optional date 2027-10-10.
2. Exact topic	W50D6: open-source contribution or technical artifact
3. Exact subtopics	Issue selection, reproduction, small fix, documentation, tests, pull request or standalone utility
4. Learning objective	Contribute a small, useful, reviewable technical artifact
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 2-3 and W13 GitHub security docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Choose one good-first issue or create a standalone detection/test utility; make a small change with tests and documentation. **2.** Standard (~2 h): Choose one good-first issue or create a standalone detection/test utility; make a small change with tests and documentation. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Choose one good-first issue or create a standalone detection/test utility; make a small change with tests and documentation. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-contribution/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What did maintainers or your own reviewer ask you to change? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Contribution is original, tested and free of secrets or personal data Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 2-3 and W13 GitHub security docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D350 · W50D7 · W50D7: research communication and recovery

Phase: Final consolidation · **Optional date:** 2027-10-11 · **Domain resources:** A08 / A11 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 350 of 365; calendar-free W50D7; optional date 2027-10-11.
2. Exact topic	W50D7: research communication and recovery
3. Exact subtopics	Abstract, problem, method, result, limitation, future work and safe disclosure of findings
4. Learning objective	Explain the week's research to a technical peer and recover
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering writing guidance and A12 paper structure examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Google Project Zero blog `https://googleprojectzero.blogspot.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a two-page technical note, record a five-minute explanation, schedule follow-up questions and rest — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a two-page technical note, record a five-minute explanation, schedule follow-up questions and rest — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a two-page technical note, record a five-minute explanation, schedule follow-up questions and rest — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-technical-note.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which sentence could mislead a reader about certainty? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Note is clear, modest and reproducible from the artifact Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering writing guidance and A12 paper structure examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Week 51 · independent integration

D351 · W51D1 · W51D1: teaching a foundational topic

Phase: Final consolidation · **Optional date:** 2027-10-12 · **Domain resources:** F01 / C20

1. Day / sequence	Global Day 351 of 365; calendar-free W51D1; optional date 2027-10-12.
2. Exact topic	W51D1: teaching a foundational topic
3. Exact subtopics	OS, networking, Linux/Windows, Python, core security, web, blue team or cloud concepts
4. Learning objective	Test mastery by teaching one topic without notes to a beginner
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x notes or N01 Course Index for the selected topic. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a 20-minute lesson with one analogy, one diagram, one demo, three retrieval questions and common misconceptions. **2.** Standard (~2 h): Create a 20-minute lesson with one analogy, one diagram, one demo, three retrieval questions and common misconceptions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a 20-minute lesson with one analogy, one diagram, one demo, three retrieval questions and common misconceptions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-lesson.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which misconception did your explanation need to prevent? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Lesson is understandable and technically correct Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x notes or N01 Course Index for the selected topic"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D352 · W51D2 · W51D2: code review and mentorship simulation

Phase: Final consolidation · **Optional date:** 2027-10-13 · **Domain resources:** P08 / W13

1. Day / sequence	Global Day 352 of 365; calendar-free W51D2; optional date 2027-10-13.
2. Exact topic	W51D2: code review and mentorship simulation
3. Exact subtopics	Diff reading, security impact, false positives, constructive feedback, tests and documentation
4. Learning objective	Review another student's sanitized code or your old Month 4 code
5. Primary lecture / reading + direct URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/` **Type:** Checklist/reference · **Access:** Free
6. Exact section / lesson locator	OWASP Secure Coding Practices and W14 Semgrep documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Perform a code review with five findings, one positive observation, suggested tests and respectful language. **2.** Standard (~2 h): Perform a code review with five findings, one positive observation, suggested tests and respectful language. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Perform a code review with five findings, one positive observation, suggested tests and respectful language. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-code-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding is a real risk and which is style preference? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Feedback is actionable, prioritized and technically justified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Secure Coding Practices and W14 Semgrep documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D353 · W51D3 · W51D3: incident report peer review

Phase: Final consolidation · **Optional date:** 2027-10-14 · **Domain resources:** C07 / B17

1. Day / sequence	Global Day 353 of 365; calendar-free W51D3; optional date 2027-10-14.
2. Exact topic	W51D3: incident report peer review
3. Exact subtopics	Facts, assumptions, confidence, timeline, root cause, impact, recommendations and executive summary
4. Learning objective	Improve a synthetic incident report through peer-style critique
5. Primary lecture / reading + direct URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-61r3 and B15 SC-200 study guide. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CISA cybersecurity advisories `https://www.cisa.gov/news-events/cybersecurity-advisories`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review a intentionally flawed report, mark ten issues and rewrite the executive summary and recommendations. **2.** Standard (~2 h): Review a intentionally flawed report, mark ten issues and rewrite the executive summary and recommendations. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review a intentionally flawed report, mark ten issues and rewrite the executive summary and recommendations. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-report-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which recommendation lacks evidence or an owner? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Revised report is concise and evidence-linked Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-61r3 and B15 SC-200 study guide"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D354 · W51D4 · W51D4: interview scenario circuit

Phase: Final consolidation · **Optional date:** 2027-10-15 · **Domain resources:** C20 / W17

1. Day / sequence	Global Day 354 of 365; calendar-free W51D4; optional date 2027-10-15.
2. Exact topic	W51D4: interview scenario circuit
3. Exact subtopics	Technical, behavioural, troubleshooting, ethical and uncertainty questions
4. Learning objective	Practice communicating competence honestly under interview conditions
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role tasks and B15 study guide for terminology. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete six timed interview scenarios; score answers for structure, evidence, safety, uncertainty and listening. **2.** Standard (~2 h): Complete six timed interview scenarios; score answers for structure, evidence, safety, uncertainty and listening. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete six timed interview scenarios; score answers for structure, evidence, safety, uncertainty and listening. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-interview-scores.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which answer needs a concrete lab example? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	At least five answers are clear and evidence-based Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role tasks and B15 study guide for terminology"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D355 · W51D5 · W51D5: CV, portfolio and job-description mapping

Phase: Final consolidation · **Optional date:** 2027-10-16 · **Domain resources:** C20 / B15

1. Day / sequence	Global Day 355 of 365; calendar-free W51D5; optional date 2027-10-16.
2. Exact topic	W51D5: CV, portfolio and job-description mapping
3. Exact subtopics	Real job postings, required/must-have/nice-to-have skills, evidence mapping and gap plan
4. Learning objective	Analyze three current job descriptions without assuming they represent the whole market
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework https://www.nist.gov/itl/applied-cybersecurity/nice **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work roles; use live job postings saved with retrieval date in the workspace if available. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-200 certification page https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: For each of three postings, map ten requirements to portfolio evidence or a learning task; note visa/remote constraints separately. **2.** Standard (~2 h): For each of three postings, map ten requirements to portfolio evidence or a learning task; note visa/remote constraints separately. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): For each of three postings, map ten requirements to portfolio evidence or a learning task; note visa/remote constraints separately. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-job-analysis.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which requirement is genuinely missing from your evidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Analysis distinguishes employer-specific wants from broad patterns and cites posting dates Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work roles; use live job postings saved with retrieval date in the workspace if available"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D356 · W51D6 · W51D6: professional network and community contribution

Phase: Final consolidation · **Optional date:** 2027-10-17 · **Domain resources:** C20 / B17

1. Day / sequence	Global Day 356 of 365; calendar-free W51D6; optional date 2027-10-17.
2. Exact topic	W51D6: professional network and community contribution
3. Exact subtopics	Local/international communities, CTFs, open-source, meetups, responsible communication and mentorship
4. Learning objective	Build a sustainable professional learning network without spam
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework community resources and E05 disclosure guidelines. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CISA cybersecurity advisories `https://www.cisa.gov/news-events/cybersecurity-advisories`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Identify five credible communities or events, draft two useful questions or contributions, and plan one monthly participation habit. **2.** Standard (~2 h): Identify five credible communities or events, draft two useful questions or contributions, and plan one monthly participation habit. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Identify five credible communities or events, draft two useful questions or contributions, and plan one monthly participation habit. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-community-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which community gives feedback from people stronger than you? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan prioritizes feedback and contribution over self-promotion Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework community resources and E05 disclosure guidelines"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D357 · W51D7 · W51D7: weekly review, rest and Year 2 research questions

Phase: Final consolidation · **Optional date:** 2027-10-18 · **Domain resources:** A08 / C20 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 357 of 365; calendar-free W51D7; optional date 2027-10-18.
2. Exact topic	W51D7: weekly review, rest and Year 2 research questions
3. Exact subtopics	Teaching, review, interviews, job analysis and community; spaced repetition and recovery
4. Learning objective	Consolidate the week and generate long-term research questions
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering book and A12 proceedings for question inspiration. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write five research questions, review 20 flashcards, update the Year 2 plan, and rest or take a light lab walkthrough — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write five research questions, review 20 flashcards, update the Year 2 plan, and rest or take a light lab walkthrough — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write five research questions, review 20 flashcards, update the Year 2 plan, and rest or take a light lab walkthrough — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which question is narrow enough for a 90-day investigation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	At least two research questions are specific and measurable Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering book and A12 proceedings for question inspiration"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Week 52 · independent integration

D358 · W52D1 · W52D1: 90-day specialization sprint design

Phase: Final consolidation · **Optional date:** 2027-10-19 · **Domain resources:** C20 / A12

1. Day / sequence	Global Day 358 of 365; calendar-free W52D1; optional date 2027-10-19.
2. Exact topic	W52D1: 90-day specialization sprint design
3. Exact subtopics	One specialization, one secondary skill, problem, metrics, weekly milestones, artifacts and review cadence
4. Learning objective	Launch the first post-12-month sprint with realistic workload
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework and B03 SigmaHQ for primary specialization evidence. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	USENIX Security conference papers `https://www.usenix.org/conferences/byname/164`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a 90-day plan with weekly deliverables, leading/lagging metrics, reading list, feedback source and stop/continue criteria. **2.** Standard (~2 h): Write a 90-day plan with weekly deliverables, leading/lagging metrics, reading list, feedback source and stop/continue criteria. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a 90-day plan with weekly deliverables, leading/lagging metrics, reading list, feedback source and stop/continue criteria. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-90-day-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which metric measures independent technical impact? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan fits the chosen 1/2/3-4 hour track and has a review date Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework and B03 SigmaHQ for primary specialization evidence"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D359 · W52D2 · W52D2: advanced lab environment rebuild

Phase: Final consolidation · **Optional date:** 2027-10-20 · **Domain resources:** CL06 / W13

1. Day / sequence	Global Day 359 of 365; calendar-free W52D2; optional date 2027-10-20.
2. Exact topic	W52D2: advanced lab environment rebuild
3. Exact subtopics	Clean snapshots, reproducible setup, documentation, backups, resource budget and isolation
4. Learning objective	Build a stable environment for Year 2 independent work
5. Primary lecture / reading + direct URL	Docker documentation `https://docs.docker.com/` **Type:** Official documentation · **Access:** Free; Docker Desktop licensing may apply in some organizations
6. Exact section / lesson locator	Docker documentation and F12 VirtualBox User Manual snapshots/networking. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rebuild or verify the lab from documentation, restore from clean snapshots, hash key artifacts and record costs/resources. **2.** Standard (~2 h): Rebuild or verify the lab from documentation, restore from clean snapshots, hash key artifacts and record costs/resources. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rebuild or verify the lab from documentation, restore from clean snapshots, hash key artifacts and record costs/resources. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-lab-rebuild.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which step still requires undocumented manual work? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Fresh setup succeeds and no active cloud bill remains Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Docker documentation and F12 VirtualBox User Manual snapshots/networking"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Docker Engine security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D360 · W52D3 · W52D3: Year 2 reading and research pipeline

Phase: Final consolidation · **Optional date:** 2027-10-21 · **Domain resources:** A08 / A11

1. Day / sequence	Global Day 360 of 365; calendar-free W52D3; optional date 2027-10-21.
2. Exact topic	W52D3: Year 2 reading and research pipeline
3. Exact subtopics	Books, official docs, conference papers, advisories, incident reports, notes and hypothesis tracking
4. Learning objective	Create a repeatable reading and note-taking workflow
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering, A12 USENIX Security, A13 arXiv cs.CR, B17 CISA advisories. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Google Project Zero blog `https://googleprojectzero.blogspot.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a reading queue with 12 sources, a note template, a hypothesis log and a monthly paper discussion goal. **2.** Standard (~2 h): Build a reading queue with 12 sources, a note template, a hypothesis log and a monthly paper discussion goal. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a reading queue with 12 sources, a note template, a hypothesis log and a monthly paper discussion goal. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-reading-pipeline.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which source is primary evidence and which is secondary commentary? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Queue is staged by prerequisite and every source has a purpose Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering, A12 USENIX Security, A13 arXiv cs.CR, B17 CISA advisories"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D361 · W52D4 · W52D4: advanced practice ladder

Phase: Final consolidation · **Optional date:** 2027-10-22 · **Domain resources:** A01 / A09

1. Day / sequence	Global Day 361 of 365; calendar-free W52D4; optional date 2027-10-22.
2. Exact topic	W52D4: advanced practice ladder
3. Exact subtopics	Guided reproduction, modified exercise, unfamiliar problem, experiment design, original tool/ research and peer defence
4. Learning objective	Map Year 2 practice from dependent to independent work
5. Primary lecture / reading + direct URL	pwn.college `https://pwn.college/` **Type:** Hands-on university platform · **Access:** Free
6. Exact section / lesson locator	pwn.college dojos and A03 OpenSecurityTraining2 for advanced progressions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Nightmare binary exploitation course `https://guyinatuxedo.github.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a six-level ladder with one concrete challenge per level for your chosen specialization. **2.** Standard (~2 h): Create a six-level ladder with one concrete challenge per level for your chosen specialization. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a six-level ladder with one concrete challenge per level for your chosen specialization. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-practice-ladder.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which level marks the transition from learning to original contribution? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Ladder has objective evidence at each level Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pwn.college dojos and A03 OpenSecurityTraining2 for advanced progressions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup pwn.college Dojo index; write a five-line correction; retest next morning with a new 10-minute problem.

D362 · W52D5 · W52D5: technical impact and peer group definition

Phase: Final consolidation · **Optional date:** 2027-10-23 · **Domain resources:** C20 / A08

1. Day / sequence	Global Day 362 of 365; calendar-free W52D5; optional date 2027-10-23.
2. Exact topic	W52D5: technical impact and peer group definition
3. Exact subtopics	Specialization scope, peer group, impact metrics, reputation versus skill, review and feedback
4. Learning objective	Define what exceptional competence would mean in a realistic peer group
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework and A12 USENIX Security for peer-reviewed impact examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a personal definition of impact, choose a peer group, list three measurable signals and three vanity metrics to ignore. **2.** Standard (~2 h): Write a personal definition of impact, choose a peer group, list three measurable signals and three vanity metrics to ignore. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a personal definition of impact, choose a peer group, list three measurable signals and three vanity metrics to ignore. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-impact-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which signal would still matter if nobody publicly praised the work? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Definition separates technical impact from popularity Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework and A12 USENIX Security for peer-reviewed impact examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D363 · W52D6 · W52D6: final archive, backup and privacy audit

Phase: Final consolidation · **Optional date:** 2027-10-24 · **Domain resources:** W13 / C24

1. Day / sequence	Global Day 363 of 365; calendar-free W52D6; optional date 2027-10-24.
2. Exact topic	W52D6: final archive, backup and privacy audit
3. Exact subtopics	All evidence, reports, repositories, snapshots, keys, credentials, personal data and retention
4. Learning objective	Preserve useful work while removing unnecessary sensitive data
5. Primary lecture / reading + direct URL	GitHub code security documentation `https://docs.github.com/en/code-security` **Type:** Official documentation · **Access:** Free features vary by repository visibility and plan
6. Exact section / lesson locator	GitHub security docs and F09 Pro Git. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GDPR full text (EUR-Lex) `https://eur-lex.europa.eu/eli/reg/2016/679/oj`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create an encrypted local archive, verify hashes, delete unnecessary secrets/data from old repositories, and write a retention schedule. **2.** Standard (~2 h): Create an encrypted local archive, verify hashes, delete unnecessary secrets/data from old repositories, and write a retention schedule. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create an encrypted local archive, verify hashes, delete unnecessary secrets/data from old repositories, and write a retention schedule. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-archive-manifest.txt; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which old artifact should be deleted rather than archived? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Restore test succeeds and no unencrypted secrets remain Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub security docs and F09 Pro Git"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Semgrep documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D364 · W52D7 · W52D7: one-year retrospective and public learning statement

Phase: Final consolidation · **Optional date:** 2027-10-25 · **Domain resources:** C06 / C20 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 364 of 365; calendar-free W52D7; optional date 2027-10-25.
2. Exact topic	W52D7: one-year retrospective and public learning statement
3. Exact subtopics	Achievements, evidence, failures, limitations, gratitude, next milestone and sustainable habits
4. Learning objective	Complete the 12-month curriculum with an honest retrospective
5. Primary lecture / reading + direct URL	NIST Cybersecurity Framework 2.0 https://www.nist.gov/cyberframework **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST CSF 2.0 outcome categories for broad reflection. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework https://www.nist.gov/itl/applied-cybersecurity/nice
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Publish a sanitized lessons-learned post, update CV/portfolio links, record the next 90-day commitment and rest — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Publish a sanitized lessons-learned post, update CV/portfolio links, record the next 90-day commitment and rest — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Publish a sanitized lessons-learned post, update CV/portfolio links, record the next 90-day commitment and rest — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-retrospective.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: What is the most important behaviour to sustain, not just the most important topic learned? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Post is honest, safe and includes at least three measurable Year 2 goals Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST CSF 2.0 outcome categories for broad reflection"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Controls Version 8; write a five-line correction; retest next morning with a new 10-minute problem.

D365 · D365 · D365: graduation checklist and next-day start

Phase: Graduation · **Optional date:** 2027-10-26 · **Domain resources:** C20 / B03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 365 of 365; calendar-free D365; optional date 2027-10-26.
2. Exact topic	D365: graduation checklist and next-day start
3. Exact subtopics	All 365 daily assignments, 15 projects, four milestone gates, practical exams, capstone defence, portfolio, certification decision and Year 2 plan
4. Learning objective	Verify that every required artifact exists and decide the very first task of Year 2
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role tasks and the master checklist in Volume 1. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run the final completion checklist, repair any missing artifact, present the master portfolio to the user, and schedule tomorrow's first 90-day sprint task — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run the final completion checklist, repair any missing artifact, present the master portfolio to the user, and schedule tomorrow's first 90-day sprint task — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run the final completion checklist, repair any missing artifact, present the master portfolio to the user, and schedule tomorrow's first 90-day sprint task — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	completion-certificate.md and final checklist; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: What evidence best proves independent problem-solving today? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All mandatory items are present, files open, and tomorrow has one specific task Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role tasks and the master checklist in Volume 1"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Day 365 · graduation and next-day start

D365 · D365 · D365: graduation checklist and next-day start

Phase: Graduation · **Optional date:** 2027-10-26 · **Domain resources:** C20 / B03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 365 of 365; calendar-free D365; optional date 2027-10-26.
2. Exact topic	D365: graduation checklist and next-day start
3. Exact subtopics	All 365 daily assignments, 15 projects, four milestone gates, practical exams, capstone defence, portfolio, certification decision and Year 2 plan
4. Learning objective	Verify that every required artifact exists and decide the very first task of Year 2
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role tasks and the master checklist in Volume 1. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run the final completion checklist, repair any missing artifact, present the master portfolio to the user, and schedule tomorrow's first 90-day sprint task — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run the final completion checklist, repair any missing artifact, present the master portfolio to the user, and schedule tomorrow's first 90-day sprint task — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run the final completion checklist, repair any missing artifact, present the master portfolio to the user, and schedule tomorrow's first 90-day sprint task — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	completion-certificate.md and final checklist; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: What evidence best proves independent problem-solving today? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All mandatory items are present, files open, and tomorrow has one specific task Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role tasks and the master checklist in Volume 1"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.



ULTIMATE CYBERSECURITY MASTER BLUEPRINT 2026-2027

Volume 5 — Resource Directory

169 entries, coverage matrix, gap analysis, playlist policy and verification register

Edition: Volume 5
Resource verification date: 2026-10-27
Plan window: 2026-10-27 to 2027-10-26 (shift dates if you start later)
System: 365 days, 12 academic months, 52 weeks + 1 graduation day
Profile: BS CS student in Pakistan; free/high-value resources; safe authorized labs

Academic-style self-study manual. No employment, salary, visa, certification or top-1% guarantee.

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Resource directory and verification **3**

Volume 5 — Resource directory, coverage matrix and verification register **3**

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Detailed resource cards (R001-R169) **21**

Resource directory and verification

169	162	7	0
verified registry entries	HTTP 200 responses	automated-blocker responses	404 links remaining

Full directory follows. Each card includes exact title, direct URL, type, topic/subtopic coverage, exact path, difficulty, time, access, prerequisites, selection rationale, post-study practice, backup, limitations, classification, age status and verification result.

Volume 5 — Resource directory, coverage matrix and verification register

Verification date: 2026-10-27. Every URL in the registry was checked with an automated HTTP request from this workspace. A `200` means the endpoint responded; it does not guarantee that a specific paid/free feature remains available in every country. A `403`, `429`, `202` or TLS issue is explicitly labelled as an automated-checker limitation and has a backup resource. No link was assumed to work simply because it looked plausible.

Selection principles

1. **Primary source bias.** Official documentation, university courses, standards and project-maintained labs outrank popularity.
2. **One primary per objective.** Supplementary material is used for alternate explanation and coverage gaps, not endless course collecting.
3. **Free first.** Paid resources are labelled and always paired with a free alternative.
4. **Current plus old-but-gold.** Older material is used only when the model/concept remains durable; dated commands and tool UIs must be checked against current docs.
5. **No invented timestamps.** For video, the PDF names the course, playlist, module or objective code. Use the chapters actually visible on the video page.
6. **Legal practice.** Lab platforms and vulnerable apps are used only in their authorized form.

Classification

- **Mandatory** — core to this curriculum.
- **Recommended** — fills important gaps or provides a better alternate explanation.
- **Optional enrichment** — useful when time/hardware allows.
- **Advanced reference** — deeper source for Year 2+; not a Year 1 mastery target.
- **Paid/alternative** — paid resource with a free replacement.

Coverage matrix and gaps

The matrix uses module letters A-K defined in Volume 1. ● = primary coverage; ○ = supporting/partial; blank = not intended. Exact supplementary links are in the resource cards.

Domain	Primary	Supplementary	Lab/practice	Authoritative reference	Assessment/ challenge	Main gap and fix
A CS/IT foundations	CS50x 2026 (F01)	OSTEP (F06), Nand2Tetris (F07)	VM setup (F12), Linux Journey (L01)	Python/Git/ MDN docs	CS50 problem sets	Hardware timing is advanced; F07 optional projects supply it
B Networking	Jeremy's IT Lab (N01)	Professor Messer	Packet Tracer (N15),	RFC 9293/791/820	Subnetting Practice (N16),	IPv6/security details need RFCs; N12 fills header gaps

Domain	Primary	Supplementary	Lab/practice	Authoritative reference	Assessment/ challenge	Main gap and fix
		N10-009 (N04)	Wireshark samples (N07)	0, IANA ports (N14)	Wireshark reports	
C Linux and Windows	Linux Journey (L01) + Microsoft docs (L07-L14)	BashGuide (L03), man7 (L04)	OverTheWire Bandit (L02), pwn.college (L15)	systemd/man-pages/ Windows Event Log docs	Bandit, event-log investigation	Full AD deployment needs 16 GB+; diagram-based alternative provided
D Programming	Python Tutorial (F08) + CS50 (F01)	Automate the Boring Stuff (P03), py4e (P01)	RegexOne (P04), pytest (P05), SQLite (P07)	Python/ Requests/Git docs	Month 4 log parser tool	Java/.NET secure coding not taught; OWASP cheat sheets are language-neutral
E Core security	CS50 Cyber (F04) + Professor Messer (C01)	NIST CSF/NIST SP docs	ISC2 free flashcards, Crypto 101 exercises	NIST CSF, NIST SP 800-63B, RFC 8446	Month 5 risk/ crypto practical	Formal proofs deferred to Stanford/graduate crypto resources
F Web/ API	PortSwigger Academy (W01/W02)	OWASP Top 10/API Top 10/ ASVS	PortSwigger labs, Juice Shop/WebGoat	MDN HTTP, OWASP Cheat Sheets	Apprentice labs + secure API project	Advanced request smuggling/GraphQL are Year 2+; W03 expert labs postponed
G Ethical testing	NIST SP 800-115 (E01)	OWASP WSTG (W06)	Local VMs, HTB Academy free modules	FIRST CVSS, CVE/NVD, HackerOne guidance	PRJ-10 scoped report	Real-world commercial scoping requires supervised experience
H Blue team	Microsoft Sentinel/KQL/ Sigma (B01/ B02/B03)	MITRE ATT&CK, NIST 800-61r3	CyberDefenders, BTLO, EVTX samples	Microsoft/ Sigma/Elastic official docs	PRJ-11/12	Paid SIEM ingestion avoided; synthetic/local logs used instead
I Cloud/ Infra	AWS Security Pillar + Docker/K8s docs	Microsoft Azure, OWASP K8s Top 10	Docker/kind/ minikube, Trivy, Checkov	AWS/Azure/ K8s/GitHub official docs	PRJ-13/14	No expensive managed cluster; local control-plane emulation has limits
J GRC	NIST CSF 2.0 (C06)	CIS Controls, ISO overview, GDPR text	Tabletop risk/IR exercises	NIST SP 800-30/34/53, EUR-Lex	M5/M12 risk register	Full ISO standard is paid; legal applicability needs a qualified adviser
K Advanced	pwn.college, OST2, MIT 6.858	Security Engineering, USENIX, Project Zero	pwn.college dojos, Malware Unicorn, Nightmare	Papers, standards, advisories	Year 2+ reproduction	Years of C/assembly/ systems prerequisites; not mastered in Year 1

Gap analysis: exact supplements

Missing/weak subtopic	Why primary course omits it	Exact supplement
CPU datapath and sequential logic	CS50 is broad and abstract	Nand2Tetris Part I Projects 1-3 (F07)
OS process/memory mechanisms	Intro courses emphasize usage	OSTEP Chapters 4-20 and 53-57 (F06)
IPv6 extension headers and modern TCP state machine	CCNA course is exam-scoped	RFC 8200 (N12) and RFC 9293 (N10)
Linux kernel/user API exact semantics	Linux Journey is beginner text	man7 pages (L04) and systemd.unit (L05)
Windows event schema and Sysmon	Intro admin courses focus on UI	Windows Event Log (L09), Sysmon (L11), EVTX samples (B05)

Missing/weak subtopic	Why primary course omits it	Exact supplement
Secure coding verification	PortSwigger is exploit-first	OWASP ASVS (C15) and Cheat Sheets (W17)
API object-level authorization	General OWASP Top 10 is broad	OWASP API Top 10 (C16) and local API project
Detection rule portability	SIEM docs are vendor-specific	SigmaHQ (B03/B04)
Cloud control catalog	AWS docs are service-specific	CSA CCM (CL16), NIST CSF (C06)
Privacy legal distinctions	Frameworks are voluntary	GDPR official text (C24) and NIST Privacy Framework (C25)
Research paper critique	Cert courses use summaries	MIT 6.858 (A06) and USENIX Security (A12)
Low-level RE prerequisites	Tool tutorials assume assembly	OST2 Arch1001 (A04) and pwn.college Linux Luminarium (A01)
AI application security	Classical web courses predate LLM apps	OWASP LLM Top 10 (A15) and NIST AI RMF (A16)

Do not waste time

- **Do not complete three beginner courses for the same topic.** Use one primary and one supplementary.
- **Do not watch entire playlists sequentially when a single objective is weak.** Use the course index by objective code.
- **Do not collect tools.** Master Wireshark, Nmap, Burp, Python, Git, Docker, KQL/Sigma and one cloud model deeply.
- **Do not buy exam vouchers for motivation.** Buy only after current objectives, free practice and a concrete job/education reason.
- **Do not buy SANS/GIAC, OSCP, BTL1 or HTB subscriptions in Month 1.** Their value depends on prior foundations and supervised practice.
- **Do not copy write-ups.** Struggle, hint, docs, then solution; always re-solve with a variation.
- **Do not deploy vulnerable apps publicly.** Use loopback/host-only/internal networks and snapshots.
- **Do not confuse framework compliance with security.** A control matrix is not evidence that a control works.

Lesser-known but credible resources

Resource	What it teaches better than a generic alternative	Prerequisite	Safe use
OSTEP (F06)	OS mechanisms, not just commands; security chapters added	C/Unix basics	Selected chapters, homework questions
OpenSecurityTraining2 (A03/A04)	x86-64 assembly from a systems-security viewpoint	C and binary arithmetic	Year 2+, disposable VM
pwn.college (A01/L15)	University-grade Linux, web, memory and exploitation challenge ladder	Linux/C/Python	Free dojo, browser workspace, no external targets
Practical Networking Subnetting Mastery (N17)	Visual subnetting method, avoids rote tricks	Binary/IP	Month 2 daily practice
Greg's Wiki BashGuide (L03)	Correct quoting and word-splitting model	Basic shell	Audit scripts only on own files
EVTX-ATTACK-SAMPLES (B05)	Realistic synthetic Windows telemetry for detection tests	Windows logs	Data only; isolated VM; do not execute binaries
SigmaHQ (B03/B04)	Vendor-neutral detection-as-code rules	Logs/ATT&CK	

Resource	What it teaches better than a generic alternative	Prerequisite	Safe use
			Write rules, test on synthetic events
Malware Traffic Analysis (B08)	Realistic PCAP exercise archive	Wireshark/TCP/IP	Never execute samples; use disposable VM
Security Engineering 3e (A08)	Technical, human and economic failure modes together	Broad security basics	One chapter/month in Year 2+
Nightmare / RPISEC MBE (A09/A10)	Progressive binary-exploitation challenge structure	C, assembly, GDB, Linux	Year 2-3 only, isolated VMs, modern mitigations checked
Project Zero / USENIX / CISA KEV	Root-cause research, peer-reviewed work, exploitation evidence	Advanced domain knowledge	Read and reproduce safe measurements only
SLSA / OpenSSF Scorecard / Semgrep	Concrete software supply-chain controls, not vague DevSecOps slogans	Git/CI	Use on sanitized public repos
Kubernetes Goat / OWASP K8s Top 10	Hands-on cluster misconfigurations without public cloud	Docker/K8s basics	Local disposable cluster, 16 GB+ RAM

YouTube publication-date policy

- Check the individual video's upload date on YouTube before citing it.
- If a course index lists an objective code/version, that version controls more than the upload date.
- **Old but gold** means core models remain valid (TCP, x86, crypto primitives, OS mechanisms), not that all commands/UI are current.
- Security+ SY0-701 videos are useful for concepts until V8/SY0-801 launches; do not use them as the sole exam prep after the version switch.
- Jeremy's IT Lab playlist originated in 2020 and has been augmented; verify current Cisco exam topics before certification.
- MIT 6.858 (2014) and Malware Unicorn RE101 (2019) are pedagogically strong but require updated tools and VM images.

Resource verification register — all 169 resources

Status is from the automated HTTP check on 2026-10-27. A 200 means reachable, not that every regional free/paid feature is guaranteed. 403/429/202 are automated-checker limits, not evidence that the official link is fake. No 404 links remain in the required registry; backups are provided.

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
F01	CS50x 2026 OpenCourseWare	https://cs50.harvard.edu/x/	A,D	Verified HTTP 200; endpoint reachable	Free; paid edX verified certificate optional	2026-10-27	Annual edition changes; use 2026 or current site	F02 — CS50x Lectures 2026 (official YouTube playlist)
F02	CS50x Lectures 2026 (official YouTube playlist)	https://www.youtube.com/playlist?list=PLhQjrBD2T380hlTqAU8HfvVepCcjCqTg6	A,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Playlist titles can be renumbered annually; match the official week page first	F01 — CS50x 2026 OpenCourseWare
F03	CS50x Sections 2026 (official YouTube playlist)	https://www.youtube.com/playlist?list=PLhQjrBD2T3839iqxqUdw7NWOCboh00vsV	A,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Older section videos may use previous tool versions	F02 — CS50x Lectures 2026 (official YouTube playlist)
F04	CS50's Introduction to Cybersecurity	https://cs50.harvard.edu/cybersecurity/	E	Verified HTTP 200; endpoint reachable	Free OCW; paid edX verified certificate optional	2026-10-27	High level; does not teach SOC tooling, exploitation, or secure coding in depth	F05 — CS50's Introduction to Cybersecurity (official YouTube playlist)
F05	CS50's Introduction to Cybersecurity (official YouTube playlist)	https://www.youtube.com/playlist?list=PLhQjrBD2T383Cqo5I1oRrbC1EKRAKGKUE	E	Verified HTTP 200; endpoint reachable	Free	2026-10-27	2023 publication date; core concepts remain valid, current product interfaces may differ	F04 — CS50's Introduction to Cybersecurity
F06	Operating Systems: Three Easy Pieces (OSTEP)	https://pages.cs.wisc.edu/~remzi/OSTEP/	A,C	Verified HTTP 200; endpoint reachable	Free online chapters; paid print optional	2026-10-27	Some examples assume Unix/C familiarity; do not attempt every chapter in Month 1	F01 — CS50x 2026 OpenCourseWare
F07	Nand2Tetris / The Elements of Computing Systems	https://www.nand2tetris.org/	A	Verified HTTP 200; endpoint reachable	Free software and course materials; book optional paid	2026-10-27	Large time commitment; use selected projects only	F06 — Operating Systems: Three Easy Pieces (OSTEP)
F08	Python 3 Official Tutorial	https://docs.python.org/3/tutorial/	A,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Reference style can be terse for absolute beginners	P02 — Learn Python - Full Course for Beginners (freeCodeCamp)
F09	Pro Git, 2nd Edition	https://git-scm.com/book/en/v2	A,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	GUI screenshots may differ from local tools	P01 — Python for Everybody
F10	MDN HTTP Overview	https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview	A,B,F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Some pages move as MDN reorganizes; use site search if a URL changes	W04 — MDN HTTP documentation
F11	SQLBolt	https://sqlbolt.com/	A,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	SQLite dialect differs slightly from	

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
							PostgreSQL or SQL Server	P08 — OWASP Secure Coding Practices Quick Reference Guide
F12	Oracle VirtualBox User Manual	https://www.virtualbox.org/manual/	A,C	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Version-specific menus may differ; use the manual matching your installed version	L07 — Microsoft Windows Server documentation
F13	Ubuntu Desktop Guide	https://help.ubuntu.com/	A,C	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Release-specific UI changes	L01 — Linux Journey
F14	Microsoft Learn Windows fundamentals	https://learn.microsoft.com/en-us/training/modules/explore-windows-client/	A,C	Verified HTTP 200; endpoint reachable	Free learning content; exams separate	2026-10-27	Product names and UI change; rely on current Microsoft Learn pages	L08 — PowerShell documentation
N01	Jeremy's IT Lab Free CCNA 200-301 Complete Course	https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ	B	Verified HTTP 200; endpoint reachable	Free on YouTube; optional paid course extras	2026-10-27	Playlist originated in 2020 and was augmented for v1.1; verify current Cisco blueprint before certification study	N02 — Jeremy's IT Lab CCNA Course Index and Files
N02	Jeremy's IT Lab CCNA Course Index and Files	https://courses.jeremysitlab.com/p/ccna	B	Verified HTTP 200; endpoint reachable	Free YouTube course; paid ad-free edition has bonus quizzes	2026-10-27	Paid edition is optional; free files are linked separately	N03 — Jeremy's IT Lab CCNA Course Files
N03	Jeremy's IT Lab CCNA Course Files	https://jitl.jp/ccna-files	B	Verified HTTP 200; endpoint reachable	Free	2026-10-27	External host may change; use N02 video description if unavailable	N02 — Jeremy's IT Lab CCNA Course Index and Files
N04	Professor Messer N10-009 Network+ Training Course	https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8	B	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Certification-oriented, not a complete packet-analysis course	N05 — Professor Messer N10-009 Course Index
N05	Professor Messer N10-009 Course Index	https://www.professormesser.com/network-plus/n10-009/n10-009-video/n10-009-training-course/	B	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Older N10-008 course also exists; ensure N10-009	N04 — Professor Messer N10-009 Network+ Training Course
N06	Wireshark User's Guide	https://www.wireshark.org/docs/wsug_html_chunked/	B,H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Version-specific UI; match your installed Wireshark version	N07 — Wireshark SampleCaptures
N07	Wireshark SampleCaptures	https://wiki.wireshark.org/SampleCaptures	B,H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Some linked external captures disappear; download from stable Wireshark-hosted mirrors where possible	N06 — Wireshark User's Guide
N08	tcpdump manual page	https://www.tcpdump.org/manpages/tcpdump.1.html	B,H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Root privileges may be needed; do not capture networks you do not own	N06 — Wireshark User's Guide

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
N09	Nmap Reference Guide	https://nmap.org/book/man.html	B,G	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Many options are not appropriate for beginners or public networks	N10 — RFC 9293: Transmission Control Protocol
N10	RFC 9293: Transmission Control Protocol	https://www.rfc-editor.org/rfc/rfc9293	B	Verified HTTP 200; endpoint reachable	Free	2026-10-27	RFC-level prose is dense; use after guided lessons	B10 — Suricata documentation
N11	RFC 791: Internet Protocol v4	https://www.rfc-editor.org/rfc/rfc791	B	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Historic RFC; use current clarifications for security specifics	N10 — RFC 9293: Transmission Control Protocol
N12	RFC 8200: Internet Protocol v6	https://www.rfc-editor.org/rfc/rfc8200	B	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Updated by later RFCs; use as baseline	N10 — RFC 9293: Transmission Control Protocol
N13	Cloudflare Learning Center: What is DNS?	https://www.cloudflare.com/learning/dns/what-is-dns/	B	Automated checker blocked (HTTP 403); official/canonical URL; use backup if browser access fails	Free	2026-10-27	Vendor educational content; cross-check record behavior in packet captures	B10 — Suricata documentation
N14	IANA Service Name and Transport Protocol Port Number Registry	https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml	B,G	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Registration is not proof a service uses that port in a particular environment	N10 — RFC 9293: Transmission Control Protocol
N15	Cisco Packet Tracer course page	https://www.netacad.com/courses/packet-tracer	B	Verified HTTP 200; endpoint reachable	Free with NetAcad account availability varies by region	2026-10-27	Registration and local availability may vary	N03 — Jeremy's IT Lab CCNA Course Files
N16	Subnetting Practice	https://subnettingpractice.com/	B	Verified HTTP 200; endpoint reachable	Free tier; optional ads/ premium features may appear	2026-10-27	Not a conceptual textbook; use only after the lesson	N17 — Practical Networking Subnetting Mastery
N17	Practical Networking Subnetting Mastery	https://www.practicalnetworking.net/stand-alone/subnetting-mastery/	B	Verified HTTP 200; endpoint reachable	Free articles/videos; optional paid resources	2026-10-27	Course packaging and optional product prices may change	N16 — Subnetting Practice
L01	Linux Journey	https://linuxjourney.com/	C	Automated checker blocked (HTTP 403); official/canonical URL; use backup if browser access fails	Free	2026-10-27	Does not replace distribution documentation or systemd internals	L02 — OverTheWire Bandit
L02	OverTheWire Bandit	https://overthewire.org/wargames/bandit/	C	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Occasional service availability; some late levels rely on Unix tricks	L01 — Linux Journey
L03	Greg's Wiki BashGuide	https://mywiki.wooledge.org/BashGuide	C,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Wiki can be terse; use as a reference while coding	P04 — RegexOne
L04	Linux man-pages project	https://man7.org/linux/man-pages/	C	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Distribution versions may differ; compare with local man pages	L01 — Linux Journey
L05	systemd.unit manual		C		Free	2026-10-27		

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
		https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html		Verified HTTP 200; endpoint reachable			Not all distributions use identical defaults	L04 — Linux man-pages project
L06	The Linux Foundation Introduction to Linux	https://www.edx.org/learn/linux/the-linux-foundation-introduction-to-linux	C	Verified HTTP 200; endpoint reachable	Free audit may be time-limited; verified certificate paid	2026-10-27	edX audit terms and course versions vary	L01 — Linux Journey
L07	Microsoft Windows Server documentation	https://learn.microsoft.com/en-us/windows-server/	C	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Broad documentation; use the daily plan's exact pages	L12 — Active Directory Domain Services overview
L08	PowerShell documentation	https://learn.microsoft.com/en-us/powershell/	C,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Windows PowerShell 5.1 and PowerShell 7 differ; check edition	L03 — Greg's Wiki BashGuide
L09	Windows Event Log documentation	https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log	C,H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Some APIs are developer-oriented; use Event Viewer for first practice	L11 — Sysmon documentation and download
L10	Microsoft Sysinternals	https://learn.microsoft.com/en-us/sysinternals/	C,H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Tools change; use current downloads and hashes	L08 — PowerShell documentation
L11	Sysmon documentation and download	https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon	H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Do not deploy on production systems without organizational approval	L09 — Windows Event Log documentation
L12	Active Directory Domain Services overview	https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview	C,E	Verified HTTP 200; endpoint reachable	Free	2026-10-27	UI and server version differ across environments	L07 — Microsoft Windows Server documentation
L13	Windows security baselines	https://learn.microsoft.com/en-us/windows/security/operating-system-security/device-management/windows-security-configuration-framework/windows-security-baselines	C,E	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Do not apply a baseline blindly to production or your primary computer	L14 — CIS Benchmarks
L14	CIS Benchmarks	https://www.cisecurity.org/cis-benchmarks	C,E,I,J	Verified HTTP 200; endpoint reachable	Free PDFs may require registration; some tools paid	2026-10-27	Registration and profile versions; not all benchmarks fit student hardware	L13 — Windows security baselines
L15	pwn.college Dojo index	https://pwn.college/dojos	A,C,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Dojo names and course instances change; use	L01 — Linux Journey

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
							current Core Material list	
P01	Python for Everybody	https://www.py4e.com/	D	Verified HTTP 200; endpoint reachable	Free site; Coursera specialization freemium	2026-10-27	Some videos are older; Python 3 syntax remains valid with minor updates	F08 — Python 3 Official Tutorial
P02	Learn Python - Full Course for Beginners (freeCodeCamp)	https://www.youtube.com/watch?v=eWRfhZUzrAc	D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Single video chapter times can change with re-uploads; use YouTube's displayed chapters	F08 — Python 3 Official Tutorial
P03	Automate the Boring Stuff with Python	https://automatetheboringstuff.com/	D	Verified HTTP 200; endpoint reachable	Free online; paid print/course optional	2026-10-27	Some third-party APIs or screenshots are dated; use current library docs	F08 — Python 3 Official Tutorial
P04	RegexOne	https://regexone.com/	D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Regex dialects differ; verify in Python re docs	F08 — Python 3 Official Tutorial
P05	pytest documentation	https://docs.pytest.org/en/stable/	D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Plugin compatibility varies	F08 — Python 3 Official Tutorial
P06	Requests: HTTP for Humans	https://requests.readthedocs.io/en/latest/	D,F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Never disable certificate verification except in a controlled local TLS lab	F10 — MDN HTTP Overview
P07	SQLite documentation	https://www.sqlite.org/docs.html	A,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	SQLite is not a substitute for learning PostgreSQL/MySQL administration	F11 — SQLBolt
P08	OWASP Secure Coding Practices Quick Reference Guide	https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/	D,F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Checklist, not a tutorial or complete standard	W01 — PortSwigger Web Security Academy
C01	Professor Messer SY0-701 Security+ Course Index	https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/	E,J	Verified HTTP 200; endpoint reachable	Free videos; paid notes/practice exams optional	2026-10-27	SY0-701 is V7; V8/SY0-801 launches Nov 17 2026, so use current objectives if certifying later	C02 — Professor Messer SY0-701 Security+ official playlist
C02	Professor Messer SY0-701 Security+ official playlist	https://www.youtube.com/playlist?list=PLG49S3nxzAnl4QDVqK-hOnoqcSKEIDDuv	E,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	V8 content will differ after Nov 2026	C01 — Professor Messer SY0-701 Security+ Course Index
C03	ISC2 Certified in Cybersecurity Exam Outline	https://www.isc2.org/certifications/cc/cc-certification-exam-outline	E,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Exam domains change; download current outline before paying	C04 — ISC2 CC self-study resources
C04	ISC2 CC self-study resources	https://www.isc2.org/certifications/cc/cc-self-study-resources	E,J	Verified HTTP 200; endpoint reachable	Free resources; official training paid	2026-10-27	Marketing pages may change; exam outline controls	C03 — ISC2 Certified in Cybersecurity Exam Outline

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
C05	ISC2 exam pricing	https://www.isc2.org/register-for-exam/isc2-exam-pricing	J	Verified HTTP 200; endpoint reachable	Free to view; exam paid	2026-10-27	Local taxes and currencies vary; price may change	C03 — ISC2 Certified in Cybersecurity Exam Outline
C06	NIST Cybersecurity Framework 2.0	https://www.nist.gov/cyberframework	E,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Framework is not law and not a certification	C21 — CIS Controls Version 8
C07	NIST SP 800-61 Revision 3	https://csrc.nist.gov/pubs/sp/800/61/r3/final	E,H,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Guidance is advisory for US federal systems; other jurisdictions may differ	C06 — NIST Cybersecurity Framework 2.0
C08	NIST SP 800-63B	https://pages.nist.gov/800-63-4/sp800-63b.html	E,F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Draft/revision status and version changes; use the page's current edition marker	C09 — Crypto 101
C09	Crypto 101	https://www.crypto101.io/	E	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Course is educational, not a substitute for vetted libraries or proofs	C10 — Stanford Cryptography I
C10	Stanford Cryptography I	https://www.coursera.org/learn/crypto	E	Verified HTTP 200; endpoint reachable	Free audit/lectures may be available; verified certificate and graded homework paid	2026-10-27	Older course; primitives remain important but verify current protocol guidance	C09 — Crypto 101
C11	A Graduate Course in Applied Cryptography	https://toc.cryptobook.us/	E	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Graduate-level; not a Month 5 mandatory read	C09 — Crypto 101
C12	Let's Encrypt documentation	https://letsencrypt.org/docs/	E,F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Vendor-specific ACME details; general PKI principles still apply	C13 — RFC 8446: TLS 1.3
C13	RFC 8446: TLS 1.3	https://www.rfc-editor.org/rfc/rfc8446	E,F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Dense standard; use simplified material first	C12 — Let's Encrypt documentation
C14	OWASP Top 10	https://owasp.org/www-project-top-ten/	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	2021 edition; check current release before citing year in interviews	W01 — PortSwigger Web Security Academy
C15	OWASP Application Security Verification Standard	https://owasp.org/www-project-application-security-verification-standard/	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Formal standard; not all levels apply to student projects	P08 — OWASP Secure Coding Practices Quick Reference Guide
C16	OWASP API Security Top 10	https://owasp.org/API-Security/editions/2023/en/0x11-t10/	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	2023 edition; verify current release later	C14 — OWASP Top 10
C17	OWASP Threat Modeling community page	https://owasp.org/www-community/Threat_Modeling	E,F,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Community page is an overview, not a complete methodology	C18 — MITRE ATT&CK
C18	MITRE ATT&CK	https://attack.mitre.org/	E,H,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Not a maturity model or proof of detection coverage	C19 — MITRE D3FEND
C19	MITRE D3FEND	https://d3fend.mitre.org/	E,H,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Research project; not all entries have turnkey implementations	C18 — MITRE ATT&CK

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C20	NIST NICE Workforce Framework	https://www.nist.gov/itl/applied-cybersecurity/nice	J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	US-origin framework; employers use many different vocabularies	C06 — NIST Cybersecurity Framework 2.0
C21	CIS Controls Version 8	https://www.cisecurity.org/controls/v8	E,J	Verified HTTP 200; endpoint reachable	Free overview; some tools paid	2026-10-27	Not regulation; legal obligations depend on jurisdiction	C06 — NIST Cybersecurity Framework 2.0
C22	NIST SP 800-53 Revision 5	https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final	J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Large catalog; use search and mappings	C06 — NIST Cybersecurity Framework 2.0
C23	ISO/IEC 27001 overview	https://www.iso.org/standard/27001	J	Automated checker blocked (HTTP 403); official/canonical URL; use backup if browser access fails	Page free; standard document paid	2026-10-27	Full standard is not freely distributable; do not rely on unauthorized copies	C06 — NIST Cybersecurity Framework 2.0
C24	GDPR full text (EUR-Lex)	https://eur-lex.europa.eu/eli/reg/2016/679/oj	J	Accepted/asynchronous response (HTTP 202); official legal text; use backup if browser fails	Free	2026-10-27	EU regulation, not global law; seek legal advice for actual obligations	C25 — NIST Privacy Framework
C25	NIST Privacy Framework	https://www.nist.gov/privacy-framework	J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Voluntary framework, not legal compliance proof	C06 — NIST Cybersecurity Framework 2.0
C26	NIST SP 800-30 Revision 1	https://csrc.nist.gov/pubs/sp/800/30/r1/final	J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	US federal guidance; terminology can be adapted	C06 — NIST Cybersecurity Framework 2.0
C27	NIST SP 800-34 Revision 1	https://csrc.nist.gov/pubs/sp/800/34/r1/final	J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	US federal guidance; adapt to local context	C06 — NIST Cybersecurity Framework 2.0
W01	PortSwigger Web Security Academy	https://portswigger.net/web-security	F	Verified HTTP 200; endpoint reachable	Free account required	2026-10-27	Labs require a free account and internet; do not target real public sites	W02 — PortSwigger Web Security Academy Learning Paths
W02	PortSwigger Web Security Academy Learning Paths	https://portswigger.net/web-security/learning-paths	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Path names and lab counts can change	W01 — PortSwigger Web Security Academy
W03	PortSwigger All Labs index	https://portswigger.net/web-security/all-labs	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Expert labs can be frustrating early; postpone them	W02 — PortSwigger Web Security Academy Learning Paths
W04	MDN HTTP documentation	https://developer.mozilla.org/en-US/docs/Web/HTTP	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Large documentation set; use site search and assigned pages	F10 — MDN HTTP Overview
W05	Burp Suite documentation	https://portswigger.net/burp/documentation	F,G	Verified HTTP 200; endpoint reachable	Community edition free; Pro paid	2026-10-27	Some features require paid Pro edition	W01 — PortSwigger Web Security Academy
W06	OWASP Web Security Testing Guide	https://owasp.org/www-project-web-security-testing-guide/	F,G	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Guide is not a step-by-step beginner tutorial	W01 — PortSwigger Web Security Academy
W07	OWASP ZAP documentation	https://www.zaproxy.org/docs/	F,G	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Automated scans do not prove exploitability or absence of bugs	W05 — Burp Suite documentation

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W08	OWASP Juice Shop	https://owasp.org/www-project-juice-shop/	F,G	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Never expose intentionally vulnerable software to the public internet	W09 — OWASP Juice Shop GitHub repository
W09	OWASP Juice Shop GitHub repository	https://github.com/juice-shop/juice-shop	F,G	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Development branch may change; use official tags or documented release	W08 — OWASP Juice Shop
W10	DVWA GitHub repository	https://github.com/digininja/DVWA	F,G	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Outdated interface and some PHP dependencies; do not expose publicly	W09 — OWASP Juice Shop GitHub repository
W11	OWASP WebGoat	https://owasp.org/www-project-webgoat/	F,G	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Requires Java/container resources; use selectively	W01 — PortSwigger Web Security Academy
W12	OWASP Dependency-Check	https://owasp.org/www-project-dependency-check/	F,I	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	NVD API rate limits and database bootstrap can be slow; use offline sample report if blocked	W15 — OpenSSF Scorecard
W13	GitHub code security documentation	https://docs.github.com/en/code-security	F,I	Verified HTTP 200; endpoint reachable	Free features vary by repository visibility and plan	2026-10-27	Feature availability differs between public/private/enterprise plans	W14 — Semgrep documentation
W14	Semgrep documentation	https://semgrep.dev/docs/	F,I	Verified HTTP 200; endpoint reachable	Open-source engine free; some platform features paid	2026-10-27	Cloud platform and engine features differ	W13 — GitHub code security documentation
W15	OpenSSF Scorecard	https://securityscorecards.dev/	F,I	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Not all checks apply to small student projects	W13 — GitHub code security documentation
W16	SLSA specification	https://slsa.dev/	F,I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Specification evolves; use current level documentation	W13 — GitHub code security documentation
W17	OWASP Cheat Sheet Series	https://cheatsheetseries.owasp.org/	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Cheat sheets can be dense; use them after hands-on practice	C15 — OWASP Application Security Verification Standard
E01	NIST SP 800-115	https://csrc.nist.gov/pubs/sp/800/115/final	G	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Published 2008; methodology remains valid but tool examples are outdated	W06 — OWASP Web Security Testing Guide
E02	TryHackMe Pre Security learning path	https://tryhackme.com/path/outline/presecurity	A,B,E,G	Rate-limited automated check (HTTP 429); official URL; retry in browser or use backup	Freemium; introductory rooms free, full access may require subscription	2026-10-27	Free-room availability and path structure changed in 2026; verify before buying	L01 — Linux Journey
E03	TryHackMe Cyber Security 101 path	https://tryhackme.com/path/outline/cybersecurity101	E,G	Rate-limited automated check (HTTP 429); official URL; retry in browser or use backup	Freemium; many rooms subscription-only	2026-10-27	Path content and free/paid status can change	E01 — NIST SP 800-115
E04	Hack The Box Academy	https://academy.hackthebox.com/	G,K	Verified HTTP 200; endpoint reachable	Freemium; some modules free, others require cubes/subscription	2026-10-27	Pricing and free-module availability vary	N09 — Nmap Reference Guide

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
E05	HackerOne Disclosure Guidelines	https://www.hackerone.com/disclosure-guidelines	G,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Platform policy is not universal law and does not grant authorization for arbitrary targets	E06 — FIRST CVSS specification
E06	FIRST CVSS specification	https://www.first.org/cvss/	G,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	CVSS is not a full risk assessment	E08 — NVD
E07	CVE Program	https://www.cve.org/	G,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	CVE assignment itself does not prove exploitability or priority	E08 — NVD
E08	NVD	https://nvd.nist.gov/	G,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	NVD enrichment can lag; API rate limits apply	N09 — Nmap Reference Guide
E09	picoCTF	https://picoctf.org/	D,G,K	Verified HTTP 200; endpoint reachable	Free competition/ picoGym account	2026-10-27	Competition archive availability and scoreboards change	L15 — pwn.college Dojo index
E10	CyLab Security Academy / picoGym	https://www.cylabacademy.org/	D,G,K	Verified HTTP 200; endpoint reachable	Free account	2026-10-27	Practice catalog may change	E09 — picoCTF
B01	Microsoft Sentinel documentation	https://learn.microsoft.com/en-us/azure/sentinel/	H	Verified HTTP 200; endpoint reachable	Docs free; Azure/ Sentinel usage may incur costs	2026-10-27	Product names and UI change; cloud cost risk	B02 — Kusto Query Language overview
B02	Kusto Query Language overview	https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric	H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Syntax and service-specific tables vary by product	B03 — SigmaHQ
B03	SigmaHQ	https://sigmahq.io/	H,K	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Rule quality varies; always validate in your own data	B18 — CISA Known Exploited Vulnerabilities Catalog
B04	Sigma rule creation guide	https://sigmahq.io/docs/guide/getting-started.html	H,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Documentation URL or schema can change; use site navigation	B03 — SigmaHQ
B05	SBousseaden EVTX-ATTACK-SAMPLES	https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES	H,K	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Third-party repository; verify contents and do not execute unknown binaries; EVTX files are data	B02 — Kusto Query Language overview
B06	CyberDefenders	https://cyberdefenders.org/	H,K	Verified HTTP 200; endpoint reachable	Freemium; some challenges free, some require subscription	2026-10-27	Free/paid status changes; some datasets are large	N07 — Wireshark SampleCaptures
B07	Blue Team Labs Online	https://blueteamlabs.online/	H,K	Verified HTTP 200; endpoint reachable	Freemium; free and paid labs	2026-10-27	Paid tier/lab catalogue changes; lab platform login required	B06 — CyberDefenders
B08	Malware Traffic Analysis	https://malware-traffic-analysis.net/	B,H,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Contains malware-related artifacts; never execute samples; some links may be old	N07 — Wireshark SampleCaptures
B09	Zeek documentation	https://docs.zeek.org/en/master/	B,H	Rate-limited automated check (HTTP 429);	Free/open-source	2026-10-27	Installation and version differences; not needed on low-end hardware	B10 — Suricata documentation

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
				official URL; retry in browser or use backup				
B10	Suricata documentation	https://docs.suricata.io/en/latest/	B,H	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Rule syntax and version-specific features; use matching docs version	B03 — SigmaHQ
B11	Autopsy documentation	https://www.autopsy.com/support/	H,K	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Older screenshots may differ from current release; never analyze personal disks without permission	B12 — SANS SIFT Workstation
B12	SANS SIFT Workstation	https://www.sans.org/tools/sift-workstation/	H,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Large download and Linux familiarity required; SANS courses themselves are paid	B11 — Autopsy documentation
B13	MISP Project	https://www.misp-project.org/	H,K	Verified HTTP 200; endpoint reachable	Free/open-source; hosting costs possible	2026-10-27	Do not upload personal or unauthorized data; local hosting can be heavy	C18 — MITRE ATT&CK
B14	Elastic detection rules documentation	https://www.elastic.co/guide/en/security/current/detection-engine-overview.html	H,K	Verified HTTP 200; endpoint reachable	Free docs/open-source rules; Elastic Stack may have paid features	2026-10-27	Product UI and rule schema change by version	B03 — SigmaHQ
B15	Microsoft SC-200 certification page	https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/	H,J	Verified HTTP 200; endpoint reachable	Free learning content; exam price region-based	2026-10-27	Exam objectives were refreshed in 2026; check current study guide	B01 — Microsoft Sentinel documentation
B16	Blue Team Level 1 official certification	https://www.centri.org/certifications/blue-team-level-1	H,J	Verified HTTP 200; endpoint reachable	Paid; GBP 399 listed on official page as checked 2026-10-27; one free resit included	2026-10-27	Price, provider brand and access terms may change; not required for this roadmap	B06 — CyberDefenders
B17	CISA cybersecurity advisories	https://www.cisa.gov/news-events/cybersecurity-advisories	H,J,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	US government source; not exhaustive global coverage	C18 — MITRE ATT&CK
B18	CISA Known Exploited Vulnerabilities Catalog	https://www.cisa.gov/known-exploited-vulnerabilities-catalog	G,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	US federal remediation deadlines do not automatically bind Pakistani or private organizations	E08 — NVD
B19	SC-200 Exam Prep YouTube playlist	https://www.youtube.com/playlist?list=PLahhVEj9XNTcHNtYH-inn7xFJactEM7zz	H,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Certification content can change; use current study guide first	B15 — Microsoft SC-200 certification page
CL01	AWS Skill Builder	https://skillbuilder.aws/	I,J	Verified HTTP 200; endpoint reachable	Freemium; many digital courses free, labs/ subscription can be paid	2026-10-27	Catalog and free/paid status changes	CL02 — AWS security documentation

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CL02	AWS security documentation	https://docs.aws.amazon.com/security/	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Very broad; use search rather than linear reading	CL03 — AWS Well-Architected Security Pillar
CL03	AWS Well-Architected Security Pillar	https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html	I,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	AWS-specific; other clouds need their own architecture guidance	CL02 — AWS security documentation
CL04	AWS Free Tier	https://aws.amazon.com/free/	I	Verified HTTP 200; endpoint reachable	Program terms; some services free temporarily, others always free	2026-10-27	Terms change; credits/expiry vary; a credit card may still be required	CL01 — AWS Skill Builder
CL05	Microsoft Azure security documentation	https://learn.microsoft.com/en-us/azure/security/	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Product names and services change	CL02 — AWS security documentation
CL06	Docker documentation	https://docs.docker.com/	C,I	Verified HTTP 200; endpoint reachable	Free; Docker Desktop licensing may apply in some organizations	2026-10-27	Menu changes and Docker Desktop license terms differ from Docker Engine	CL07 — Docker Engine security documentation
CL07	Docker Engine security documentation	https://docs.docker.com/engine/security/	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Host kernel distribution affects defaults	CL06 — Docker documentation
CL08	Kubernetes official documentation	https://kubernetes.io/docs/home/	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Version-specific APIs; use supported version docs	CL09 — Kubernetes security documentation
CL09	Kubernetes security documentation	https://kubernetes.io/docs/concepts/security/	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Some features require cluster configuration that kind/minikube may not fully emulate	CL08 — Kubernetes official documentation
CL10	OWASP Kubernetes Top 10	https://owasp.org/www-project-kubernetes-top-ten/	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Checklist, not a deployment tutorial	CL09 — Kubernetes security documentation
CL11	Kubernetes Goat	https://madhuakula.com/kubernetes-goat/	I,K	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Can consume significant RAM; skip if hardware is below 16 GB	CL10 — OWASP Kubernetes Top 10
CL12	Trivy documentation	https://trivy.dev/docs/	I	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Vulnerability DB freshness and false positives; no scanner proves security	W12 — OWASP Dependency-Check
CL13	Checkov documentation	https://www.checkov.io/1.Welcome/What%20is%20Checkov.html	I	Verified HTTP 200; endpoint reachable	Free/open-source; Prisma Cloud features paid	2026-10-27	Policy coverage varies by framework and version	CL12 — Trivy documentation
CL14	GitHub Actions security hardening	https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Features differ for public/private/enterprise repos	W13 — GitHub code security documentation

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
		hardening-for-github-actions						
CL15	Terraform documentation	https://developer.hashicorp.com/terraform/docs	I	Verified HTTP 200; endpoint reachable	Free/open-source; HCP Terraform has paid tiers	2026-10-27	State files can contain secrets; never commit them	CL13 — Checkov documentation
CL16	Cloud Security Alliance Cloud Controls Matrix	https://cloudsecurityalliance.org/research/cloud-controls-matrix	I,J	Verified HTTP 200; endpoint reachable	Free registration may be required for some downloads	2026-10-27	Registration and version changes; not law	CL03 — AWS Well-Architected Security Pillar
CL17	AWS Certified Security - Specialty official page	https://aws.amazon.com/certification/certified-security-specialty/	I,J	Verified HTTP 200; endpoint reachable	Exam US$300 listed 2026-10-27; training optional/paid	2026-10-27	Region/language and recertification policies change; not entry-level	CL01 — AWS Skill Builder
CL18	AWS recertification page	https://aws.amazon.com/certification/recertification/	J	Verified HTTP 200; endpoint reachable	Free to read; exam/maintenance may cost	2026-10-27	Two official AWS pages describe policy in different detail; verify in your certification account	CL17 — AWS Certified Security - Specialty official page
A01	pwn.college	https://pwn.college/	A,C,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Course instance names change yearly; use Core Material	L15 — pwn.college Dojo index
A02	pwn.college dojos index	https://pwn.college/dojos	A,C,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Dojo catalogue changes; links may require login	A01 — pwn.college
A03	OpenSecurityTraining2	https://ost2.fyi/	A,K	Verified HTTP 200; endpoint reachable	Free/open-source course materials	2026-10-27	Many classes are long and assume C; do not attempt in Month 1	A01 — pwn.college
A04	OST2 Architecture 1001: x86-64 Assembly	https://p.ost2.fyi/courses/course-v1:OpenSecurityTraining2+Arch1001_x86-64_Asm+2021_v1/about	A,K	Verified HTTP 200; endpoint reachable	Free; optional certificate mechanics may vary	2026-10-27	Older recordings but durable architecture concepts; verify tools on current OS	A03 — OpenSecurityTraining2
A05	Malware Unicorn Reverse Engineering 101	https://malwareunicorn.org/workshops/re101.html	K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	2019 VM images and tools are old; use current official installers and isolated networking	A04 — OST2 Architecture 1001: x86-64 Assembly
A06	MIT 6.858 Computer Systems Security OCW	https://ocw.mit.edu/courses/6-858-computer-systems-security-fall-2014/	E,F,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	2014 material; timeless foundations but obsolete tool/browser specifics	A07 — MIT 6.858 official lecture playlist
A07	MIT 6.858 official lecture playlist	https://www.youtube.com/playlist?list=PLUl4u3cNGP62K2DjQLRxDNRi0z2IRWnNh	E,F,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	2014 publication; update all practical tool details	A06 — MIT 6.858 Computer Systems Security OCW
A08	Security Engineering, Third Edition (free online chapters)	https://www.cl.cam.ac.uk/~rja14/book.html	E,J,K	Verified HTTP 200; endpoint reachable	Free online draft; print edition paid	2026-10-27	Third-edition draft chapters may be updated; not a certification cram book	C06 — NIST Cybersecurity Framework 2.0

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
A09	Nightmare binary exploitation course	https://guyinatuxedo.github.io/	K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Advanced and sometimes terse; not a Year 1 mastery target	A01 — pwn.college
A10	RPISEC Modern Binary Exploitation	https://github.com/RPISEC/MBE	K	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Older VM images and mitigations; modern mitigations require updated setup	A01 — pwn.college
A11	Google Project Zero blog	https://googleprojectzero.blogspot.com/	K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Advanced; many posts require specialized background	A12 — USENIX Security conference papers
A12	USENIX Security conference papers	https://www.usenix.org/conferences/byname/164	K	Verified HTTP 200; endpoint reachable	Many papers freely available; conference attendance paid	2026-10-27	Not every paper includes code; reproduction may be hard	A11 — Google Project Zero blog
A13	arXiv cs.CR recent papers	https://arxiv.org/list/cs.CR/recent	K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Preprints are not necessarily peer reviewed; quality varies	A12 — USENIX Security conference papers
A14	xairy/linux-kernel-exploitation	https://github.com/xairy/linux-kernel-exploitation	K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Dangerous material if misunderstood; use isolated research VMs only	A01 — pwn.college
A15	OWASP Top 10 for LLM Applications	https://genai.owasp.org/llm-top-10/	F,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Versioned document; verify current edition before citing	C16 — OWASP API Security Top 10
A16	NIST AI Risk Management Framework	https://www.nist.gov/itl/ai-risk-management-framework	J,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Not law and not an appsec testing checklist	A15 — OWASP Top 10 for LLM Applications
A17	Cisco CCST Cybersecurity official exam page	https://www.cisco.com/site/us/en/learn/training-certifications/exams/ccst-cybersecurity.html	B,E,J	Verified HTTP 200; endpoint reachable	Exam paid US$125 listed; free Cisco training linked	2026-10-27	Cisco certification names/pathways are changing in 2026; verify current page and 5-year renewal rules	E01 — NIST SP 800-115
A18	CompTIA Security+ V7 and V8 official pages	https://www.comptia.org/en-us/certifications/security/	E,J	Verified HTTP 200; endpoint reachable	Paid exam; V7 US$404 from CompTIA blog dated January 2025; verify current store	2026-10-27	V7 retirement English 2027-06-11; V8 launch November 2026; do not buy stale practice material	C01 — Professor Messer SY0-701 Security+ Course Index
A19	ISC2 Certified in Cybersecurity official page	https://www.isc2.org/Certifications/CC	E,J	Verified HTTP 200; endpoint reachable	Exam paid; US$199 listed on official pricing page 2026-10-27; occasional free promotions may exist	2026-10-27	Annual maintenance fee after certification; exact local price/tax at checkout	C03 — ISC2 Certified in Cybersecurity Exam Outline
A20	Microsoft SC-900 official certification page	https://learn.microsoft.com/en-us/credentials/certifications/security-compliance-and-identity-fundamentals/	E,J	Verified HTTP 200; endpoint reachable	Free Learn training; exam price region-based	2026-10-27	Price at checkout depends on country; event vouchers are not guaranteed	B15 — Microsoft SC-200 certification page
A21			G,K			2026-10-27		E01 — NIST SP 800-115

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
	OffSec PEN-200 / OSCP+ official page	https://www.offsec.com/courses/pen-200/		Verified HTTP 200; endpoint reachable	Paid; official page lists US$1749 course+exam bundle and US$1699 standalone exam on 2026-10-27		High cost, renewal/maintenance conditions and exam format can change; free alternatives exist	
A22	INE eJPT official certification page	https://ine.com/security/certifications/ejpt-certification	G,K	Verified HTTP 200; endpoint reachable	Paid; official page states an INE subscription and exam voucher are needed; checkout lists voucher/bundle prices	2026-10-27	Official pricing is split between subscription and voucher; checkout promotions change; certification valid three years	E01 — NIST SP 800-115
A23	KQL Essentials official Azure Data Explorer playlist	https://www.youtube.com/playlist?list=PLKHIhZJiQ5AkOMEC54mQQcAkfg4UXT8uK	H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Individual publication dates vary; services and UI can change	B02 — Kusto Query Language overview
A24	SANS DFIR playlist	https://www.youtube.com/playlist?list=PLrY_AbzZGqt9MJiLjyH-zTrypGLs154e4	H,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Some SANS courses are paid; webcasts may be older; do not confuse a webcast with certification	B11 — Autopsy documentation

Detailed resource cards (R001-R169)

F01 · CS50x 2026 OpenCourseWare

Exact title	CS50x 2026 OpenCourseWare
Direct URL	https://cs50.harvard.edu/x/
Resource type / domains	University course · Modules A,D
Topic and subtopic coverage	Best single academic bridge from programming and CS fundamentals to Python, SQL, web and systems thinking
Exact path / module / chapter / playlist	Weeks 0-10; problem sets; final project
Recommended difficulty / estimated time	Beginner to intermediate · 11 weeks part-time
Access status	Free; paid edX verified certificate optional
Prerequisites	Basic computer literacy
Why selected	Best single academic bridge from programming and CS fundamentals to Python, SQL, web and systems thinking
What to practise after	Submit selected problem sets from Weeks 0,1,4,6,7,9 as scheduled in the daily plan
Backup if inaccessible	F02 — CS50x Lectures 2026 (official YouTube playlist)
Known limitations	Annual edition changes; use 2026 or current site
Classification / age status	Mandatory · Current 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F02 · CS50x Lectures 2026 (official YouTube playlist)

Exact title	CS50x Lectures 2026 (official YouTube playlist)
Direct URL	https://www.youtube.com/playlist?list=PLhQjrBD2T380hlTqAU8HfvVepCcjCqTg6
Resource type / domains	YouTube playlist · Modules A,D
Topic and subtopic coverage	Harvard's current lecture sequence with chapters and transcripts on the course site
Exact path / module / chapter / playlist	Lecture 0 through Lecture 10; use exact lecture titles shown on YouTube
Recommended difficulty / estimated time	Beginner to intermediate · 25 hours video
Access status	Free
Prerequisites	None
Why selected	Harvard's current lecture sequence with chapters and transcripts on the course site
What to practise after	Watch one lecture segment, then reproduce every demonstrated command or code snippet
Backup if inaccessible	F01 — CS50x 2026 OpenCourseWare
Known limitations	Playlist titles can be renumbered annually; match the official week page first
Classification / age status	Recommended · 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F03 · CS50x Sections 2026 (official YouTube playlist)

Exact title	CS50x Sections 2026 (official YouTube playlist)
Direct URL	https://www.youtube.com/playlist?list=PLhQjrBD2T3839iqxqUdw7NWOCboh00vsV
Resource type / domains	YouTube playlist · Modules A,D
Topic and subtopic coverage	Short worked examples make abstract lectures easier to translate into code
Exact path / module / chapter / playlist	Sections: C, Arrays, Algorithms, Memory, Data Structures, Python, SQL, HTML/CSS/JS, Flask
Recommended difficulty / estimated time	Beginner to intermediate · 12-18 hours
Access status	Free
Prerequisites	CS50 lecture or equivalent
Why selected	Short worked examples make abstract lectures easier to translate into code
What to practise after	Code along without looking, then delete and recreate from memory
Backup if inaccessible	F02 — CS50x Lectures 2026 (official YouTube playlist)
Known limitations	Older section videos may use previous tool versions
Classification / age status	Optional enrichment · 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F04 · CS50's Introduction to Cybersecurity

Exact title	CS50's Introduction to Cybersecurity
Direct URL	https://cs50.harvard.edu/cybersecurity/
Resource type / domains	University course · Modules E
Topic and subtopic coverage	Risk-based, technically honest introduction to security trade-offs
Exact path / module / chapter / playlist	Weeks 0-4: Securing Accounts, Data, Systems, Software, Privacy; final project
Recommended difficulty / estimated time	Beginner · 5 weeks at 2-6 h/week
Access status	Free OCW; paid edX verified certificate optional
Prerequisites	None
Why selected	Risk-based, technically honest introduction to security trade-offs
What to practise after	Complete each week's assignment and one final project; do not use it as a substitute for hands-on labs
Backup if inaccessible	F05 — CS50's Introduction to Cybersecurity (official YouTube playlist)
Known limitations	High level; does not teach SOC tooling, exploitation, or secure coding in depth
Classification / age status	Mandatory · Current site; 2023 lecture recordings
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F05 · CS50's Introduction to Cybersecurity (official YouTube playlist)

Exact title	CS50's Introduction to Cybersecurity (official YouTube playlist)
Direct URL	https://www.youtube.com/playlist?list=PLhQjrBD2T383Cqo5I1oRrbC1EKRAKGKUE
Resource type / domains	YouTube playlist · Modules E
Topic and subtopic coverage	Easy to watch in segments and cross-reference with official notes
Exact path / module / chapter / playlist	Five lectures corresponding to Weeks 0-4
Recommended difficulty / estimated time	Beginner · 8-10 hours
Access status	Free
Prerequisites	None
Why selected	Easy to watch in segments and cross-reference with official notes
What to practise after	After each lecture, write a one-page risk trade-off analysis
Backup if inaccessible	F04 — CS50's Introduction to Cybersecurity
Known limitations	2023 publication date; core concepts remain valid, current product interfaces may differ
Classification / age status	Recommended · Old but gold (2023)
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F06 · Operating Systems: Three Easy Pieces (OSTEP)

Exact title	Operating Systems: Three Easy Pieces (OSTEP)
Direct URL	https://pages.cs.wisc.edu/~remzi/OSTEP/
Resource type / domains	Free textbook · Modules A,C
Topic and subtopic coverage	University-grade explanation of processes, virtualization, concurrency, persistence and security fundamentals
Exact path / module / chapter / playlist	Chapters on Processes, Process API, Direct Execution, Scheduling, Address Spaces, Memory API, Translation, Files and Directories, Security chapters
Recommended difficulty / estimated time	Intermediate · Selected chapters 15-25 h
Access status	Free online chapters; paid print optional
Prerequisites	Basic C/Python and computer architecture
Why selected	University-grade explanation of processes, virtualization, concurrency, persistence and security fundamentals
What to practise after	Draw mechanism diagrams and answer end-of-chapter questions; reproduce tiny C examples where available
Backup if inaccessible	F01 — CS50x 2026 OpenCourseWare
Known limitations	Some examples assume Unix/C familiarity; do not attempt every chapter in Month 1
Classification / age status	Recommended · v1.10, 2023
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F07 · Nand2Tetris / The Elements of Computing Systems

Exact title	Nand2Tetris / The Elements of Computing Systems
Direct URL	https://www.nand2tetris.org/
Resource type / domains	Free course and software · Modules A
Topic and subtopic coverage	Builds a concrete mental model of how logic gates become a CPU
Exact path / module / chapter / playlist	Part I projects 1-5: Boolean logic, Boolean arithmetic, sequential logic, machine language, computer architecture
Recommended difficulty / estimated time	Intermediate · 20-40 h for Part I
Access status	Free software and course materials; book optional paid
Prerequisites	Basic programming and binary arithmetic
Why selected	Builds a concrete mental model of how logic gates become a CPU
What to practise after	Complete Projects 1-3 or a software simulation if hardware simulators are blocked
Backup if inaccessible	F06 — Operating Systems: Three Easy Pieces (OSTEP)
Known limitations	Large time commitment; use selected projects only
Classification / age status	Optional enrichment · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F08 · Python 3 Official Tutorial

Exact title	Python 3 Official Tutorial
Direct URL	https://docs.python.org/3/tutorial/
Resource type / domains	Official documentation · Modules A,D
Topic and subtopic coverage	Canonical, current Python syntax and standard library semantics
Exact path / module / chapter / playlist	Sections 1-10: interpreter, control flow, functions, data structures, modules, input/output, errors, classes, standard library
Recommended difficulty / estimated time	Beginner to intermediate · 12-18 h
Access status	Free
Prerequisites	None
Why selected	Canonical, current Python syntax and standard library semantics
What to practise after	Type each example locally, then write a variation without copying
Backup if inaccessible	P02 — Learn Python - Full Course for Beginners (freeCodeCamp)
Known limitations	Reference style can be terse for absolute beginners
Classification / age status	Mandatory · Current documentation
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F09 · Pro Git, 2nd Edition

Exact title	Pro Git, 2nd Edition
Direct URL	https://git-scm.com/book/en/v2
Resource type / domains	Free book / official reference · Modules A,D
Topic and subtopic coverage	Authoritative Git model rather than memorized button clicks
Exact path / module / chapter / playlist	Chapters 1-3 and 7: getting started, basics, branching, selected tools
Recommended difficulty / estimated time	Beginner to intermediate · 8-12 h
Access status	Free
Prerequisites	Command line basics
Why selected	Authoritative Git model rather than memorized button clicks
What to practise after	Create a repository, branch, merge, resolve a deliberate conflict, and publish a clean README
Backup if inaccessible	P01 — Python for Everybody
Known limitations	GUI screenshots may differ from local tools
Classification / age status	Mandatory · Classic, current model
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F10 · MDN HTTP Overview

Exact title	MDN HTTP Overview
Direct URL	https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview
Resource type / domains	Official web documentation · Modules A,B,F
Topic and subtopic coverage	Clear browser-vendor documentation for HTTP semantics and security-relevant headers
Exact path / module / chapter / playlist	HTTP basics, requests, responses, methods, headers, status codes, cookies, caching
Recommended difficulty / estimated time	Beginner · 2-4 h
Access status	Free
Prerequisites	Basic networking vocabulary
Why selected	Clear browser-vendor documentation for HTTP semantics and security-relevant headers
What to practise after	Inspect requests in browser DevTools and map each header to MDN
Backup if inaccessible	W04 — MDN HTTP documentation
Known limitations	Some pages move as MDN reorganizes; use site search if a URL changes
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F11 · SQLBolt

Exact title	SQLBolt
Direct URL	https://sqlbolt.com/
Resource type / domains	Interactive lessons · Modules A,D
Topic and subtopic coverage	Short, interactive SQL drills with immediate feedback suitable for a first database foundation
Exact path / module / chapter / playlist	Lessons 1-18: SELECT, filtering, joins, aggregation, order, inserts, updates, deletes, schema
Recommended difficulty / estimated time	Beginner · 4-8 h
Access status	Free
Prerequisites	None
Why selected	Short, interactive SQL drills with immediate feedback suitable for a first database foundation
What to practise after	Build a synthetic security-events table and write SELECT, WHERE, GROUP BY, JOIN and UPDATE exercises
Backup if inaccessible	P08 — OWASP Secure Coding Practices Quick Reference Guide
Known limitations	SQLite dialect differs slightly from PostgreSQL or SQL Server
Classification / age status	Recommended · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F12 · Oracle VirtualBox User Manual

Exact title	Oracle VirtualBox User Manual
Direct URL	https://www.virtualbox.org/manual/
Resource type / domains	Official documentation · Modules A,C
Topic and subtopic coverage	Authoritative setup and isolation reference for a no-cost lab
Exact path / module / chapter / playlist	Chapters 1-3, 5-6, 8: introduction, installation, basic configuration, virtual storage, networking, snapshots
Recommended difficulty / estimated time	Beginner · Selected sections 4-6 h
Access status	Free
Prerequisites	Desktop OS administration basics
Why selected	Authoritative setup and isolation reference for a no-cost lab
What to practise after	Create an Ubuntu VM with NAT plus an isolated internal adapter and take a clean snapshot
Backup if inaccessible	L07 — Microsoft Windows Server documentation
Known limitations	Version-specific menus may differ; use the manual matching your installed version
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F13 · Ubuntu Desktop Guide

Exact title	Ubuntu Desktop Guide
Direct URL	https://help.ubuntu.com/
Resource type / domains	Official documentation · Modules A,C
Topic and subtopic coverage	Stable first Linux desktop reference
Exact path / module / chapter / playlist	Installation, desktop, files, networking, users, software, troubleshooting
Recommended difficulty / estimated time	Beginner · Selected sections 3-6 h
Access status	Free
Prerequisites	None
Why selected	Stable first Linux desktop reference
What to practise after	Configure users, package updates, firewall and logs in the lab VM
Backup if inaccessible	L01 — Linux Journey
Known limitations	Release-specific UI changes
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F14 · Microsoft Learn Windows fundamentals

Exact title	Microsoft Learn Windows fundamentals
Direct URL	https://learn.microsoft.com/en-us/training/modules/explore-windows-client/
Resource type / domains	Official learning hub · Modules A,C
Topic and subtopic coverage	Structured Microsoft-authored introduction before Active Directory
Exact path / module / chapter / playlist	Windows client and server administration learning paths; select modules on accounts, processes, services and PowerShell
Recommended difficulty / estimated time	Beginner · 8-20 h
Access status	Free learning content; exams separate
Prerequisites	A Windows lab VM or Windows host
Why selected	Structured Microsoft-authored introduction before Active Directory
What to practise after	Document processes, services, permissions and event IDs from your own VM
Backup if inaccessible	L08 — PowerShell documentation
Known limitations	Product names and UI change; rely on current Microsoft Learn pages
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N01 · Jeremy's IT Lab Free CCNA 200-301 Complete Course

Exact title	Jeremy's IT Lab Free CCNA 200-301 Complete Course
Direct URL	https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ
Resource type / domains	YouTube playlist · Modules B
Topic and subtopic coverage	Exceptional depth and free Packet Tracer labs for real networking fundamentals
Exact path / module / chapter / playlist	Days 1-63; begin with Network Devices, Interfaces and Cables, TCP/IP Model, CLI, Ethernet LAN Switching, IPv4 Addressing
Recommended difficulty / estimated time	Beginner to intermediate · 63 study days; selected first 45 days in this roadmap
Access status	Free on YouTube; optional paid course extras
Prerequisites	None
Why selected	Exceptional depth and free Packet Tracer labs for real networking fundamentals
What to practise after	Watch one day, complete the matching Packet Tracer lab and Anki cards where accessible
Backup if inaccessible	N02 — Jeremy's IT Lab CCNA Course Index and Files
Known limitations	Playlist originated in 2020 and was augmented for v1.1; verify current Cisco blueprint before certification study
Classification / age status	Mandatory · Old but gold, updated additions
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N02 · Jeremy's IT Lab CCNA Course Index and Files

Exact title	Jeremy's IT Lab CCNA Course Index and Files
Direct URL	https://courses.jeremysitlab.com/p/ccna
Resource type / domains	Course index and lab files · Modules B
Topic and subtopic coverage	The index supplies verifiable lesson titles and lab names without inventing timestamps
Exact path / module / chapter / playlist	Course Content: Days 1-63; lecture titles, durations, labs; free files link via video descriptions
Recommended difficulty / estimated time	Beginner to intermediate · Same as N01
Access status	Free YouTube course; paid ad-free edition has bonus quizzes
Prerequisites	None
Why selected	The index supplies verifiable lesson titles and lab names without inventing timestamps
What to practise after	Use the matching Day lecture title and Packet Tracer lab; download files only from the official link
Backup if inaccessible	N03 — Jeremy's IT Lab CCNA Course Files
Known limitations	Paid edition is optional; free files are linked separately
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N03 · Jeremy's IT Lab CCNA Course Files

Exact title	Jeremy's IT Lab CCNA Course Files
Direct URL	https://jitl.jp/ccna-files
Resource type / domains	Downloadable labs/flashcards · Modules B
Topic and subtopic coverage	Official free practice material accompanying N01
Exact path / module / chapter / playlist	Anki decks and Packet Tracer lab files by course day
Recommended difficulty / estimated time	Beginner to intermediate · Varies
Access status	Free
Prerequisites	Packet Tracer and Anki
Why selected	Official free practice material accompanying N01
What to practise after	Complete the current Day's lab, then reconfigure it from a blank topology
Backup if inaccessible	N02 — Jeremy's IT Lab CCNA Course Index and Files
Known limitations	External host may change; use N02 video description if unavailable
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N04 · Professor Messer N10-009 Network+ Training Course

Exact title	Professor Messer N10-009 Network+ Training Course
Direct URL	https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8
Resource type / domains	YouTube playlist · Modules B
Topic and subtopic coverage	Objective-indexed short videos are ideal for targeted review and gap filling
Exact path / module / chapter / playlist	Official 87-video N10-009 course; use exact numbered objective titles
Recommended difficulty / estimated time	Beginner to intermediate · 12 h 55 m total video
Access status	Free
Prerequisites	Basic IT vocabulary
Why selected	Objective-indexed short videos are ideal for targeted review and gap filling
What to practise after	After a Jeremy's IT Lab lesson, watch one Messer objective video as an alternate explanation
Backup if inaccessible	N05 — Professor Messer N10-009 Course Index
Known limitations	Certification-oriented, not a complete packet-analysis course
Classification / age status	Recommended · 2024 onward
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N05 · Professor Messer N10-009 Course Index

Exact title	Professor Messer N10-009 Course Index
Direct URL	https://www.professormesser.com/network-plus/n10-009/n10-009-video/n10-009-training-course/
Resource type / domains	Course index · Modules B
Topic and subtopic coverage	More reliable navigation than a channel homepage; exact video titles and run times
Exact path / module / chapter / playlist	All N10-009 video titles grouped by exam objective
Recommended difficulty / estimated time	Beginner to intermediate · 12 h 55 m total video
Access status	Free
Prerequisites	None
Why selected	More reliable navigation than a channel homepage; exact video titles and run times
What to practise after	Find the exact objective from the daily plan and open the linked video
Backup if inaccessible	N04 — Professor Messer N10-009 Network+ Training Course
Known limitations	Older N10-008 course also exists; ensure N10-009
Classification / age status	Recommended · 2024 onward
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N06 · Wireshark User's Guide

Exact title	Wireshark User's Guide
Direct URL	https://www.wireshark.org/docs/wsug_html_chunked/
Resource type / domains	Official documentation · Modules B,H
Topic and subtopic coverage	Primary reference for legally captured packets and repeatable analysis
Exact path / module / chapter / playlist	Chapters 3-7: user interface, capturing, display filters, coloring, protocol dissection, statistics
Recommended difficulty / estimated time	Beginner to intermediate · 8-12 h selected
Access status	Free
Prerequisites	Basic TCP/IP
Why selected	Primary reference for legally captured packets and repeatable analysis
What to practise after	Use display filters, follow streams, export objects and annotate packet fields in assigned sample captures
Backup if inaccessible	N07 — Wireshark SampleCaptures
Known limitations	Version-specific UI; match your installed Wireshark version
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N07 · Wireshark SampleCaptures

Exact title	Wireshark SampleCaptures
Direct URL	https://wiki.wireshark.org/SampleCaptures
Resource type / domains	Packet capture repository · Modules B,H
Topic and subtopic coverage	Provides legal, public captures for packet analysis without sniffing other people's traffic
Exact path / module / chapter / playlist	Sample captures by protocol; DNS, HTTP, TCP, TLS, ARP, ICMP and malware traffic
Recommended difficulty / estimated time	Intermediate · 1-3 h per capture
Access status	Free
Prerequisites	Wireshark fundamentals
Why selected	Provides legal, public captures for packet analysis without sniffing other people's traffic
What to practise after	Write a structured report: endpoints, conversations, protocol layers, anomaly, limitations
Backup if inaccessible	N06 — Wireshark User's Guide
Known limitations	Some linked external captures disappear; download from stable Wireshark-hosted mirrors where possible
Classification / age status	Mandatory · Maintained wiki with older captures
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N08 · tcpdump manual page

Exact title	tcpdump manual page
Direct URL	https://www.tcpdump.org/manpages/tcpdump.1.html
Resource type / domains	Official command reference · Modules B,H
Topic and subtopic coverage	Canonical syntax for command-line capture filters and safe packet collection
Exact path / module / chapter / playlist	Options, expression syntax, output formats, examples
Recommended difficulty / estimated time	Intermediate · 2-4 h
Access status	Free
Prerequisites	Linux command line and TCP/IP
Why selected	Canonical syntax for command-line capture filters and safe packet collection
What to practise after	Capture only loopback or own lab traffic, then convert a BPF filter into a Wireshark display filter
Backup if inaccessible	N06 — Wireshark User's Guide
Known limitations	Root privileges may be needed; do not capture networks you do not own
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N09 · Nmap Reference Guide

Exact title	Nmap Reference Guide
Direct URL	https://nmap.org/book/man.html
Resource type / domains	Official documentation · Modules B,G
Topic and subtopic coverage	The definitive source for scan semantics and responsible interpretation
Exact path / module / chapter / playlist	Host discovery, port scanning, service/version detection, OS detection, timing, output
Recommended difficulty / estimated time	Intermediate · 6-10 h selected
Access status	Free
Prerequisites	TCP/IP and authorization
Why selected	The definitive source for scan semantics and responsible interpretation
What to practise after	Scan only an isolated lab target with written scope; compare default, SYN and version scans
Backup if inaccessible	N10 — RFC 9293: Transmission Control Protocol
Known limitations	Many options are not appropriate for beginners or public networks
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N10 · RFC 9293: Transmission Control Protocol

Exact title	RFC 9293: Transmission Control Protocol
Direct URL	https://www.rfc-editor.org/rfc/rfc9293
Resource type / domains	Authoritative standard · Modules B
Topic and subtopic coverage	Primary standard for TCP behavior; removes myths from secondary tutorials
Exact path / module / chapter / playlist	Sections 3.1-3.10: header format, connection states, sequence numbers, handshake, retransmission, flow control
Recommended difficulty / estimated time	Advanced reference · 5-8 h selected
Access status	Free
Prerequisites	IPv4 and protocol layering
Why selected	Primary standard for TCP behavior; removes myths from secondary tutorials
What to practise after	Map every field in a real SYN/SYN-ACK/ACK capture to the RFC
Backup if inaccessible	B10 — Suricata documentation
Known limitations	RFC-level prose is dense; use after guided lessons
Classification / age status	Advanced reference · 2022, current TCP standard
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N11 · RFC 791: Internet Protocol v4

Exact title	RFC 791: Internet Protocol v4
Direct URL	https://www.rfc-editor.org/rfc/rfc791
Resource type / domains	Authoritative standard · Modules B
Topic and subtopic coverage	Primary IPv4 source for header analysis and fragmentation theory
Exact path / module / chapter / playlist	Addressing, fragmentation, header fields, TTL and protocol fields
Recommended difficulty / estimated time	Advanced reference · 3-5 h selected
Access status	Free
Prerequisites	Binary and CIDR
Why selected	Primary IPv4 source for header analysis and fragmentation theory
What to practise after	Inspect an IPv4 header in Wireshark and explain every field
Backup if inaccessible	N10 — RFC 9293: Transmission Control Protocol
Known limitations	Historic RFC; use current clarifications for security specifics
Classification / age status	Advanced reference · 1981, old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N12 · RFC 8200: Internet Protocol v6

Exact title	RFC 8200: Internet Protocol v6
Direct URL	https://www.rfc-editor.org/rfc/rfc8200
Resource type / domains	Authoritative standard · Modules B
Topic and subtopic coverage	Primary source for IPv6 differences and extension-header reasoning
Exact path / module / chapter / playlist	IPv6 header, extension headers, MTU, fragmentation and addressing model
Recommended difficulty / estimated time	Advanced reference · 4-7 h selected
Access status	Free
Prerequisites	IPv4/CIDR
Why selected	Primary source for IPv6 differences and extension-header reasoning
What to practise after	Compare an IPv6 packet capture with IPv4 and list security implications
Backup if inaccessible	N10 — RFC 9293: Transmission Control Protocol
Known limitations	Updated by later RFCs; use as baseline
Classification / age status	Advanced reference · 2017, old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N13 · Cloudflare Learning Center: What is DNS?

Exact title	Cloudflare Learning Center: What is DNS?
Direct URL	https://www.cloudflare.com/learning/dns/what-is-dns/
Resource type / domains	Educational documentation · Modules B
Topic and subtopic coverage	Clear visual explanation of the resolution chain before packet-level DNS analysis
Exact path / module / chapter / playlist	DNS resolution, recursive resolvers, root/TLD/authoritative servers, common records
Recommended difficulty / estimated time	Beginner · 1-3 h
Access status	Free
Prerequisites	Basic networking
Why selected	Clear visual explanation of the resolution chain before packet-level DNS analysis
What to practise after	Use dig or nslookup for one owned domain/example domain and trace each answer section
Backup if inaccessible	B10 — Suricata documentation
Known limitations	Vendor educational content; cross-check record behavior in packet captures
Classification / age status	Recommended · Maintained
Verification status / date	Automated checker blocked (HTTP 403); official/canonical URL; use backup if browser access fails · checked 2026-10-27

N14 · IANA Service Name and Transport Protocol Port Number Registry

Exact title	IANA Service Name and Transport Protocol Port Number Registry
Direct URL	https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml
Resource type / domains	Authoritative registry · Modules B,G
Topic and subtopic coverage	Avoid memorizing misleading port lists from random guides
Exact path / module / chapter / playlist	Assigned ports and service names for TCP, UDP, SCTP and DCCP
Recommended difficulty / estimated time	Reference · 1 h lookup
Access status	Free
Prerequisites	TCP/UDP fundamentals
Why selected	Avoid memorizing misleading port lists from random guides
What to practise after	Build a personal port table with service, protocol, purpose and security caveat
Backup if inaccessible	N10 — RFC 9293: Transmission Control Protocol
Known limitations	Registration is not proof a service uses that port in a particular environment
Classification / age status	Recommended · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N15 · Cisco Packet Tracer course page

Exact title	Cisco Packet Tracer course page
Direct URL	https://www.netacad.com/courses/packet-tracer
Resource type / domains	Network simulator · Modules B
Topic and subtopic coverage	Low-resource alternative to physical Cisco hardware for switching and routing labs
Exact path / module / chapter / playlist	Introductory Packet Tracer course and download access
Recommended difficulty / estimated time	Beginner · 4-8 h
Access status	Free with NetAcad account availability varies by region
Prerequisites	None
Why selected	Low-resource alternative to physical Cisco hardware for switching and routing labs
What to practise after	Build the exact topologies assigned in N01/N03 and save annotated .pkt files
Backup if inaccessible	N03 — Jeremy's IT Lab CCNA Course Files
Known limitations	Registration and local availability may vary
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N16 · Subnetting Practice

Exact title	Subnetting Practice
Direct URL	https://subnettingpractice.com/
Resource type / domains	Interactive practice · Modules B
Topic and subtopic coverage	Fast, randomized drills with immediate feedback create subnetting fluency
Exact path / module / chapter / playlist	CIDR, network/broadcast/host calculations, VLSM and masks
Recommended difficulty / estimated time	Beginner to intermediate · 10-15 min daily in Month 2
Access status	Free tier; optional ads/premium features may appear
Prerequisites	Binary and IPv4 structure
Why selected	Fast, randomized drills with immediate feedback create subnetting fluency
What to practise after	Complete 20 timed questions, write error explanations for every miss
Backup if inaccessible	N17 — Practical Networking Subnetting Mastery
Known limitations	Not a conceptual textbook; use only after the lesson
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N17 · Practical Networking Subnetting Mastery

Exact title	Practical Networking Subnetting Mastery
Direct URL	https://www.practicalnetworking.net/stand-alone/subnetting-mastery/
Resource type / domains	Course / articles · Modules B
Topic and subtopic coverage	Visual method for subnetting rather than rote tricks
Exact path / module / chapter / playlist	Seven-part subnetting series: CIDR, masks, FLSM, VLSM, shortcuts, route summarization
Recommended difficulty / estimated time	Beginner to intermediate · 6-12 h
Access status	Free articles/videos; optional paid resources
Prerequisites	Binary notation
Why selected	Visual method for subnetting rather than rote tricks
What to practise after	Draw the address block for each assigned problem and verify with ipcalc
Backup if inaccessible	N16 — Subnetting Practice
Known limitations	Course packaging and optional product prices may change
Classification / age status	Recommended · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L01 · Linux Journey

Exact title	Linux Journey
Direct URL	https://linuxjourney.com/
Resource type / domains	Interactive text lessons · Modules C
Topic and subtopic coverage	Short, structured lessons that are easy to convert into terminal practice
Exact path / module / chapter / playlist	Grasshopper, Ostrich, Panda and Tortoise sections: command line, filesystem, processes, packages, users, permissions, networking, shell scripting
Recommended difficulty / estimated time	Beginner · 15-25 h
Access status	Free
Prerequisites	A Linux VM or WSL
Why selected	Short, structured lessons that are easy to convert into terminal practice
What to practise after	After every lesson, reproduce commands in the VM and note one failure mode
Backup if inaccessible	L02 — OverTheWire Bandit
Known limitations	Does not replace distribution documentation or systemd internals
Classification / age status	Mandatory · Maintained; older design
Verification status / date	Automated checker blocked (HTTP 403); official/canonical URL; use backup if browser access fails · checked 2026-10-27

L02 · OverTheWire Bandit

Exact title	OverTheWire Bandit
Direct URL	https://overthewire.org/wargames/bandit/
Resource type / domains	CTF / command-line lab · Modules C
Topic and subtopic coverage	Forces independent command-line problem-solving, not passive video watching
Exact path / module / chapter / playlist	Levels 0-34: SSH, files, hidden files, permissions, pipes, grep, find, cron, git, networking, shells
Recommended difficulty / estimated time	Beginner to intermediate · 10-20 h
Access status	Free
Prerequisites	SSH client and Linux basics
Why selected	Forces independent command-line problem-solving, not passive video watching
What to practise after	Keep a concept journal for each level; do not publish level passwords or copy walkthroughs verbatim
Backup if inaccessible	L01 — Linux Journey
Known limitations	Occasional service availability; some late levels rely on Unix tricks
Classification / age status	Mandatory · Classic
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L03 · Greg's Wiki BashGuide

Exact title	Greg's Wiki BashGuide
Direct URL	https://mywiki.wooledge.org/BashGuide
Resource type / domains	Free scripting guide · Modules C,D
Topic and subtopic coverage	Explains Bash correctly, including quoting and word splitting pitfalls common in security automation
Exact path / module / chapter / playlist	Commands and arguments, quoting, expansions, tests, conditionals, loops, functions, arrays, redirection
Recommended difficulty / estimated time	Intermediate · 10-15 h
Access status	Free
Prerequisites	Basic Linux CLI
Why selected	Explains Bash correctly, including quoting and word splitting pitfalls common in security automation
What to practise after	Refactor one script with shellcheck and write tests for empty strings and spaces in filenames
Backup if inaccessible	P04 — RegexOne
Known limitations	Wiki can be terse; use as a reference while coding
Classification / age status	Recommended · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L04 · Linux man-pages project

Exact title	Linux man-pages project
Direct URL	https://man7.org/linux/man-pages/
Resource type / domains	Official reference · Modules C
Topic and subtopic coverage	Primary local/online reference for exact command and kernel API behavior
Exact path / module / chapter / playlist	Sections 1, 2, 3, 4, 5, 7 and 8; start with ls, chmod, ps, mount, namespaces, capabilities, proc, syscalls
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free
Prerequisites	Linux basics
Why selected	Primary local/online reference for exact command and kernel API behavior
What to practise after	Replace memorized commands with a man-page lookup; cite section numbers in notes
Backup if inaccessible	L01 — Linux Journey
Known limitations	Distribution versions may differ; compare with local man pages
Classification / age status	Advanced reference · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L05 · systemd.unit manual

Exact title	systemd.unit manual
Direct URL	https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html
Resource type / domains	Official reference · Modules C
Topic and subtopic coverage	Authoritative service configuration and troubleshooting reference
Exact path / module / chapter / playlist	Unit file structure, sections, dependencies, targets and drop-ins
Recommended difficulty / estimated time	Intermediate · 2-4 h
Access status	Free
Prerequisites	Linux services
Why selected	Authoritative service configuration and troubleshooting reference
What to practise after	Create a benign timer/service pair, inspect journal logs, then disable and remove it
Backup if inaccessible	L04 — Linux man-pages project
Known limitations	Not all distributions use identical defaults
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L06 · The Linux Foundation Introduction to Linux

Exact title	The Linux Foundation Introduction to Linux
Direct URL	https://www.edx.org/learn/linux/the-linux-foundation-introduction-to-linux
Resource type / domains	MOOC · Modules C
Topic and subtopic coverage	University-like sequence by the organization behind Linux training
Exact path / module / chapter / playlist	Filesystem, command line, processes, packages, users, permissions, shells, scripting
Recommended difficulty / estimated time	Beginner · 40-60 h; audit selected modules
Access status	Free audit may be time-limited; verified certificate paid
Prerequisites	None
Why selected	University-like sequence by the organization behind Linux training
What to practise after	Use selected chapters to fill Linux Journey gaps; do all terminal exercises locally
Backup if inaccessible	L01 — Linux Journey
Known limitations	edX audit terms and course versions vary
Classification / age status	Optional enrichment · Maintained with periodic updates
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L07 · Microsoft Windows Server documentation

Exact title	Microsoft Windows Server documentation
Direct URL	https://learn.microsoft.com/en-us/windows-server/
Resource type / domains	Official documentation · Modules C
Topic and subtopic coverage	Primary source for Windows Server and Active Directory concepts
Exact path / module / chapter / playlist	Installation, administration, identity, networking, security, PowerShell, failover and containers
Recommended difficulty / estimated time	Beginner to intermediate · Reference use
Access status	Free
Prerequisites	Windows client basics
Why selected	Primary source for Windows Server and Active Directory concepts
What to practise after	Create a small virtual domain only if your hardware allows; otherwise diagram it and use a local Windows event lab
Backup if inaccessible	L12 — Active Directory Domain Services overview
Known limitations	Broad documentation; use the daily plan's exact pages
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L08 · PowerShell documentation

Exact title	PowerShell documentation
Direct URL	https://learn.microsoft.com/en-us/powershell/
Resource type / domains	Official documentation · Modules C,D
Topic and subtopic coverage	Primary source for current PowerShell syntax and security controls
Exact path / module / chapter / playlist	Getting started, scripting, modules, remoting, security and cmdlet reference
Recommended difficulty / estimated time	Beginner to intermediate · 8-15 h
Access status	Free
Prerequisites	Windows command line
Why selected	Primary source for current PowerShell syntax and security controls
What to practise after	Write a benign inventory script and export structured JSON/CSV
Backup if inaccessible	L03 — Greg's Wiki BashGuide
Known limitations	Windows PowerShell 5.1 and PowerShell 7 differ; check edition
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L09 · Windows Event Log documentation

Exact title	Windows Event Log documentation
Direct URL	https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log
Resource type / domains	Official reference · Modules C,H
Topic and subtopic coverage	Definitive event-log architecture reference for blue-team work
Exact path / module / chapter / playlist	Event logs, channels, records, queries, subscriptions and security
Recommended difficulty / estimated time	Intermediate · 3-6 h
Access status	Free
Prerequisites	Windows administration
Why selected	Definitive event-log architecture reference for blue-team work
What to practise after	Export a sample EVTX, identify Channel, EventID, Provider and RecordId
Backup if inaccessible	L11 — Sysmon documentation and download
Known limitations	Some APIs are developer-oriented; use Event Viewer for first practice
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L10 · Microsoft Sysinternals

Exact title	Microsoft Sysinternals
Direct URL	https://learn.microsoft.com/en-us/sysinternals/
Resource type / domains	Official tools/docs · Modules C,H
Topic and subtopic coverage	Industry-standard, free Windows internals tools with official documentation
Exact path / module / chapter / playlist	Process Explorer, Process Monitor, Autoruns, Sysmon and PsTools
Recommended difficulty / estimated time	Beginner to intermediate · 5-10 h
Access status	Free
Prerequisites	Windows fundamentals
Why selected	Industry-standard, free Windows internals tools with official documentation
What to practise after	Investigate a benign process launch and document process tree, loaded modules, handles and autoruns
Backup if inaccessible	L08 — PowerShell documentation
Known limitations	Tools change; use current downloads and hashes
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L11 · Sysmon documentation and download

Exact title	Sysmon documentation and download
Direct URL	https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon
Resource type / domains	Official tool/doc · Modules H
Topic and subtopic coverage	Adds high-value process, network and file events for detection engineering in an authorized lab
Exact path / module / chapter / playlist	Installation, configuration, event IDs and schema
Recommended difficulty / estimated time	Intermediate · 3-6 h
Access status	Free
Prerequisites	Windows event logs
Why selected	Adds high-value process, network and file events for detection engineering in an authorized lab
What to practise after	Install Sysmon in a disposable VM and generate benign process/network telemetry
Backup if inaccessible	L09 — Windows Event Log documentation
Known limitations	Do not deploy on production systems without organizational approval
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L12 · Active Directory Domain Services overview

Exact title	Active Directory Domain Services overview
Direct URL	https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview
Resource type / domains	Official documentation · Modules C,E
Topic and subtopic coverage	Authoritative identity architecture reference before AD security topics
Exact path / module / chapter / playlist	Forest, domain, OU, users, groups, DNS, trusts and authentication
Recommended difficulty / estimated time	Beginner to intermediate · 4-8 h
Access status	Free
Prerequisites	Windows Server basics
Why selected	Authoritative identity architecture reference before AD security topics
What to practise after	Create a diagram of a fictional domain and label every identity boundary
Backup if inaccessible	L07 — Microsoft Windows Server documentation
Known limitations	UI and server version differ across environments
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L13 · Windows security baselines

Exact title	Windows security baselines
Direct URL	https://learn.microsoft.com/en-us/windows/security/operating-system-security/device-management/windows-security-configuration-framework/windows-security-baselines
Resource type / domains	Official configuration guidance · Modules C,E
Topic and subtopic coverage	Real enterprise hardening baselines rather than ad hoc tweaks
Exact path / module / chapter / playlist	Microsoft security baseline packages, policy analyzer, deployment guidance and versioning
Recommended difficulty / estimated time	Intermediate · 4-8 h
Access status	Free
Prerequisites	Group Policy and Windows administration
Why selected	Real enterprise hardening baselines rather than ad hoc tweaks
What to practise after	Compare one baseline setting against a clean VM, document compatibility impact
Backup if inaccessible	L14 — CIS Benchmarks
Known limitations	Do not apply a baseline blindly to production or your primary computer
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L14 · CIS Benchmarks

Exact title	CIS Benchmarks
Direct URL	https://www.cisecurity.org/cis-benchmarks
Resource type / domains	Consensus benchmarks · Modules C,E,I,J
Topic and subtopic coverage	Detailed consensus controls with rationale and audit/remediation steps
Exact path / module / chapter / playlist	Benchmarks by OS, cloud, database, container and network platform
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free PDFs may require registration; some tools paid
Prerequisites	Hardening basics
Why selected	Detailed consensus controls with rationale and audit/remediation steps
What to practise after	Select five controls for Ubuntu or Windows, test in a snapshot, record before/after evidence
Backup if inaccessible	L13 — Windows security baselines
Known limitations	Registration and profile versions; not all benchmarks fit student hardware
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L15 · pwn.college Dojo index

Exact title	pwn.college Dojo index
Direct URL	https://pwn.college/dojos
Resource type / domains	Hands-on challenges · Modules A,C,K
Topic and subtopic coverage	University-grade challenge sequence with a browser workspace; excellent low-hardware option
Exact path / module / chapter / playlist	Linux Luminarium, Computing 101, Intro to Cybersecurity, Program Security, System Security, Software Exploitation
Recommended difficulty / estimated time	Beginner to advanced · Varies by dojo
Access status	Free
Prerequisites	Command line and programming
Why selected	University-grade challenge sequence with a browser workspace; excellent low-hardware option
What to practise after	Use Computing 101 and Linux Fundamentals in Months 1-4; defer exploitation dojos until prerequisites are met
Backup if inaccessible	L01 — Linux Journey
Known limitations	Dojo names and course instances change; use current Core Material list
Classification / age status	Recommended · Maintained 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P01 · Python for Everybody

Exact title	Python for Everybody
Direct URL	https://www.py4e.com/
Resource type / domains	University course · Modules D
Topic and subtopic coverage	Very gentle programming sequence with abundant exercises
Exact path / module / chapter / playlist	Chapters 1-16: variables, conditionals, functions, loops, strings, files, lists, dictionaries, tuples, regex, networking, databases, automation
Recommended difficulty / estimated time	Beginner · 30-50 h; selected chapters in Month 4
Access status	Free site; Coursera specialization freemium
Prerequisites	None
Why selected	Very gentle programming sequence with abundant exercises
What to practise after	Complete selected chapter exercises and adapt one to parse a security log
Backup if inaccessible	F08 — Python 3 Official Tutorial
Known limitations	Some videos are older; Python 3 syntax remains valid with minor updates
Classification / age status	Recommended · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P02 · Learn Python - Full Course for Beginners (freeCodeCamp)

Exact title	Learn Python - Full Course for Beginners (freeCodeCamp)
Direct URL	https://www.youtube.com/watch?v=eWRfhZUzrAc
Resource type / domains	Single full-length video course · Modules D
Topic and subtopic coverage	One coherent video session avoids jumping among dozens of tutorials
Exact path / module / chapter / playlist	Chapters shown in video; variables, strings, lists, tuples, dictionaries, conditionals, loops, functions, classes and projects
Recommended difficulty / estimated time	Beginner · 4.5 h
Access status	Free
Prerequisites	None
Why selected	One coherent video session avoids jumping among dozens of tutorials
What to practise after	Watch a 25-minute segment, code all examples locally, then write a variation
Backup if inaccessible	F08 — Python 3 Official Tutorial
Known limitations	Single video chapter times can change with re-uploads; use YouTube's displayed chapters
Classification / age status	Recommended · 2022
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P03 · Automate the Boring Stuff with Python

Exact title	Automate the Boring Stuff with Python
Direct URL	https://automatetheboringstuff.com/
Resource type / domains	Free online book · Modules D
Topic and subtopic coverage	Practical automation tasks map directly to defensive utility building
Exact path / module / chapter / playlist	Chapters 1-18; focus on files, regex, CSV/JSON, web scraping, requests, Excel, email and input validation
Recommended difficulty / estimated time	Beginner to intermediate · 20-40 h selected
Access status	Free online; paid print/course optional
Prerequisites	Basic Python
Why selected	Practical automation tasks map directly to defensive utility building
What to practise after	Build the Month 4 log parser and audit utility from the book's patterns
Backup if inaccessible	F08 — Python 3 Official Tutorial
Known limitations	Some third-party APIs or screenshots are dated; use current library docs
Classification / age status	Recommended · Classic, current Python edition
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P04 · RegexOne

Exact title	RegexOne
Direct URL	https://regexone.com/
Resource type / domains	Interactive lessons · Modules D
Topic and subtopic coverage	Short, interactive regex practice with immediate feedback
Exact path / module / chapter / playlist	Lessons 1-15: literals, wildcards, character classes, anchors, groups, quantifiers, backreferences and practical examples
Recommended difficulty / estimated time	Beginner · 3-6 h
Access status	Free
Prerequisites	None
Why selected	Short, interactive regex practice with immediate feedback
What to practise after	Write regexes for three synthetic log formats and document false positives
Backup if inaccessible	F08 — Python 3 Official Tutorial
Known limitations	Regex dialects differ; verify in Python re docs
Classification / age status	Recommended · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P05 · pytest documentation

Exact title	pytest documentation
Direct URL	https://docs.pytest.org/en/stable/
Resource type / domains	Official documentation · Modules D
Topic and subtopic coverage	Reliable testing habits for security scripts and parsers
Exact path / module / chapter / playlist	Getting started, fixtures, parametrization, assertions, marking, plugins and CI examples
Recommended difficulty / estimated time	Intermediate · 5-10 h
Access status	Free
Prerequisites	Python functions and modules
Why selected	Reliable testing habits for security scripts and parsers
What to practise after	Write unit tests for normal, malformed and adversarial inputs to the log parser
Backup if inaccessible	F08 — Python 3 Official Tutorial
Known limitations	Plugin compatibility varies
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P06 · Requests: HTTP for Humans

Exact title	Requests: HTTP for Humans
Direct URL	https://requests.readthedocs.io/en/latest/
Resource type / domains	Official library docs · Modules D,F
Topic and subtopic coverage	Safe, documented HTTP client usage with timeouts and TLS verification
Exact path / module / chapter / playlist	Quickstart, advanced usage, authentication, sessions, SSL verification, timeouts and streaming
Recommended difficulty / estimated time	Beginner to intermediate · 3-6 h
Access status	Free
Prerequisites	Python and HTTP basics
Why selected	Safe, documented HTTP client usage with timeouts and TLS verification
What to practise after	Query a mock API or local test server; record status codes, headers, errors and retries
Backup if inaccessible	F10 — MDN HTTP Overview
Known limitations	Never disable certificate verification except in a controlled local TLS lab
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P07 · SQLite documentation

Exact title	SQLite documentation
Direct URL	https://www.sqlite.org/docs.html
Resource type / domains	Official documentation · Modules A,D
Topic and subtopic coverage	Lightweight local database for capstone and log-analysis projects
Exact path / module / chapter / playlist	SQL syntax, datatypes, indexes, transactions, command-line shell and security considerations
Recommended difficulty / estimated time	Beginner to intermediate · Reference use
Access status	Free
Prerequisites	SQLBolt basics
Why selected	Lightweight local database for capstone and log-analysis projects
What to practise after	Create a normalized schema for synthetic events and write parameterized queries
Backup if inaccessible	F11 — SQLBolt
Known limitations	SQLite is not a substitute for learning PostgreSQL/MySQL administration
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P08 · OWASP Secure Coding Practices Quick Reference Guide

Exact title	OWASP Secure Coding Practices Quick Reference Guide
Direct URL	https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/
Resource type / domains	Checklist/reference · Modules D,F
Topic and subtopic coverage	Maps secure implementation decisions to concrete verification checks
Exact path / module / chapter / playlist	Input validation, output encoding, authentication, session management, access control, cryptography, error handling, data protection, communications and system configuration
Recommended difficulty / estimated time	Intermediate · 4-8 h
Access status	Free
Prerequisites	Programming and web basics
Why selected	Maps secure implementation decisions to concrete verification checks
What to practise after	Apply ten relevant checklist items to your own secure API project
Backup if inaccessible	W01 — PortSwigger Web Security Academy
Known limitations	Checklist, not a tutorial or complete standard
Classification / age status	Mandatory · Community-maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C01 · Professor Messer SY0-701 Security+ Course Index

Exact title	Professor Messer SY0-701 Security+ Course Index
Direct URL	https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/
Resource type / domains	Course index · Modules E,J
Topic and subtopic coverage	Objective-indexed, concise, fully free core security theory coverage
Exact path / module / chapter / playlist	121 videos grouped by SY0-701 objectives 1.1-5.6
Recommended difficulty / estimated time	Beginner to intermediate · 15 h 11 m total
Access status	Free videos; paid notes/practice exams optional
Prerequisites	None
Why selected	Objective-indexed, concise, fully free core security theory coverage
What to practise after	Use one video per assigned objective to supplement CS50 and hands-on labs
Backup if inaccessible	C02 — Professor Messer SY0-701 Security+ official playlist
Known limitations	SY0-701 is V7; V8/SY0-801 launches Nov 17 2026, so use current objectives if certifying later
Classification / age status	Mandatory · 2023-2024, exam-relevant until stated retirement
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C02 · Professor Messer SY0-701 Security+ official playlist

Exact title	Professor Messer SY0-701 Security+ official playlist
Direct URL	https://www.youtube.com/playlist?list=PLG49S3nxzAnl4QDVqK-hOnoqcSKEIDDuv
Resource type / domains	YouTube playlist · Modules E,J
Topic and subtopic coverage	Exact playlist rather than channel home; short videos suit retrieval practice
Exact path / module / chapter / playlist	Official 121-video SY0-701 playlist; match objective code in video title
Recommended difficulty / estimated time	Beginner to intermediate · 15 h 11 m total
Access status	Free
Prerequisites	None
Why selected	Exact playlist rather than channel home; short videos suit retrieval practice
What to practise after	Watch only the objective in the daily plan; do not binge entire playlists
Backup if inaccessible	C01 — Professor Messer SY0-701 Security+ Course Index
Known limitations	V8 content will differ after Nov 2026
Classification / age status	Recommended · 2023-2024
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C03 · ISC2 Certified in Cybersecurity Exam Outline

Exact title	ISC2 Certified in Cybersecurity Exam Outline
Direct URL	https://www.isc2.org/certifications/cc/cc-certification-exam-outline
Resource type / domains	Official exam outline · Modules E,J
Topic and subtopic coverage	Current official scope for the free/low-cost entry-level certification option
Exact path / module / chapter / playlist	Five domains; outline effective September 1 2026; AI concepts integrated across domains
Recommended difficulty / estimated time	Beginner · 2-4 h
Access status	Free
Prerequisites	None
Why selected	Current official scope for the free/low-cost entry-level certification option
What to practise after	Map each Month 5 topic to an outline domain and create flashcards
Backup if inaccessible	C04 — ISC2 CC self-study resources
Known limitations	Exam domains change; download current outline before paying
Classification / age status	Recommended · Effective 2026-09-01
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C04 · ISC2 CC self-study resources

Exact title	ISC2 CC self-study resources
Direct URL	https://www.isc2.org/certifications/cc/cc-self-study-resources
Resource type / domains	Official study resources · Modules E,J
Topic and subtopic coverage	Verified official free preparation options for CC
Exact path / module / chapter / playlist	Exam outline, flash cards, official training links and community resources
Recommended difficulty / estimated time	Beginner · 10-20 h
Access status	Free resources; official training paid
Prerequisites	None
Why selected	Verified official free preparation options for CC
What to practise after	Use flashcards only after practical exercises; write original examples for each term
Backup if inaccessible	C03 — ISC2 Certified in Cybersecurity Exam Outline
Known limitations	Marketing pages may change; exam outline controls
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C05 · ISC2 exam pricing

Exact title	ISC2 exam pricing
Direct URL	https://www.isc2.org/register-for-exam/isc2-exam-pricing
Resource type / domains	Official pricing · Modules J
Topic and subtopic coverage	Provides verified current price instead of relying on outdated blogs
Exact path / module / chapter / playlist	CC exam price by region; rescheduling and cancellation fees
Recommended difficulty / estimated time	Reference · 10 min
Access status	Free to view; exam paid
Prerequisites	None
Why selected	Provides verified current price instead of relying on outdated blogs
What to practise after	Check Pakistan-inclusive region at checkout before budgeting
Backup if inaccessible	C03 — ISC2 Certified in Cybersecurity Exam Outline
Known limitations	Local taxes and currencies vary; price may change
Classification / age status	Recommended · Checked 2026-10-27; CC listed US$199 in listed regions
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C06 · NIST Cybersecurity Framework 2.0

Exact title	NIST Cybersecurity Framework 2.0
Direct URL	https://www.nist.gov/cyberframework
Resource type / domains	Framework · Modules E,J
Topic and subtopic coverage	Current, vendor-neutral framework for organizing a security program
Exact path / module / chapter / playlist	Govern, Identify, Protect, Detect, Respond, Recover; implementation examples and quick-start guides
Recommended difficulty / estimated time	Beginner to intermediate · 6-12 h
Access status	Free
Prerequisites	Core security vocabulary
Why selected	Current, vendor-neutral framework for organizing a security program
What to practise after	Build a mini CSF profile for your home lab and map evidence to each function
Backup if inaccessible	C21 — CIS Controls Version 8
Known limitations	Framework is not law and not a certification
Classification / age status	Mandatory · CSF 2.0 current
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C07 · NIST SP 800-61 Revision 3

Exact title	NIST SP 800-61 Revision 3
Direct URL	https://csrc.nist.gov/pubs/sp/800/61/r3/final
Resource type / domains	Official guidance · Modules E,H,J
Topic and subtopic coverage	Current authoritative IR guidance connecting technical work to governance
Exact path / module / chapter / playlist	Incident response life cycle, CSF alignment, preparation, detection, analysis, containment, eradication, recovery and improvement
Recommended difficulty / estimated time	Intermediate · 5-8 h
Access status	Free
Prerequisites	Logging and networking basics
Why selected	Current authoritative IR guidance connecting technical work to governance
What to practise after	Write an IR plan and runbook for one simulated lab incident
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	Guidance is advisory for US federal systems; other jurisdictions may differ
Classification / age status	Mandatory · 2025
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C08 · NIST SP 800-63B

Exact title	NIST SP 800-63B
Direct URL	https://pages.nist.gov/800-63-4/sp800-63b.html
Resource type / domains	Official guidance · Modules E,F
Topic and subtopic coverage	Current digital identity guidance used in modern authentication decisions
Exact path / module / chapter / playlist	Authentication and authenticator management; passwords, MFA, phishing resistance, lifecycle and assurance
Recommended difficulty / estimated time	Intermediate · 5-10 selected pages
Access status	Free
Prerequisites	Identity basics
Why selected	Current digital identity guidance used in modern authentication decisions
What to practise after	Review a sample login design and list five control gaps
Backup if inaccessible	C09 — Crypto 101
Known limitations	Draft/revision status and version changes; use the page's current edition marker
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C09 · Crypto 101

Exact title	Crypto 101
Direct URL	https://www.crypto101.io/
Resource type / domains	Free course/book · Modules E
Topic and subtopic coverage	Concise, implementation-aware cryptography introduction with exercises
Exact path / module / chapter / playlist	XOR, block ciphers, stream ciphers, hash functions, MACs, key exchange, public-key encryption, signatures and TLS concepts
Recommended difficulty / estimated time	Intermediate · 10-20 h
Access status	Free
Prerequisites	Python and basic algebra
Why selected	Concise, implementation-aware cryptography introduction with exercises
What to practise after	Implement only toy demonstrations in Python; never build production crypto
Backup if inaccessible	C10 — Stanford Cryptography I
Known limitations	Course is educational, not a substitute for vetted libraries or proofs
Classification / age status	Recommended · Classic
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C10 · Stanford Cryptography I

Exact title	Stanford Cryptography I
Direct URL	https://www.coursera.org/learn/crypto
Resource type / domains	University course · Modules E
Topic and subtopic coverage	Rigorous cryptography taught by a leading researcher; excellent second exposure after CS50
Exact path / module / chapter / playlist	Week 1-6 lecture list: stream ciphers, block ciphers, message integrity, authenticated encryption, key exchange, public-key encryption and signatures
Recommended difficulty / estimated time	Intermediate to advanced · 20-40 h
Access status	Free audit/lectures may be available; verified certificate and graded homework paid
Prerequisites	Discrete math and programming
Why selected	Rigorous cryptography taught by a leading researcher; excellent second exposure after CS50
What to practise after	Watch one selected lecture, then explain the security goal and attack model in a notebook
Backup if inaccessible	C09 — Crypto 101
Known limitations	Older course; primitives remain important but verify current protocol guidance
Classification / age status	Optional enrichment · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C11 · A Graduate Course in Applied Cryptography

Exact title	A Graduate Course in Applied Cryptography
Direct URL	https://toc.cryptobook.us/
Resource type / domains	Free textbook · Modules E
Topic and subtopic coverage	Free modern cryptography reference when you need exact definitions and proof intuition
Exact path / module / chapter / playlist	Symmetric ciphers, MACs, hash functions, authenticated encryption, public-key crypto, digital signatures and proofs
Recommended difficulty / estimated time	Advanced · Reference use
Access status	Free
Prerequisites	Discrete math, probability and algorithms
Why selected	Free modern cryptography reference when you need exact definitions and proof intuition
What to practise after	Look up one theorem or construction from the lecture and write a plain-English summary
Backup if inaccessible	C09 — Crypto 101
Known limitations	Graduate-level; not a Month 5 mandatory read
Classification / age status	Advanced reference · Maintained draft
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C12 · Let's Encrypt documentation

Exact title	Let's Encrypt documentation
Direct URL	https://letsencrypt.org/docs/
Resource type / domains	Official documentation · Modules E,F
Topic and subtopic coverage	Current practical documentation for certificate issuance and lifecycle
Exact path / module / chapter / playlist	Certificates, ACME, chain of trust, validation, renewal, certificate transparency and best practices
Recommended difficulty / estimated time	Beginner to intermediate · 3-6 h
Access status	Free
Prerequisites	DNS and HTTP
Why selected	Current practical documentation for certificate issuance and lifecycle
What to practise after	Inspect a certificate chain in a browser and explain each field; do not deploy on public sites without authorization
Backup if inaccessible	C13 — RFC 8446: TLS 1.3
Known limitations	Vendor-specific ACME details; general PKI principles still apply
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C13 · RFC 8446: TLS 1.3

Exact title	RFC 8446: TLS 1.3
Direct URL	https://www.rfc-editor.org/rfc/rfc8446
Resource type / domains	Authoritative standard · Modules E,F
Topic and subtopic coverage	Primary TLS 1.3 reference for advanced protocol understanding
Exact path / module / chapter / playlist	Handshake, key schedule, record layer, certificates, resumption, 0-RTT and security considerations
Recommended difficulty / estimated time	Advanced · 8-15 selected pages
Access status	Free
Prerequisites	Cryptography and TCP/IP
Why selected	Primary TLS 1.3 reference for advanced protocol understanding
What to practise after	Annotate a TLS 1.3 handshake capture and identify the client/server hello extensions
Backup if inaccessible	C12 — Let's Encrypt documentation
Known limitations	Dense standard; use simplified material first
Classification / age status	Advanced reference · 2018, current baseline
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C14 · OWASP Top 10

Exact title	OWASP Top 10
Direct URL	https://owasp.org/www-project-top-ten/
Resource type / domains	Reference · Modules F
Topic and subtopic coverage	Industry-recognized risk categories; best as an index, not a complete course
Exact path / module / chapter / playlist	A01-A10 web application security risks, mappings and prevention guidance
Recommended difficulty / estimated time	Beginner to intermediate · 4-8 h
Access status	Free
Prerequisites	HTTP and web app basics
Why selected	Industry-recognized risk categories; best as an index, not a complete course
What to practise after	For each category, point to a PortSwigger lab and a secure control in your own project
Backup if inaccessible	W01 — PortSwigger Web Security Academy
Known limitations	2021 edition; check current release before citing year in interviews
Classification / age status	Mandatory · Community-maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C15 · OWASP Application Security Verification Standard

Exact title	OWASP Application Security Verification Standard
Direct URL	https://owasp.org/www-project-application-security-verification-standard/
Resource type / domains	Standard/reference · Modules F
Topic and subtopic coverage	Concrete verification requirements for secure engineering rather than vague advice
Exact path / module / chapter / playlist	V1 architecture, V2 authentication, V3 session, V4 access control, V5 validation, V6 cryptography, V7 errors/logging, V8 data protection, V9 communications, V10 malicious code and more
Recommended difficulty / estimated time	Intermediate · 6-12 h selected
Access status	Free
Prerequisites	Web security fundamentals
Why selected	Concrete verification requirements for secure engineering rather than vague advice
What to practise after	Select Level 1 requirements and write pass/fail tests for your secure API
Backup if inaccessible	P08 — OWASP Secure Coding Practices Quick Reference Guide
Known limitations	Formal standard; not all levels apply to student projects
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C16 · OWASP API Security Top 10

Exact title	OWASP API Security Top 10
Direct URL	https://owasp.org/API-Security/editions/2023/en/0x11-t10/
Resource type / domains	Reference · Modules F
Topic and subtopic coverage	API-specific risk taxonomy missing from general web courses
Exact path / module / chapter / playlist	API1-API10: BOLA, authentication, object property authorization, resources, BFLA, business flows, SSRF, misconfiguration, inventory and unsafe APIs
Recommended difficulty / estimated time	Intermediate · 4-8 h
Access status	Free
Prerequisites	HTTP, APIs and authorization
Why selected	API-specific risk taxonomy missing from general web courses
What to practise after	Review an OpenAPI spec or sample API and assign each risk a test case
Backup if inaccessible	C14 — OWASP Top 10
Known limitations	2023 edition; verify current release later
Classification / age status	Mandatory · 2023
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C17 · OWASP Threat Modeling community page

Exact title	OWASP Threat Modeling community page
Direct URL	https://owasp.org/www-community/Threat_Modeling
Resource type / domains	Reference/process · Modules E,F,K
Topic and subtopic coverage	Short, practical entry point before more formal methodologies
Exact path / module / chapter / playlist	Threat modeling process, assets, trust boundaries, data flows, threats and mitigations
Recommended difficulty / estimated time	Intermediate · 3-6 h
Access status	Free
Prerequisites	System design basics
Why selected	Short, practical entry point before more formal methodologies
What to practise after	Draw a DFD for your capstone and enumerate STRIDE threats per element
Backup if inaccessible	C18 — MITRE ATT&CK
Known limitations	Community page is an overview, not a complete methodology
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C18 · MITRE ATT&CK

Exact title	MITRE ATT&CK
Direct URL	https://attack.mitre.org/
Resource type / domains	Knowledge base · Modules E,H,K
Topic and subtopic coverage	Common language for adversary behavior and detection coverage
Exact path / module / chapter / playlist	Tactics, techniques, sub-techniques, procedures, mitigations, detections, groups and software
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free
Prerequisites	Logs, OS and networking
Why selected	Common language for adversary behavior and detection coverage
What to practise after	Pick one technique, read its procedure examples and map synthetic telemetry to data sources
Backup if inaccessible	C19 — MITRE D3FEND
Known limitations	Not a maturity model or proof of detection coverage
Classification / age status	Mandatory · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C19 · MITRE D3FEND

Exact title	MITRE D3FEND
Direct URL	https://d3fend.mitre.org/
Resource type / domains	Knowledge base · Modules E,H,K
Topic and subtopic coverage	Balances attacker-centric thinking with concrete defensive engineering options
Exact path / module / chapter / playlist	Defensive countermeasure taxonomy mapped to ATT&CK techniques
Recommended difficulty / estimated time	Intermediate to advanced · Reference use
Access status	Free
Prerequisites	ATT&CK basics
Why selected	Balances attacker-centric thinking with concrete defensive engineering options
What to practise after	For one ATT&CK technique, propose two D3FEND countermeasures and evidence needed
Backup if inaccessible	C18 — MITRE ATT&CK
Known limitations	Research project; not all entries have turnkey implementations
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C20 · NIST NICE Workforce Framework

Exact title	NIST NICE Workforce Framework
Direct URL	https://www.nist.gov/itl/applied-cybersecurity/nice
Resource type / domains	Framework · Modules J
Topic and subtopic coverage	Useful international reference for analyzing job descriptions and planning skill evidence
Exact path / module / chapter / playlist	Work roles, knowledge, skills, abilities and tasks; NICE Framework resources
Recommended difficulty / estimated time	Beginner to intermediate · 3-8 h
Access status	Free
Prerequisites	None
Why selected	Useful international reference for analyzing job descriptions and planning skill evidence
What to practise after	Map your capstone evidence to three NICE work-role task statements
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	US-origin framework; employers use many different vocabularies
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C21 · CIS Controls Version 8

Exact title	CIS Controls Version 8
Direct URL	https://www.cisecurity.org/controls/v8
Resource type / domains	Best-practice controls · Modules E,J
Topic and subtopic coverage	Practical prioritized baseline for small environments and labs
Exact path / module / chapter / playlist	18 safeguard categories, implementation groups and asset/control mappings
Recommended difficulty / estimated time	Beginner to intermediate · 6-10 h
Access status	Free overview; some tools paid
Prerequisites	Basic IT asset concepts
Why selected	Practical prioritized baseline for small environments and labs
What to practise after	Create a before/after implementation-group 1 checklist for your VMs
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	Not regulation; legal obligations depend on jurisdiction
Classification / age status	Recommended · Version 8
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C22 · NIST SP 800-53 Revision 5

Exact title	NIST SP 800-53 Revision 5
Direct URL	https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final
Resource type / domains	Official controls catalog · Modules J
Topic and subtopic coverage	Authoritative control catalog for understanding enterprise security requirements
Exact path / module / chapter / playlist	Control families and baselines for information systems and organizations
Recommended difficulty / estimated time	Advanced reference · Reference use
Access status	Free
Prerequisites	Security architecture and GRC vocabulary
Why selected	Authoritative control catalog for understanding enterprise security requirements
What to practise after	Look up AC, AU, CM, IR and SI controls relevant to your lab; do not try to read linearly
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	Large catalog; use search and mappings
Classification / age status	Advanced reference · Current revision
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C23 · ISO/IEC 27001 overview

Exact title	ISO/IEC 27001 overview
Direct URL	https://www.iso.org/standard/27001
Resource type / domains	Standards page · Modules J
Topic and subtopic coverage	Distinguishes a management-system standard from technical best practices
Exact path / module / chapter / playlist	Requirements for an information security management system; standard text is normally paid
Recommended difficulty / estimated time	Beginner to intermediate · 2-4 h overview
Access status	Page free; standard document paid
Prerequisites	Basic governance vocabulary
Why selected	Distinguishes a management-system standard from technical best practices
What to practise after	Write a fictional ISMS scope, asset inventory and risk treatment plan
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	Full standard is not freely distributable; do not rely on unauthorized copies
Classification / age status	Optional enrichment · Current ISO catalog entry
Verification status / date	Automated checker blocked (HTTP 403); official/canonical URL; use backup if browser access fails · checked 2026-10-27

C24 · GDPR full text (EUR-Lex)

Exact title	GDPR full text (EUR-Lex)
Direct URL	https://eur-lex.europa.eu/eli/reg/2016/679/oj
Resource type / domains	Regulation/legal text · Modules J
Topic and subtopic coverage	Primary legal text for privacy-by-design basics
Exact path / module / chapter / playlist	Principles, lawful basis, data subject rights, controller/processor duties, breach notification, transfers and accountability
Recommended difficulty / estimated time	Intermediate · Selected articles 4-6 h
Access status	Free
Prerequisites	Privacy vocabulary
Why selected	Primary legal text for privacy-by-design basics
What to practise after	Extract ten technical privacy requirements from Articles 5, 25, 32, 33 and 35
Backup if inaccessible	C25 — NIST Privacy Framework
Known limitations	EU regulation, not global law; seek legal advice for actual obligations
Classification / age status	Recommended · Official consolidated text
Verification status / date	Accepted/asynchronous response (HTTP 202); official legal text; use backup if browser fails · checked 2026-10-27

C25 · NIST Privacy Framework

Exact title	NIST Privacy Framework
Direct URL	https://www.nist.gov/privacy-framework
Resource type / domains	Framework · Modules J
Topic and subtopic coverage	Complements CSF with privacy risk management without presenting itself as law
Exact path / module / chapter / playlist	Identify-P, Govern-P, Control-P, Communicate-P and Protect-P functions
Recommended difficulty / estimated time	Intermediate · 3-6 h
Access status	Free
Prerequisites	Basic data governance
Why selected	Complements CSF with privacy risk management without presenting itself as law
What to practise after	Add privacy controls to the capstone data-flow diagram
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	Voluntary framework, not legal compliance proof
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C26 · NIST SP 800-30 Revision 1

Exact title	NIST SP 800-30 Revision 1
Direct URL	https://csrc.nist.gov/pubs/sp/800/30/r1/final
Resource type / domains	Official risk guidance · Modules J
Topic and subtopic coverage	Methodical risk assessment vocabulary for project reports
Exact path / module / chapter / playlist	Threat sources, vulnerabilities, likelihood, impact, risk assessment process and risk registers
Recommended difficulty / estimated time	Intermediate · 5-8 h
Access status	Free
Prerequisites	Security fundamentals
Why selected	Methodical risk assessment vocabulary for project reports
What to practise after	Complete a small risk register with likelihood, impact, owner, treatment and residual risk
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	US federal guidance; terminology can be adapted
Classification / age status	Recommended · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C27 · NIST SP 800-34 Revision 1

Exact title	NIST SP 800-34 Revision 1
Direct URL	https://csrc.nist.gov/pubs/sp/800/34/r1/final
Resource type / domains	Official continuity guidance · Modules J
Topic and subtopic coverage	Evidence-based continuity and disaster recovery planning
Exact path / module / chapter / playlist	Business impact analysis, contingency planning, backup, recovery and testing concepts
Recommended difficulty / estimated time	Intermediate · 4-7 h
Access status	Free
Prerequisites	Asset inventory and risk basics
Why selected	Evidence-based continuity and disaster recovery planning
What to practise after	Write a backup/restore test plan for the capstone and prove one restore
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	US federal guidance; adapt to local context
Classification / age status	Recommended · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W01 · PortSwigger Web Security Academy

Exact title	PortSwigger Web Security Academy
Direct URL	https://portswigger.net/web-security
Resource type / domains	Free interactive academy · Modules F
Topic and subtopic coverage	Best free web-security lab sequence; each concept has a deliberately vulnerable safe lab
Exact path / module / chapter / playlist	All topics and 270+ labs; start with learning paths and Apprentice labs
Recommended difficulty / estimated time	Beginner to advanced · 100+ h available; selected 40-60 h in roadmap
Access status	Free account required
Prerequisites	HTTP, browser DevTools, basic Python
Why selected	Best free web-security lab sequence; each concept has a deliberately vulnerable safe lab
What to practise after	Read topic, solve Apprentice lab without a walkthrough, then write a remediation note
Backup if inaccessible	W02 — PortSwigger Web Security Academy Learning Paths
Known limitations	Labs require a free account and internet; do not target real public sites
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W02 · PortSwigger Web Security Academy Learning Paths

Exact title	PortSwigger Web Security Academy Learning Paths
Direct URL	https://portswigger.net/web-security/learning-paths
Resource type / domains	Structured learning paths · Modules F
Topic and subtopic coverage	Prevents random lab hopping and provides prerequisite-aware order
Exact path / module / chapter / playlist	Curated paths such as Server-side vulnerabilities and SQL injection; choose current paths shown on site
Recommended difficulty / estimated time	Beginner to intermediate · 20-40 h selected
Access status	Free
Prerequisites	W01 basics
Why selected	Prevents random lab hopping and provides prerequisite-aware order
What to practise after	Complete the first two assigned paths in Month 6 and log each lab's root cause and fix
Backup if inaccessible	W01 — PortSwigger Web Security Academy
Known limitations	Path names and lab counts can change
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W03 · PortSwigger All Labs index

Exact title	PortSwigger All Labs index
Direct URL	https://portswigger.net/web-security/all-labs
Resource type / domains	Lab index · Modules F
Topic and subtopic coverage	Direct lab navigation for assessments and retesting
Exact path / module / chapter / playlist	All Apprentice, Practitioner and Expert labs by topic
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free
Prerequisites	W01 account
Why selected	Direct lab navigation for assessments and retesting
What to practise after	Select only the exact lab names listed in the daily plan; stop if a lab requires unlearned prerequisites
Backup if inaccessible	W02 — PortSwigger Web Security Academy Learning Paths
Known limitations	Expert labs can be frustrating early; postpone them
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W04 · MDN HTTP documentation

Exact title	MDN HTTP documentation
Direct URL	https://developer.mozilla.org/en-US/docs/Web/HTTP
Resource type / domains	Official documentation · Modules F
Topic and subtopic coverage	Authoritative browser behavior and security-header reference
Exact path / module / chapter / playlist	Methods, status codes, headers, cookies, CORS, CSP, authentication, caching, redirects and connection management
Recommended difficulty / estimated time	Beginner to intermediate · Reference use
Access status	Free
Prerequisites	Basic web development
Why selected	Authoritative browser behavior and security-header reference
What to practise after	Inspect each response header in a local app and explain its security effect
Backup if inaccessible	F10 — MDN HTTP Overview
Known limitations	Large documentation set; use site search and assigned pages
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W05 · Burp Suite documentation

Exact title	Burp Suite documentation
Direct URL	https://portswigger.net/burp/documentation
Resource type / domains	Official tool documentation · Modules F,G
Topic and subtopic coverage	Primary documentation for the main web-security lab tool
Exact path / module / chapter / playlist	Proxy, target/scope, Repeater, Intruder, Scanner, browser, extensions and TLS configuration
Recommended difficulty / estimated time	Beginner to intermediate · 6-10 h selected
Access status	Community edition free; Pro paid
Prerequisites	HTTP and browser proxy basics
Why selected	Primary documentation for the main web-security lab tool
What to practise after	Configure scope for only localhost and PortSwigger lab domains; use Repeater to compare requests
Backup if inaccessible	W01 — PortSwigger Web Security Academy
Known limitations	Some features require paid Pro edition
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W06 · OWASP Web Security Testing Guide

Exact title	OWASP Web Security Testing Guide
Direct URL	https://owasp.org/www-project-web-security-testing-guide/
Resource type / domains	Testing methodology · Modules F,G
Topic and subtopic coverage	Structured professional test methodology emphasizing evidence and scope
Exact path / module / chapter / playlist	Information gathering, configuration, identity, authentication, authorization, session, input validation, error handling, cryptography, business logic and client-side testing
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free
Prerequisites	W01
Why selected	Structured professional test methodology emphasizing evidence and scope
What to practise after	Use the relevant WSTG section as a checklist for the local assessment project
Backup if inaccessible	W01 — PortSwigger Web Security Academy
Known limitations	Guide is not a step-by-step beginner tutorial
Classification / age status	Mandatory · Community-maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W07 · OWASP ZAP documentation

Exact title	OWASP ZAP documentation
Direct URL	https://www.zaproxy.org/docs/
Resource type / domains	Official tool docs · Modules F,G
Topic and subtopic coverage	Free alternative to Burp Pro for automated scanning and local security testing
Exact path / module / chapter / playlist	Getting started, proxy, spider, active scan, passive scan, authentication, API and automation
Recommended difficulty / estimated time	Beginner to intermediate · 4-8 h
Access status	Free/open-source
Prerequisites	HTTP and proxy fundamentals
Why selected	Free alternative to Burp Pro for automated scanning and local security testing
What to practise after	Run a passive scan against a local Juice Shop container only and manually validate one finding
Backup if inaccessible	W05 — Burp Suite documentation
Known limitations	Automated scans do not prove exploitability or absence of bugs
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W08 · OWASP Juice Shop

Exact title	OWASP Juice Shop
Direct URL	https://owasp.org/www-project-juice-shop/
Resource type / domains	Vulnerable-by-design app · Modules F,G
Topic and subtopic coverage	Provides a realistic application to test and then remediate through configuration/code review
Exact path / module / chapter / playlist	OWASP Top 10 challenges with hints, score board and deployment options
Recommended difficulty / estimated time	Intermediate · 10-20 h
Access status	Free/open-source
Prerequisites	Docker or Node.js; isolated lab
Why selected	Provides a realistic application to test and then remediate through configuration/code review
What to practise after	Deploy only on localhost or isolated VM; complete five challenges and write defensive fixes
Backup if inaccessible	W09 — OWASP Juice Shop GitHub repository
Known limitations	Never expose intentionally vulnerable software to the public internet
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W09 · OWASP Juice Shop GitHub repository

Exact title	OWASP Juice Shop GitHub repository
Direct URL	https://github.com/juice-shop/juice-shop
Resource type / domains	Source code / setup · Modules F,G
Topic and subtopic coverage	Primary installation and source repository; safer than random third-party copies
Exact path / module / chapter / playlist	README, Docker deployment, configuration, challenge tracking and source code
Recommended difficulty / estimated time	Intermediate · Setup 1-2 h
Access status	Free/open-source
Prerequisites	Docker basics
Why selected	Primary installation and source repository; safer than random third-party copies
What to practise after	Clone an official release/tag, verify checksums where available, run with loopback binding only
Backup if inaccessible	W08 — OWASP Juice Shop
Known limitations	Development branch may change; use official tags or documented release
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W10 · DVWA GitHub repository

Exact title	DVWA GitHub repository
Direct URL	https://github.com/digininja/DVWA
Resource type / domains	Vulnerable-by-design app · Modules F,G
Topic and subtopic coverage	Simple local environment for comparing low/medium/high security levels
Exact path / module / chapter / playlist	Docker/local setup, security levels, SQLi, XSS, CSRF, file inclusion and upload challenges
Recommended difficulty / estimated time	Beginner to intermediate · 5-12 h
Access status	Free/open-source
Prerequisites	Docker or local web stack
Why selected	Simple local environment for comparing low/medium/high security levels
What to practise after	Deploy in host-only VM, complete two vulnerabilities and inspect the corresponding source code
Backup if inaccessible	W09 — OWASP Juice Shop GitHub repository
Known limitations	Outdated interface and some PHP dependencies; do not expose publicly
Classification / age status	Optional enrichment · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W11 · OWASP WebGoat

Exact title	OWASP WebGoat
Direct URL	https://owasp.org/www-project-webgoat/
Resource type / domains	Vulnerable-by-design lessons · Modules F,G
Topic and subtopic coverage	Lesson-oriented alternative when PortSwigger's exploit-first labs feel too abrupt
Exact path / module / chapter / playlist	Lessons on HTTP basics, injection, authentication, access control, CSRF, SSRF and cryptography
Recommended difficulty / estimated time	Beginner to intermediate · 8-15 h
Access status	Free/open-source
Prerequisites	Java or Docker; isolated lab
Why selected	Lesson-oriented alternative when PortSwigger's exploit-first labs feel too abrupt
What to practise after	Complete selected lessons and explain the vulnerable code path and fix
Backup if inaccessible	W01 — PortSwigger Web Security Academy
Known limitations	Requires Java/container resources; use selectively
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W12 · OWASP Dependency-Check

Exact title	OWASP Dependency-Check
Direct URL	https://owasp.org/www-project-dependency-check/
Resource type / domains	Open-source SCA tool · Modules F,I
Topic and subtopic coverage	Introduces supply-chain risk through a real scanner rather than theory
Exact path / module / chapter / playlist	Dependency scanning, NVD data, reports and CI integration concepts
Recommended difficulty / estimated time	Intermediate · 3-6 h
Access status	Free/open-source
Prerequisites	Build system basics
Why selected	Introduces supply-chain risk through a real scanner rather than theory
What to practise after	Scan a small synthetic project with a known-vulnerable dependency in a sandboxed environment, then triage false positives
Backup if inaccessible	W15 — OpenSSF Scorecard
Known limitations	NVD API rate limits and database bootstrap can be slow; use offline sample report if blocked
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W13 · GitHub code security documentation

Exact title	GitHub code security documentation
Direct URL	https://docs.github.com/en/code-security
Resource type / domains	Official documentation · Modules F,I
Topic and subtopic coverage	Directly relevant to publishing a professional portfolio
Exact path / module / chapter / playlist	Dependabot, secret scanning, code scanning, security advisories, supply-chain security and repository hardening
Recommended difficulty / estimated time	Beginner to intermediate · 5-10 h
Access status	Free features vary by repository visibility and plan
Prerequisites	Git basics
Why selected	Directly relevant to publishing a professional portfolio
What to practise after	Enable available free protections on a public sanitized repo and document residual risks
Backup if inaccessible	W14 — Semgrep documentation
Known limitations	Feature availability differs between public/private/enterprise plans
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W14 · Semgrep documentation

Exact title	Semgrep documentation
Direct URL	https://semgrep.dev/docs/
Resource type / domains	Official docs / open-source engine · Modules F,I
Topic and subtopic coverage	Teaches how static analysis rules are written, not just how to run a scanner
Exact path / module / chapter / playlist	Rule syntax, local scans, CI, secrets, supply chain and custom rules
Recommended difficulty / estimated time	Intermediate · 5-10 h
Access status	Open-source engine free; some platform features paid
Prerequisites	Programming and regex
Why selected	Teaches how static analysis rules are written, not just how to run a scanner
What to practise after	Write one simple custom rule for a deliberately insecure local test file and document false positives
Backup if inaccessible	W13 — GitHub code security documentation
Known limitations	Cloud platform and engine features differ
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W15 · OpenSSF Scorecard

Exact title	OpenSSF Scorecard
Direct URL	https://securityscorecards.dev/
Resource type / domains	Open-source project / checks · Modules F,I
Topic and subtopic coverage	Concrete, automated supply-chain hygiene checks for repositories
Exact path / module / chapter / playlist	Checks for branch protection, pinned dependencies, signed releases, tokens, CI and vulnerability disclosure
Recommended difficulty / estimated time	Intermediate · 2-5 h
Access status	Free/open-source
Prerequisites	Git and CI basics
Why selected	Concrete, automated supply-chain hygiene checks for repositories
What to practise after	Run scorecard against a public non-sensitive repository and improve two failing checks
Backup if inaccessible	W13 — GitHub code security documentation
Known limitations	Not all checks apply to small student projects
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W16 · SLSA specification

Exact title	SLSA specification
Direct URL	https://slsa.dev/
Resource type / domains	Supply-chain framework · Modules F,I
Topic and subtopic coverage	Primary framework for reasoning about build provenance and software supply-chain integrity
Exact path / module / chapter / playlist	Build levels, provenance, source, build and dependency threats
Recommended difficulty / estimated time	Advanced · 3-6 h
Access status	Free
Prerequisites	CI/CD and software delivery
Why selected	Primary framework for reasoning about build provenance and software supply-chain integrity
What to practise after	Map one GitHub Actions workflow to SLSA threats and identify missing controls
Backup if inaccessible	W13 — GitHub code security documentation
Known limitations	Specification evolves; use current level documentation
Classification / age status	Advanced reference · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W17 · OWASP Cheat Sheet Series

Exact title	OWASP Cheat Sheet Series
Direct URL	https://cheatsheetseries.owasp.org/
Resource type / domains	Official checklist library · Modules F
Topic and subtopic coverage	Short, authoritative remediation guidance that pairs well with PortSwigger offense labs
Exact path / module / chapter / playlist	Authentication, session, access control, XSS, SQL injection, CSRF, SSRF, file upload, secrets, logging, TLS and API security cheat sheets
Recommended difficulty / estimated time	Beginner to advanced · Reference use
Access status	Free
Prerequisites	Web security fundamentals
Why selected	Short, authoritative remediation guidance that pairs well with PortSwigger offense labs
What to practise after	After each lab, read the matching cheat sheet and implement one control in the secure app
Backup if inaccessible	C15 — OWASP Application Security Verification Standard
Known limitations	Cheat sheets can be dense; use them after hands-on practice
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E01 · NIST SP 800-115

Exact title	NIST SP 800-115
Direct URL	https://csrc.nist.gov/pubs/sp/800/115/final
Resource type / domains	Official testing guidance · Modules G
Topic and subtopic coverage	Authoritative methodology for safe security testing and evidence handling
Exact path / module / chapter / playlist	Planning, discovery, vulnerability scanning, validation, reporting and post-test activities
Recommended difficulty / estimated time	Intermediate · 6-10 h
Access status	Free
Prerequisites	Networking and written authorization
Why selected	Authoritative methodology for safe security testing and evidence handling
What to practise after	Create a one-page rules-of-engagement document for your isolated lab before any scanning
Backup if inaccessible	W06 — OWASP Web Security Testing Guide
Known limitations	Published 2008; methodology remains valid but tool examples are outdated
Classification / age status	Mandatory · Old but gold, update tool commands
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E02 · TryHackMe Pre Security learning path

Exact title	TryHackMe Pre Security learning path
Direct URL	https://tryhackme.com/path/outline/presecurity
Resource type / domains	Interactive path · Modules A,B,E,G
Topic and subtopic coverage	Browser-based guided labs for low-end hardware and early confidence
Exact path / module / chapter / playlist	Computer basics, networking, web fundamentals, Linux and Windows basics, introductory offensive/defensive rooms
Recommended difficulty / estimated time	Beginner · 30-40 h
Access status	Freemium; introductory rooms free, full access may require subscription
Prerequisites	None
Why selected	Browser-based guided labs for low-end hardware and early confidence
What to practise after	Complete only assigned free rooms; use local backups for locked material
Backup if inaccessible	L01 — Linux Journey
Known limitations	Free-room availability and path structure changed in 2026; verify before buying
Classification / age status	Optional enrichment · Revamped 2026
Verification status / date	Rate-limited automated check (HTTP 429); official URL; retry in browser or use backup · checked 2026-10-27

E03 · TryHackMe Cyber Security 101 path

Exact title	TryHackMe Cyber Security 101 path
Direct URL	https://tryhackme.com/path/outline/cybersecurity101
Resource type / domains	Interactive path · Modules E,G
Topic and subtopic coverage	Structured bridge from basics to junior analyst topics; useful only when free rooms are available
Exact path / module / chapter / playlist	Networks, cryptography, Windows/AD, Linux, web, defensive tools and junior roles
Recommended difficulty / estimated time	Beginner · 60-80 h
Access status	Freemium; many rooms subscription-only
Prerequisites	Pre Security foundations
Why selected	Structured bridge from basics to junior analyst topics; useful only when free rooms are available
What to practise after	Use selected free intro rooms as extra practice; do not let subscription replace local projects
Backup if inaccessible	E01 — NIST SP 800-115
Known limitations	Path content and free/paid status can change
Classification / age status	Optional enrichment · Current 2026 path
Verification status / date	Rate-limited automated check (HTTP 429); official URL; retry in browser or use backup · checked 2026-10-27

E04 · Hack The Box Academy

Exact title	Hack The Box Academy
Direct URL	https://academy.hackthebox.com/
Resource type / domains	Interactive learning platform · Modules G,K
Topic and subtopic coverage	High-quality guided modules with downloadable resources and hands-on practice
Exact path / module / chapter / playlist	Introductory modules on networking, Linux, Windows, web, privilege escalation and security fundamentals
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Freemium; some modules free, others require cubes/subscription
Prerequisites	Basic Linux and networking
Why selected	High-quality guided modules with downloadable resources and hands-on practice
What to practise after	Filter for free modules; record local commands and concepts rather than collecting badges
Backup if inaccessible	N09 — Nmap Reference Guide
Known limitations	Pricing and free-module availability vary
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E05 · HackerOne Disclosure Guidelines

Exact title	HackerOne Disclosure Guidelines
Direct URL	https://www.hackerone.com/disclosure-guidelines
Resource type / domains	Responsible disclosure guidance · Modules G,J
Topic and subtopic coverage	Useful practical introduction to professional disclosure expectations
Exact path / module / chapter / playlist	Safe harbor, authorized testing, reporting expectations and coordinated disclosure concepts
Recommended difficulty / estimated time	Beginner to intermediate · 1-2 h
Access status	Free
Prerequisites	Ethics and scope basics
Why selected	Useful practical introduction to professional disclosure expectations
What to practise after	Draft a disclosure report for a synthetic finding; verify any real program's policy before testing
Backup if inaccessible	E06 — FIRST CVSS specification
Known limitations	Platform policy is not universal law and does not grant authorization for arbitrary targets
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E06 · FIRST CVSS specification

Exact title	FIRST CVSS specification
Direct URL	https://www.first.org/cvss/
Resource type / domains	Scoring standard · Modules G,J
Topic and subtopic coverage	Standardized severity language; necessary but insufficient for risk prioritization
Exact path / module / chapter / playlist	Base, temporal and environmental metrics; calculator and examples
Recommended difficulty / estimated time	Intermediate · 2-4 h
Access status	Free
Prerequisites	Vulnerability basics
Why selected	Standardized severity language; necessary but insufficient for risk prioritization
What to practise after	Score one synthetic vulnerability, then explain why business context can override the numeric score
Backup if inaccessible	E08 — NVD
Known limitations	CVSS is not a full risk assessment
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E07 · CVE Program

Exact title	CVE Program
Direct URL	https://www.cve.org/
Resource type / domains	Vulnerability identifier program · Modules G,J
Topic and subtopic coverage	Primary source for identifier process and terminology
Exact path / module / chapter / playlist	CVE IDs, CNAs, records and program information
Recommended difficulty / estimated time	Beginner · 1-2 h
Access status	Free
Prerequisites	Vulnerability concepts
Why selected	Primary source for identifier process and terminology
What to practise after	Take one synthetic finding and explain why it would or would not deserve a CVE
Backup if inaccessible	E08 — NVD
Known limitations	CVE assignment itself does not prove exploitability or priority
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E08 · NVD

Exact title	NVD
Direct URL	https://nvd.nist.gov/
Resource type / domains	Vulnerability database · Modules G,J
Topic and subtopic coverage	Authoritative public vulnerability-data source for triage exercises
Exact path / module / chapter / playlist	CVE records, CVSS, CPE, references and search APIs
Recommended difficulty / estimated time	Intermediate · 2-5 h
Access status	Free
Prerequisites	CVE/CVSS basics
Why selected	Authoritative public vulnerability-data source for triage exercises
What to practise after	Compare an NVD record with the vendor advisory and note data freshness limitations
Backup if inaccessible	N09 — Nmap Reference Guide
Known limitations	NVD enrichment can lag; API rate limits apply
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E09 · picoCTF

Exact title	picoCTF
Direct URL	https://picoctf.org/
Resource type / domains	CTF platform · Modules D,G,K
Topic and subtopic coverage	Carnegie Mellon-affordable CTF with approachable early challenges
Exact path / module / chapter / playlist	Beginner cryptography, web exploitation, reverse engineering, forensics and general skills
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Free competition/picoGym account
Prerequisites	Programming and Linux basics
Why selected	Carnegie Mellon-affordable CTF with approachable early challenges
What to practise after	Solve five general-skills or forensics challenges and write independent solution notes
Backup if inaccessible	L15 — pwn.college Dojo index
Known limitations	Competition archive availability and scoreboards change
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E10 · CyLab Security Academy / picoGym

Exact title	CyLab Security Academy / picoGym
Direct URL	https://www.cylabacademy.org/
Resource type / domains	Practice platform · Modules D,G,K
Topic and subtopic coverage	Allows non-competition practice without waiting for the annual contest
Exact path / module / chapter / playlist	Practice challenges from prior picoCTF competitions
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Free account
Prerequisites	CTF basics
Why selected	Allows non-competition practice without waiting for the annual contest
What to practise after	Choose beginner picoGym problems assigned in the daily plan; do not publish current contest flags
Backup if inaccessible	E09 — picoCTF
Known limitations	Practice catalog may change
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B01 · Microsoft Sentinel documentation

Exact title	Microsoft Sentinel documentation
Direct URL	https://learn.microsoft.com/en-us/azure/sentinel/
Resource type / domains	Official documentation · Modules H
Topic and subtopic coverage	Leading SIEM documentation with free learning paths; use simulated data before cloud deployment
Exact path / module / chapter / playlist	Architecture, data connectors, workspaces, analytics rules, incidents, hunting, KQL, SOAR and content hub
Recommended difficulty / estimated time	Intermediate · 15-30 h selected
Access status	Docs free; Azure/Sentinel usage may incur costs
Prerequisites	Logs and cloud basics
Why selected	Leading SIEM documentation with free learning paths; use simulated data before cloud deployment
What to practise after	Use Microsoft free training data or local sample logs; do not enable paid ingestion without budgets
Backup if inaccessible	B02 — Kusto Query Language overview
Known limitations	Product names and UI change; cloud cost risk
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B02 · Kusto Query Language overview

Exact title	Kusto Query Language overview
Direct URL	https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric
Resource type / domains	Official language docs · Modules H
Topic and subtopic coverage	KQL is essential for modern Microsoft SOC and threat-hunting work
Exact path / module / chapter / playlist	KQL operators, tabular expressions, joins, summarize, let, time-series, parsing and best practices
Recommended difficulty / estimated time	Beginner to intermediate · 10-20 h
Access status	Free
Prerequisites	SQL/log analysis basics
Why selected	KQL is essential for modern Microsoft SOC and threat-hunting work
What to practise after	Write five queries against synthetic security event tables and explain each operator
Backup if inaccessible	B03 — SigmaHQ
Known limitations	Syntax and service-specific tables vary by product
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B03 · SigmaHQ

Exact title	SigmaHQ
Direct URL	https://sigmahq.io/
Resource type / domains	Open-source detection rules · Modules H,K
Topic and subtopic coverage	Vendor-neutral detection engineering with readable, reviewable rules
Exact path / module / chapter / playlist	Generic SIEM detection rule format, rule repository and tooling links
Recommended difficulty / estimated time	Intermediate · 5-10 h
Access status	Free/open-source
Prerequisites	Logs and ATT&CK basics
Why selected	Vendor-neutral detection engineering with readable, reviewable rules
What to practise after	Convert one Sigma rule to a local query and test it against synthetic true/false positive events
Backup if inaccessible	B18 — CISA Known Exploited Vulnerabilities Catalog
Known limitations	Rule quality varies; always validate in your own data
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B04 · Sigma rule creation guide

Exact title	Sigma rule creation guide
Direct URL	https://sigmahq.io/docs/guide/getting-started.html
Resource type / domains	Official rule reference · Modules H,K
Topic and subtopic coverage	Exact syntax reference for writing and reviewing detection rules
Exact path / module / chapter / playlist	Title, ID, status, description, author, date, logsource, detection, condition, falsepositives, level and tags
Recommended difficulty / estimated time	Intermediate · 2-4 h
Access status	Free
Prerequisites	Sigma basics
Why selected	Exact syntax reference for writing and reviewing detection rules
What to practise after	Author one original Sigma rule for a benign lab event and include test data
Backup if inaccessible	B03 — SigmaHQ
Known limitations	Documentation URL or schema can change; use site navigation
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B05 · SBousseaden EVTX-ATTACK-SAMPLES

Exact title	SBousseaden EVTX-ATTACK-SAMPLES
Direct URL	https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES
Resource type / domains	Synthetic/educational log samples · Modules H,K
Topic and subtopic coverage	Rare, credible public event-log dataset for offline detection practice without cloud costs
Exact path / module / chapter / playlist	Windows EVTX samples organized by ATT&CK technique, useful for detection testing
Recommended difficulty / estimated time	Intermediate · 3-8 h
Access status	Free/open-source
Prerequisites	Windows event logs and safe VM
Why selected	Rare, credible public event-log dataset for offline detection practice without cloud costs
What to practise after	Download one sample in an isolated VM, hash it, identify fields, and write a local query
Backup if inaccessible	B02 — Kusto Query Language overview
Known limitations	Third-party repository; verify contents and do not execute unknown binaries; EVTX files are data
Classification / age status	Recommended · Community-maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B06 · CyberDefenders

Exact title	CyberDefenders
Direct URL	https://cyberdefenders.org/
Resource type / domains	Blue-team challenges · Modules H,K
Topic and subtopic coverage	Defensive labs built around realistic evidence and question sets
Exact path / module / chapter / playlist	Digital forensics, incident response, threat hunting, network traffic, endpoint and malware traffic challenges
Recommended difficulty / estimated time	Intermediate to advanced · Varies
Access status	Freemium; some challenges free, some require subscription
Prerequisites	Wireshark, logs and IR basics
Why selected	Defensive labs built around realistic evidence and question sets
What to practise after	Complete one free beginner challenge and write an incident report, not just answers
Backup if inaccessible	N07 — Wireshark SampleCaptures
Known limitations	Free/paid status changes; some datasets are large
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B07 · Blue Team Labs Online

Exact title	Blue Team Labs Online
Direct URL	https://blueteamlabs.online/
Resource type / domains	Blue-team labs · Modules H,K
Topic and subtopic coverage	Short scenario-based labs suitable for weekly assessments
Exact path / module / chapter / playlist	Investigations, phishing, endpoint, SIEM, threat intel and DFIR labs
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Freemium; free and paid labs
Prerequisites	Basic blue-team concepts
Why selected	Short scenario-based labs suitable for weekly assessments
What to practise after	Complete assigned free labs only; substitute local synthetic data if unavailable
Backup if inaccessible	B06 — CyberDefenders
Known limitations	Paid tier/lab catalogue changes; lab platform login required
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B08 · Malware Traffic Analysis

Exact title	Malware Traffic Analysis
Direct URL	https://malware-traffic-analysis.net/
Resource type / domains	Packet capture exercises · Modules B,H,K
Topic and subtopic coverage	Long-running specialist site with realistic traffic-analysis practice
Exact path / module / chapter / playlist	Realistic network captures and malware traffic exercises with questions; use datasets only in isolated analysis
Recommended difficulty / estimated time	Intermediate to advanced · 2-8 h per exercise
Access status	Free
Prerequisites	Wireshark and network fundamentals
Why selected	Long-running specialist site with realistic traffic-analysis practice
What to practise after	Analyze one specified capture in a disposable VM and report IOCs with confidence levels
Backup if inaccessible	N07 — Wireshark SampleCaptures
Known limitations	Contains malware-related artifacts; never execute samples; some links may be old
Classification / age status	Recommended · Old but gold with maintained exercises
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B09 · Zeek documentation

Exact title	Zeek documentation
Direct URL	https://docs.zeek.org/en/master/
Resource type / domains	Official documentation · Modules B,H
Topic and subtopic coverage	Produces structured network telemetry and teaches protocol-aware logging
Exact path / module / chapter / playlist	Architecture, logs, scripting, frameworks, signatures, packet analysis and cluster use
Recommended difficulty / estimated time	Advanced · 10-20 selected pages
Access status	Free/open-source
Prerequisites	Linux and networking
Why selected	Produces structured network telemetry and teaches protocol-aware logging
What to practise after	Install in a low-resource VM or use sample logs; explain one conn.log field
Backup if inaccessible	B10 — Suricata documentation
Known limitations	Installation and version differences; not needed on low-end hardware
Classification / age status	Advanced reference · Maintained
Verification status / date	Rate-limited automated check (HTTP 429); official URL; retry in browser or use backup · checked 2026-10-27

B10 · Suricata documentation

Exact title	Suricata documentation
Direct URL	https://docs.suricata.io/en/latest/
Resource type / domains	Official documentation · Modules B,H
Topic and subtopic coverage	Free IDS/NSM platform for network detection engineering
Exact path / module / chapter / playlist	Installation, configuration, rules, outputs, performance and Lua; Alert and EVE JSON logs
Recommended difficulty / estimated time	Intermediate · 8-15 h
Access status	Free/open-source
Prerequisites	TCP/IP and Linux
Why selected	Free IDS/NSM platform for network detection engineering
What to practise after	Write one benign local alert rule, generate synthetic traffic and validate EVE JSON
Backup if inaccessible	B03 — SigmaHQ
Known limitations	Rule syntax and version-specific features; use matching docs version
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B11 · Autopsy documentation

Exact title	Autopsy documentation
Direct URL	https://www.autopsy.com/support/
Resource type / domains	Official tool docs · Modules H,K
Topic and subtopic coverage	Accessible entry into digital forensics using disk images the student creates
Exact path / module / chapter / playlist	Case creation, disk images, ingest modules, timeline, keyword search, artifacts and reports
Recommended difficulty / estimated time	Beginner to intermediate · 5-10 h
Access status	Free/open-source
Prerequisites	Filesystems and evidence integrity basics
Why selected	Accessible entry into digital forensics using disk images the student creates
What to practise after	Create a small FAT/exFAT or ext4 test image, add benign files, delete one, and recover evidence
Backup if inaccessible	B12 — SANS SIFT Workstation
Known limitations	Older screenshots may differ from current release; never analyze personal disks without permission
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B12 · SANS SIFT Workstation

Exact title	SANS SIFT Workstation
Direct URL	https://www.sans.org/tools/sift-workstation/
Resource type / domains	Forensic toolkit · Modules H,K
Topic and subtopic coverage	Well-known free forensic environment consolidating many analysis utilities
Exact path / module / chapter / playlist	Free Ubuntu-based DFIR workstation, tools, download and documentation
Recommended difficulty / estimated time	Intermediate to advanced · Setup 2-4 h
Access status	Free
Prerequisites	Linux and DFIR basics
Why selected	Well-known free forensic environment consolidating many analysis utilities
What to practise after	Install only if hardware allows; otherwise use Autopsy plus native tools on a disposable VM
Backup if inaccessible	B11 — Autopsy documentation
Known limitations	Large download and Linux familiarity required; SANS courses themselves are paid
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B13 · MISP Project

Exact title	MISP Project
Direct URL	https://www.misp-project.org/
Resource type / domains	Threat intelligence platform · Modules H,K
Topic and subtopic coverage	Widely used platform for structured threat intelligence and indicator context
Exact path / module / chapter / playlist	Indicators, events, attributes, taxonomies, sharing and API concepts
Recommended difficulty / estimated time	Intermediate to advanced · 5-12 h
Access status	Free/open-source; hosting costs possible
Prerequisites	Logging and networking
Why selected	Widely used platform for structured threat intelligence and indicator context
What to practise after	Use official documentation or a local demo; create a synthetic event and map an indicator to an ATT&CK technique
Backup if inaccessible	C18 — MITRE ATT&CK
Known limitations	Do not upload personal or unauthorized data; local hosting can be heavy
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B14 · Elastic detection rules documentation

Exact title	Elastic detection rules documentation
Direct URL	https://www.elastic.co/guide/en/security/current/detection-engine-overview.html
Resource type / domains	Official docs / open-source rules · Modules H,K
Topic and subtopic coverage	Real detection-rule implementation patterns and open-source rule content
Exact path / module / chapter / playlist	Detection engine, rule types, exceptions, timelines, alerts and prebuilt rule repository
Recommended difficulty / estimated time	Intermediate · 5-10 h
Access status	Free docs/open-source rules; Elastic Stack may have paid features
Prerequisites	SIEM basics
Why selected	Real detection-rule implementation patterns and open-source rule content
What to practise after	Read one prebuilt rule, reproduce its logic locally on synthetic logs
Backup if inaccessible	B03 — SigmaHQ
Known limitations	Product UI and rule schema change by version
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B15 · Microsoft SC-200 certification page

Exact title	Microsoft SC-200 certification page
Direct URL	https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/
Resource type / domains	Official certification page · Modules H,J
Topic and subtopic coverage	Optional credential if you specialize in Microsoft security operations
Exact path / module / chapter / playlist	Security operations, incident response and threat hunting objectives using Microsoft Defender/ Sentinel; annual free renewal assessment
Recommended difficulty / estimated time	Intermediate · 20-40 h prep
Access status	Free learning content; exam price region-based
Prerequisites	SOC fundamentals and KQL
Why selected	Optional credential if you specialize in Microsoft security operations
What to practise after	Use free study guide and practice assessment before considering the paid exam
Backup if inaccessible	B01 — Microsoft Sentinel documentation
Known limitations	Exam objectives were refreshed in 2026; check current study guide
Classification / age status	Optional enrichment · Updated 2026-10-21
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B16 · Blue Team Level 1 official certification

Exact title	Blue Team Level 1 official certification
Direct URL	https://www.centri.org/certifications/blue-team-level-1
Resource type / domains	Paid course/certification · Modules H,J
Topic and subtopic coverage	Practical blue-team credential if you later have a training budget and already finished free projects
Exact path / module / chapter / playlist	Phishing, threat intelligence, DFIR, SIEM and incident response with a practical exam
Recommended difficulty / estimated time	Intermediate · 80-120 h
Access status	Paid; GBP 399 listed on official page as checked 2026-10-27; one free resit included
Prerequisites	Networking, logs and Wireshark
Why selected	Practical blue-team credential if you later have a training budget and already finished free projects
What to practise after	Do not buy in Year 1; recreate equivalent labs locally first
Backup if inaccessible	B06 — CyberDefenders
Known limitations	Price, provider brand and access terms may change; not required for this roadmap
Classification / age status	Paid/alternative · Current 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B17 · CISA cybersecurity advisories

Exact title	CISA cybersecurity advisories
Direct URL	https://www.cisa.gov/news-events/cybersecurity-advisories
Resource type / domains	Official advisories · Modules H,J,K
Topic and subtopic coverage	Primary-source reading habit that teaches how incidents are reported in practice
Exact path / module / chapter / playlist	Current technical alerts, TTPs, IOCs and mitigation guidance
Recommended difficulty / estimated time	Intermediate · Ongoing 15 min/week
Access status	Free
Prerequisites	Threat intelligence basics
Why selected	Primary-source reading habit that teaches how incidents are reported in practice
What to practise after	Select one advisory and map two TTPs to ATT&CK and two mitigations to lab evidence
Backup if inaccessible	C18 — MITRE ATT&CK
Known limitations	US government source; not exhaustive global coverage
Classification / age status	Recommended · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B18 · CISA Known Exploited Vulnerabilities Catalog

Exact title	CISA Known Exploited Vulnerabilities Catalog
Direct URL	https://www.cisa.gov/known-exploited-vulnerabilities-catalog
Resource type / domains	Official vulnerability catalog · Modules G,J
Topic and subtopic coverage	Evidence-based prioritization data that complements CVSS
Exact path / module / chapter / playlist	KEV entries with CVE, vendor/product, due date, name, description and required action
Recommended difficulty / estimated time	Intermediate · 2-5 h
Access status	Free
Prerequisites	CVE basics
Why selected	Evidence-based prioritization data that complements CVSS
What to practise after	Choose one KEV and write a risk-based patch priority note using local context
Backup if inaccessible	E08 — NVD
Known limitations	US federal remediation deadlines do not automatically bind Pakistani or private organizations
Classification / age status	Recommended · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B19 · SC-200 Exam Prep YouTube playlist

Exact title	SC-200 Exam Prep YouTube playlist
Direct URL	https://www.youtube.com/playlist?list=PLahhVEj9XNTcHNtYH-inn7xFJactEM7zz
Resource type / domains	Official Microsoft exam-prep video series · Modules H,J
Topic and subtopic coverage	Structured Microsoft-produced video supplement to official Learn paths
Exact path / module / chapter / playlist	SC-200 exam preparation episodes mapped to current skills measured
Recommended difficulty / estimated time	Intermediate · 4-8 h
Access status	Free
Prerequisites	SOC and Microsoft security basics
Why selected	Structured Microsoft-produced video supplement to official Learn paths
What to practise after	Watch selected episodes during Month 8 and Month 10, then complete the linked Learn module
Backup if inaccessible	B15 — Microsoft SC-200 certification page
Known limitations	Certification content can change; use current study guide first
Classification / age status	Optional enrichment · Current playlist, individual dates vary
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL01 · AWS Skill Builder

Exact title	AWS Skill Builder
Direct URL	https://skillbuilder.aws/
Resource type / domains	Official training platform · Modules I,J
Topic and subtopic coverage	Official AWS training hub with current exam prep and foundational security material
Exact path / module / chapter / playlist	Digital courses, exam prep plans, Builder Labs, Cloud Quest and AWS Jam; free and paid content
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Freemium; many digital courses free, labs/subscription can be paid
Prerequisites	Cloud basics
Why selected	Official AWS training hub with current exam prep and foundational security material
What to practise after	Use only free digital courses in Month 9; avoid paid labs until you set a strict budget
Backup if inaccessible	CL02 — AWS security documentation
Known limitations	Catalog and free/paid status changes
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL02 · AWS security documentation

Exact title	AWS security documentation
Direct URL	https://docs.aws.amazon.com/security/
Resource type / domains	Official documentation · Modules I
Topic and subtopic coverage	Primary source for AWS security configuration and service behavior
Exact path / module / chapter / playlist	Security guidance by service, identity, logging, encryption, incident response and best practices
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free
Prerequisites	Cloud fundamentals
Why selected	Primary source for AWS security configuration and service behavior
What to practise after	Look up the exact service used in a daily lab and verify security defaults
Backup if inaccessible	CL03 — AWS Well-Architected Security Pillar
Known limitations	Very broad; use search rather than linear reading
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL03 · AWS Well-Architected Security Pillar

Exact title	AWS Well-Architected Security Pillar
Direct URL	https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html
Resource type / domains	Official architecture guide · Modules I,J
Topic and subtopic coverage	Design-level security guidance with review questions
Exact path / module / chapter / playlist	Security foundations, IAM, detection, infrastructure protection, data protection, incident response and application security
Recommended difficulty / estimated time	Intermediate · 8-15 h
Access status	Free
Prerequisites	Cloud fundamentals
Why selected	Design-level security guidance with review questions
What to practise after	Perform a paper review of a fictional small AWS architecture and document ten risks
Backup if inaccessible	CL02 — AWS security documentation
Known limitations	AWS-specific; other clouds need their own architecture guidance
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL04 · AWS Free Tier

Exact title	AWS Free Tier
Direct URL	https://aws.amazon.com/free/
Resource type / domains	Official program page · Modules I
Topic and subtopic coverage	Critical source for avoiding accidental cloud spending
Exact path / module / chapter / playlist	Free-tier offer types, service limits, expiry and account controls
Recommended difficulty / estimated time	Beginner · 1-2 h
Access status	Program terms; some services free temporarily, others always free
Prerequisites	Cloud account basics
Why selected	Critical source for avoiding accidental cloud spending
What to practise after	Read current offer types and configure billing alerts before any hands-on cloud lab
Backup if inaccessible	CL01 — AWS Skill Builder
Known limitations	Terms change; credits/expiry vary; a credit card may still be required
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL05 · Microsoft Azure security documentation

Exact title	Microsoft Azure security documentation
Direct URL	https://learn.microsoft.com/en-us/azure/security/
Resource type / domains	Official documentation · Modules I
Topic and subtopic coverage	Alternative cloud primary source if Azure fits available opportunities better than AWS
Exact path / module / chapter / playlist	Security architecture, identity, network, data, logging, incident response and compliance
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free
Prerequisites	Cloud fundamentals
Why selected	Alternative cloud primary source if Azure fits available opportunities better than AWS
What to practise after	Compare one AWS and one Azure control for identity and logging
Backup if inaccessible	CL02 — AWS security documentation
Known limitations	Product names and services change
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL06 · Docker documentation

Exact title	Docker documentation
Direct URL	https://docs.docker.com/
Resource type / domains	Official documentation · Modules C,I
Topic and subtopic coverage	Authoritative container foundation before Kubernetes security
Exact path / module / chapter / playlist	Get started, engine, images, containers, networks, volumes, Compose, build and security
Recommended difficulty / estimated time	Beginner to intermediate · 15-25 h selected
Access status	Free; Docker Desktop licensing may apply in some organizations
Prerequisites	Linux command line
Why selected	Authoritative container foundation before Kubernetes security
What to practise after	Build a tiny image, run isolated container, inspect layers, networks and volumes
Backup if inaccessible	CL07 — Docker Engine security documentation
Known limitations	Menu changes and Docker Desktop license terms differ from Docker Engine
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL07 · Docker Engine security documentation

Exact title	Docker Engine security documentation
Direct URL	https://docs.docker.com/engine/security/
Resource type / domains	Official security docs · Modules I
Topic and subtopic coverage	Primary source for container threat model and hardening controls
Exact path / module / chapter / playlist	Daemon attack surface, namespaces, capabilities, seccomp, AppArmor, rootless mode, images and secrets
Recommended difficulty / estimated time	Intermediate · 6-12 h
Access status	Free
Prerequisites	Docker basics
Why selected	Primary source for container threat model and hardening controls
What to practise after	Run a container as a non-root user, drop capabilities, inspect seccomp and record before/after metadata
Backup if inaccessible	CL06 — Docker documentation
Known limitations	Host kernel distribution affects defaults
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL08 · Kubernetes official documentation

Exact title	Kubernetes official documentation
Direct URL	https://kubernetes.io/docs/home/
Resource type / domains	Official documentation · Modules I
Topic and subtopic coverage	Primary source for cluster architecture and current API behavior
Exact path / module / chapter / playlist	Concepts, tasks, tutorials, reference, kubectl, workloads, services, storage, configuration and security
Recommended difficulty / estimated time	Intermediate · 20-40 h selected
Access status	Free
Prerequisites	Containers and networking
Why selected	Primary source for cluster architecture and current API behavior
What to practise after	Use kind/minikube locally for basic objects; do not pay for a managed cluster in Year 1
Backup if inaccessible	CL09 — Kubernetes security documentation
Known limitations	Version-specific APIs; use supported version docs
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL09 · Kubernetes security documentation

Exact title	Kubernetes security documentation
Direct URL	https://kubernetes.io/docs/concepts/security/
Resource type / domains	Official documentation · Modules I
Topic and subtopic coverage	Primary security reference after architecture fundamentals
Exact path / module / chapter / playlist	Cloud-native security, pod security, RBAC, network policies, secrets, audit, multi-tenancy and supply chain
Recommended difficulty / estimated time	Intermediate · 8-15 h
Access status	Free
Prerequisites	Kubernetes basics
Why selected	Primary security reference after architecture fundamentals
What to practise after	Create a local RBAC role, service account, NetworkPolicy and Pod Security Admission example
Backup if inaccessible	CL08 — Kubernetes official documentation
Known limitations	Some features require cluster configuration that kind/minikube may not fully emulate
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL10 · OWASP Kubernetes Top 10

Exact title	OWASP Kubernetes Top 10
Direct URL	https://owasp.org/www-project-kubernetes-top-ten/
Resource type / domains	Risk reference · Modules I
Topic and subtopic coverage	Risk-oriented checklist for reviewing local manifests and cluster settings
Exact path / module / chapter / playlist	K01-K10: insecure workloads, supply chain, RBAC, policy enforcement, logging, secrets, network, config, vulnerabilities and components
Recommended difficulty / estimated time	Intermediate · 4-8 h
Access status	Free
Prerequisites	Kubernetes basics
Why selected	Risk-oriented checklist for reviewing local manifests and cluster settings
What to practise after	Map ten findings from a synthetic manifest review to K01-K10 and remediation
Backup if inaccessible	CL09 — Kubernetes security documentation
Known limitations	Checklist, not a deployment tutorial
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL11 · Kubernetes Goat

Exact title	Kubernetes Goat
Direct URL	https://madhuakula.com/kubernetes-goat/
Resource type / domains	Vulnerable-by-design K8s lab · Modules I,K
Topic and subtopic coverage	Hands-on local cluster security practice without cloud spending
Exact path / module / chapter / playlist	Scenarios on insecure RBAC, container escape, secrets, supply chain, network policies and observability
Recommended difficulty / estimated time	Advanced · 8-20 h
Access status	Free/open-source
Prerequisites	Kubernetes and Docker basics
Why selected	Hands-on local cluster security practice without cloud spending
What to practise after	Run only in a disposable local cluster; complete two beginner scenarios and remediation notes
Backup if inaccessible	CL10 — OWASP Kubernetes Top 10
Known limitations	Can consume significant RAM; skip if hardware is below 16 GB
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL12 · Trivy documentation

Exact title	Trivy documentation
Direct URL	https://trivy.dev/docs/
Resource type / domains	Official scanner docs · Modules I
Topic and subtopic coverage	Single free scanner for images, dependencies and IaC misconfigurations
Exact path / module / chapter / playlist	Container image, filesystem, repository, IaC, Kubernetes and SBOM scanning; installation and CI examples
Recommended difficulty / estimated time	Beginner to intermediate · 3-8 h
Access status	Free/open-source
Prerequisites	Docker basics
Why selected	Single free scanner for images, dependencies and IaC misconfigurations
What to practise after	Scan a local Dockerfile/image, triage one finding, rebuild and rescan
Backup if inaccessible	W12 — OWASP Dependency-Check
Known limitations	Vulnerability DB freshness and false positives; no scanner proves security
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL13 · Checkov documentation

Exact title	Checkov documentation
Direct URL	https://www.checkov.io/1.Welcome/What%20is%20Checkov.html
Resource type / domains	Official IaC scanner docs · Modules I
Topic and subtopic coverage	Shows how to codify preventive infrastructure controls and validate fixes
Exact path / module / chapter / playlist	Terraform, CloudFormation, Kubernetes, Dockerfile and GitHub Actions policy scanning; custom policies and CI
Recommended difficulty / estimated time	Intermediate · 3-8 h
Access status	Free/open-source; Prisma Cloud features paid
Prerequisites	IaC basics
Why selected	Shows how to codify preventive infrastructure controls and validate fixes
What to practise after	Scan a synthetic Terraform/Kubernetes manifest, remediate three findings and rescan
Backup if inaccessible	CL12 — Trivy documentation
Known limitations	Policy coverage varies by framework and version
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL14 · GitHub Actions security hardening

Exact title	GitHub Actions security hardening
Direct URL	https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions
Resource type / domains	Official guidance · Modules I
Topic and subtopic coverage	Prevents supply-chain mistakes in the very platform used for the portfolio
Exact path / module / chapter / playlist	Secrets, GITHUB_TOKEN permissions, third-party actions, OIDC, dependency pinning, runners and fork safety
Recommended difficulty / estimated time	Intermediate · 4-8 h
Access status	Free
Prerequisites	Git and CI/CD basics
Why selected	Prevents supply-chain mistakes in the very platform used for the portfolio
What to practise after	Harden one workflow: least permissions, pinned action SHAs, no secrets in logs and dependency scanning
Backup if inaccessible	W13 — GitHub code security documentation
Known limitations	Features differ for public/private/enterprise repos
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL15 · Terraform documentation

Exact title	Terraform documentation
Direct URL	https://developer.hashicorp.com/terraform/docs
Resource type / domains	Official documentation · Modules I
Topic and subtopic coverage	Common IaC platform for reproducible cloud labs and configuration review
Exact path / module / chapter / playlist	Configuration language, providers, resources, state, modules, CLI and policy guidance
Recommended difficulty / estimated time	Intermediate · 8-20 h selected
Access status	Free/open-source; HCP Terraform has paid tiers
Prerequisites	Infrastructure and Git basics
Why selected	Common IaC platform for reproducible cloud labs and configuration review
What to practise after	Create a local-only Terraform plan for a mock resource; do not apply to a paid cloud account
Backup if inaccessible	CL13 — Checkov documentation
Known limitations	State files can contain secrets; never commit them
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL16 · Cloud Security Alliance Cloud Controls Matrix

Exact title	Cloud Security Alliance Cloud Controls Matrix
Direct URL	https://cloudsecurityalliance.org/research/cloud-controls-matrix
Resource type / domains	Framework/reference · Modules I,J
Topic and subtopic coverage	Cloud-specific control catalog for comparing AWS/Azure implementations
Exact path / module / chapter / playlist	Cloud control domains, control IDs, implementation guidance and mappings
Recommended difficulty / estimated time	Intermediate · 5-10 h
Access status	Free registration may be required for some downloads
Prerequisites	Cloud basics and GRC vocabulary
Why selected	Cloud-specific control catalog for comparing AWS/Azure implementations
What to practise after	Select five CCM controls and map them to AWS/Azure documentation
Backup if inaccessible	CL03 — AWS Well-Architected Security Pillar
Known limitations	Registration and version changes; not law
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL17 · AWS Certified Security - Specialty official page

Exact title	AWS Certified Security - Specialty official page
Direct URL	https://aws.amazon.com/certification/certified-security-specialty/
Resource type / domains	Official certification page · Modules I,J
Topic and subtopic coverage	Advanced exam reference for Year 2+; not a Year 1 target
Exact path / module / chapter / playlist	SCS-C03 exam guide, domains, cost, format, recommended experience and exam prep plan
Recommended difficulty / estimated time	Advanced · 40-80 h prep
Access status	Exam US$300 listed 2026-10-27; training optional/paid
Prerequisites	Five years IT security and two years AWS security recommended
Why selected	Advanced exam reference for Year 2+; not a Year 1 target
What to practise after	Complete free AWS Skill Builder exam prep only after Month 9 foundations and cloud experience
Backup if inaccessible	CL01 — AWS Skill Builder
Known limitations	Region/language and recertification policies change; not entry-level
Classification / age status	Paid/alternative · Current 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL18 · AWS recertification page

Exact title	AWS recertification page
Direct URL	https://aws.amazon.com/certification/recertification/
Resource type / domains	Official policy · Modules J
Topic and subtopic coverage	Clarifies renewal obligations before buying an advanced certification
Exact path / module / chapter / playlist	Three-year validity; renewal and maintenance pathways by certification level
Recommended difficulty / estimated time	Reference · 1 h
Access status	Free to read; exam/maintenance may cost
Prerequisites	Certification basics
Why selected	Clarifies renewal obligations before buying an advanced certification
What to practise after	Check current pathway before paying for any AWS exam
Backup if inaccessible	CL17 — AWS Certified Security - Specialty official page
Known limitations	Two official AWS pages describe policy in different detail; verify in your certification account
Classification / age status	Recommended · Updated 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A01 · pwn.college

Exact title	pwn.college
Direct URL	https://pwn.college/
Resource type / domains	Hands-on university platform · Modules A,C,K
Topic and subtopic coverage	University-grade free systems-security curriculum from Arizona State University
Exact path / module / chapter / playlist	Core dojos and current CSE courses; lectures, challenges and browser workspaces
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Free
Prerequisites	Linux, C, Python and assembly depending on dojo
Why selected	University-grade free systems-security curriculum from Arizona State University
What to practise after	Use Computing 101/Linux Luminarium in Year 1; defer Program Security and Software Exploitation to Year 2+
Backup if inaccessible	L15 — pwn.college Dojo index
Known limitations	Course instance names change yearly; use Core Material
Classification / age status	Mandatory for advanced optional track · Maintained 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A02 · pwn.college dojos index

Exact title	pwn.college dojos index
Direct URL	https://pwn.college/dojos
Resource type / domains	Dojo catalog · Modules A,C,K
Topic and subtopic coverage	Best navigation page to pick prerequisites in order
Exact path / module / chapter / playlist	Core Material, Community Material and The Courses list with modules and challenge counts
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Free
Prerequisites	Browser and Linux basics
Why selected	Best navigation page to pick prerequisites in order
What to practise after	Select one module from the assigned dojo and record challenge evidence
Backup if inaccessible	A01 — pwn.college
Known limitations	Dojo catalogue changes; links may require login
Classification / age status	Recommended · Current 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A03 · OpenSecurityTraining2

Exact title	OpenSecurityTraining2
Direct URL	https://ost2.fyi/
Resource type / domains	Free technical training · Modules A,K
Topic and subtopic coverage	University-level low-level security training that is deeper than typical tool tutorials
Exact path / module / chapter / playlist	Architecture, debuggers, reverse engineering, vulnerabilities and exploitation learning paths
Recommended difficulty / estimated time	Intermediate to advanced · 20-100 h
Access status	Free/open-source course materials
Prerequisites	Programming, Linux and architecture
Why selected	University-level low-level security training that is deeper than typical tool tutorials
What to practise after	Use selected Arch1001 videos only after C basics and pwn.college Linux modules
Backup if inaccessible	A01 — pwn.college
Known limitations	Many classes are long and assume C; do not attempt in Month 1
Classification / age status	Advanced reference · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A04 · OST2 Architecture 1001: x86-64 Assembly

Exact title	OST2 Architecture 1001: x86-64 Assembly
Direct URL	https://p.ost2.fyi/courses/course-v1:OpenSecurityTraining2+Arch1001_x86-64_Asm+2021_v1/about
Resource type / domains	Free course · Modules A,K
Topic and subtopic coverage	Best structured x86-64 assembly foundation for later RE, malware analysis and vulnerability research
Exact path / module / chapter / playlist	Registers, stack, functions, control flow, Boolean logic, shifts, repeat instructions, debugging and binary bomb lab
Recommended difficulty / estimated time	Advanced · ~28.5 h
Access status	Free; optional certificate mechanics may vary
Prerequisites	C programming
Why selected	Best structured x86-64 assembly foundation for later RE, malware analysis and vulnerability research
What to practise after	Complete the first four modules and a GDB observation exercise in Year 2+
Backup if inaccessible	A03 — OpenSecurityTraining2
Known limitations	Older recordings but durable architecture concepts; verify tools on current OS
Classification / age status	Advanced reference · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A05 · Malware Unicorn Reverse Engineering 101

Exact title	Malware Unicorn Reverse Engineering 101
Direct URL	https://malwareunicorn.org/workshops/re101.html
Resource type / domains	Free workshop · Modules K
Topic and subtopic coverage	Classic hands-on RE workshop with a strong safety-first lab workflow
Exact path / module / chapter / playlist	Safe VM setup, OS/assembly concepts, malware classes, static/dynamic analysis, disassemblers, debuggers and indicators
Recommended difficulty / estimated time	Advanced · 15-30 h
Access status	Free
Prerequisites	Windows/Linux VMs and assembly basics
Why selected	Classic hands-on RE workshop with a strong safety-first lab workflow
What to practise after	In Year 3+, recreate the lab using current tools and synthetic samples; do not download malware casually
Backup if inaccessible	A04 — OST2 Architecture 1001: x86-64 Assembly
Known limitations	2019 VM images and tools are old; use current official installers and isolated networking
Classification / age status	Advanced reference · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A06 · MIT 6.858 Computer Systems Security OCW

Exact title	MIT 6.858 Computer Systems Security OCW
Direct URL	https://ocw.mit.edu/courses/6-858-computer-systems-security-fall-2014/
Resource type / domains	Graduate university course · Modules E,F,K
Topic and subtopic coverage	Research-paper-driven graduate course that teaches security architecture reasoning
Exact path / module / chapter / playlist	Threat models, OS security, capabilities, information flow, language security, network protocols, web security and research papers
Recommended difficulty / estimated time	Advanced · One semester
Access status	Free
Prerequisites	OS, networking, C and systems programming
Why selected	Research-paper-driven graduate course that teaches security architecture reasoning
What to practise after	Read one assigned lecture note and summarize its threat model, design and evaluation
Backup if inaccessible	A07 — MIT 6.858 official lecture playlist
Known limitations	2014 material; timeless foundations but obsolete tool/browser specifics
Classification / age status	Advanced reference · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A07 · MIT 6.858 official lecture playlist

Exact title	MIT 6.858 official lecture playlist
Direct URL	https://www.youtube.com/playlist?list=PLUl4u3cNGP62K2DjQLRxDNRi0z2IRWnNh
Resource type / domains	YouTube playlist · Modules E,F,K
Topic and subtopic coverage	Direct playlist for the graduate lecture sequence; no invented timestamps
Exact path / module / chapter / playlist	MIT OCW 6.858 Fall 2014 lecture sequence; use exact lecture titles on YouTube
Recommended difficulty / estimated time	Advanced · 20+ lectures
Access status	Free
Prerequisites	Advanced CS fundamentals
Why selected	Direct playlist for the graduate lecture sequence; no invented timestamps
What to practise after	Watch one selected lecture at 1.25x with the matching OCW reading
Backup if inaccessible	A06 — MIT 6.858 Computer Systems Security OCW
Known limitations	2014 publication; update all practical tool details
Classification / age status	Optional enrichment · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A08 · Security Engineering, Third Edition (free online chapters)

Exact title	Security Engineering, Third Edition (free online chapters)
Direct URL	https://www.cl.cam.ac.uk/~rja14/book.html
Resource type / domains	Free book draft · Modules E,J,K
Topic and subtopic coverage	Ross Anderson's systems view connects technical controls to human, economic and organizational failure
Exact path / module / chapter / playlist	Security architecture, psychology, protocols, access control, cryptography, distributed systems, economics, policy and assurance
Recommended difficulty / estimated time	Advanced · Reference/selected chapters
Access status	Free online draft; print edition paid
Prerequisites	Broad technical maturity
Why selected	Ross Anderson's systems view connects technical controls to human, economic and organizational failure
What to practise after	Read one chapter per month in Years 2-3 and write a case analysis
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	Third-edition draft chapters may be updated; not a certification cram book
Classification / age status	Advanced reference · Maintained draft
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A09 · Nightmare binary exploitation course

Exact title	Nightmare binary exploitation course
Direct URL	https://guyinatuxedo.github.io/
Resource type / domains	Free course · Modules K
Topic and subtopic coverage	Large set of progressive exploitation challenges with explanations, useful for true specialization
Exact path / module / chapter / playlist	Linux/Windows exploitation fundamentals, assembly, stack overflows, ROP, heap and challenge walkthroughs
Recommended difficulty / estimated time	Advanced · 30-80 h
Access status	Free
Prerequisites	C, assembly, Linux, GDB and pwn.college prerequisites
Why selected	Large set of progressive exploitation challenges with explanations, useful for true specialization
What to practise after	Attempt only after pwn.college Program Security; use isolated VMs
Backup if inaccessible	A01 — pwn.college
Known limitations	Advanced and sometimes terse; not a Year 1 mastery target
Classification / age status	Advanced reference · Classic
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A10 · RPISEC Modern Binary Exploitation

Exact title	RPISEC Modern Binary Exploitation
Direct URL	https://github.com/RPISEC/MBE
Resource type / domains	University course materials · Modules K
Topic and subtopic coverage	Detailed university lab archive with downloadable materials
Exact path / module / chapter / playlist	Course syllabus, labs and lecture slides on memory corruption and binary exploitation
Recommended difficulty / estimated time	Advanced · One semester
Access status	Free/open-source
Prerequisites	C, Linux and x86
Why selected	Detailed university lab archive with downloadable materials
What to practise after	Use one lab in Year 2+ with a disposable 32/64-bit VM and compare with current mitigations
Backup if inaccessible	A01 — pwn.college
Known limitations	Older VM images and mitigations; modern mitigations require updated setup
Classification / age status	Advanced reference · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A11 · Google Project Zero blog

Exact title	Google Project Zero blog
Direct URL	https://googleprojectzero.blogspot.com/
Resource type / domains	Technical research blog · Modules K
Topic and subtopic coverage	Examples of rigorous vulnerability research methodology and technical writing
Exact path / module / chapter / playlist	Vulnerability research write-ups, root-cause analysis, exploit primitives and mitigation discussions
Recommended difficulty / estimated time	Advanced · Ongoing reading
Access status	Free
Prerequisites	Systems, memory safety and browser/OS internals
Why selected	Examples of rigorous vulnerability research methodology and technical writing
What to practise after	Read one post, extract hypothesis, method, evidence and limitations
Backup if inaccessible	A12 — USENIX Security conference papers
Known limitations	Advanced; many posts require specialized background
Classification / age status	Advanced reference · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A12 · USENIX Security conference papers

Exact title	USENIX Security conference papers
Direct URL	https://www.usenix.org/conferences/byname/164
Resource type / domains	Conference proceedings · Modules K
Topic and subtopic coverage	Strong source for current peer-reviewed security research with open proceedings
Exact path / module / chapter / playlist	Peer-reviewed security research papers and talks by year
Recommended difficulty / estimated time	Advanced · Ongoing reading
Access status	Many papers freely available; conference attendance paid
Prerequisites	Research methods and domain background
Why selected	Strong source for current peer-reviewed security research with open proceedings
What to practise after	Choose one paper, reproduce a non-dangerous measurement or analysis on synthetic data
Backup if inaccessible	A11 — Google Project Zero blog
Known limitations	Not every paper includes code; reproduction may be hard
Classification / age status	Advanced reference · Annual
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A13 · arXiv cs.CR recent papers

Exact title	arXiv cs.CR recent papers
Direct URL	https://arxiv.org/list/cs.CR/recent
Resource type / domains	Preprint archive · Modules K
Topic and subtopic coverage	Fast way to follow emerging claims, including AI security and systems security
Exact path / module / chapter / playlist	Cryptography and security preprints; use to discover topics, not as peer-reviewed truth
Recommended difficulty / estimated time	Advanced · Ongoing reading
Access status	Free
Prerequisites	Research literacy
Why selected	Fast way to follow emerging claims, including AI security and systems security
What to practise after	Compare one preprint with a later peer-reviewed version or critique its assumptions
Backup if inaccessible	A12 — USENIX Security conference papers
Known limitations	Preprints are not necessarily peer reviewed; quality varies
Classification / age status	Advanced reference · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A14 · xairy/linux-kernel-exploitation

Exact title	xairy/linux-kernel-exploitation
Direct URL	https://github.com/xairy/linux-kernel-exploitation
Resource type / domains	Curated research index · Modules K
Topic and subtopic coverage	Highly technical index maintained by a practitioner; useful only after years of foundations
Exact path / module / chapter / playlist	Linux kernel exploitation techniques, CVEs, papers, talks and proof-of-concept references
Recommended difficulty / estimated time	Advanced · Reference use
Access status	Free
Prerequisites	Kernel, C and exploitation fundamentals
Why selected	Highly technical index maintained by a practitioner; useful only after years of foundations
What to practise after	Read one concept page and summarize required prerequisites; do not run code on host
Backup if inaccessible	A01 — pwn.college
Known limitations	Dangerous material if misunderstood; use isolated research VMs only
Classification / age status	Advanced reference · Maintained community index
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A15 · OWASP Top 10 for LLM Applications

Exact title	OWASP Top 10 for LLM Applications
Direct URL	https://genai.owasp.org/llm-top-10/
Resource type / domains	Risk reference · Modules F,K
Topic and subtopic coverage	Current application-security risk taxonomy for AI-enabled systems
Exact path / module / chapter / playlist	Prompt injection, sensitive information disclosure, supply chain, data/model poisoning, improper output handling, excessive agency and related AI application risks
Recommended difficulty / estimated time	Intermediate to advanced · 4-8 h
Access status	Free
Prerequisites	Web/API security and LLM basics
Why selected	Current application-security risk taxonomy for AI-enabled systems
What to practise after	Threat-model a simple local RAG app and propose one control for each applicable risk
Backup if inaccessible	C16 — OWASP API Security Top 10
Known limitations	Versioned document; verify current edition before citing
Classification / age status	Advanced reference · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A16 · NIST AI Risk Management Framework

Exact title	NIST AI Risk Management Framework
Direct URL	https://www.nist.gov/itl/ai-risk-management-framework
Resource type / domains	Framework · Modules J,K
Topic and subtopic coverage	Authoritative, voluntary framework for AI risk that complements technical appsec controls
Exact path / module / chapter / playlist	Govern, Map, Measure, Manage; trustworthy AI characteristics and supporting resources
Recommended difficulty / estimated time	Intermediate · 6-12 h
Access status	Free
Prerequisites	Risk and system design basics
Why selected	Authoritative, voluntary framework for AI risk that complements technical appsec controls
What to practise after	Add AI-specific risks to a capstone risk register and map them to NIST AI RMF functions
Backup if inaccessible	A15 — OWASP Top 10 for LLM Applications
Known limitations	Not law and not an appsec testing checklist
Classification / age status	Advanced reference · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A17 · Cisco CCST Cybersecurity official exam page

Exact title	Cisco CCST Cybersecurity official exam page
Direct URL	https://www.cisco.com/site/us/en/learn/training-certifications/exams/ccst-cybersecurity.html
Resource type / domains	Official certification page · Modules B,E,J
Topic and subtopic coverage	Lower-cost optional first certification with a bounded syllabus, but less portable technical proof than a project portfolio
Exact path / module / chapter / playlist	Exam 100-160: essential security principles, network security, endpoint security, vulnerability/risk management and incident handling; US$125 and 50 minutes listed 2026-10-27
Recommended difficulty / estimated time	Beginner · 30-60 h prep
Access status	Exam paid US$125 listed; free Cisco training linked
Prerequisites	Networking and security basics
Why selected	Lower-cost optional first certification with a bounded syllabus, but less portable technical proof than a project portfolio
What to practise after	Use free Cisco training first; consider only after Month 5 if an employer or university requires it
Backup if inaccessible	E01 — NIST SP 800-115
Known limitations	Cisco certification names/pathways are changing in 2026; verify current page and 5-year renewal rules
Classification / age status	Optional certification · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A18 · CompTIA Security+ V7 and V8 official pages

Exact title	CompTIA Security+ V7 and V8 official pages
Direct URL	https://www.comptia.org/en-us/certifications/security/
Resource type / domains	Official certification page · Modules E,J
Topic and subtopic coverage	Widely recognized foundational exam, useful as a structured knowledge check but not proof of job-ready skill
Exact path / module / chapter / playlist	V7 SY0-701 current until stated retirement; V8 SY0-801 expected 2026-11-17
Recommended difficulty / estimated time	Beginner to intermediate · Exam 90 min; max 90 questions
Access status	Paid exam; V7 US$404 from CompTIA blog dated January 2025; verify current store
Prerequisites	Core security vocabulary
Why selected	Widely recognized foundational exam, useful as a structured knowledge check but not proof of job-ready skill
What to practise after	Use free Professor Messer course first; consider V8 only after the current objectives are published and Month 12 review
Backup if inaccessible	C01 — Professor Messer SY0-701 Security+ Course Index
Known limitations	V7 retirement English 2027-06-11; V8 launch November 2026; do not buy stale practice material
Classification / age status	Optional certification · Checked 2026-10-27
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A19 · ISC2 Certified in Cybersecurity official page

Exact title	ISC2 Certified in Cybersecurity official page
Direct URL	https://www.isc2.org/Certifications/CC
Resource type / domains	Official certification page · Modules E,J
Topic and subtopic coverage	Low-experience, globally recognized entry-level credential; useful if a scholarship/voucher is available
Exact path / module / chapter / playlist	Five exam domains; no work experience required; current exam outline effective 2026-09-01
Recommended difficulty / estimated time	Beginner · 20-40 h prep
Access status	Exam paid; US$199 listed on official pricing page 2026-10-27; occasional free promotions may exist
Prerequisites	None
Why selected	Low-experience, globally recognized entry-level credential; useful if a scholarship/voucher is available
What to practise after	Use free official flashcards and this roadmap's Month 5 gate before paying
Backup if inaccessible	C03 — ISC2 Certified in Cybersecurity Exam Outline
Known limitations	Annual maintenance fee after certification; exact local price/tax at checkout
Classification / age status	Optional certification · Maintained 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A20 · Microsoft SC-900 official certification page

Exact title	Microsoft SC-900 official certification page
Direct URL	https://learn.microsoft.com/en-us/credentials/certifications/security-compliance-and-identity-fundamentals/
Resource type / domains	Official certification page · Modules E,J
Topic and subtopic coverage	Good free learning path even if you never buy the exam; fundamentals certification does not expire per current Microsoft policy
Exact path / module / chapter / playlist	Security, compliance, identity, Microsoft Entra, Microsoft security and compliance solutions; free practice assessment
Recommended difficulty / estimated time	Beginner · 10-20 h prep
Access status	Free Learn training; exam price region-based
Prerequisites	Cloud and identity basics
Why selected	Good free learning path even if you never buy the exam; fundamentals certification does not expire per current Microsoft policy
What to practise after	Finish all four Learn paths before considering the exam; do not assume Microsoft-specific knowledge equals general security skill
Backup if inaccessible	B15 — Microsoft SC-200 certification page
Known limitations	Price at checkout depends on country; event vouchers are not guaranteed
Classification / age status	Optional certification · Updated 2026-10-21
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A21 · OffSec PEN-200 / OSCP+ official page

Exact title	OffSec PEN-200 / OSCP+ official page
Direct URL	https://www.offsec.com/courses/pen-200/
Resource type / domains	Paid advanced course/certification · Modules G,K
Topic and subtopic coverage	Recognized practical penetration-testing credential for a later specialization, not a beginner goal
Exact path / module / chapter / playlist	PEN-200 modules, challenge labs and 24-hour proctored OSCP+ exam; OSCP lifetime and OSCP+ three-year designation
Recommended difficulty / estimated time	Advanced · 321 h content; 90-day course bundle
Access status	Paid; official page lists US$1749 course+exam bundle and US$1699 standalone exam on 2026-10-27
Prerequisites	Strong Linux, Windows, networking, scripting and ethical testing experience
Why selected	Recognized practical penetration-testing credential for a later specialization, not a beginner goal
What to practise after	Defer until Year 2+ and only after independent authorized assessments and report writing
Backup if inaccessible	E01 — NIST SP 800-115
Known limitations	High cost, renewal/maintenance conditions and exam format can change; free alternatives exist
Classification / age status	Paid/alternative · Maintained 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A22 · INE eJPT official certification page

Exact title	INE eJPT official certification page
Direct URL	https://ine.com/security/certifications/ejpt-certification
Resource type / domains	Official certification page · Modules G,K
Topic and subtopic coverage	More affordable hands-on pentesting exam than OSCP; still not necessary in Year 1
Exact path / module / chapter / playlist	Assessment methodologies, host/network penetration testing, auditing and web application testing objectives
Recommended difficulty / estimated time	Beginner to intermediate · 40-100 h prep
Access status	Paid; official page states an INE subscription and exam voucher are needed; checkout lists voucher/ bundle prices
Prerequisites	Networking, Linux, Windows and basic web security
Why selected	More affordable hands-on pentesting exam than OSCP; still not necessary in Year 1
What to practise after	Consider after Month 7 only if you can demonstrate equivalent skills with local projects and have funds
Backup if inaccessible	E01 — NIST SP 800-115
Known limitations	Official pricing is split between subscription and voucher; checkout promotions change; certification valid three years
Classification / age status	Paid/alternative · Maintained 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A23 · KQL Essentials official Azure Data Explorer playlist

Exact title	KQL Essentials official Azure Data Explorer playlist
Direct URL	https://www.youtube.com/playlist?list=PLKHIhZJiQ5AkOMEC54mQQcAkfg4UXT8uK
Resource type / domains	Official YouTube playlist · Modules H
Topic and subtopic coverage	Short official Microsoft KQL videos complement the textual KQL docs
Exact path / module / chapter / playlist	KQL overview and query operators; use exact episode titles shown on YouTube
Recommended difficulty / estimated time	Beginner to intermediate · 3-8 h
Access status	Free
Prerequisites	Logs and SQL-like query concepts
Why selected	Short official Microsoft KQL videos complement the textual KQL docs
What to practise after	Watch one episode, then reproduce each query on synthetic tables
Backup if inaccessible	B02 — Kusto Query Language overview
Known limitations	Individual publication dates vary; services and UI can change
Classification / age status	Recommended · Current playlist
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A24 · SANS DFIR playlist

Exact title	SANS DFIR playlist
Direct URL	https://www.youtube.com/playlist?list=PLrY_AbzZGqt9MJiLjyH-zTrypGLs154e4
Resource type / domains	YouTube playlist · Modules H,K
Topic and subtopic coverage	Free specialist lectures from a reputable training organization; good supplement to official docs
Exact path / module / chapter / playlist	Windows and Linux forensic artifacts, collection and investigation concepts
Recommended difficulty / estimated time	Intermediate to advanced · 4-10 h
Access status	Free
Prerequisites	Windows/Linux internals and evidence integrity
Why selected	Free specialist lectures from a reputable training organization; good supplement to official docs
What to practise after	Select one video, list the artifacts discussed, and verify them on a disposable VM
Backup if inaccessible	B11 — Autopsy documentation
Known limitations	Some SANS courses are paid; webcasts may be older; do not confuse a webcast with certification
Classification / age status	Optional enrichment · Older but technically useful
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

ULTIMATE CYBERSECURITY MASTER BLUEPRINT 2026-2027

Volume 6 — Lab, Projects, Assessments and Career

Home lab, 15 projects, scorecards, certification plan, GitHub and international preparation

Edition: Volume 6
Resource verification date: 2026-10-27
Plan window: 2026-10-27 to 2027-10-26 (shift dates if you start later)
System: 365 days, 12 academic months, 52 weeks + 1 graduation day
Profile: BS CS student in Pakistan; free/high-value resources; safe authorized labs

Academic-style self-study manual. No employment, salary, visa, certification or top-1% guarantee.

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Volume 6 — Lab, projects, assessments, certifications, portfolio and career

1. Affordable home lab

Minimum hardware

- CPU: 64-bit dual-core with virtualization (Intel VT-x/AMD-V); 4 threads recommended.
- RAM: **8 GB minimum** (host + one 2 GB Linux VM + browser labs). 16 GB strongly recommended; 32 GB for Windows Server/AD/Kubernetes.
- Storage: **80 GB free minimum**, 256 GB SSD recommended for snapshots.
- OS: Windows, Linux or macOS host; VirtualBox/KVM/Hyper-V.
- Network: intermittent internet for downloads; keep labs isolated offline.

Recommended hardware

- 4-8 core CPU, 16-32 GB RAM, 512 GB SSD, external backup drive.
- One Ubuntu LTS VM (2-4 GB RAM), one Windows evaluation VM (4 GB RAM), Docker Engine on Linux, optional kind/minikube cluster.
- Browser-based labs when RAM is low: PortSwigger, TryHackMe free rooms, pwn.college workspace, CyberDefenders datasets.

Official installation sources

- VirtualBox manual: https://www.virtualbox.org/manual/
- Ubuntu help: https://help.ubuntu.com/
- Docker Engine docs: https://docs.docker.com/
- Kubernetes docs / kind / minikube: https://kubernetes.io/docs/home/, https://kind.sigs.k8s.io/, https://minikube.sigs.k8s.io/docs/
- Windows evaluation / developer VMs: use only Microsoft's official pages https://www.microsoft.com/en-us/evalcenter/ and https://developer.microsoft.com/en-us/windows/downloads/virtual-machines/
- Wireshark: https://www.wireshark.org/docs/wsug_html_chunked/
- Packet Tracer: https://www.netacad.com/courses/packet-tracer
- OWASP Juice Shop: https://github.com/juice-shop/juice-shop
- DVWA: https://github.com/digininja/DVWA
- Kubernetes Goat: https://madhuakula.com/kubernetes-goat/

Network isolation

Use NAT for updates, host-only/internal adapters for vulnerable services, and never bridge intentionally vulnerable VMs to a university/hostel/public LAN. Bind local web apps to `127.0.0.1` or an internal adapter only. Do not expose lab ports on your router. Keep host firewall/AV enabled; do not disable essential security controls as routine practice. Take a clean snapshot before each risky lab and write down the restore point. Use synthetic data, not personal files.

VM vs container vs browser lab

Need	Best tool	Why
OS internals, Active Directory, Windows event logs, snapshots, malware-like artifacts	Full VM	Kernel/identity state, snapshot/rollback, stronger isolation
Stateless vulnerable web app, scanner, parser	Container	Fast, reproducible, lower RAM
Low-end laptop / public Wi-Fi / no admin rights	Browser lab	No install; platform-provided isolation
Multi-node Kubernetes	kind/minikube on 16 GB+ host	Local control plane, no cloud bill

Need	Best tool	Why
Real cloud API behaviour	Free-tier cloud account with budgets	Only when local emulation cannot show the service

Avoid real malware on the host. Malware-traffic datasets are PCAPs/artifacts, not permission to execute suspicious binaries. For any RE/malware lab in Year 2+, use a disposable VM with no shared clipboard, no bridged adapter and no personal credentials.

Cloud cost safety

1. Read current Free Tier terms (https://aws.amazon.com/free/) before account creation.
2. Enable MFA on root/owner accounts; never commit access keys.
3. Create billing alerts/budgets before deploying resources.
4. Tag every resource with project and expiry date.
5. Prefer local/synthetic cloud reviews. If you deploy, capture evidence then tear down the same day.
6. Check the billing page and delete unused volumes, snapshots, IPs and clusters; these often cost money even when VMs are stopped.
7. Never paste cloud credentials into GitHub, screenshots or PDFs.

Common setup problems

Symptom	Likely cause	Fix
VM will not start	Virtualization disabled in BIOS/UEFI or Hyper-V conflict	Enable VT-x/AMD-V; check host hypervisor settings; use VirtualBox manual network chapter
No internet in VM	Wrong adapter/DNS/DHCP	NAT for updates; verify `ip addr`, `/etc/resolv.conf`, DHCP lease; do not disable firewall globally
SSH timeout	Guest IP changed, service not running, host-only route	Check `ip addr`, `systemctl status ssh`, `ss -ltn`, and VM adapter
Port conflict on host	Another service bound to 8080/3000	Use `ss -ltnp` or `netstat -ano`, choose another loopback port
Docker permission denied	User not in docker group or daemon not running	Use official docs; avoid `curl | sh` scripts from random sources; consider rootless Docker
Windows events missing	Audit policy/Sysmon config not applied	Verify Event Viewer, audit policy and Sysmon service; reapply in a clean snapshot
Kubernetes cluster high RAM	Too many nodes/control-plane components	Use single-node kind/minikube, 2-4 GB RAM, stop cluster after lab
Packet Tracer file missing	Registration/download issue	Use official NetAcad course; backup with network diagrams and Wireshark local captures
Cloud surprise bill	Free-tier expiry, orphaned storage/IPs	Budget alerts, tags, daily teardown, billing dashboard check

2. Project system

Every month has at least one deliverable. All projects use synthetic data, isolated labs and systems you own or are explicitly authorized to test. Every project section includes problem, objective, prerequisites, architecture, tools, milestones, expected results, validation, common mistakes, troubleshooting, evidence, report template, GitHub structure, README, security boundaries, rubric and extension challenges.

Shared GitHub structure

```
project-name/
  README.md
  LICENSE
```

```
CHANGELOG.md
.gitignore
docs/
  problem.md
  architecture.md
  threat-model.md
  report.md
  limitations.md
src/ or manifests/
tests/
data/synthetic/
evidence/sanitized/
scripts/setup.sh
```

README template

1. One-sentence problem statement; 2. Scope/authorization statement; 3. Architecture diagram; 4. exact setup commands; 5. usage examples; 6. tests/validation; 7. findings/results; 8. limitations and lessons learned; 9. sanitization notes; 10. licence/acknowledgements.

Report template

Executive summary → scope and rules → architecture/data → methodology → findings with evidence → risk/priority → remediation → retest → residual risk → timeline → appendices. Clearly label facts, interpretations and assumptions. Never publish secrets, private personal data, sensitive logs or unauthorized exploit material.

Common project rubric (100 points)

Conceptual understanding 20; practical execution 25; independent problem-solving 15; troubleshooting 15; documentation 15; safety/ethics 10. Passing = 70, with safety ≥9 and practical ≥18. Project-specific validation tests in each card override generic claims.

3. Assessments and mastery gates

Monthly practicals

Month	Timed practical	Critical prerequisite gate
M1	Configure clean VM, users, local service, SQL query, Git evidence	CLI, permissions, VM isolation
M2	20 subnetting questions, unknown PCAP, broken topology, scope judgement	OSI/TCP/IP, CIDR, Wireshark, authorized Nmap
M3	Linux/Windows hardening, event-log timeline, Bash/PowerShell script	Users/processes/logs, identity boundaries
M4	Unseen log format parser, tests, normalized schema, short report	Python, regex, files, Git, testing
M5	Risk register, crypto primitive selection, IAM design, IR decision	CIA/AAA, crypto purpose, risk model
M6	Find/fix two local web bugs, API authorization design, remediation report	HTTP, sessions, server-side controls
M7	Scoped local assessment, evidence, validation, report, retest	Written scope, no unauthorized targets
M8	Synthetic alert investigation, KQL/Sigma rule, incident report	Log normalization, ATT&CK, confidence
M9	Cloud/K8s/Docker/CI/IaC misconfiguration review	IAM, shared responsibility, isolation
M10	Unfamiliar specialization problem + technical defence	Primary docs, independent debugging
M11	Capstone implementation, failure injection, investigation, report	Full stack, threat model, evidence
M12	Three final practicals + capstone defence	All gates

Six-step failure repair

1. **Identify the gap:** score each rubric category and write the exact failed claim/command.
2. **Revisit primary lesson:** return to today's exact section; do not rewatch the whole course.
3. **Use alternate explanation:** use the supplementary resource or a different medium (diagram, RFC, man page).
4. **Repeat exercise:** same objective, new input; record prediction before result.
5. **New exercise:** solve a modified problem with no worked solution and write a short postmortem.
6. **Retest:** new 10-question or 30-minute practical; require ≥80% and no safety error before proceeding.

If two consecutive attempts fail, reduce scope to a prerequisite module for three days and notify your mentor/peer. Do not hide failed attempts; error logs are evidence of learning.

4. Certification planning (verified 2026-10-27)

Prices are official list prices where displayed; taxes, currency, region and promotions can change. Always re-open the official page before paying. No proprietary exam questions are reproduced.

Certification	Current official title/ version	Prerequisites	Exam cost shown	Training cost	Renewal/ maintenance	Consider	What it does not prove	Free alternative
ISC2 CC	Certified in Cybersecurity; outline effective 2026-09-01	No work experience; application/ endorsement process	US$199 in official pricing table for listed regions incl. Asia Pacific; local tax/ currency at Pearson	Official self-paced/ live training is paid; free flashcards/ quiz/ outline exist	Annual maintenance/ CPE obligations apply; verify current CPE handbook	Month 12 if a voucher or an employer/ university requests it	Hands-on SOC/ AppSec ability; real incident judgement	CS50 Cyber, Professor Messer, ISC2 free flashcards, this roadmap
CompTIA Security+	V7 SY0-701 current; V8 SY0-801 expected 2026-11-17; V7 English retirement listed 2027-06-11	Recommended Network+ plus ~2 years admin/ security experience, but no formal prerequisite	V7 voucher shown US$404 on CompTIA price page/ blog; verify current store and V8 price	CertMaster Learn/ Labs/ Practice paid; free Professor Messer videos	3-year cycle, 50 CEUs or higher cert/ CertMaster CE; official CE fee listed US$150 over 3 years	Only if a job/degree filter explicitly asks; prefer V8 after objectives settle	Deep technical execution; practical exams are limited	Professor Messer SY0-701, NIST/ISC2 free resources, local labs
Cisco CCST Cybersecurity	100-160 CCST Cybersecurity	None	US$125, 50 minutes	Free Cisco Junior Cybersecurity Analyst career path; paid instructor options exist	Cisco states 5-year validity for certifications earned from 2025-07-15; verify current rules	Month 5-12 if local employer values Cisco credentials	Broad hands-on engineering	Cisco free career path + this roadmap's networking/ security labs
Microsoft SC-900	Security, Compliance and Identity Fundamentals	Beginner; Azure/M365 familiarity helpful	Region-based; US price commonly listed as US$99 but verify at checkout	Microsoft Learn learning paths free	Fundamentals certification currently does not expire per Microsoft policy; verify	Optional after M5/ M9 if targeting Microsoft ecosystem	General security skill; product-specific knowledge	Four free SC-900 Learn paths, free practice assessment

Certification	Current official title/ version	Prerequisites	Exam cost shown	Training cost	Renewal/ maintenance	Consider	What it does not prove	Free alternative
Microsoft SC-200	Security Operations Analyst Associate	SOC/KQL/ Sentinel/ Defender experience recommended	Region-based; US associate price typically US$165 but verify official schedule page	Microsoft Learn free; instructor/ practice options paid	Annual free renewal assessment on Microsoft Learn	Year 2 if you choose Microsoft-aligned detection engineering	Vendor-independent detection engineering	KQL docs, Sigma, CyberDefenders, BTLO, local SIEM-like labs
AWS Security Specialty	SCS-C03 AWS Certified Security - Specialty	Official recommendation: ~5 years IT security and 2+ years AWS security	US$300 on official exam page	AWS Skill Builder free/paid; labs may cost	3-year validity; official recertification now lists exam or Skill Builder maintenance path—verify in your account	Year 2+ after real AWS work, not Month 12	General cloud security or hands-on skill	AWS Security Pillar, free Skill Builder digital courses, local IaC reviews
BTL1	Blue Team Level 1 (Centri, formerly Security Blue Team)	0-2 years recommended	£399 GBP training/exam bundle, one free resit included (official page)	Included in bundle	Lifetime certification per official page	Year 2 if employer funds it and you already completed free blue-team projects	Production SOC judgement	CyberDefenders, BTLO free labs, Autopsy, Sigma/ KQL, PRJ-11/12
eJPT	INE Junior Penetration Tester	Basic networking/ Linux/ Windows; no formal prerequisite	Official page says INE subscription + voucher required; checkout lists voucher/ bundle prices; verify current total	INE subscription/bundle paid	Valid three years per official page	Optional after M7 if you can afford it and have local reports	Professional scoping / reporting under real clients	NIST SP 800-115, local VMs, HTB Academy free modules, PRJ-10
OSCP + / PEN-200	OffSec Certified Professional / OSCP+	Strong TCP/ IP, Linux/ Windows, scripting recommended	Official page: US$1,699 standalone exam; US$1,749 course+exam bundle	Included in PEN-200 bundles; Learn One US$2,749/ year	OSCP lifetime; OSCP+ expires after 3 years unless maintained	Year 2-3 only with consistent independent pentest practice	Vulnerability research or elite status	pwn.college, HTB, local AD labs, report writing
CompTIA CySA+ / PenTest+ / GIAC / CISSP	Various intermediate/advanced credentials	Depends on credential; CISSP requires experience	Check each official page; not recommended this year	Often high	Varies	Not Year 1 priorities	Job-ready skill by themselves	Free blue/ red projects and primary documentation

Readiness checklist: current official objectives downloaded; ≥80% on two free/self-generated practice exams; portfolio evidence mapped to each domain; budget includes retake/renewal; reason for exam is written; practical projects already demonstrate skill. Practice strategy: closed-book timed attempt → mark confidence on each answer → error taxonomy (concept, wording, arithmetic, fatigue) → targeted revision → new self-generated questions → spaced retest after 3 and 7 days. Never use brain dumps.

5. GitHub and professional portfolio

A strong portfolio is not a folder of screenshots. It shows a problem, a decision, evidence, a failed attempt, a fix, a test and a limitation. Use meaningful commits, issues, tags and release notes. Add architecture diagrams as SVG and sanitized screenshots; redact IPs, MACs, usernames, hostnames, tokens, private paths and personal data. Include an explicit scope statement on security work. Never publish credentials, secrets, private logs or unauthorized exploit material.

Superficial project: copied Dockerfile, no README, one happy-path screenshot, no tests, no limitations. **Original project:** states a problem, threat model, 2+ design alternatives, implementation rationale, reproducible setup, negative tests, false-positive analysis, before/after evidence, lessons learned and a future-work section.

Recommended public index: one `portfolio` README linking 5-8 best projects (not all 15 if quality is uneven), a CV, a lab safety statement, and contact information. Keep private evidence encrypted.

6. International career preparation

Broad market patterns, not employer guarantees

Entry-level security work in US/European markets commonly appears under titles such as SOC analyst, security analyst, junior detection engineer, security engineer, vulnerability analyst, GRC analyst, application security engineer, cloud security analyst and junior penetration tester. Many "entry-level" listings still ask for foundational IT, networking, scripting, cloud and incident exposure. This roadmap provides project evidence but cannot promise employment, salary, visa sponsorship or relocation. Remote work from Pakistan depends on the employer's legal entity, tax, data-access, time-zone and export-control rules.

Use official frameworks to analyze roles: NIST NICE (https://www.nist.gov/itl/applied-cybersecurity/nice) for task/skill language; NIST CSF 2.0 for control vocabulary; MITRE ATT&CK for detection roles. No live job listing is used in this PDF as proof of demand. If you analyze a real listing, save the URL and retrieval date, then distinguish must-have skills from preferred tools.

Job-description analysis

Make a table with columns: requirement, evidence in your portfolio, gap level (0-3), proof task, interview story. Ignore inflated years-of-experience claims only after you can show equivalent evidence. Classify product names separately from transferable concepts (e.g., "Sentinel" vs "SIEM correlation").

CV and LinkedIn

- One page for an early-career CV; no photo/personal identifiers unnecessary for the target country.
- Top: degree, expected graduation, GitHub/portfolio, location/time-zone, work authorization if relevant.
- Project bullets use problem → action → evidence → result, e.g., "Built a Python parser for 3 synthetic log formats; added 25 pytest cases; reduced malformed-line crashes to zero; documented limitations."
- LinkedIn headline should name target role and evidence, not "ethical hacker" buzzwords.
- Ask for feedback from a teacher, OSS maintainer or working analyst; track changes.

Interview practice

Prepare five stories: one networking/OS troubleshooting, one secure coding fix, one detection/investigation, one cloud/container misconfiguration, one ethical disagreement. For each, state context, hypothesis, evidence, decision, failure, lesson. Practice explaining uncertainty and what data you would request. Do not claim tool mastery from a badge.

Internships and first applications

Apply to internships, university security labs, open-source security projects, CTF teams, local non-profits with written authorization, and junior roles that allow learning. Build an application tracker with date, role, source, requirements, evidence mapping, follow-up date and outcome. Seek feedback, not just offers. For international roles, check official visa/work-authorization sources (e.g., US State Department https://travel.state.gov/content/travel/en/us-visas.html and EU immigration https://home-affairs.ec.europa.eu/), and never assume remote work bypasses work authorization.

Specialization comparison (Month 8-10 decision aid)

Score each row 1-5 for demonstrated interest, project results, programming comfort, local opportunity and available feedback. No path is "best" without your evidence.

Decision framework: evidence weight 35%, genuine interest 20%, programming/technical fit 20%, available mentorship/opportunity 15%, opportunity cost 10%. Require at least two strong project results before selecting. Review after 90 days; switching early because a topic is hard is different from switching because evidence shows poor fit.

Recommendation for this profile: primary = Detection Engineering & Security Automation; secondary = Cloud/Container Security. If Month 8-10 sprints contradict this, choose the path with stronger independent problem-solving evidence and better feedback access.

Project catalogue — 15 required deliverables

Every project includes all required fields. Shared templates T1/T2/T3 and rubric R1 are defined in the preceding text; project rows reference and extend them.

PRJ-01 — Personal cybersecurity learning lab

Problem statement	A beginner needs a safe, reproducible environment but has modest hardware and no paid software license.
Objective	Build a documented, isolated, snapshot-backed lab capable of hosting Linux, Windows, local web apps and browser labs without exposing vulnerable services.
Prerequisites	Virtualization basics, host OS administration, CLI, Git
Architecture	Host with VirtualBox/KVM; Ubuntu VM on NAT plus internal adapter; optional Windows evaluation VM; browser labs; evidence repository
Tools and installation links	VirtualBox https://www.virtualbox.org/manual/; Ubuntu https://help.ubuntu.com/; Git https://git-scm.com/book/en/v2; Wireshark https://www.wireshark.org/docs/wsug_html_chunked/
Step-by-step milestones	1 Inventory hardware; 2 enable virtualization; 3 install hypervisor; 4 create Ubuntu VM; 5 configure NAT/internal adapters; 6 snapshot; 7 optional Windows VM; 8 document setup
Expected results	A clean VM with internet via NAT, an isolated internal adapter, a restore point, and a reproducible README
Validation tests	Boot VM; ping gateway; verify no bridged vulnerable adapter; restore snapshot; compare configuration hash; confirm no host firewall disabled
Common mistakes	Bridging vulnerable VMs; storing personal files in lab; no snapshot; copying random ISO images
Troubleshooting guidance	No boot: enable VT-x/AMD-V; no network: check NAT/DHCP; slow VM: reduce RAM to 2 GB; snapshot corruption: keep exported appliance and backup
Evidence to collect	VM settings export, sanitized screenshots, network diagram, command transcript, snapshot identifier, setup log
Report template	T1 with scope, architecture, setup, validation, limitations, recovery test
GitHub structure	lab-baseline/README.md, docs/, scripts/, evidence/sanitized/
README requirements	Purpose, hardware, exact versions, setup commands, network isolation, restore test, limitations
Security boundaries	Only owned VMs; no public exposure; no personal credentials; host security stays enabled
Grading rubric	R1 standard rubric; special weight on isolation and recovery evidence
Extension challenges	Add Ansible/Vagrant provisioning; add segmented Windows VM; add offline package mirror

PRJ-02 — Security control baseline and risk register

Problem statement	The lab has assets and configurations but no systematic risk-based control selection.
Objective	Apply NIST CSF and CIS-style controls to the student's own lab and measure whether each control works.
Prerequisites	M1-M4 foundations, basic risk vocabulary, VM administration
Architecture	Inventory of VMs, GitHub repo, local web service, synthetic data; controls mapped to Govern/Identify/Protect/Detect/Respond/Recover
Tools and installation links	NIST CSF https://www.nist.gov/cyberframework; CIS Controls https://www.cisecurity.org/controls/v8; NIST SP 800-30 https://csrc.nist.gov/pubs/sp/800/30/r1/final

Step-by-step milestones	1 Asset inventory; 2 data classification; 3 five threats; 4 risk register; 5 select ten controls; 6 implement low-risk controls; 7 test; 8 write exceptions
Expected results	Ten tested controls, a risk register with owners/treatments, and one 90-day improvement plan
Validation tests	Before/after evidence for each control; negative test where control should block; restore test; peer-style review of residual risk
Common mistakes	Confusing policy with working control; no rollback; selecting controls only from a checklist; claiming zero risk
Troubleshooting guidance	If control breaks usability, snapshot and rollback; if evidence is missing, define a measurable test; if risk ranking is inconsistent, state assumptions
Evidence to collect	Asset inventory, risk register, control matrix, test results, screenshots, change log, rollback notes
Report template	T1 plus risk methodology, control-by-control results, exceptions, residual risk
GitHub structure	security-baseline/README.md, docs/threat-model.md, controls/, tests/, evidence/
README requirements	Scope, methodology, how to reproduce each test, limitations, no compliance claim
Security boundaries	Controls applied only to owned lab; no production system; no unauthorized scanning
Grading rubric	R1 with conceptual 25, practical 25, documentation 20, troubleshooting 10, safety 10, independence 10
Extension challenges	Add quantitative risk scoring; automate control tests; compare two frameworks

PRJ-03 — Network diagram and troubleshooting report

Problem statement	A small lab network has connectivity faults and no documented topology or diagnostic method.
Objective	Produce a professional topology and a repeatable OSI-layer troubleshooting report for an isolated network.
Prerequisites	OSI/TCP/IP, CIDR, ARP, routing, Wireshark, Packet Tracer
Architecture	Three-node virtual network with client, server, router/switch simulation; NAT for updates; internal adapter for tests
Tools and installation links	Packet Tracer https://www.netacad.com/courses/packet-tracer; Wireshark https://www.wireshark.org/docs/wsug_html_chunked/; JITL labs https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ
Step-by-step milestones	1 Draw physical/logical topology; 2 assign addresses; 3 verify ARP and routes; 4 inject three faults; 5 diagnose layer by layer; 6 fix; 7 retest; 8 report
Expected results	Correct diagram, IP plan, fault timeline, packet evidence and verification tests
Validation tests	Ping/traceroute success; ARP table matches; route table checks; packet capture confirms each fault/fix; diagram matches observed state
Common mistakes	Confusing default gateway and DNS; changing multiple variables at once; ignoring time/ARP cache; no rollback
Troubleshooting guidance	No route: check subnet/default gateway; DNS failure: use IP to isolate; ARP stale: clear cache safely; capture empty: use correct interface
Evidence to collect	Topology diagram, .pkt file or VM config, PCAP excerpts, command logs, before/after test matrix
Report template	T1 plus OSI-layer analysis and evidence table
GitHub structure	network-troubleshooting/README.md, diagrams/, captures/, reports/
README requirements	Exact topology, reproduction commands, sanitized addresses, assumptions, limitations
Security boundaries	Only isolated lab addresses; no campus/public network scanning

Grading rubric	R1 with special weight on troubleshooting and independent hypotheses
Extension challenges	Add IPv6, VLANs, OSPF, or a simulated firewall failure

PRJ-04 — Wireshark traffic-analysis report

Problem statement	Network traffic contains security-relevant evidence, but the student cannot yet distinguish observation from inference.
Objective	Analyze a legal sample PCAP and write a defensible report with confidence levels.
Prerequisites	TCP/IP, DNS, HTTP/TLS, Wireshark display filters
Architecture	Single analysis VM with Wireshark and sample captures; no live user traffic
Tools and installation links	Wireshark Guide https://www.wireshark.org/docs/wsug_html_chunked/; SampleCaptures https://wiki.wireshark.org/SampleCaptures; RFC 9293 https://www.rfc-editor.org/rfc/rfc9293
Step-by-step milestones	1 Hash capture; 2 identify endpoints/conversations; 3 protocol hierarchy; 4 follow streams; 5 identify anomalies; 6 form hypotheses; 7 test with filters; 8 report
Expected results	A report with endpoints, timeline, protocol fields, evidence for each claim and explicit limitations
Validation tests	PCAP hash unchanged; five display filters documented; at least three claims tied to frame numbers; alternative explanation considered
Common mistakes	Treating encrypted payload as plaintext; ignoring time zones; confusing capture and display filters; overclaiming malware
Troubleshooting guidance	No capture: verify interface/permissions; TLS hides fields: use metadata; noisy file: filter by conversation; missing DNS: check external resolver
Evidence to collect	PCAP hash, filter list, annotated screenshots, frame table, timeline, final report
Report template	T1 plus packet-evidence appendix and confidence scale
GitHub structure	traffic-analysis/README.md, captures/, reports/, evidence/
README requirements	Capture source, hash, Wireshark version, filters, findings, limitations
Security boundaries	Use only public sample captures or traffic from owned systems; no third-party interception
Grading rubric	R1 with conceptual 25, practical 25, troubleshooting 15, documentation 15, safety 10, independence 10
Extension challenges	Add Zeek/Suricata comparison; analyze TLS 1.3 metadata; create a synthetic beaconing dataset

PRJ-05 — Linux system-audit script

Problem statement	Manual Linux checks are slow, inconsistent and risky when copied from random blogs.
Objective	Write a read-only Bash/Python audit utility that collects security-relevant system state from an owned Linux VM.
Prerequisites	Linux permissions, processes, services, Bash, Python, JSON
Architecture	Audit script runs with least privilege; outputs common JSON schema; no network calls and no destructive commands
Tools and installation links	BashGuide https://mywiki.wooledge.org/BashGuide; man7 https://man7.org/linux/man-pages/; Python docs https://docs.python.org/3/; CIS Benchmarks https://www.cisecurity.org/cis-benchmarks

Step-by-step milestones	1 Define schema; 2 collect users/groups; 3 permissions; 4 services/listening ports; 5 patch state; 6 logs; 7 JSON output; 8 tests; 9 review
Expected results	A deterministic, read-only tool with tests, sample output and a design document
Validation tests	Run on clean VM; compare output against manual commands; test empty fields, filenames with spaces, missing permissions; shellcheck; no modifications
Common mistakes	Running as root unnecessarily; parsing human-readable text; unquoted variables; exposing /etc/shadow; destructive commands
Troubleshooting guidance	Permission denied: handle gracefully; `ss` missing: use fallback or document; JSON invalid: validate; locale changes: normalize output
Evidence to collect	Script, schema, tests, sample output, manual verification table, shellcheck report, README
Report template	T1 plus control-to-command mapping and limitations
GitHub structure	linux-audit/README.md, src/, tests/, data/synthetic/, docs/
README requirements	Scope, read-only guarantee, setup, usage, schema, limitations
Security boundaries	Run only on owned VM; never publish real user hashes, hostnames or sensitive logs
Grading rubric	R1 with secure coding and least-privilege checks
Extension challenges	Add PowerShell counterpart; add config drift test; add CI and signed release

PRJ-06 — Windows event-log investigation

Problem statement	A synthetic Windows incident needs a defensible timeline, but event IDs are easy to misinterpret.
Objective	Investigate benign simulated authentication/process/service events and write an incident report with confidence levels.
Prerequisites	Windows fundamentals, Event Viewer, PowerShell, Sysmon, evidence integrity
Architecture	Disposable Windows VM; Security/System/Sysmon logs; benign event generator; no real personal data
Tools and installation links	Windows Event Log https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log; Sysmon https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon; Sysinternals https://learn.microsoft.com/en-us/sysinternals/
Step-by-step milestones	1 Clean snapshot; 2 configure audit/Sysmon; 3 generate logon/service/process events; 4 export EVTX; 5 hash; 6 query fields; 7 timeline; 8 report
Expected results	A reproducible timeline of at least five events with EventID, provider, time, user, process and interpretation
Validation tests	Hash manifest matches; XPath/PowerShell queries reproduce timeline; at least one alternative explanation rejected by evidence
Common mistakes	Ignoring logon type; mixing local and UTC time; relying on EventID alone; editing original EVTX
Troubleshooting guidance	No events: check audit policy; Sysmon absent: verify service; time skew: configure NTP; access denied: use admin shell in lab only
Evidence to collect	Sanitized EVTX, hash manifest, query file, timeline, screenshots, incident report
Report template	T2 incident form with evidence references and confidence labels
GitHub structure	windows-investigation/README.md, events/, queries/, reports/, evidence/
README requirements	Lab scope, event-generation steps, sanitization, tool versions, query instructions
Security boundaries	Disposable VM only; no personal accounts; no live malware; preserve original evidence

Grading rubric	R1 with evidence integrity and false-positive analysis
Extension challenges	Add Sysmon config comparison; add KQL/Sigma rule; add memory acquisition concepts in Year 2

PRJ-07 — Python security-log parser and defensive automation tool

Problem statement	Security logs arrive in multiple formats and contain malformed lines; manual grep is not enough.
Objective	Build a tested parser and analyzer that normalizes synthetic logs and produces defensible summaries.
Prerequisites	Python, regex, files, exceptions, JSON/CSV, SQL, testing, Git
Architecture	CLI tool with parser/analyzer modules; SQLite storage; synthetic inputs; common JSON schema; no external network
Tools and installation links	Python Tutorial https://docs.python.org/3/tutorial/; pytest https://docs.pytest.org/en/stable/; RegexOne https://regexone.com/; SQLite https://www.sqlite.org/docs.html
Step-by-step milestones	1 Define schema; 2 stream input; 3 parse three formats; 4 normalize; 5 store; 6 query top talkers; 7 structured errors; 8 tests; 9 docs
Expected results	Working CLI, normalized dataset, SQL queries, Markdown report and full test suite
Validation tests	20+ tests; 100 MB synthetic file under documented memory cap; malformed lines rejected with reasons; secrets redacted; no shell interpolation
Common mistakes	Catastrophic regex; reading whole file into memory; logging secrets; string-built SQL; no time zone handling
Troubleshooting guidance	Timeout: bound regex and line length; Unicode errors: explicit encoding; DB lock: transaction handling; query slow: index fields
Evidence to collect	Code, schema, sample logs, test results, performance note, redacted output, changelog
Report template	T1 plus data schema, query logic, false-positive discussion, performance
GitHub structure	security-log-parser/README.md, src/, tests/, data/synthetic/, reports/
README requirements	Problem, formats, exact CLI, schema, tests, redaction, limitations
Security boundaries	Synthetic logs only; no production logs; no credential collection
Grading rubric	R1 with secure coding and test quality
Extension challenges	Add property-based tests; add Sigma export; add notebook visualization; add streaming benchmark

PRJ-08 — Secure web application / API

Problem statement	A small notes API must support multiple users without exposing one user's data to another.
Objective	Implement a secure local API with authentication, object ownership, validation, parameterized queries, CSRF protection and structured logging.
Prerequisites	HTTP, SQL, Python, threat modelling, OWASP ASVS, testing
Architecture	Local web app, SQLite/PostgreSQL-like database, session cookies, server-side authz, test suite, CI
Tools and installation links	OWASP ASVS https://owasp.org/www-project-application-security-verification-standard/; Cheat Sheets https://cheatsheetseries.owasp.org/; Python docs https://docs.python.org/3/; SQLite https://www.sqlite.org/docs.html

Step-by-step milestones	1 Requirements; 2 threat model; 3 schema; 4 registration/login/logout; 5 object ownership; 6 anti-CSRF; 7 rate limits; 8 errors/logs; 9 tests; 10 CI
Expected results	Working API with secure defaults, 20+ tests, threat model and reproducible setup
Validation tests	Cross-user access denied; SQL injection strings do not change queries; XSS payload escaped; expired session rejected; CSRF token mismatch rejected; rate limit triggers
Common mistakes	Client-side authz only; plaintext passwords; string SQL; hiding IDs as security; verbose stack traces; secrets in repo
Troubleshooting guidance	Login loop: cookie attributes/domain; 403 unexpectedly: check object owner; CSRF failing: token/session binding; DB locked: transaction scope
Evidence to collect	Source, migrations, threat model, test report, CI log, sanitized request/response examples
Report template	T1 plus security requirements and verification matrix
GitHub structure	secure-api/README.md, src/, migrations/, tests/, docs/, .github/workflows/
README requirements	Scope, setup, roles, endpoints, test commands, threat model, limitations
Security boundaries	Loopback/host-only only; no public deployment; synthetic users only
Grading rubric	R1 with special weight on server-side authorization and negative tests
Extension challenges	Add JWT/OIDC carefully; add CSP/Trusted Types; add SAST/DAST and SBOM

PRJ-09 — Local web application security assessment with remediation

Problem statement	A secure app still needs independent validation and defensive recommendations, not just passing tests.
Objective	Perform an authorized assessment of PRJ-08 on loopback, validate findings, fix root causes and retest.
Prerequisites	PRJ-08, HTTP, OWASP WSTG, PortSwigger basics, secure coding
Architecture	Assessor VM/container, local app, Burp/ZAP proxy, written scope, evidence repository
Tools and installation links	OWASP WSTG https://owasp.org/www-project-web-security-testing-guide/; Burp docs https://portswigger.net/burp/documentation; ZAP docs https://www.zaproxy.org/docs/; Cheat Sheets https://cheatsheetseries.owasp.org/
Step-by-step milestones	1 Scope/RoE; 2 asset inventory; 3 five test areas; 4 validate findings; 5 risk rank; 6 fix; 7 retest; 8 report
Expected results	Three evidence-backed findings, one rejected false positive, before/after tests and a closure memo
Validation tests	Scope contains only 127.0.0.1/internal host; original/variant/regression retests; raw evidence hashes; no destructive action
Common mistakes	Testing without written scope; WAF-only fix; over-ranking scanner alerts; leaving sessions/cookies in report
Troubleshooting guidance	Proxy not intercepting: check browser proxy/cert; app down: verify container; finding not reproducing: compare environment; regression after fix: isolate commit
Evidence to collect	RoE, request/response evidence, scan report, remediation diff, retest results
Report template	T2 assessment report with executive summary, findings, evidence, remediation, retest
GitHub structure	web-assessment/README.md, scope/, evidence/sanitized/, reports/, fixes/
README requirements	Explicit ownership/authorization, sanitized evidence, reproduction steps, limitations
Security boundaries	Only own local app; no third-party target; no public exploit material

Grading rubric	R1 with safety/ethics double weight
Extension challenges	Add authenticated scanning; add business-logic abuse cases; add CVSS/KEV contextual ranking

PRJ-10 — Authorized vulnerability assessment report for an isolated lab

Problem statement	A realistic assessment needs planning, discovery, validation, communication and closure under a written scope.
Objective	Execute an end-to-end assessment of two owned VMs and one local vulnerable web app, then report and retest.
Prerequisites	M1-M7 prerequisites, networking, Linux/Windows, web security, ethics
Architecture	Two local VMs, one vulnerable app, NAT/internal network, Nmap/Burp/Wireshark, evidence repo
Tools and installation links	NIST SP 800-115 https://csrc.nist.gov/pubs/sp/800/115/final; Nmap Reference https://nmap.org/book/man.html; WSTG https://owasp.org/www-project-web-security-testing-guide/; FIRST CVSS https://www.first.org/cvss/
Step-by-step milestones	1 Charter; 2 RoE; 3 inventory; 4 threat model; 5 discovery; 6 validation; 7 evidence; 8 report; 9 remediation; 10 retest
Expected results	A professional report with prioritized findings, false-positive analysis, remediation tickets and closure evidence
Validation tests	All targets in scope; three findings with raw evidence; CVSS plus business context; no persistence/lateral movement beyond minimal proof; retest closes root cause
Common mistakes	Scope creep; destructive proof; scanner dependence; missing evidence hash; confusing version with vulnerability
Troubleshooting guidance	Host down: stop and verify; scanner crash: reduce rate; ambiguous banner: second validation; report disagreement: cite raw evidence
Evidence to collect	Scope, asset inventory, scan outputs, PCAPs, screenshots, evidence manifest, final report
Report template	T2 full assessment template with appendices
GitHub structure	vuln-assessment/README.md, scope/, evidence/sanitized/, reports/, remediation/
README requirements	Scope statement, sanitized artifacts, tool versions, limitations
Security boundaries	Isolated owned systems only; no public IPs; no real user data
Grading rubric	R1 with safety/ethics threshold 10/10
Extension challenges	Add credentialed scanning; add AD lab; add CVSS environmental scoring; add retest automation

PRJ-11 — SIEM alert investigation and incident report

Problem statement	A synthetic alert must be triaged, investigated and communicated without overclaiming from limited logs.
Objective	Build a reproducible incident investigation from synthetic Windows/Linux/cloud logs and write a professional report.
Prerequisites	Log normalization, KQL/SQL, ATT&CK, IR lifecycle, evidence integrity
Architecture	Local dataset or free sample workspace; normalized event table; KQL-style queries; timeline; case management notes

Tools and installation links	KQL docs https://learn.microsoft.com/en-us/kusto/query/; Sentinel docs https://learn.microsoft.com/en-us/azure/sentinel/; NIST 800-61r3 https://csrc.nist.gov/pubs/sp/800/61/r3/final; ATT&CK https://attack.mitre.org/
Step-by-step milestones	1 Define alert; 2 generate/download data; 3 normalize; 4 triage; 5 eight queries; 6 timeline; 7 ATT&CK map; 8 response; 9 report
Expected results	Incident report with timeline, confidence levels, root-cause hypothesis, evidence gaps and response plan
Validation tests	All timestamps normalized; eight queries reproducible; one benign alternative rejected; evidence hashes; no cloud overspend
Common mistakes	Assuming an IOC is an incident; mixing time zones; query joins without cardinality check; ignoring data gaps
Troubleshooting guidance	No tables: use local SQLite equivalent; high false positives: narrow entity/time; missing host: state limitation; cost alert: stop ingestion
Evidence to collect	Data manifest, queries, timeline, screenshots, incident report, executive summary, detection-gap notes
Report template	T2 incident report with known/likely/unknown sections
GitHub structure	siem-investigation/README.md, data/synthetic/, queries/, reports/, evidence/
README requirements	Dataset provenance, time zone, schema, query setup, sanitization
Security boundaries	Synthetic data only; if cloud used, budgets and teardown documented
Grading rubric	R1 with uncertainty/documentation weight
Extension challenges	Add SOAR-style playbook; add KQL to Sigma conversion; add tabletop exercise

PRJ-12 — Detection rule with test data

Problem statement	A detection idea is not credible until it is rule-formatted, tested on true/false positives and maintained as code.
Objective	Write an original Sigma rule for a lab behaviour, convert it to a query and validate on synthetic events.
Prerequisites	ATT&CK, logs, KQL/SQL, Sigma, Git, testing
Architecture	Detection-as-code repo; synthetic event generator; rule; conversion script; tests; CI-like local validation
Tools and installation links	SigmaHQ https://sigmahq.io/; Sigma guide https://sigmahq.io/docs/guide/getting-started.html; Elastic detections https://www.elastic.co/guide/en/security/current/detection-engine-overview.html; ATT&CK https://attack.mitre.org/
Step-by-step milestones	1 Choose technique; 2 data requirements; 3 rule; 4 ten positives/ten negatives; 5 convert; 6 backtest; 7 tune; 8 document bypasses
Expected results	A rule, labelled dataset, conversion output, metrics, false-positive notes and version history
Validation tests	Rule validates; all assigned true positives detected; false positives below documented threshold; at least three bypasses listed; CI test fails on broken rule
Common mistakes	Blanket suppression; untested field names; selecting only easy positives; no rule owner; confusing precision with value
Troubleshooting guidance	Conversion error: check backend/pipeline; no matches: verify field normalization; too many alerts: prevalence analysis; rule valid but useless: revisit hypothesis
Evidence to collect	Rule YAML, test events, conversion output, metrics, changelog, design document, ATT&CK mapping
Report template	T1 plus detection design document with hypothesis, data, tests, bypasses

GitHub structure	detection-rule/README.md, rules/, tests/data/, docs/, scripts/, evidence/
README requirements	Technique, logsource, setup, test commands, metrics, limitations
Security boundaries	Synthetic/authorized telemetry only; no deployment on production systems
Grading rubric	R1 with special weight on false-positive evidence
Extension challenges	Add correlation rule; add multi-backend conversion; add ATT&CK Navigator layer; add 10,000-event backtest

PRJ-13 — Cloud configuration security review

Problem statement	A fictional small cloud environment has IAM/network/storage/logging weaknesses but no budget for live deployment.
Objective	Review synthetic architecture and configuration artifacts, prioritize risks and provide verification steps.
Prerequisites	Cloud shared responsibility, IAM, storage, logging, risk, documentation
Architecture	Fictional AWS/Azure account structure with ten resources; JSON policy samples; architecture diagram; no billable resources
Tools and installation links	AWS Security Pillar https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html; AWS security docs https://docs.aws.amazon.com/security/; Azure security https://learn.microsoft.com/en-us/azure/security/; CSA CCM https://cloudsecurityalliance.org/research/cloud-controls-matrix
Step-by-step milestones	1 Inventory; 2 classify data; 3 IAM review; 4 network review; 5 storage; 6 logging; 7 risk register; 8 remediation; 9 verification
Expected results	Twelve findings with blast-radius analysis, owner, verification command/query and residual risk
Validation tests	All findings supported by synthetic config lines; policy logic checked; no live cloud resource created; risk priorities include business context
Common mistakes	Assuming encryption fixes public access; treating NAT as firewall; ignoring control-plane logs; copy-pasting another cloud's defaults
Troubleshooting guidance	Policy ambiguous: draw trust graph; missing logs: state visibility gap; service names changed: verify current docs; disputed risk: define scenario
Evidence to collect	Architecture diagram, synthetic policies, risk register, review checklist, remediation plan, references
Report template	T2 assessment report plus control-mapping appendix
GitHub structure	cloud-review/README.md, architecture/, configs/synthetic/, reports/, checks/
README requirements	Explicit fictional scope, no live account, assumptions, verification steps
Security boundaries	No real customer data; no cloud credentials; no public deployment
Grading rubric	R1 with conceptual/architecture weight
Extension challenges	Add Checkov/Terraform scan; add AWS/Azure comparison; add detection hypothesis for each high risk

PRJ-14 — Docker image and container security review

Problem statement	A container image may contain vulnerable packages, secrets, excessive privileges and insecure networking.

Objective	Build/review a local image and runtime configuration, fix findings and rescan with reproducible evidence.
Prerequisites	Docker basics, Linux permissions, CVE/SBOM, CI, scripting
Architecture	Local Docker/Podman host; intentionally weak image; Trivy scanner; runtime metadata; isolated bridge network
Tools and installation links	Docker docs https://docs.docker.com/; Docker security https://docs.docker.com/engine/security/; Trivy https://trivy.dev/docs/; OWASP K8s Top 10 https://owasp.org/www-project-kubernetes-top-ten/
Step-by-step milestones	1 Snapshot baseline; 2 inspect layers; 3 scan; 4 Dockerfile review; 5 runtime review; 6 fix; 7 rescan; 8 report
Expected results	Ten findings, five fixes, before/after scans, runtime tests and residual-risk notes
Validation tests	Hardened image runs non-root; no secret in history; capabilities dropped; read-only root where feasible; network isolation verified
Common mistakes	Using latest tag blindly; privileged container; secrets in ENV/ARG; scanning without context; leaving volumes world-writable
Troubleshooting guidance	Build fails: pin base image; app needs write: use tmpfs; scanner DB blocked: use offline/ documented limitation; container exits: inspect logs
Evidence to collect	Dockerfiles, image IDs/digests, scan JSON, runtime inspect output, test log, diff
Report template	T2 assessment report with image inventory and finding-by-finding evidence
GitHub structure	container-review/README.md, docker/, scans/, tests/, reports/evidence/
README requirements	Local-only scope, image versions, reproduction commands, sanitized outputs
Security boundaries	No public registry push; no host socket mount; no production daemon
Grading rubric	R1 with runtime-isolation and vulnerability-triage weight
Extension challenges	Add SBOM/provenance; add rootless Docker; add Kubernetes admission policy; add CI gate

PRJ-15 — Final integrated capstone

Problem statement	The student needs one coherent project that combines secure engineering, telemetry, detection, incident response, cloud/container controls and independent research.
Objective	Design, build, attack safely, detect, respond, document and defend an original system aligned with the chosen specialization.
Prerequisites	M1-M11, chosen specialization, primary docs, technical writing
Architecture	Local three-tier app/API, structured logging, three detections, hardened container, synthetic cloud/ IaC representation, CI and evidence repo
Tools and installation links	OWASP ASVS https://owasp.org/www-project-application-security-verification-standard/; SigmaHQ https://sigmahq.io/; AWS Security Pillar https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html; GitHub Actions hardening https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions; NIST 800-61r3 https://csrc.nist.gov/pubs/sp/800/61/r3/final
Step-by-step milestones	1 Charter; 2 threat model; 3 architecture; 4 reproducible setup; 5 implementation; 6 detections; 7 failure injection; 8 authorized assessment; 9 remediation; 10 report/demo
Expected results	Working system, labelled benign/malicious data, tested detections, a security assessment, incident report, reproducible setup, demo and limitations
Validation tests	Clean install succeeds; 25+ tests; three detections with true/false-positive data; failure-injection tests; assessment retests; report passes 75+
Common mistakes	

	Oversized scope; unlabelled synthetic data; no negative results; tutorial-replica architecture; overclaiming production readiness
Troubleshooting guidance	Performance: reduce components; missing telemetry: instrument first; false positives: labelled baseline; report too broad: narrow claim; demo fails: use tagged release
Evidence to collect	Source, manifests, data dictionary, test outputs, evidence manifest, threat model, report, demo script, changelog
Report template	T3 capstone report: abstract, problem, related work, design, implementation, evaluation, security assessment, IR, limitations, future work
GitHub structure	capstone/README.md, docs/, src/, detections/, data/synthetic/, tests/, evidence/sanitized/, .github/workflows/
README requirements	One-sentence contribution, architecture, exact setup, demo, tests, limitations, ethics statement
Security boundaries	All targets owned/local; synthetic data; no public vulnerable service; no secrets; no unauthorized exploit material
Grading rubric	R1 plus capstone defence: 20 concept, 25 practical, 15 independence, 15 troubleshooting, 15 documentation, 10 safety; pass 75
Extension challenges	Publish a dataset; add multi-cloud comparison; add a research reproduction; submit a conference-style extended abstract

Printable trackers

Daily checklist (copy one per day)

#	Item	Done	Evidence / note
1	Read today's card and objective	[ ]	
2	Primary resource section completed	[ ]	
3	Three facts / two questions / one command	[ ]	
4	Practical task completed	[ ]	
5	One error observed or injected	[ ]	
6	Retrieval answer written without notes	[ ]	
7	Evidence file saved and sanitized	[ ]	
8	Journal / Git commit	[ ]	
9	Track used: Core / Standard / Intensive	[ ]	
10	Weak cards scheduled 1/3/7/30/90 days	[ ]	

Weekly tracker

Monthly scorecard (100 points)

Error log template

Command notebook template

Command / API	Platform	Purpose	Safe example	Dangerous assumption	Verification	Source URL

Project log template

Date	Decision	Alternatives	Evidence	Risk	Result	Follow-up

Final completion checklist

- [ ] 365 daily cards have evidence and retrieval notes.
- [ ] 12 monthly scorecards complete; M3/M6/M9/M12 gates passed.
- [ ] 15 projects have a problem, architecture, tests, evidence, report, README, boundaries and limitations.
- [ ] Final capstone has a clean-install test, threat model, detections, incident report, assessment, remediation and demo.
- [ ] All public artifacts are sanitized; no secrets, private logs, personal data or unauthorized exploit material.
- [ ] Resource links rechecked on 2026-10-27; backup links documented.
- [ ] Certification decision uses current official pages and a written reason.
- [ ] CV, LinkedIn and interview stories map to portfolio evidence.
- [ ] Year 2 plan has a primary specialization, secondary skill, peer group, 90-day sprint and review date.
- [ ] Lab is clean, isolated, backed up and has no active billable cloud resources.

Landscape reference table

Path	Foundational knowledge	Daily work	Technical depth / programming	Common tools	Strong free resources	Portfolio expectation	Possible entry roles	Long-term opportunities	Typical prerequisites	12-month limitation
Blue Team / SOC Analysis	Networking, OS, logs, ATT&CK	Alert triage, investigation, escalation, reporting	Medium; scripting helpful	SIEM, KQL, EDR, ticketing	Microsoft Learn, SigmaHQ, CyberDefenders, BTLO	Alert investigation, incident report, detection tests	SOC analyst, security analyst	Detection lead, IR, threat hunting	M1-M8 basics	No real shift experience or production scale
Detection Engineering	Logs, OS internals, ATT&CK, SQL/KQL	Hypothesis, rule design, testing, tuning, deployment	High; Python/YAML/SQL	Sigma, KQL, Elastic rules, Git, CI	SigmaHQ, KQL docs, Elastic rules, ATT&CK	Rule repo with labelled true/false positives and metrics	Junior detection engineer, SOC analyst	Detection platform/ architecture, research	Strong logs and programming	Lab detections may not generalize to enterprise data
DFIR	Filesystems, Windows/Linux artifacts, evidence integrity	Acquisition, timeline, artifact analysis, reporting	High; scripting and OS internals	Autopsy, KAPE-like tools, Volatility, Wireshark	Autopsy, SANS webcasts, NIST 800-61r3, EVTX samples	Disk/network/ memory case with hash manifest	Junior DFIR analyst, IR analyst	Senior IR, forensic research, expert witness	Evidence handling and OS internals	No legal/casework or real incident pressure
Web / API Security	HTTP, auth, databases, secure coding	Code review, testing, remediation, threat modelling	High; one backend language	Burp/ZAP, Semgrep, OWASP ASVS, CI	PortSwigger, OWASP, MDN, GitHub security	Secure app plus authorized assessment/ remediation	AppSec analyst, secure software engineer	AppSec architecture, research, product security	M1-M6 plus testing	Advanced vulnerability classes need years
Authorized Penetration Testing	Networking, OS, web, scripting, ethics	Scoped discovery, validation, reporting, retest	Medium-high; scripting and OS	Nmap, Burp, AD tools, reporting templates	NIST 800-115, HTB free, pwn.college, local AD lab	Professional scoped report and remediation retest	Junior pentester, security consultant	Senior pentest, red team, security research	Strong M1-M7 + report writing	Certification or CTF badges do not equal client judgement
Cloud Security	IAM, networking, logging, containers, shared responsibility	Architecture review, policy, detection, incident response	High; IaC and automation	AWS/Azure docs, Trivy, Checkov, K8s	AWS Security Pillar, Azure security, K8s docs	Synthetic cloud review + IaC/CI hardening	Cloud security analyst, DevSecOps junior	Cloud security architect, platform engineering	M1-M9 and Linux/ Docker	No production multi-account experience in Year 1
Security Engineering	Secure design, coding, identity, systems, automation	Build controls, review architecture, write tools	Very high; software engineering	Git, CI, SAST/SCA, cloud, observability	OWASP ASVS, GitHub docs, NIST, Security Engineering	Maintained tool/ library with tests and design doc	Security engineer, platform security engineer	Staff/principal security architect	Strong CS and programming	Breadth and production trade-offs take years
GRC	Risk, policy, controls, audits, privacy	Assessments, evidence, exceptions, reports	Low-medium technically; analysis-heavy	GRC platforms, spreadsheets, NIST/ISO	NIST CSF/SP, CIS Controls, GDPR text	Risk register, control baseline, audit-style report	GRC analyst, risk analyst, compliance associate	Security governance lead, privacy, audit	Business/legal context and strong writing	Framework knowledge without operations is shallow
Vulnerability Research	C/assembly, OS, memory, protocols, fuzzing	Root-cause analysis, PoC in authorized lab, disclosure	Very high; C/ Python/assembly	Ghidra, GDB, pwntools, fuzzers, VMs	pwn.college, OST2, MIT 6.858, Project Zero	Reproduced bug class, detailed root-cause write-up	Rare junior roles; often via internships/OSS	Researcher, product security, exploit engineering	Several years of low-level foundations	Intro exposure is not research capability
Malware Analysis / RE	x86/x64, Windows internals, networking, scripting	Triage, static/ dynamic analysis, indicator extraction	Very high; C/ Python/assembly	Ghidra, x64dbg, Wireshark, sandboxes	OST2, Malware Unicorn RE101, pwn.college	Safe synthetic sample analysis with isolation evidence	Junior malware analyst, SOC/IR specialist	Threat research, reverse engineering	C/assembly and strict lab safety	Real malware handling requires supervised environment

Landscape reference table

Day	Sequence	Hours	Evidence link	Concept /5	Practical /5	Troubleshooting /5	Safety pass	Error log ID	Next action
1							[ ]		
2							[ ]		
3							[ ]		
4							[ ]		
5							[ ]		
6							[ ]		
7							[ ]		

Landscape reference table

Category	Max	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Conceptual understanding	20												
Practical execution	25												
Independent problem-solving	15												
Troubleshooting	15												
Documentation	15												
Safety and ethics	10												
Total	**100**												
Gate result				**M3**			**M6**			**M9**			**M12**

Landscape reference table

ID	Date	Symptom	Hypothesis	Test	Result	Root cause	Fix	New test	Retest



ULTIMATE CYBERSECURITY MASTER BLUEPRINT 2026-2027

Volume 7 — Long-Term Blueprint

Years 2-3, years 3-5, beyond year 5, research pathway and specialization decision framework

Edition: Volume 7
Resource verification date: 2026-10-27
Plan window: 2026-10-27 to 2027-10-26 (shift dates if you start later)
System: 365 days, 12 academic months, 52 weeks + 1 graduation day
Profile: BS CS student in Pakistan; free/high-value resources; safe authorized labs

Academic-style self-study manual. No employment, salary, visa, certification or top-1% guarantee.

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Volume 7 — Multi-year exceptional-competence blueprint

1. The target: narrow specialization, real peer group

"Top 1%" is meaningful only after defining (a) a narrow field, (b) a credible peer group, and (c) evidence of impact. A reasonable peer group for Year 2-5 might be "practitioners publishing and reviewing detection engineering work for cloud-native environments". Another might be "researchers reproducing and publishing web/API vulnerability classes". Do not compare yourself to all cybersecurity professionals; compare your work to people doing the same technical job.

Impact signals: an OSS rule/tool used by others; a reproduced finding that changes a design; an advisory with root cause; a detection with measured false-positive reduction; a conference paper or technical article with code/data; a maintained project with issues from strangers; an architecture decision adopted by a team; mentorship that produces independent work. **Vanity signals:** follower count, number of certificates, collection of badges, conference selfies, tool lists.

Reputation can amplify genuine skill but does not prove it. Ask: what problem was solved, what evidence supports it, what failed, who independently checked it, and did the result change practice?

2. Stage 1 — Years 2-3: deepen one specialization

Outcomes

- Move from guided labs to original tools/detections/experiments.
- Read primary docs and research papers every week.
- Reproduce one published result safely, including negative results.
- Build 2-3 substantial projects with users or credible reviewers.
- Participate in legal CTFs/competitions and one open-source community.
- Seek an internship, part-time security work, university lab or supervised volunteer work.
- Learn technical writing, code review, incident communication and threat modelling.
- Develop a repeatable troubleshooting method: observe → hypothesize → test → measure → document.

90-day cycles

Each cycle has one measurable technical question, one primary-source reading queue, one artifact, one feedback source and one public/private write-up. Example for detection engineering: "Can I detect suspicious service-account behaviour in synthetic Kubernetes audit logs with <5% false positives over 10,000 events?" Artifact: rules + test data + report. Feedback: OSS maintainer or working SOC analyst.

Year 2-3 milestones

Quarter	Focus	Evidence
Q1	Specialization foundations and primary docs	12 annotated sources, one original tool/rule, baseline metrics
Q2	Reproduction	Reproduced paper/advisory with deviations and limitations
Q3	Original project v1	Code, tests, threat model, external feedback, changelog
Q4	Impact and communication	Maintained release, technical talk/write-up, internship/application evidence
Year 3	Advanced failure modes	New experiment, peer critique, supervised real-world exposure, revised research method

3. Stage 2 — Years 3-5: independent investigation and engineering

- Investigate unfamiliar problems where the answer is unknown to you and your mentor.
- Build tools/detections/architectures that others can use and critique.
- Study implementation details and failure modes: parsers, protocols, kernels, identity protocols, build systems, cloud control planes.
- Develop a repeatable research method and a literature matrix.
- Evaluate claims critically: source, assumptions, data, baseline, confounders, reproducibility, applicability.
- Document original findings with confidence levels, alternative explanations and negative results.
- Gain feedback from skilled practitioners through internships, jobs, OSS, conferences and CTFs.
- Begin mentoring beginners; teaching reveals weak mental models.

Research notebook template

```
Question | Hypothesis | Prior work | Data/source | Method | Prediction | Result | Alternative explanations | Confidence | Reproduction steps | Next experiment | Ethical/legal constraints
```

How to read a technical paper

1. Read abstract, introduction, threat model and evaluation figures first.
2. Write the main claim in one sentence and list assumptions.
3. Identify the dataset, baseline and metric; ask whether the metric matches the real goal.
4. Re-read method; mark anything needed for reproduction.
5. Check limitations and ethics; distinguish demonstrated result from speculation.
6. Summarize in your own words before reading commentary.
7. Reproduce the smallest experiment on synthetic/authorized data; record every deviation.
8. Write findings, including failures. A negative reproduction with evidence is useful.

4. Stage 3 — Beyond Year 5: exceptional practice

Exceptional practitioners continue through specialized research, advanced security engineering, architecture, original vulnerability research where legally authorized, high-quality detection/defensive engineering, open-source maintenance, responsible disclosure, peer review/publication, mentorship, real-incident learning and continuous adaptation. No fixed number of years guarantees top 1% status. Many people with ten years of repeated tasks do not develop deep expertise; some with five years of focused, reviewed, original work become highly influential.

Ways to increase impact:

- Choose a problem that recurs across organizations, not a one-off trick.
- Build reusable artifacts: datasets, rules, libraries, test harnesses, benchmarks.
- Invite adversarial review; publish limitations and corrections.
- Contribute to standards/frameworks responsibly.
- Practice responsible disclosure with written authorization and safe-harbour clarity.
- Lead or review post-incident engineering changes, not just detection content.
- Mentor others and build a local/international community of practice.

5. Advanced practice ladder

Level	Behaviour	Detection-engineering example	Web/API example
1. Following instructions	Reproduce exact steps	Run a provided Sigma rule	Solve a lab with a walkthrough
2. Reproducing known examples	Rebuild from memory	Rewrite rule from a technique description	Rebuild an XSS proof of concept locally

Level	Behaviour	Detection-engineering example	Web/API example
3. Modified exercises	Change one variable	New log source, same behaviour	Different encoding/context
4. Unfamiliar problems	No known answer	Design a rule for a new cloud audit event	Find an authz bug in unseen local code
5. Designed experiments	Measure outcomes	10,000 synthetic events, labelled true/false positives, baseline comparison	Build a test harness for parser/ authorization bypasses
6. Original tools/ research	Generalizable contribution	Detection-as-code framework with tests and public dataset	New secure library pattern or vulnerability class analysis
7. Defending conclusions	Peer critique	Publish method, limitations, reviewer Q&A, revision history	Responsible disclosure, patch, regression tests, postmortem

6. Staged reading pathway

Stage 0 — foundations (Months 1-12)

- CS50x and CS50 Cybersecurity notes; OSTEP selected chapters.
- Python, MDN HTTP, Pro Git, Wireshark and Linux/Windows official docs.
- NIST CSF 2.0, NIST SP 800-61r3, OWASP Top 10/API Top 10/ASVS, MITRE ATT&CK.
- Strategy: one section at a time; every reading maps to a lab and an evidence file.

Stage 1 — core books and durable technical references (Years 2-3)

- Operating Systems: Three Easy Pieces — https://pages.cs.wisc.edu/~remzi/OSTEP/
- Security Engineering, 3rd ed. — https://www.cl.cam.ac.uk/~rja14/book.html
- A Graduate Course in Applied Cryptography — https://toc.cryptobook.us/
- RFC 9293 TCP, RFC 8446 TLS 1.3, RFC 8200 IPv6.
- MIT 6.858 lecture notes/papers — https://ocw.mit.edu/courses/6-858-computer-systems-security-fall-2014/
- AWS/Azure/Kubernetes/Docker/GitHub official security docs.
- Strategy: one chapter/paper per week, annotated notes, one practical mapping.

Stage 2 — papers, advisories and incident reports (Years 3-5)

- USENIX Security proceedings: https://www.usenix.org/conferences/byname/164
- IEEE S&P / ACM CCS official proceedings pages: verify current conference site before citing.
- arXiv cs.CR preprints: https://arxiv.org/list/cs.CR/recent (preprints are not peer-reviewed truth).
- Google Project Zero: https://googleprojectzero.blogspot.com/
- CISA advisories: https://www.cisa.gov/news-events/cybersecurity-advisories
- CISA KEV: https://www.cisa.gov/known-exploited-vulnerabilities-catalog
- Vendor security advisories and post-incident reports from primary organizations.
- Strategy: compare at least two sources, extract claim/evidence/limitations, reproduce one safe measurement.

Stage 3 — standards, engineering repositories and community review (Year 5+)

- NIST SP 800-53r5, SP 800-63, SP 800-30, SP 800-34, SP 800-61r3, CSF 2.0.
- ISO/IEC 27001 official overview and legally obtained standard text; GDPR official text.
- OWASP ASVS, WSTG, Cheat Sheets, SAMM, Kubernetes Top 10, LLM Top 10.
- SLSA, OpenSSF Scorecard, SigmaHQ rule repository, Elastic detection rules, Suricata/Zeek rules, osquery, Falco, Trivy, Semgrep, Checkov.
- Strategy: read implementation issues, pull requests and release notes—not just README files.

7. Safe reproduction and responsible disclosure

Reproduce only with synthetic data, local systems you own, or explicitly authorized environments. Keep a written scope, abort condition and cleanup plan. Hash inputs, record versions and avoid executing unknown binaries on a host with personal data. For responsible disclosure: read the vendor/program policy first, keep evidence proportional, protect user data, avoid service disruption, agree on a timeline where possible, and publish only after a fix or coordinated disclosure. CVE/CVSS are communication tools, not proof of importance.

8. Sustainable excellence

Deep expertise requires health and feedback. Schedule recovery weeks, keep a maximum backlog rule, and reduce intensity during exams without breaking all study habits. Track leading indicators (hours on hard problems, experiments run, pages read, reviews received) and lagging indicators (working artifact, accepted contribution, reproducible finding). Ask one stronger person each quarter: "What is the weakest part of my method?" Record the answer and act on it.

9. Reference list (primary links used throughout)

All resources are listed with full metadata in Volume 5. Core references: Harvard CS50 https://cs50.harvard.edu/, CS50 Cybersecurity https://cs50.harvard.edu/cybersecurity/, OSTEP https://pages.cs.wisc.edu/~remzi/OSTEP/, Jeremy's IT Lab https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ, Professor Messer https://www.professormesser.com/, Linux Journey https://linuxjourney.com/, OverTheWire https://overthewire.org/wargames/bandit/, Python docs https://docs.python.org/3/, Pro Git https://git-scm.com/book/en/v2, PortSwigger Academy https://portswigger.net/web-security, OWASP https://owasp.org/, pwn.college https://pwn.college/, picoCTF/CyLab Academy https://www.cylabacademy.org/, MITRE ATT&CK https://attack.mitre.org/, NIST CSF https://www.nist.gov/cyberframework, NIST NICE https://www.nist.gov/itl/applied-cybersecurity/nice, Microsoft Learn https://learn.microsoft.com/, AWS Skill Builder https://skillbuilder.aws/, Docker docs https://docs.docker.com/, Kubernetes docs https://kubernetes.io/docs/, GitHub security docs https://docs.github.com/en/code-security, SigmaHQ https://sigmahq.io/, USENIX Security https://www.usenix.org/conferences/byname/164, Project Zero https://googleprojectzero.blogspot.com/, CISA KEV https://www.cisa.gov/known-exploited-vulnerabilities-catalog.

10. Final quality and ethical statement

This curriculum was generated on 2026-10-27. It uses free/official resources wherever possible, flags paid resources, distinguishes older durable material from current official documentation, avoids invented timestamps/prices, and confines practical work to synthetic, owned or explicitly authorized environments. It makes no promise of top 1% status, employment, salary, immigration or certification success. Those outcomes depend on sustained deliberate practice, evidence, feedback, health, opportunity and ethical conduct.

Specialization comparison (Month 8-10 decision aid)

Score each row 1-5 for demonstrated interest, project results, programming comfort, local opportunity and available feedback. No path is "best" without your evidence.

Decision framework: evidence weight 35%, genuine interest 20%, programming/technical fit 20%, available mentorship/opportunity 15%, opportunity cost 10%. Require at least two strong project results before selecting. Review after 90 days; switching early because a topic is hard is different from switching because evidence shows poor fit.

Recommendation for this profile: primary = Detection Engineering & Security Automation; secondary = Cloud/Container Security. If Month 8-10 sprints contradict this, choose the path with stronger independent problem-solving evidence and better feedback access.

Printable trackers and final checklist (compact edition)

Printable trackers

Daily checklist (copy one per day)

#	Item	Done	Evidence / note
1	Read today's card and objective	[ ]	
2	Primary resource section completed	[ ]	
3	Three facts / two questions / one command	[ ]	
4	Practical task completed	[ ]	
5	One error observed or injected	[ ]	
6	Retrieval answer written without notes	[ ]	
7	Evidence file saved and sanitized	[ ]	
8	Journal / Git commit	[ ]	
9	Track used: Core / Standard / Intensive	[ ]	
10	Weak cards scheduled 1/3/7/30/90 days	[ ]	

Weekly tracker

Monthly scorecard (100 points)

Error log template

Command notebook template

Command / API	Platform	Purpose	Safe example	Dangerous assumption	Verification	Source URL

Project log template

Date	Decision	Alternatives	Evidence	Risk	Result	Follow-up

Final completion checklist

- [ ] 365 daily cards have evidence and retrieval notes.
- [ ] 12 monthly scorecards complete; M3/M6/M9/M12 gates passed.
- [ ] 15 projects have a problem, architecture, tests, evidence, report, README, boundaries and limitations.
- [ ] Final capstone has a clean-install test, threat model, detections, incident report, assessment, remediation and demo.
- [ ] All public artifacts are sanitized; no secrets, private logs, personal data or unauthorized exploit material.
- [ ] Resource links rechecked on 2026-10-27; backup links documented.
- [ ] Certification decision uses current official pages and a written reason.
- [ ] CV, LinkedIn and interview stories map to portfolio evidence.
- [ ] Year 2 plan has a primary specialization, secondary skill, peer group, 90-day sprint and review date.
- [ ] Lab is clean, isolated, backed up and has no active billable cloud resources.

Landscape reference table

Path	Foundational knowledge	Daily work	Technical depth / programming	Common tools	Strong free resources	Portfolio expectation	Possible entry roles	Long-term opportunities	Typical prerequisites	12-month limitation
Blue Team / SOC Analysis	Networking, OS, logs, ATT&CK	Alert triage, investigation, escalation, reporting	Medium; scripting helpful	SIEM, KQL, EDR, ticketing	Microsoft Learn, SigmaHQ, CyberDefenders, BTLO	Alert investigation, incident report, detection tests	SOC analyst, security analyst	Detection lead, IR, threat hunting	M1-M8 basics	No real shift experience or production scale
Detection Engineering	Logs, OS internals, ATT&CK, SQL/KQL	Hypothesis, rule design, testing, tuning, deployment	High; Python/YAML/SQL	Sigma, KQL, Elastic rules, Git, CI	SigmaHQ, KQL docs, Elastic rules, ATT&CK	Rule repo with labelled true/false positives and metrics	Junior detection engineer, SOC analyst	Detection platform/architecture, research	Strong logs and programming	Lab detections may not generalize to enterprise data
DFIR	Filesystems, Windows/Linux artifacts, evidence integrity	Acquisition, timeline, artifact analysis, reporting	High; scripting and OS internals	Autopsy, KAPE-like tools, Volatility, Wireshark	Autopsy, SANS webcasts, NIST 800-61r3, EVTX samples	Disk/network/memory case with hash manifest	Junior DFIR analyst, IR analyst	Senior IR, forensic research, expert witness	Evidence handling and OS internals	No legal/casework or real incident pressure
Web / API Security	HTTP, auth, databases, secure coding	Code review, testing, remediation, threat modelling	High; one backend language	Burp/ZAP, Semgrep, OWASP ASVS, CI	PortSwigger, OWASP, MDN, GitHub security	Secure app plus authorized assessment/remediation	AppSec analyst, secure software engineer	AppSec architecture, research, product security	M1-M6 plus testing	Advanced vulnerability classes need years
Authorized Penetration Testing	Networking, OS, web, scripting, ethics	Scoped discovery, validation, reporting, retest	Medium-high; scripting and OS	Nmap, Burp, AD tools, reporting templates	NIST 800-115, HTB free, pwn.college, local AD lab	Professional scoped report and remediation retest	Junior pentester, security consultant	Senior pentest, red team, security research	Strong M1-M7 + report writing	Certification or CTF badges do not equal client judgement
Cloud Security	IAM, networking, logging, containers, shared responsibility	Architecture review, policy, detection, incident response	High; IaC and automation	AWS/Azure docs, Trivy, Checkov, K8s	AWS Security Pillar, Azure security, K8s docs	Synthetic cloud review + IaC/CI hardening	Cloud security analyst, DevSecOps junior	Cloud security architect, platform engineering	M1-M9 and Linux/Docker	No production multi-account experience in Year 1
Security Engineering	Secure design, coding, identity, systems, automation	Build controls, review architecture, write tools	Very high; software engineering	Git, CI, SAST/SCA, cloud, observability	OWASP ASVS, GitHub docs, NIST, Security Engineering	Maintained tool/library with tests and design doc	Security engineer, platform security engineer	Staff/principal security architect	Strong CS and programming	Breadth and production trade-offs take years
GRC	Risk, policy, controls, audits, privacy	Assessments, evidence, exceptions, reports	Low-medium technically; analysis-heavy	GRC platforms, spreadsheets, NIST/ISO	NIST CSF/SP, CIS Controls, GDPR text	Risk register, control baseline, audit-style report	GRC analyst, risk analyst, compliance associate	Security governance lead, privacy, audit	Business/legal context and strong writing	Framework knowledge without operations is shallow
Vulnerability Research	C/assembly, OS, memory, protocols, fuzzing	Root-cause analysis, PoC in authorized lab, disclosure	Very high; C/Python/assembly	Ghidra, GDB, pwntools, fuzzers, VMs	pwn.college, OST2, MIT 6.858, Project Zero	Reproduced bug class, detailed root-cause write-up	Rare junior roles; often via internships/OSS	Researcher, product security, exploit engineering	Several years of low-level foundations	Intro exposure is not research capability
Malware Analysis / RE	x86/x64, Windows internals, networking, scripting	Triage, static/dynamic analysis, indicator extraction	Very high; C/Python/assembly	Ghidra, x64dbg, Wireshark, sandboxes	OST2, Malware Unicorn RE101, pwn.college	Safe synthetic sample analysis with isolation evidence	Junior malware analyst, SOC/IR specialist	Threat research, reverse engineering	C/assembly and strict lab safety	Real malware handling requires supervised environment

Landscape reference table

Day	Sequence	Hours	Evidence link	Concept /5	Practical /5	Troubleshooting /5	Safety pass	Error log ID	Next action
1							[ ]		
2							[ ]		
3							[ ]		
4							[ ]		
5							[ ]		
6							[ ]		
7							[ ]		

Landscape reference table

Category	Max	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Conceptual understanding	20												
Practical execution	25												
Independent problem-solving	15												
Troubleshooting	15												
Documentation	15												
Safety and ethics	10												
Total	**100**												
Gate result				**M3**			**M6**			**M9**			**M12**

Landscape reference table

ID	Date	Symptom	Hypothesis	Test	Result	Root cause	Fix	New test	Retest